Fluid nerve atlas

106 systems: 35 atom-based (heavy-atom sites), 36 backbone-based (Cα sites), 35 paired heavy-atom representations

Joseph Campagna, PSM — pronerv fluid-nerve pipeline — October 22, 2026

How to read the plates

Every plate shows one system in six panels. **(A)** the biophysical object: the recorded conformation nearest the occupancy peak of the chart, drawn from its sites (heavy atoms with element colours, or the Cα trace coloured from the N- to the C-terminus), with the deformation field drawn on the molecule as arrows: the mean rigid-aligned displacement of every site over the reference lag, conditional on the chart bin of that conformation. **(B)** the chart (backbone dihedrals on a torus, or the two leading principal components of the aligned sites) with the occupancy, the 18 cover balls and the drawn nerve; one composable pair is annotated in the language of the legend below. **(C)** the deformation field on the chart: mobility h (colour) and drift v (arrows). **(D)** the fluid filtration: Betti numbers of the nerve of the cover thickened by $\pm\tau$ frames along each run. **(E)** the flooding filtration of the occupancy landscape (sublevel sets of $-\log p$), the geometric analogue of metadynamics. **(F)** the categorical reading: the sizes of the category drawn in (B), the Betti numbers of the witnessed complex, the ambient-time invariants and the field summaries. All constructions are descriptive: no equilibrium free energy, rate constant or configuration-space nerve theorem is claimed.

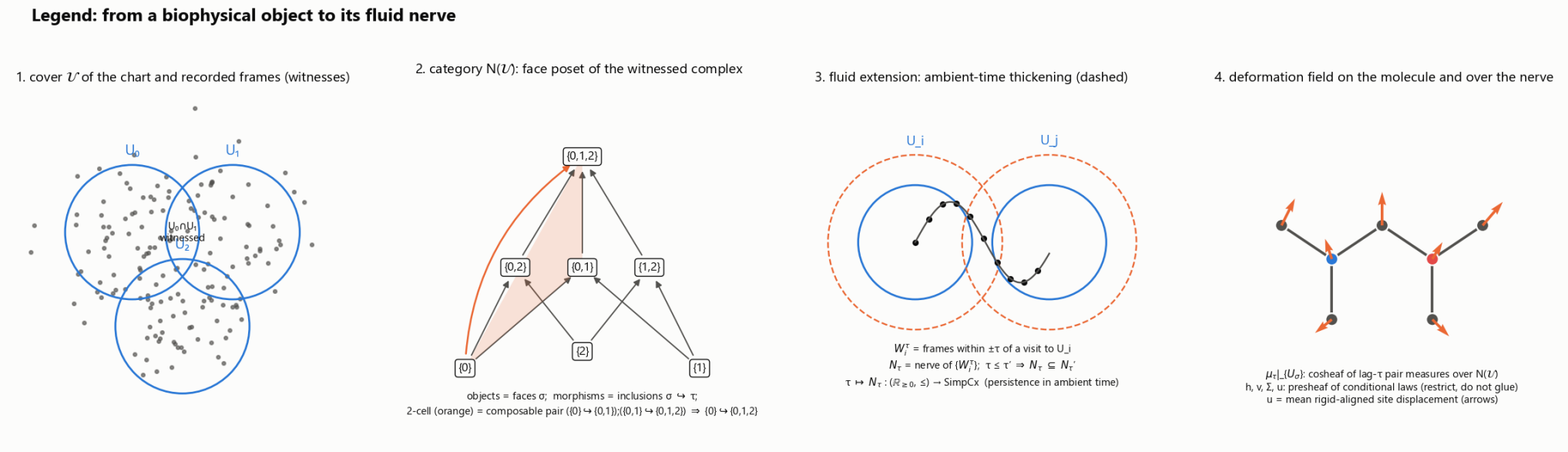

Figure 1: Legend in categorical terms. (1) A cover $\mathcal{U} = \{U_i\}$ of the chart; the recorded frames are the witnesses. (2) The witnessed complex $K(\mathcal{U}; S)$ and its face poset $N(\mathcal{U})$ seen as a category: *objects* are faces σ (the witnessed intersections $U_\sigma = \bigcap_{i\in\sigma} U_i$), *morphisms* are inclusions $\sigma \hookrightarrow \tau$ with composition given by transitivity and implicit identities, *2-cells* are composable pairs $(\sigma \hookrightarrow \tau); (\tau \hookrightarrow \rho) \Rightarrow \sigma \hookrightarrow \rho$ (Street's orientals; exported to homotopy.io together with the 3-cells of length-three chains and of quadruple overlaps). (3) The *fluid extension*: thickening every cover set by $\pm\tau$ frames along the recorded runs gives W_i^τ and the nerve N_τ; $\tau \mapsto N_\tau$ is a functor from the poset $(\mathbb{R}_{\geq 0}, \leq)$ to simplicial complexes, whose homology is a persistence module in ambient time. (4) The *deformation field*: the lag-τ pair measure μ_τ restricted to the objects U_σ is a cosheaf over $N(\mathcal{U})$; the derived statistics (mobility h, drift v, spread Σ, site displacement u) form a presheaf of conditional laws, drawn on the molecule in panel (A) and on the chart in panel (C). The theory PDF (`fluid-nerve-theory.pdf`) proves that nerves of formally open covers are rigid under infinitesimal transformations, so the ambient-time behaviour seen in (D) is a finite-time, finite-witness phenomenon.

Catalog

Table 1: All systems. Nerve counts are objects/morphisms/2-cells of the drawn face-poset nerve; β are GF(2) Betti numbers of the witnessed complex; h the median supported mobility at the reference lag; τ_c the ambient-time connection; r_c the cover-radius factor at which $\beta_0 = 1$.

id	system	group	repr.	sites	frames	runs	nerve	$\beta_{0,1,2}$	τ_c	h (nm)
dipeptide-ala	Ace-ALA-NMe dipeptide (OpenMM)	atom	heavy	10	30,000	3	196/792/654	1,0,0	0 fs	0.0416
dipeptide-arg	Ace-ARG-NMe dipeptide (OpenMM)	atom	heavy	16	30,000	3	133/486/384	2,0,0	1 ps	0.0717
dipeptide-asn	Ace-ASN-NMe dipeptide (OpenMM)	atom	heavy	13	30,000	3	289/1278/1104	1,0,0	0 fs	0.0568
dipeptide-asp	Ace-ASP-NMe dipeptide (OpenMM)	atom	heavy	13	30,000	3	213/878/732	1,0,0	0 fs	0.0538
dipeptide-cys	Ace-CYS-NMe dipeptide (OpenMM)	atom	heavy	11	30,000	3	255/1106/948	1,0,0	0 fs	0.0408
dipeptide-gln	Ace-GLN-NMe dipeptide (OpenMM)	atom	heavy	14	30,000	3	236/1000/846	1,0,0	0 fs	0.0631
dipeptide-glu	Ace-GLU-NMe dipeptide (OpenMM)	atom	heavy	14	30,000	3	235/994/840	1,0,0	0 fs	0.0653
dipeptide-gly	Ace-GLY-NMe dipeptide (OpenMM)	atom	heavy	9	30,000	3	87/246/162	1,4,0	0 fs	0.0461
dipeptide-his	Ace-HIS-NMe dipeptide (OpenMM)	atom	heavy	15	30,000	3	230/972/822	1,0,0	0 fs	0.055
dipeptide-ile	Ace-ILE-NMe dipeptide (OpenMM)	atom	heavy	13	30,000	3	262/1148/990	1,0,0	0 fs	0.0361
dipeptide-leu	Ace-LEU-NMe dipeptide (OpenMM)	atom	heavy	13	30,000	3	270/1180/1014	1,0,0	0 fs	0.0457
dipeptide-lys	Ace-LYS-NMe dipeptide (OpenMM)	atom	heavy	14	30,000	3	179/722/600	2,0,0	1 ps	0.0568
dipeptide-met	Ace-MET-NMe dipeptide (OpenMM)	atom	heavy	13	30,000	3	232/972/816	1,0,0	0 fs	0.0573
dipeptide-phe	Ace-PHE-NMe dipeptide (OpenMM)	atom	heavy	16	30,000	3	261/1138/978	1,0,0	0 fs	0.0468
dipeptide-pro	Ace-PRO-NMe dipeptide (OpenMM)	atom	heavy	12	30,000	3	258/1152/1008	2,0,0	1 ps	0.0187
dipeptide-ser	Ace-SER-NMe dipeptide (OpenMM)	atom	heavy	11	30,000	3	185/734/600	1,0,0	0 fs	0.0408
dipeptide-thr	Ace-THR-NMe dipeptide (OpenMM)	atom	heavy	12	30,000	3	214/888/744	1,0,0	0 fs	0.035
dipeptide-trp	Ace-TRP-NMe dipeptide (OpenMM)	atom	heavy	19	30,000	3	249/1078/924	1,0,0	0 fs	0.0466
dipeptide-tyr	Ace-TYR-NMe dipeptide (OpenMM)	atom	heavy	17	30,000	3	272/1196/1032	1,0,0	0 fs	0.0479
dipeptide-val	Ace-VAL-NMe dipeptide (OpenMM)	atom	heavy	12	30,000	3	223/930/780	1,0,0	0 fs	0.0336
sgdml-md17-aspirin	aspirin (MD17 ab initio MD)	atom	heavy	13	211,762	1	158/592/468	1,0,0	0 fs	0.0198
sgdml-md17-azobenzene	azobenzene (MD17 ab initio MD)	atom	heavy	14	99,999	1	63/146/84	1,1,0	0 fs	0.0149
sgdml-md17-ethanol	ethanol (MD17 ab initio MD)	atom	heavy	3	555,092	1	24/12/0	12,0,0	1.5 fs	0.00824
sgdml-md17-malonaldehyde	malonaldehyde (MD17 ab initio MD)	atom	heavy	5	993,237	1	79/210/132	1,0,0	0 fs	0.0189
sgdml-md17-naphthalene	naphthalene (MD17 ab initio MD)	atom	heavy	10	326,250	1	41/58/18	2,1,0	0.5 fs	0.00821
sgdml-md17-paracetamol	paracetamol (MD17 ab initio MD)	atom	heavy	11	106,490	1	72/188/120	1,1,0	0 fs	0.0148
sgdml-md17-salicylic	salicylic (MD17 ab initio MD)	atom	heavy	10	320,231	1	25/14/0	11,0,0	11 fs	0.0097
sgdml-md17-toluene	toluene (MD17 ab initio MD)	atom	heavy	7	442,790	1	28/20/0	8,0,0	1.5 fs	0.00765

id	system	group	repr.	sites	frames	runs	nerve	$\beta_{0,1,2}$	τ_c	h (nm)
sgdml-md17-uracil	uracil (MD17 ab initio MD)	atom	heavy	8	133,770	1	31/34/12	9,0,0	8.5 fs	0.00852
sgdml-md22-ac-ala3-nhme	Ac-Ala3-NHMe (MD22 ab initio MD)	atom	heavy	20	85,109	1	64/140/72	1,6,0	0 fs	0.0276
sgdml-md22-at-at	AT-AT (MD22 ab initio MD)	atom	heavy	38	20,001	1	118/412/318	1,0,0	0 fs	–
sgdml-md22-at-at-cg-cg	AT-AT-CG-CG (MD22 ab initio MD)	atom	heavy	76	10,153	1	135/490/384	1,0,0	0 fs	–
sgdml-md22-dha	DHA (MD22 ab initio MD)	atom	heavy	24	69,753	1	88/248/162	1,3,0	0 fs	0.0339
sgdml-md22-stachyose	stachyose (MD22 ab initio MD)	atom	heavy	45	27,272	1	144/536/426	1,0,0	0 fs	–
ala2	Alanine dipeptide (mdshare)	atom	heavy	10	750,000	3	78/212/138	3,3,0	1 ps	0.0383
cln025	Chignolin CLN025 (128-run subset)	backbone	Cα	10	63,999	128	84/228/144	1,2,0	0 fr	0.197
mdcath-1a92a00-ca	CATH domain 1a92A00 (mdCATH), C-alpha	backbone	Cα	50	10,980	25	89/254/168	1,0,0	0 fs	0.697
mdcath-1b33n01-ca	CATH domain 1b33N01 (mdCATH), C-alpha	backbone	Cα	55	11,090	25	80/216/138	2,3,0	1 ns	0.316
mdcath-1bdha01-ca	CATH domain 1bdhA01 (mdCATH), C-alpha	backbone	Cα	57	11,110	25	76/196/120	1,1,0	0 fs	0.635
mdcath-1e5wa04-ca	CATH domain 1e5wA04 (mdCATH), C-alpha	backbone	Cα	55	10,910	25	96/288/198	1,1,0	0 fs	0.848
mdcath-1ffsb00-ca	CATH domain 1ffsB00 (mdCATH), C-alpha	backbone	Cα	68	11,141	25	139/510/402	2,0,0	1 ns	0.419
mdcath-1gjsa00-ca	CATH domain 1gjsA00 (mdCATH), C-alpha	backbone	Cα	65	10,970	25	86/244/162	1,1,0	0 fs	0.604
mdcath-1ha8a00-ca	CATH domain 1ha8A00 (mdCATH), C-alpha	backbone	Cα	51	11,150	25	78/208/132	1,1,0	0 fs	0.21
mdcath-1hcra00-ca	CATH domain 1hcrA00 (mdCATH), C-alpha	backbone	Cα	52	11,160	25	65/146/78	1,4,0	0 fs	0.691
mdcath-1i5xa00-ca	CATH domain 1i5xA00 (mdCATH), C-alpha	backbone	Cα	61	11,080	25	106/336/240	1,0,0	0 fs	0.606
mdcath-1ib8a02-ca	CATH domain 1ib8A02 (mdCATH), C-alpha	backbone	Cα	67	11,090	25	86/244/162	1,2,0	0 fs	0.409
mdcath-1klil00-ca	CATH domain 1kliL00 (mdCATH), C-alpha	backbone	Cα	61	11,100	25	80/216/138	1,1,0	0 fs	0.552
mdcath-1l6ha00-ca	CATH domain 1l6hA00 (mdCATH), C-alpha	backbone	Cα	69	11,090	25	77/198/120	1,2,0	0 fs	0.344
mdcath-1r8ob01-ca	CATH domain 1r8oB01 (mdCATH), C-alpha	backbone	Cα	66	11,100	25	87/246/162	1,1,0	0 fs	0.908
mdcath-1s6la01-ca	CATH domain 1s6lA01 (mdCATH), C-alpha	backbone	Cα	52	11,170	25	100/308/216	1,0,0	0 fs	0.685
mdcath-1v1hf01-ca	CATH domain 1v1hF01 (mdCATH), C-alpha	backbone	Cα	73	10,806	25	99/306/216	1,0,0	0 fs	0.788
mdcath-1wfwa00-ca	CATH domain 1wfwA00 (mdCATH), C-alpha	backbone	Cα	74	11,190	25	72/176/102	1,4,0	0 fs	0.347
mdcath-1wpbg01-ca	CATH domain 1wpbG01 (mdCATH), C-alpha	backbone	Cα	54	10,980	25	102/320/228	1,0,0	0 fs	0.549

id	system	group	repr.	sites	frames	runs	nerve	$\beta_{0,1,2}$	τ_c	h (nm)
mdcath-1x4pa00-ca	CATH domain 1x4pA00 (mdCATH), C-alpha	backbone	Cα	66	11,130	25	76/196/120	1,2,0	0 fs	0.799
mdcath-1zbqa02-ca	CATH domain 1zbqA02 (mdCATH), C-alpha	backbone	Cα	58	11,190	25	81/218/138	1,3,0	0 fs	0.799
mdcath-2ckka02-ca	CATH domain 2ckkA02 (mdCATH), C-alpha	backbone	Cα	57	12,500	25	118/392/288	1,1,0	0 fs	0.239
mdcath-2d46a00-ca	CATH domain 2d46A00 (mdCATH), C-alpha	backbone	Cα	61	11,390	25	90/260/174	1,0,0	0 fs	1.02
mdcath-2dmpa01-ca	CATH domain 2dmpA01 (mdCATH), C-alpha	backbone	Cα	55	12,500	25	76/196/120	1,2,0	0 fs	0.886
mdcath-2e3tb03-ca	CATH domain 2e3tB03 (mdCATH), C-alpha	backbone	Cα	61	11,390	25	116/384/282	1,0,0	0 fs	0.489
mdcath-2j8ga03-ca	CATH domain 2j8gA03 (mdCATH), C-alpha	backbone	Cα	58	12,130	25	91/266/180	1,0,0	0 fs	0.732
mdcath-2jrra01-ca	CATH domain 2jrrA01 (mdCATH), C-alpha	backbone	Cα	53	12,500	25	93/282/198	1,0,0	0 fs	0.292
mdcath-2kssa01-ca	CATH domain 2kssA01 (mdCATH), C-alpha	backbone	Cα	63	12,500	25	50/92/42	2,2,0	1 ns	0.764
mdcath-2lrda00-ca	CATH domain 2lrdA00 (mdCATH), C-alpha	backbone	Cα	61	12,500	25	74/188/114	1,2,0	0 fs	0.21
mdcath-2xepa03-ca	CATH domain 2xepA03 (mdCATH), C-alpha	backbone	Cα	53	11,390	25	77/198/120	1,4,0	0 fs	0.6
mdcath-2yila02-ca	CATH domain 2yilA02 (mdCATH), C-alpha	backbone	Cα	51	12,500	25	96/300/216	1,0,0	0 fs	0.247
mdcath-3g5ca02-ca	CATH domain 3g5cA02 (mdCATH), C-alpha	backbone	Cα	55	12,500	25	85/238/156	1,0,0	0 fs	0.476
mdcath-3pmqa02-ca	CATH domain 3pmqA02 (mdCATH), C-alpha	backbone	Cα	50	12,130	25	68/160/90	1,4,0	0 fs	0.872
mdcath-4a0ta01-ca	CATH domain 4a0tA01 (mdCATH), C-alpha	backbone	Cα	61	11,390	25	75/190/114	1,3,0	0 fs	0.688
mdcath-4dcza00-ca	CATH domain 4dczA00 (mdCATH), C-alpha	backbone	Cα	55	11,020	25	81/218/138	1,1,0	0 fs	0.777
mdcath-5b1aj00-ca	CATH domain 5b1aJ00 (mdCATH), C-alpha	backbone	Cα	58	11,950	25	76/196/120	1,3,0	0 fs	0.58
mdcath-5daea00-ca	CATH domain 5daeA00 (mdCATH), C-alpha	backbone	Cα	57	11,390	25	72/172/96	1,5,0	0 fs	0.281
mdcath-1a92a00-heavy	CATH domain 1a92A00 (mdCATH), heavy atoms	paired	heavy	427	10,980	25	102/316/222	1,1,0	0 fs	0.669
mdcath-1b33n01-heavy	CATH domain 1b33N01 (mdCATH), heavy atoms	paired	heavy	471	11,090	25	68/156/84	1,6,0	0 fs	0.47
mdcath-1bdha01-heavy	CATH domain 1bdhA01 (mdCATH), heavy atoms	paired	heavy	446	11,110	25	77/202/126	1,0,0	0 fs	0.623
mdcath-1e5wa04-heavy	CATH domain 1e5wA04 (mdCATH), heavy atoms	paired	heavy	483	10,910	25	96/288/198	1,0,0	0 fs	0.913

id	system	group	repr.	sites	frames	runs	nerve	$\beta_{0,1,2}$	τ_c	h (nm)
mdcath-1ffsb00-heavy	CATH domain 1ffsB00 (mdCATH), heavy atoms	paired	heavy	518	11,141	25	128/456/354	2,0,0	1 ns	0.698
mdcath-1gjsa00-heavy	CATH domain 1gjsA00 (mdCATH), heavy atoms	paired	heavy	504	10,970	25	99/302/210	1,1,0	0 fs	0.648
mdcath-1ha8a00-heavy	CATH domain 1ha8A00 (mdCATH), heavy atoms	paired	heavy	352	11,150	25	78/204/126	1,3,0	0 fs	0.242
mdcath-1hcra00-heavy	CATH domain 1hcrA00 (mdCATH), heavy atoms	paired	heavy	430	11,160	25	75/190/114	1,4,0	0 fs	0.72
mdcath-1i5xa00-heavy	CATH domain 1i5xA00 (mdCATH), heavy atoms	paired	heavy	499	11,080	25	94/280/192	1,2,0	0 fs	0.665
mdcath-1ib8a02-heavy	CATH domain 1ib8A02 (mdCATH), heavy atoms	paired	heavy	534	11,090	25	96/296/210	1,0,0	0 fs	0.215
mdcath-1klil00-heavy	CATH domain 1kliL00 (mdCATH), heavy atoms	paired	heavy	472	11,100	25	77/202/126	1,3,0	0 fs	0.633
mdcath-1l6ha00-heavy	CATH domain 1l6hA00 (mdCATH), heavy atoms	paired	heavy	489	11,090	25	94/280/192	1,1,0	0 fs	0.401
mdcath-1r8ob01-heavy	CATH domain 1r8oB01 (mdCATH), heavy atoms	paired	heavy	523	11,100	25	85/234/150	1,1,0	0 fs	0.899
mdcath-1s6la01-heavy	CATH domain 1s6lA01 (mdCATH), heavy atoms	paired	heavy	401	11,170	25	118/396/294	1,0,0	0 fs	0.727
mdcath-1v1hf01-heavy	CATH domain 1v1hF01 (mdCATH), heavy atoms	paired	heavy	532	10,806	25	90/260/174	1,1,0	0 fs	0.807
mdcath-1wfwa00-heavy	CATH domain 1wfwA00 (mdCATH), heavy atoms	paired	heavy	538	11,190	25	80/216/138	1,1,0	0 fs	0.376
mdcath-1wpbg01-heavy	CATH domain 1wpbG01 (mdCATH), heavy atoms	paired	heavy	456	10,980	25	102/320/228	1,0,0	0 fs	0.638
mdcath-1x4pa00-heavy	CATH domain 1x4pA00 (mdCATH), heavy atoms	paired	heavy	517	11,130	25	76/192/114	1,5,0	0 fs	0.761
mdcath-1zbqa02-heavy	CATH domain 1zbqA02 (mdCATH), heavy atoms	paired	heavy	466	11,190	25	65/146/78	1,4,0	0 fs	0.894
mdcath-2ckka02-heavy	CATH domain 2ckkA02 (mdCATH), heavy atoms	paired	heavy	438	12,500	25	154/560/432	1,0,0	0 fs	0.29
mdcath-2d46a00-heavy	CATH domain 2d46A00 (mdCATH), heavy atoms	paired	heavy	486	11,390	25	92/268/180	1,0,0	0 fs	1.03
mdcath-2dmpa01-heavy	CATH domain 2dmpA01 (mdCATH), heavy atoms	paired	heavy	470	12,500	25	95/282/192	1,0,0	0 fs	0.92
mdcath-2e3tb03-heavy	CATH domain 2e3tB03 (mdCATH), heavy atoms	paired	heavy	498	11,390	25	143/526/414	1,0,0	0 fs	0.501
mdcath-2j8ga03-heavy	CATH domain 2j8gA03 (mdCATH), heavy atoms	paired	heavy	478	12,130	25	70/172/102	2,4,0	1 ns	0.834
mdcath-2jrra01-heavy	CATH domain 2jrrA01 (mdCATH), heavy atoms	paired	heavy	416	12,500	25	123/418/312	1,1,0	0 fs	0.312
mdcath-2kssa01-heavy	CATH domain 2kssA01 (mdCATH), heavy atoms	paired	heavy	489	12,500	25	63/146/84	2,2,0	1 ns	0.871

id	system	group	repr.	sites	frames	runs	nerve	$\beta_{0,1,2}$	τ_c	h (nm)
mdcath-2lrda00-heavy	CATH domain 2lrdA00 (mdCATH), heavy atoms	paired	heavy	413	12,500	25	105/334/240	1,0,0	0 fs	0.242
mdcath-2xepa03-heavy	CATH domain 2xepA03 (mdCATH), heavy atoms	paired	heavy	424	11,390	25	84/232/150	1,3,0	0 fs	0.687
mdcath-2yila02-heavy	CATH domain 2yilA02 (mdCATH), heavy atoms	paired	heavy	390	12,500	25	73/190/120	1,0,0	0 fs	0.336
mdcath-3g5ca02-heavy	CATH domain 3g5cA02 (mdCATH), heavy atoms	paired	heavy	409	12,500	25	78/204/126	1,2,0	0 fs	0.514
mdcath-3pmqa02-heavy	CATH domain 3pmqA02 (mdCATH), heavy atoms	paired	heavy	385	12,130	25	72/176/102	1,4,0	0 fs	0.964
mdcath-4a0ta01-heavy	CATH domain 4a0tA01 (mdCATH), heavy atoms	paired	heavy	490	11,390	25	91/266/180	1,1,0	0 fs	0.712
mdcath-4dcza00-heavy	CATH domain 4dczA00 (mdCATH), heavy atoms	paired	heavy	466	11,020	25	84/236/156	1,0,0	0 fs	0.865
mdcath-5b1aj00-heavy	CATH domain 5b1aJ00 (mdCATH), heavy atoms	paired	heavy	465	11,950	25	86/244/162	1,1,0	0 fs	0.641
mdcath-5daea00-heavy	CATH domain 5daeA00 (mdCATH), heavy atoms	paired	heavy	438	11,390	25	80/216/138	1,2,0	0 fs	0.329

Overview figures

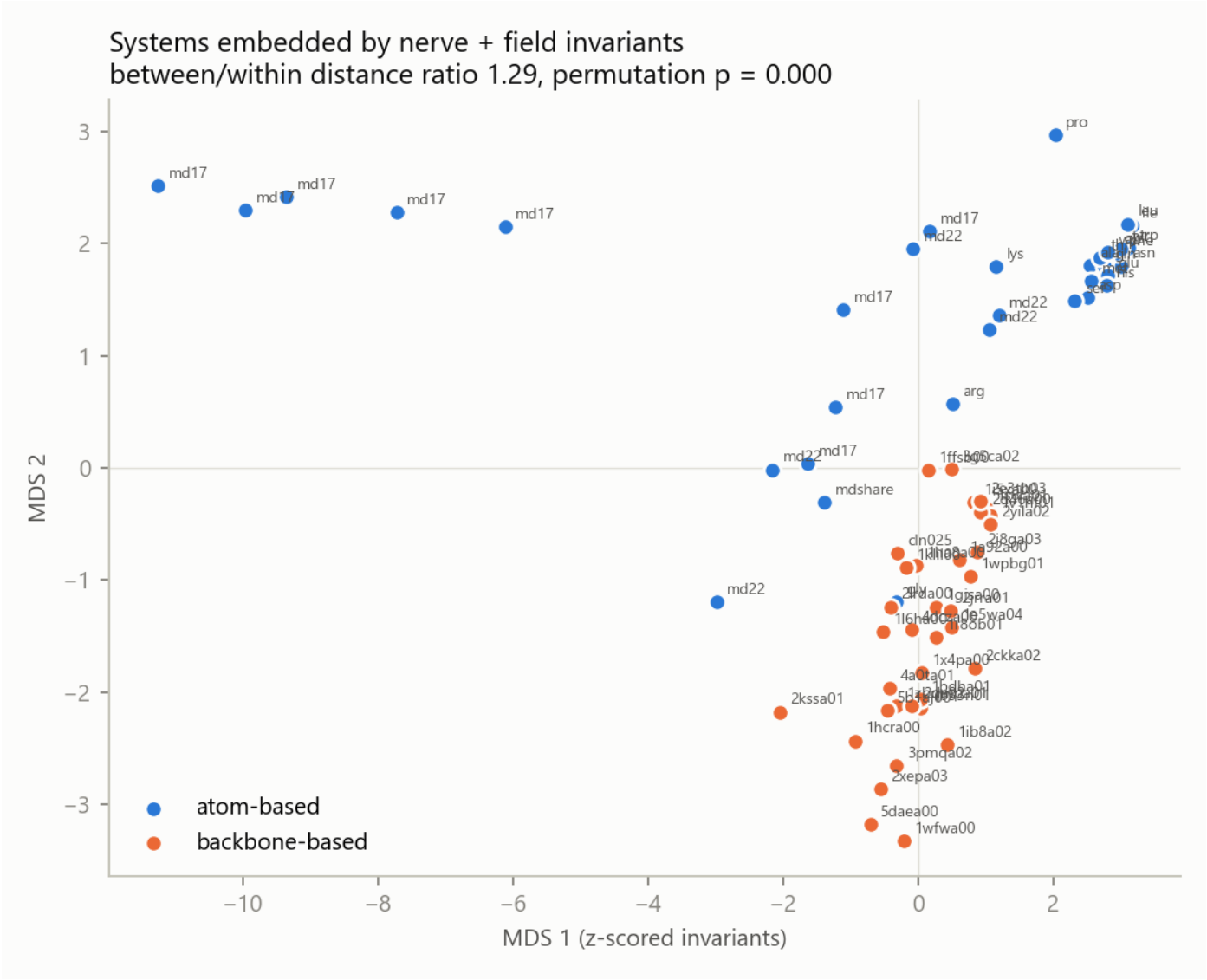

Figure 2: Systems embedded by nerve and field invariants (classical MDS of z-scored invariants).

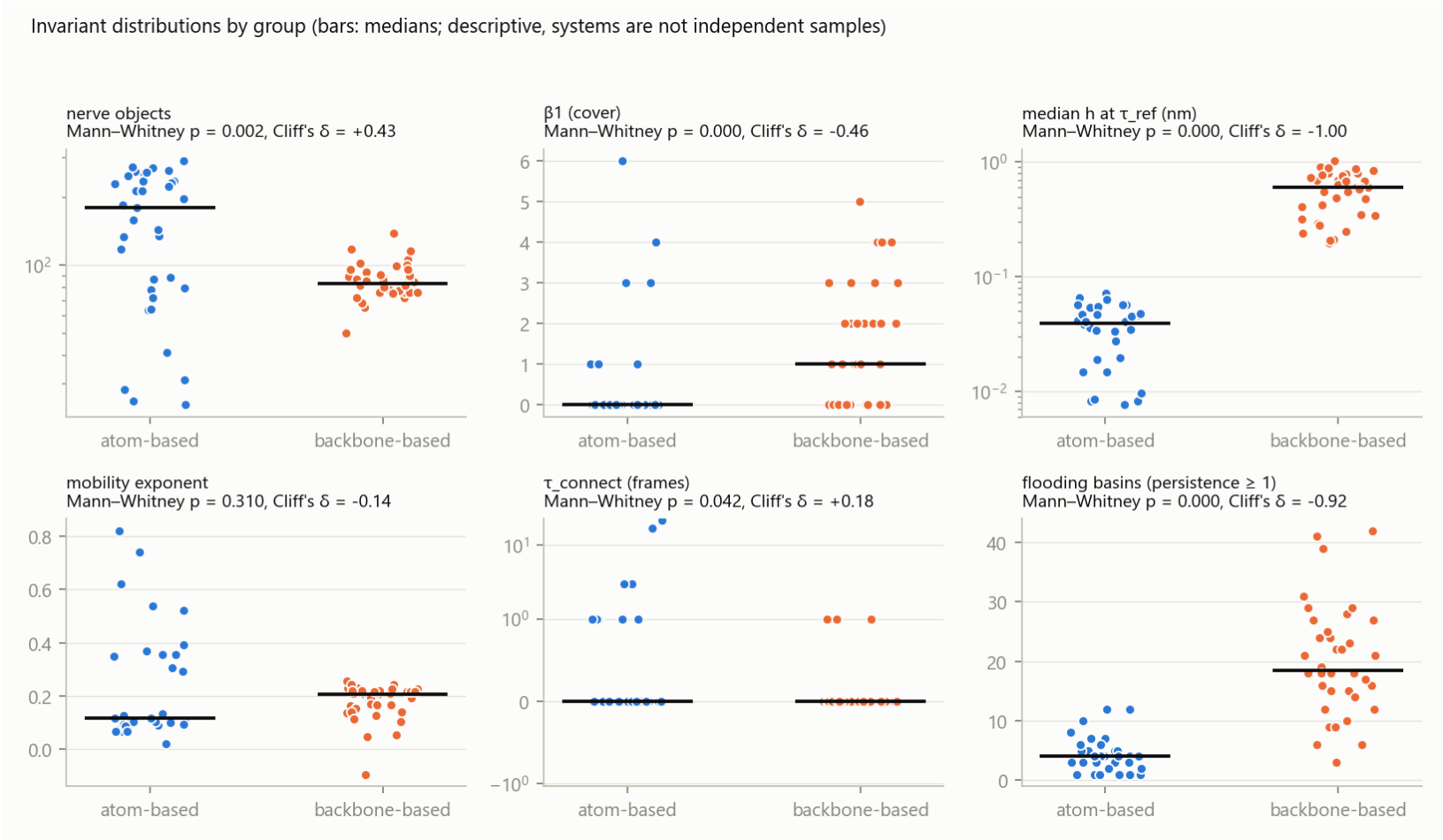

Figure 3: Invariant distributions by group with descriptive tests.

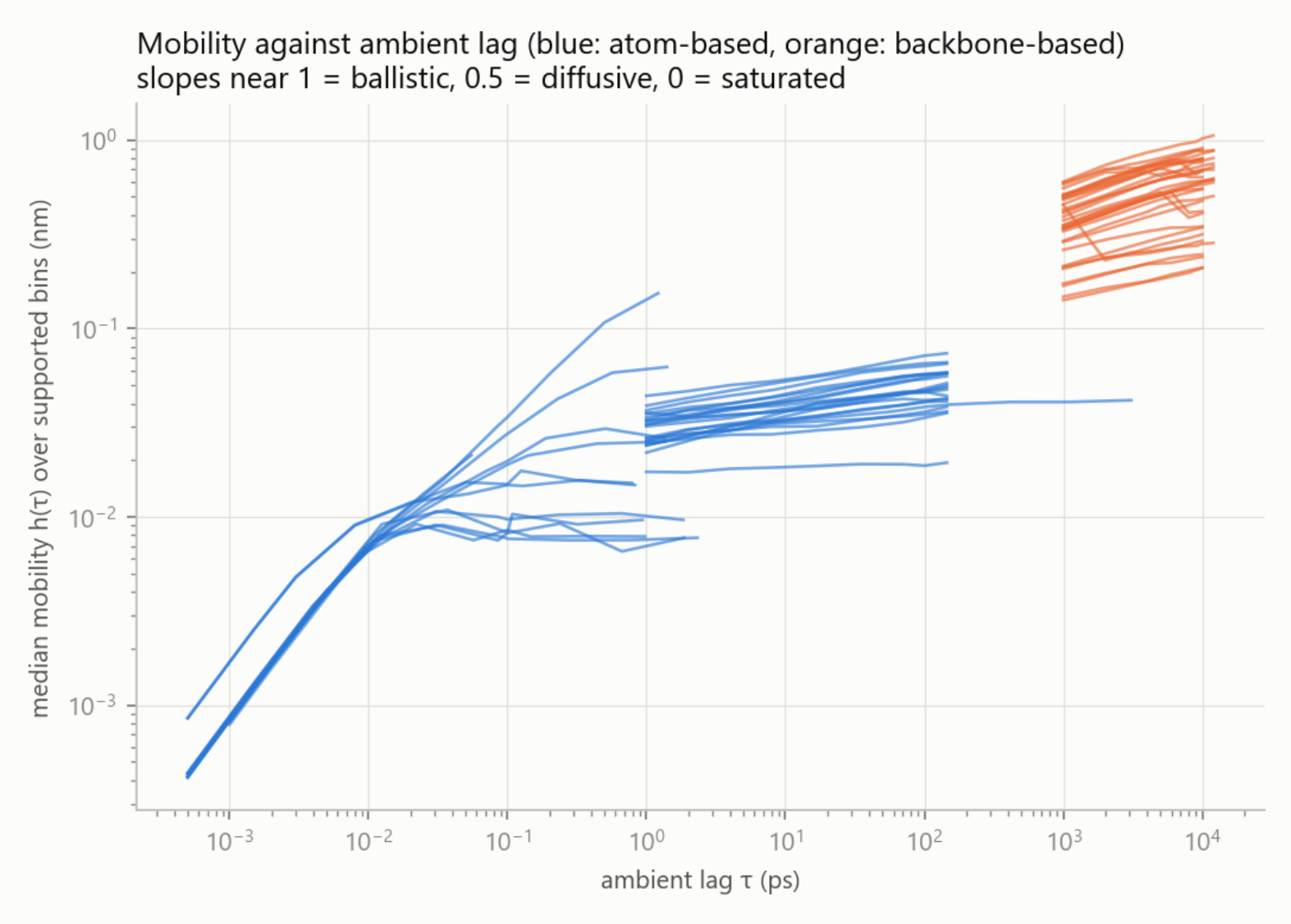

Figure 4: Median mobility against ambient lag; the slope is the mobility exponent.

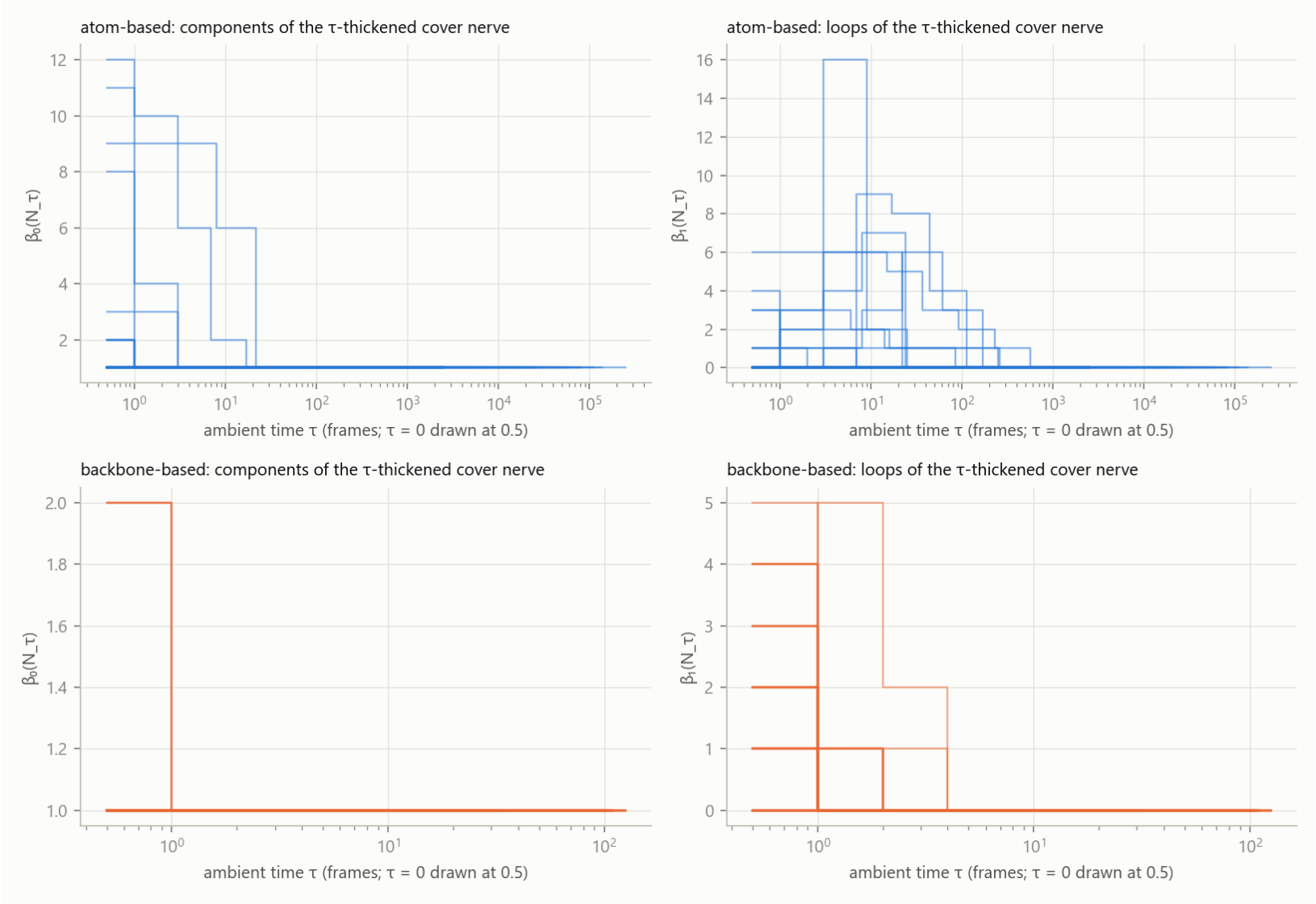

Figure 5: Fluid filtration: Betti numbers of the thickened cover nerve against ambient time.

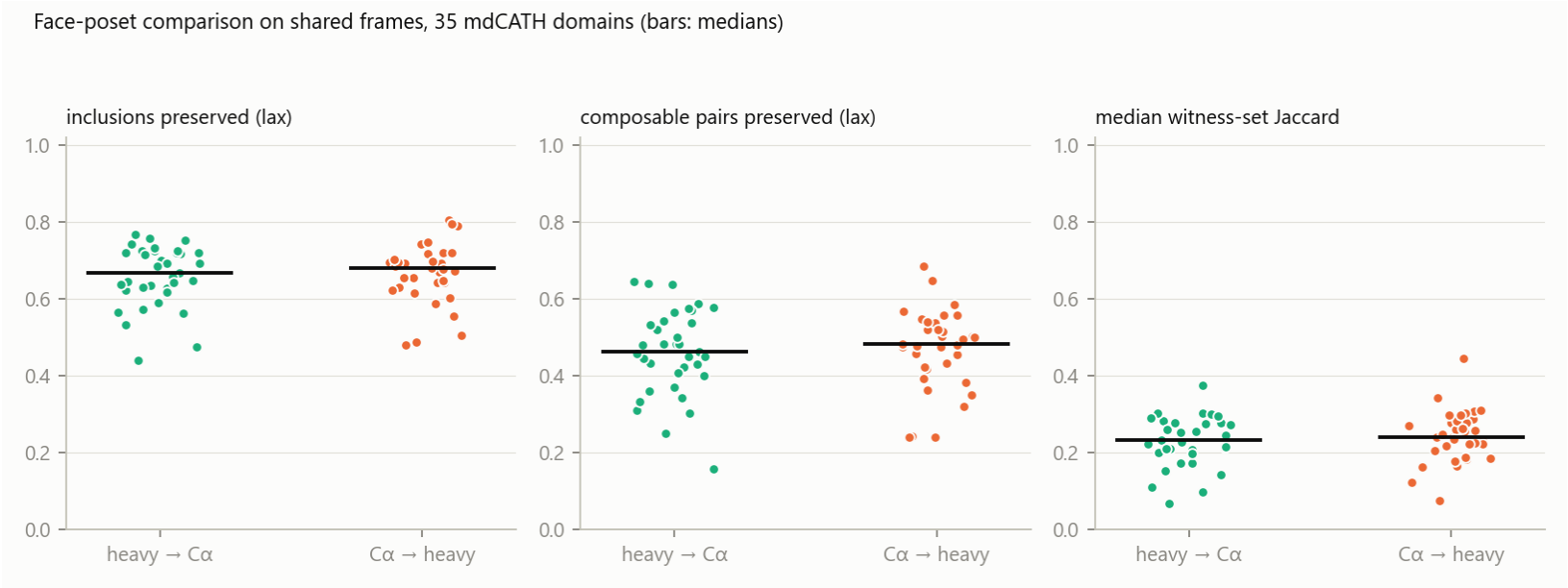

Figure 6: Comparison of face posets from heavy atoms and from Cα sites on identical frames (mdCATH).

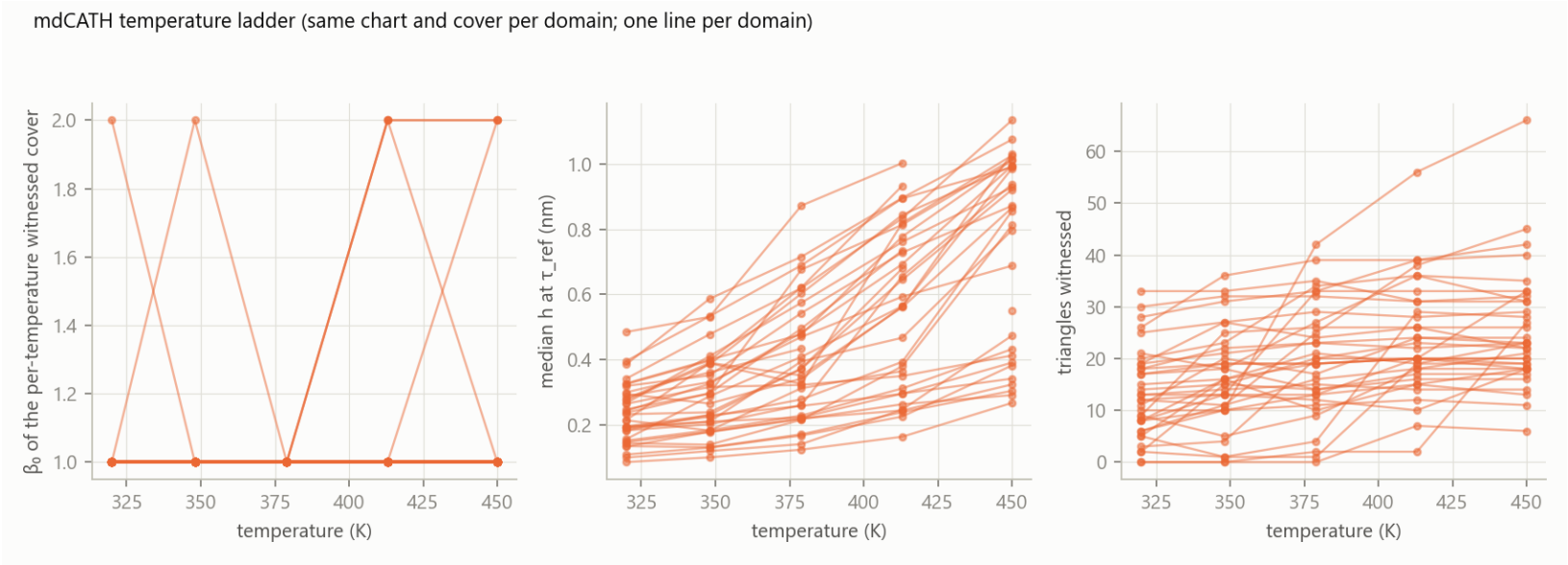

Figure 7: mdCATH temperature ladder.

Plates

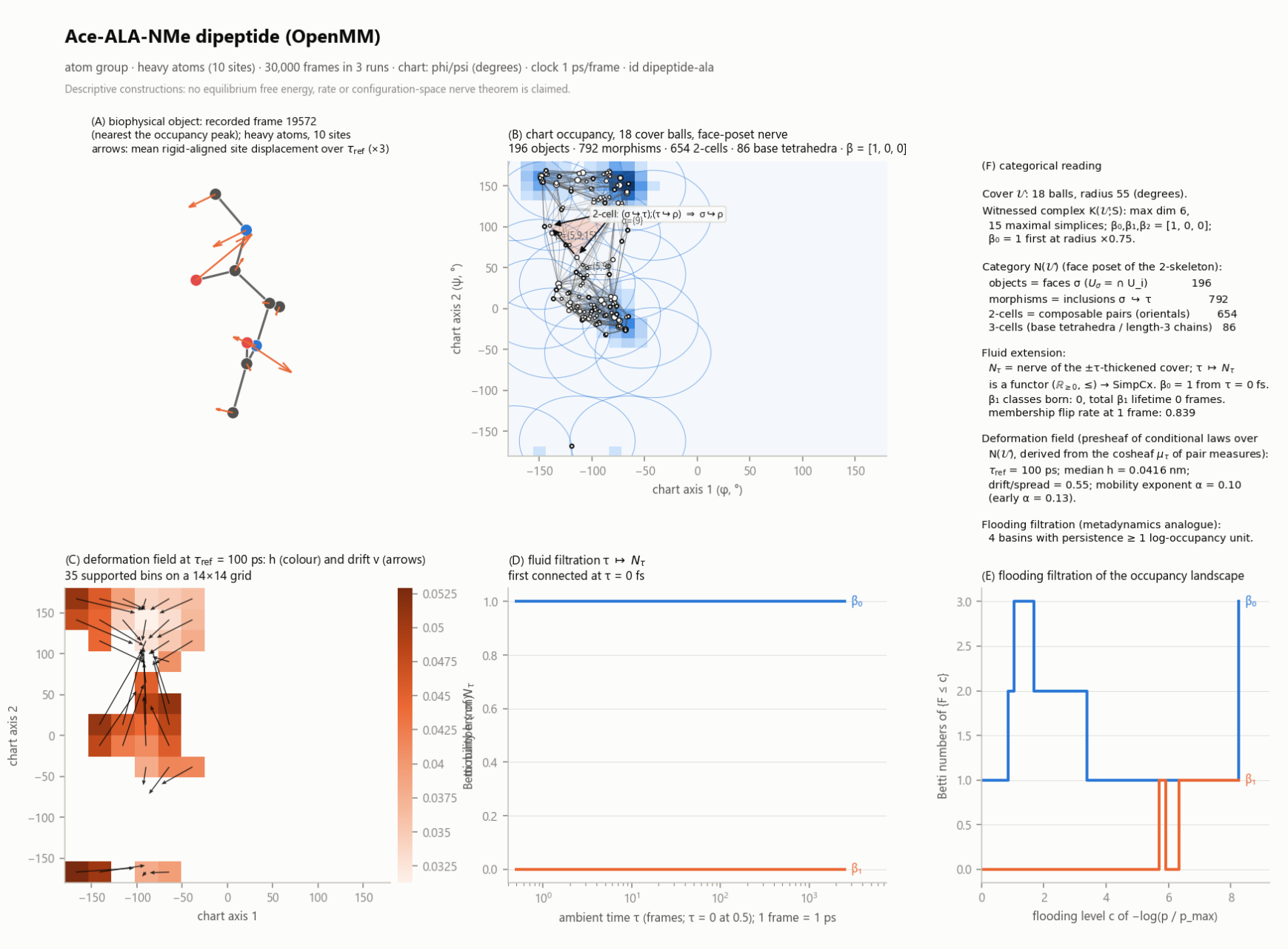

Figure 8: Ace-ALA-NMe dipeptide (OpenMM) (`dipeptide-ala`). Source: generate_dipeptides.py (this repository).

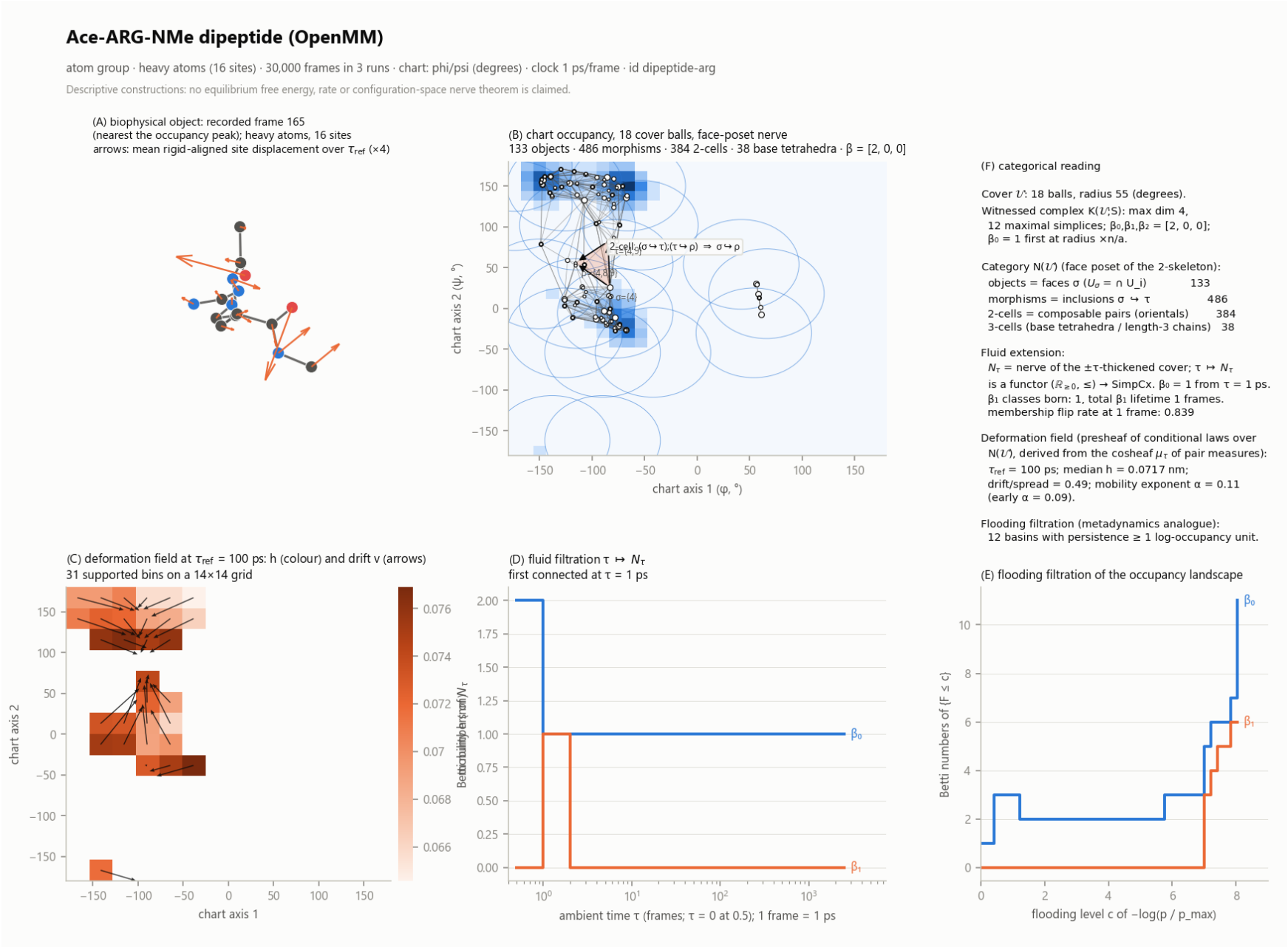

Figure 9: Ace-ARG-NMe dipeptide (OpenMM) (`dipeptide-arg`). Source: generate_dipeptides.py (this repository).

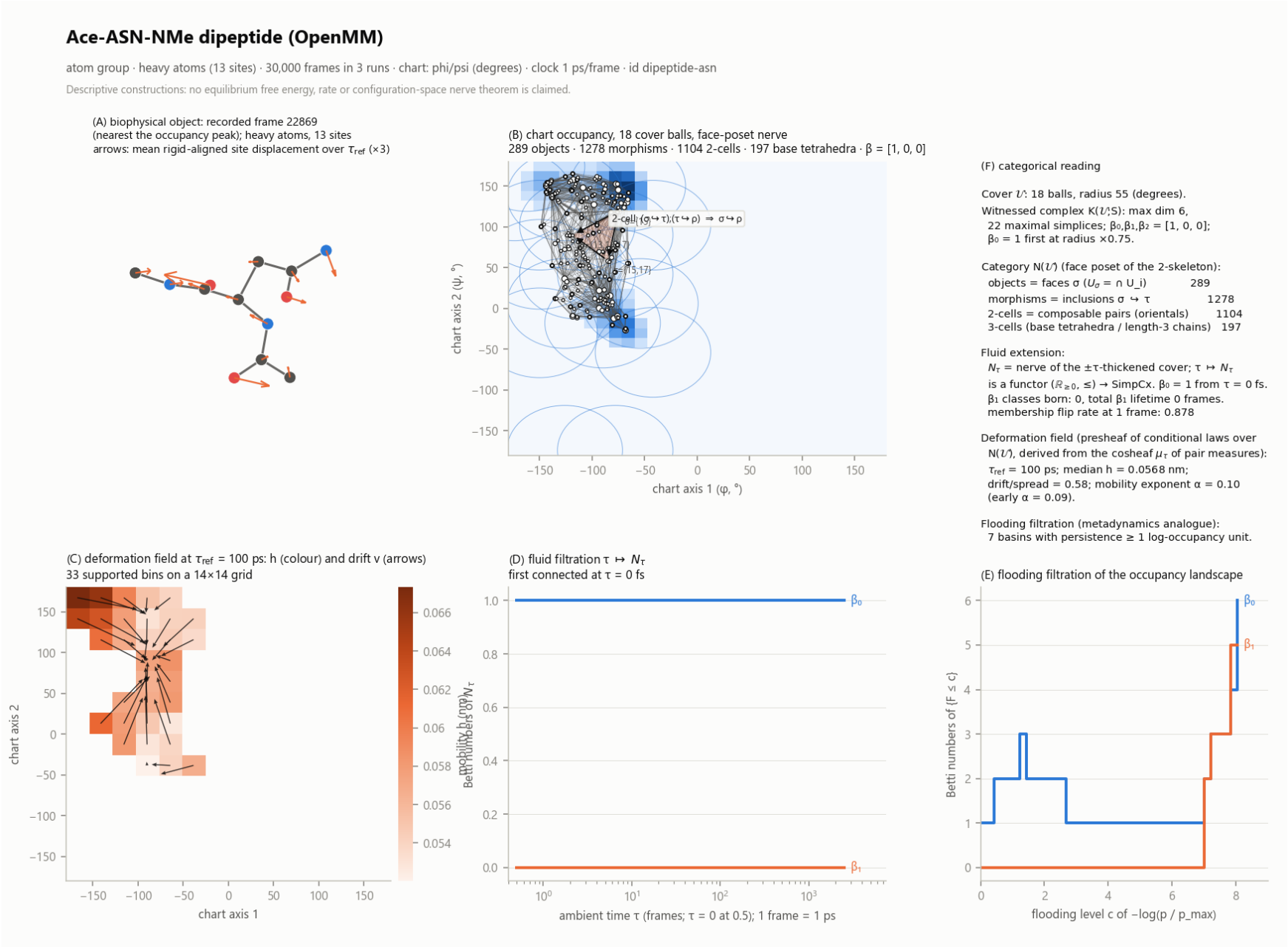

Figure 10: Ace-ASN-NMe dipeptide (OpenMM) (`dipeptide-asn`). Source: generate_dipeptides.py (this repository).

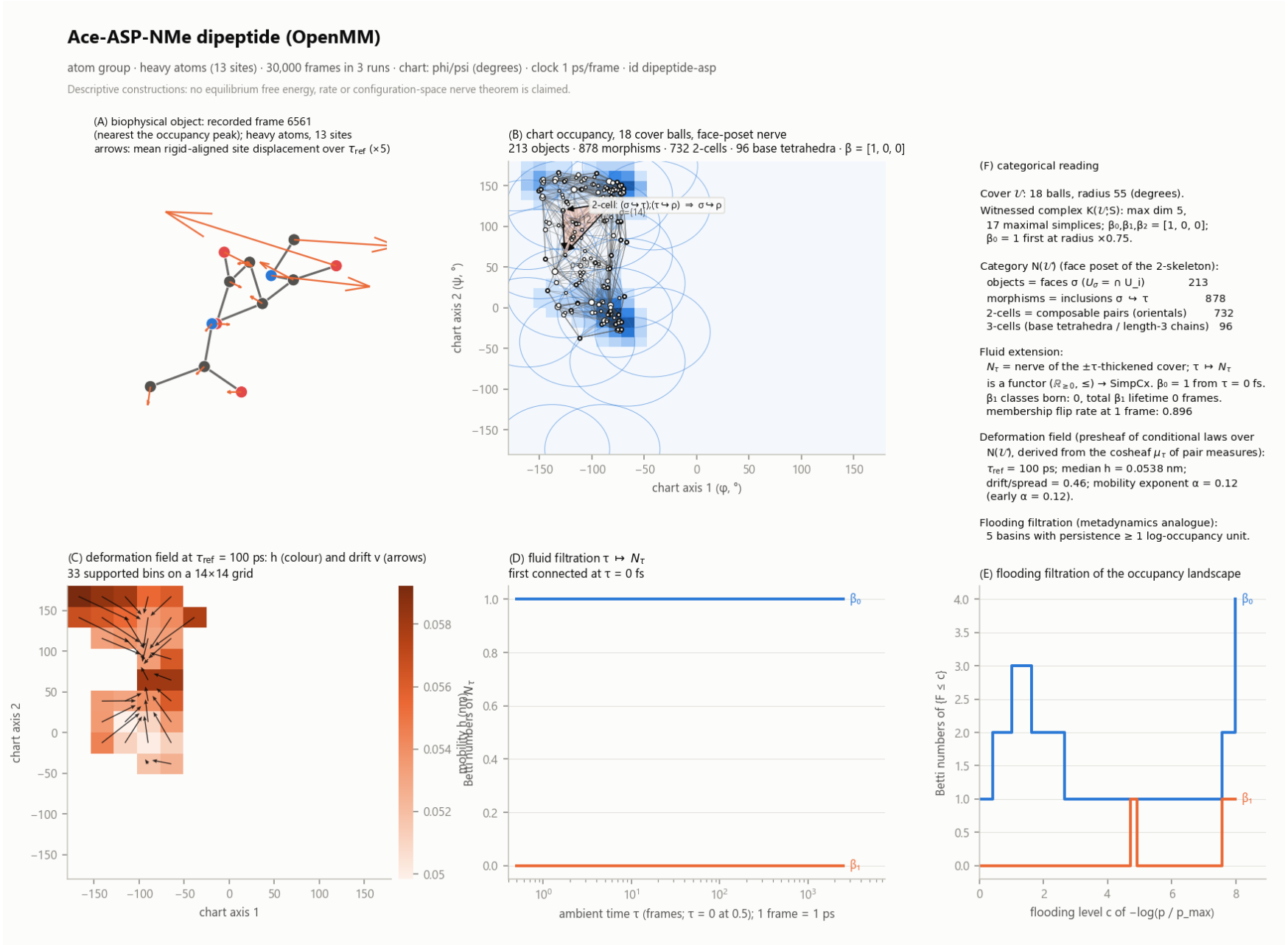

Figure 11: Ace-ASP-NMe dipeptide (OpenMM) (`dipeptide-asp`). Source: generate_dipeptides.py (this repository).

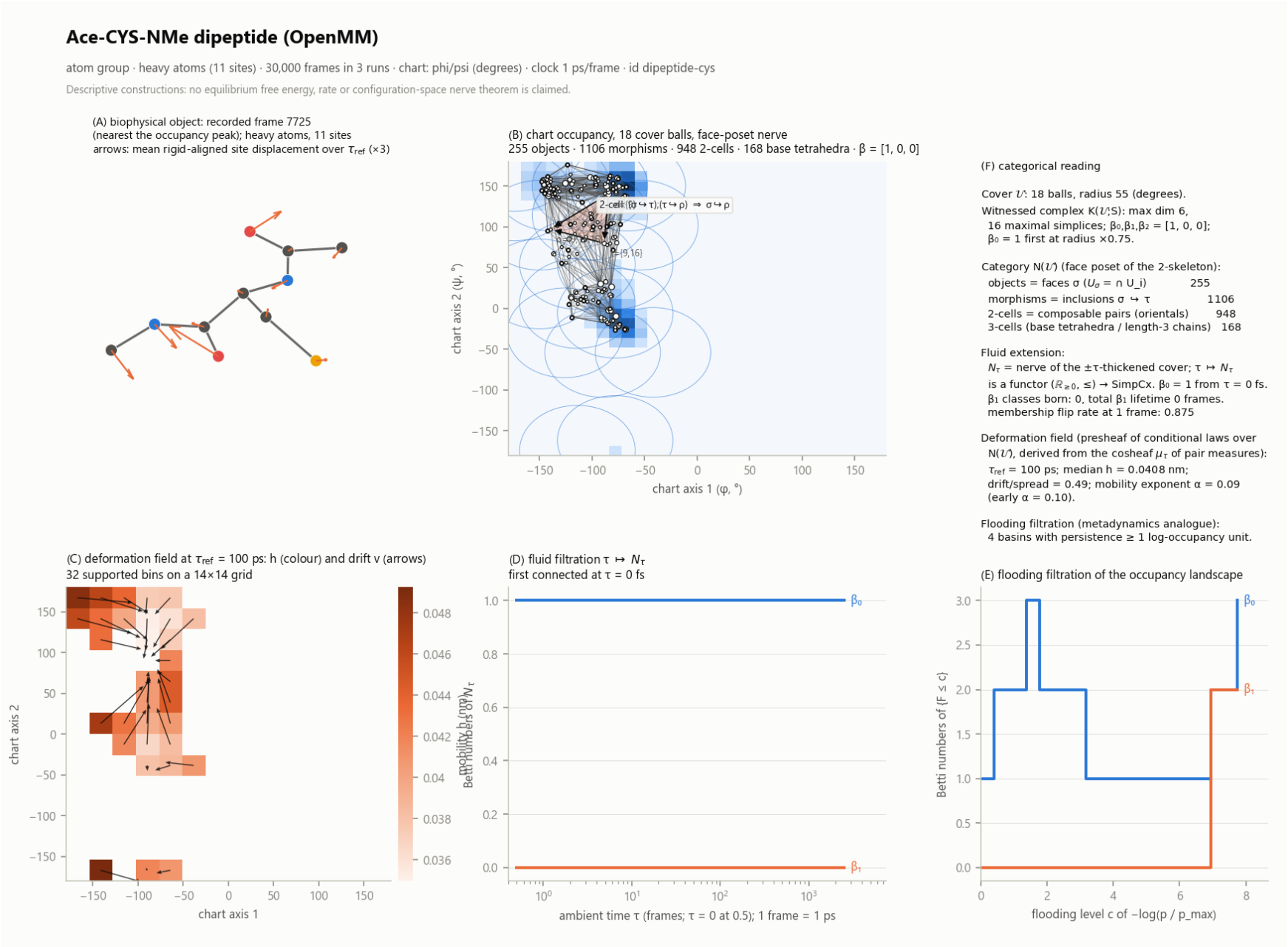

Figure 12: Ace-CYS-NMe dipeptide (OpenMM) (`dipeptide-cys`). Source: generate_dipeptides.py (this repository).

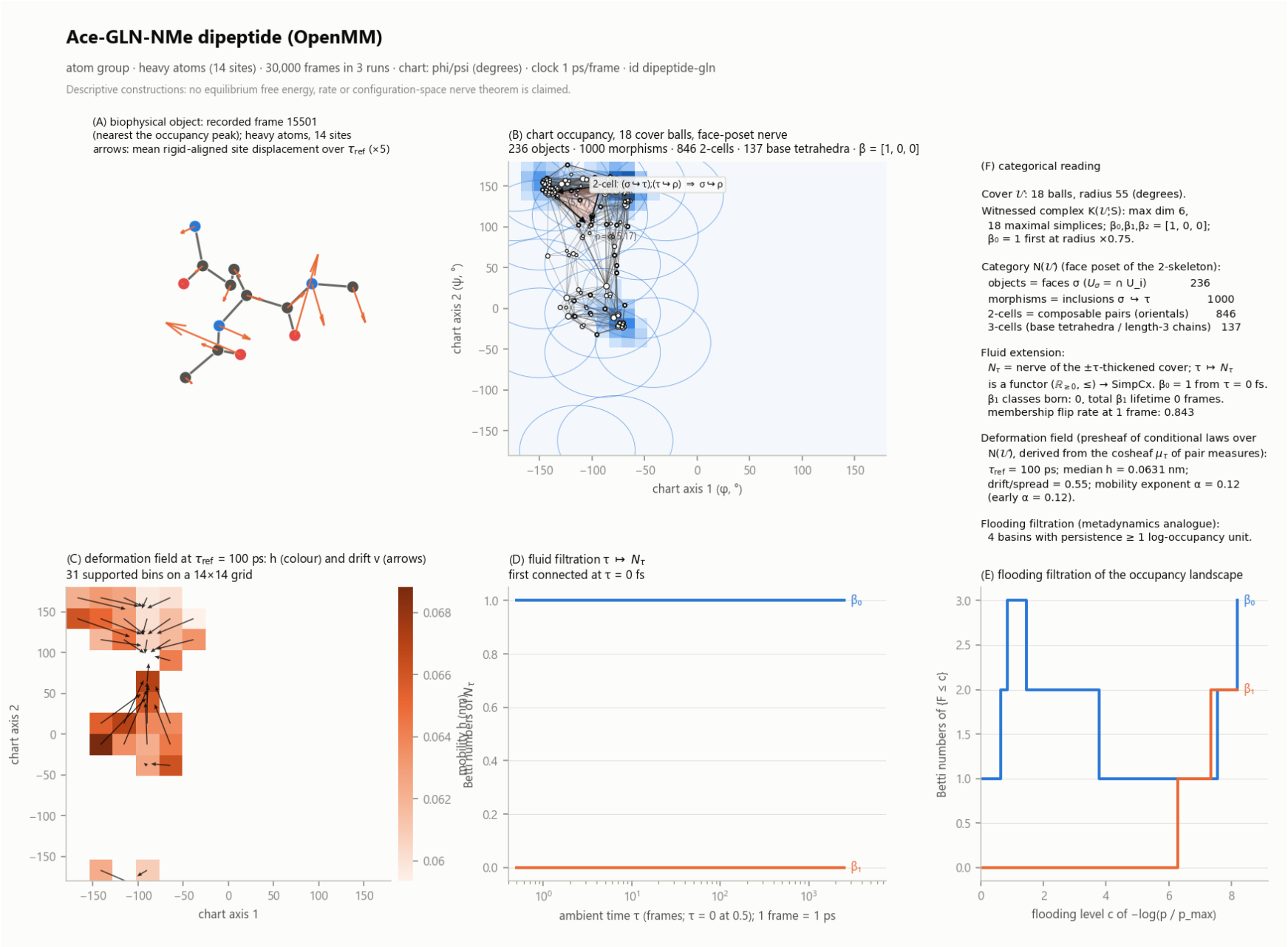

Figure 13: Ace-GLN-NMe dipeptide (OpenMM) (`dipeptide-gln`). Source: generate_dipeptides.py (this repository).

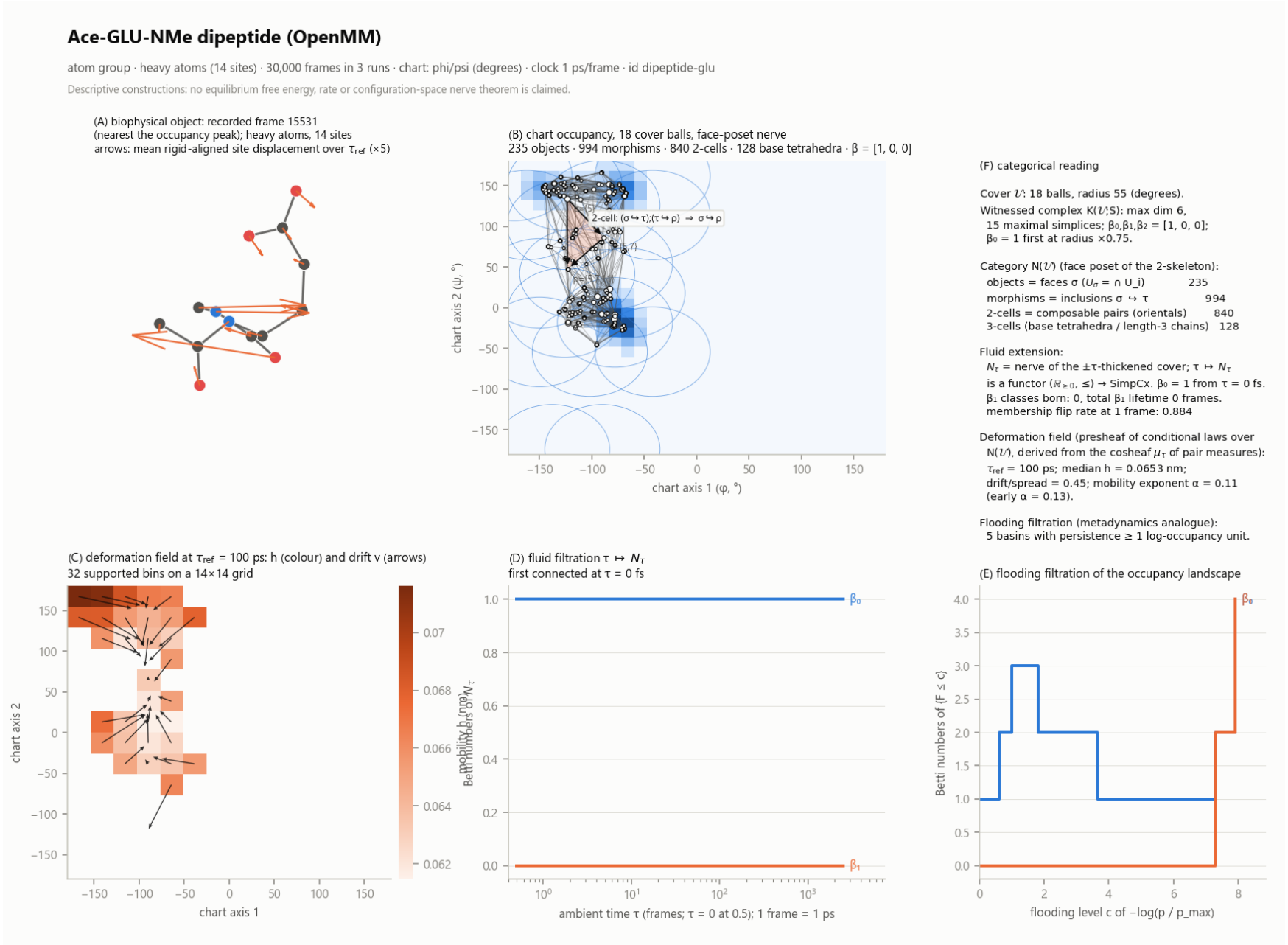

Figure 14: Ace-GLU-NMe dipeptide (OpenMM) (`dipeptide-glu`). Source: generate_dipeptides.py (this repository).

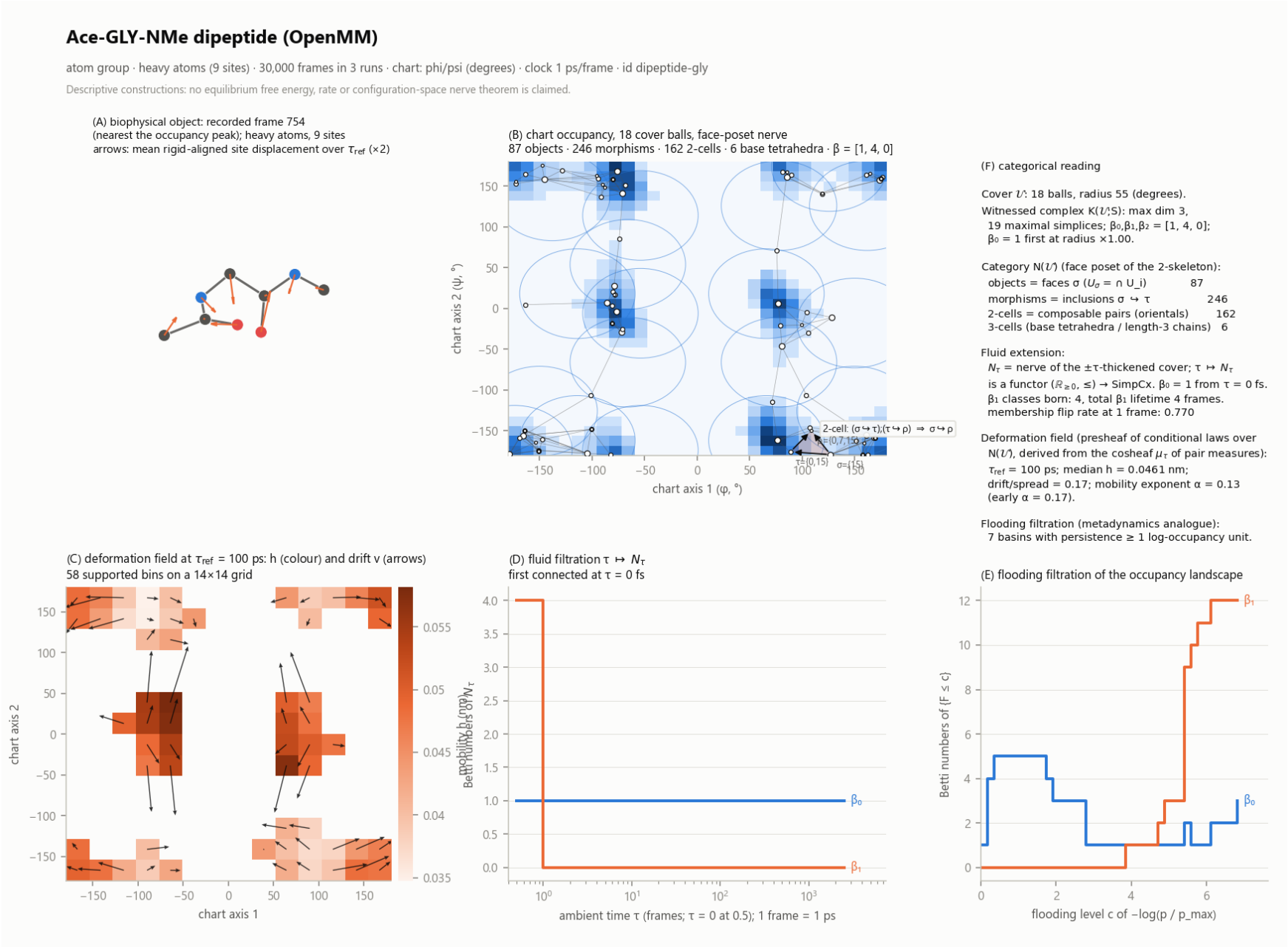

Figure 15: Ace-GLY-NMe dipeptide (OpenMM) (`dipeptide-gly`). Source: generate_dipeptides.py (this repository).

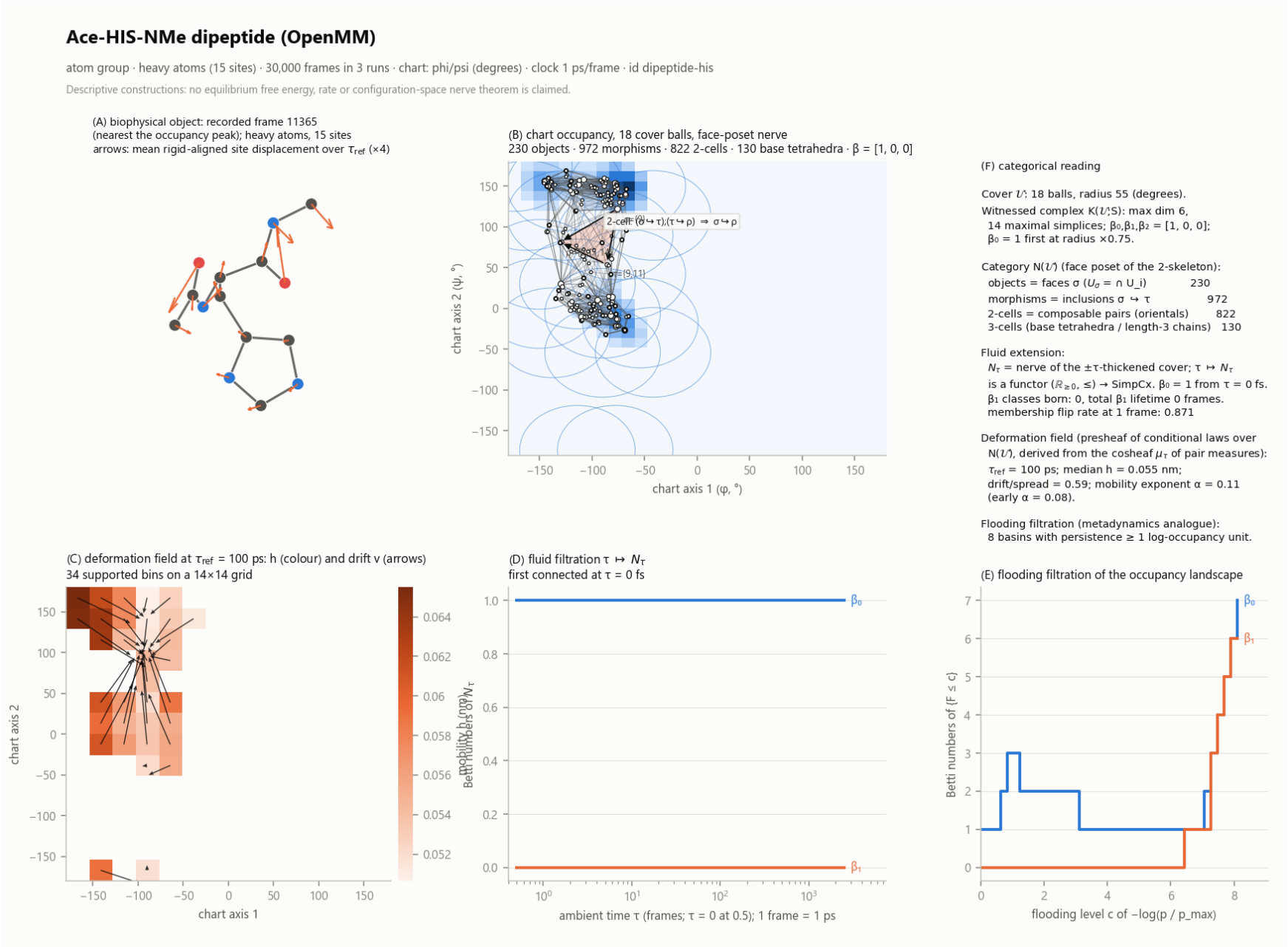

Figure 16: Ace-HIS-NMe dipeptide (OpenMM) (`dipeptide-his`). Source: generate_dipeptides.py (this repository).

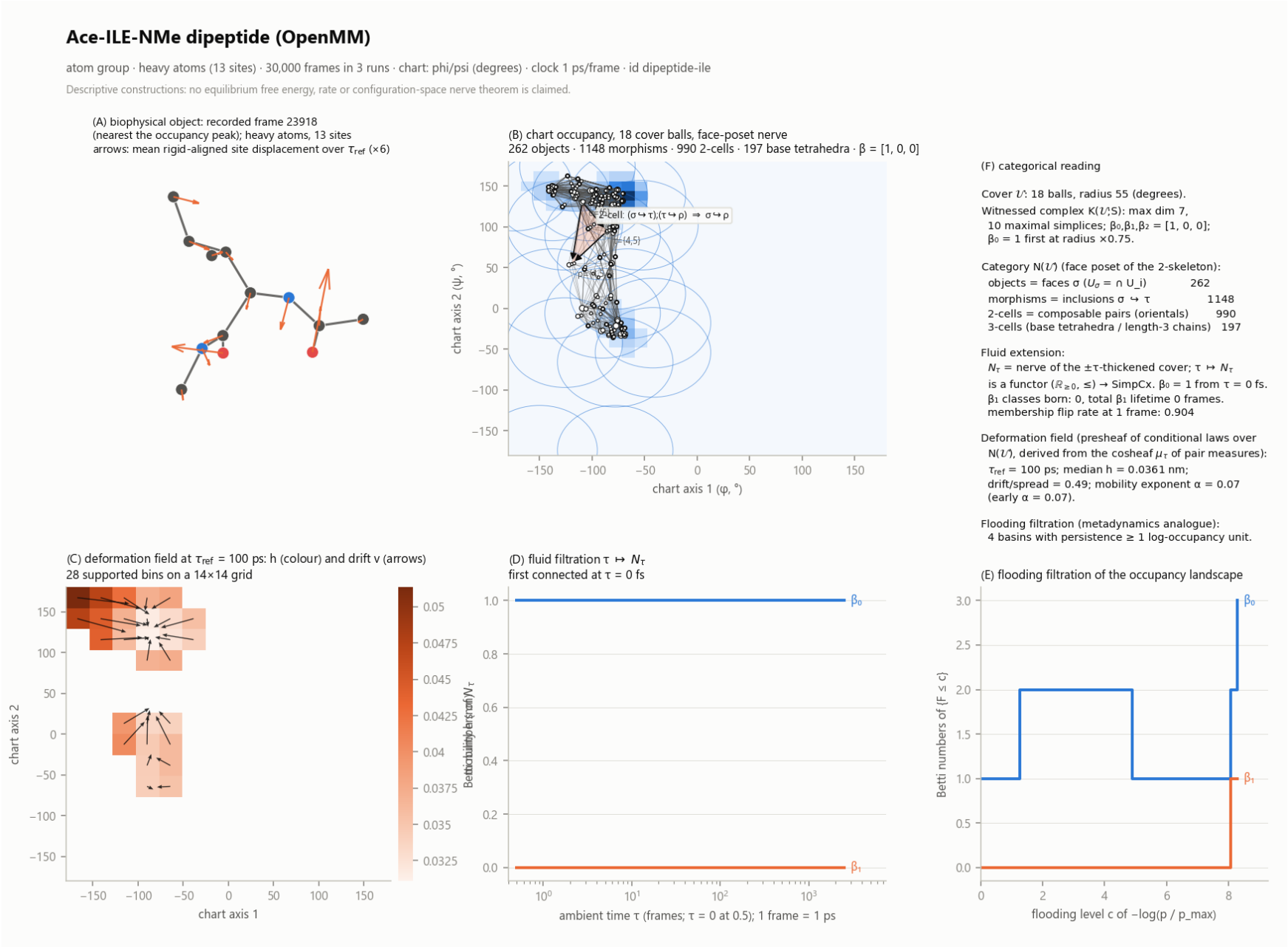

Figure 17: Ace-ILE-NMe dipeptide (OpenMM) (`dipeptide-ile`). Source: generate_dipeptides.py (this repository).

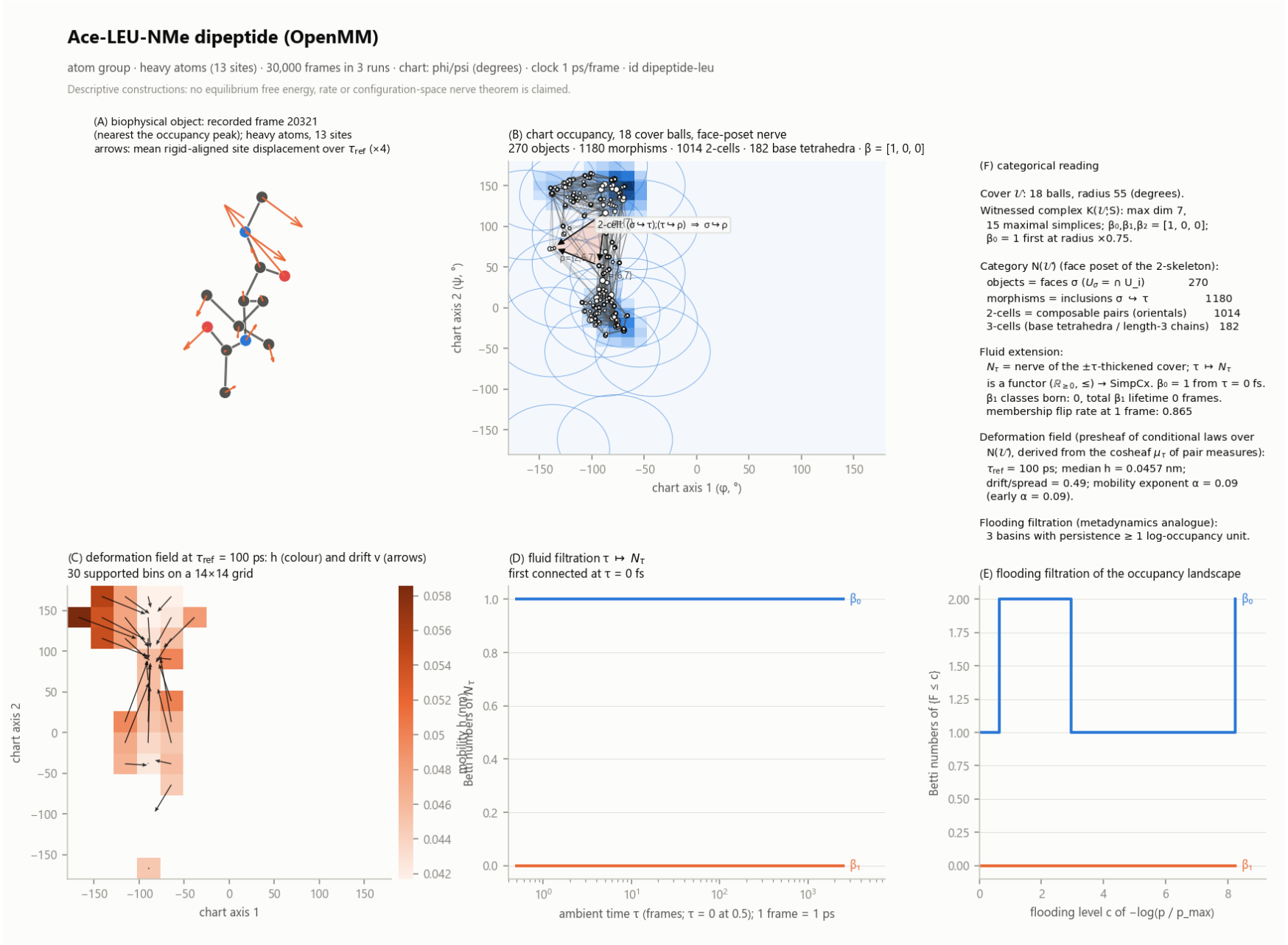

Figure 18: Ace-LEU-NMe dipeptide (OpenMM) (`dipeptide-leu`). Source: generate_dipeptides.py (this repository).

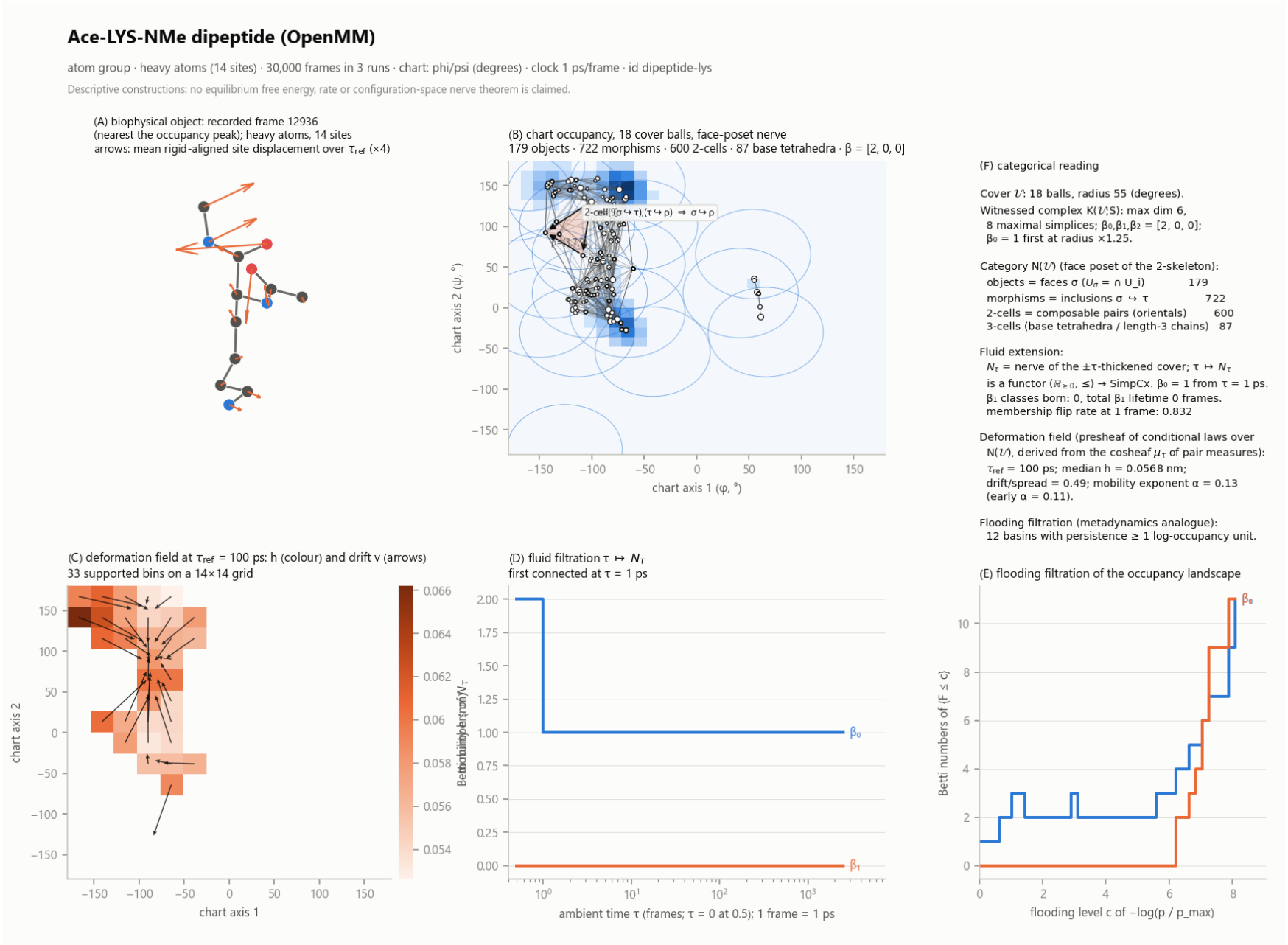

Figure 19: Ace-LYS-NMe dipeptide (OpenMM) (`dipeptide-lys`). Source: generate_dipeptides.py (this repository).

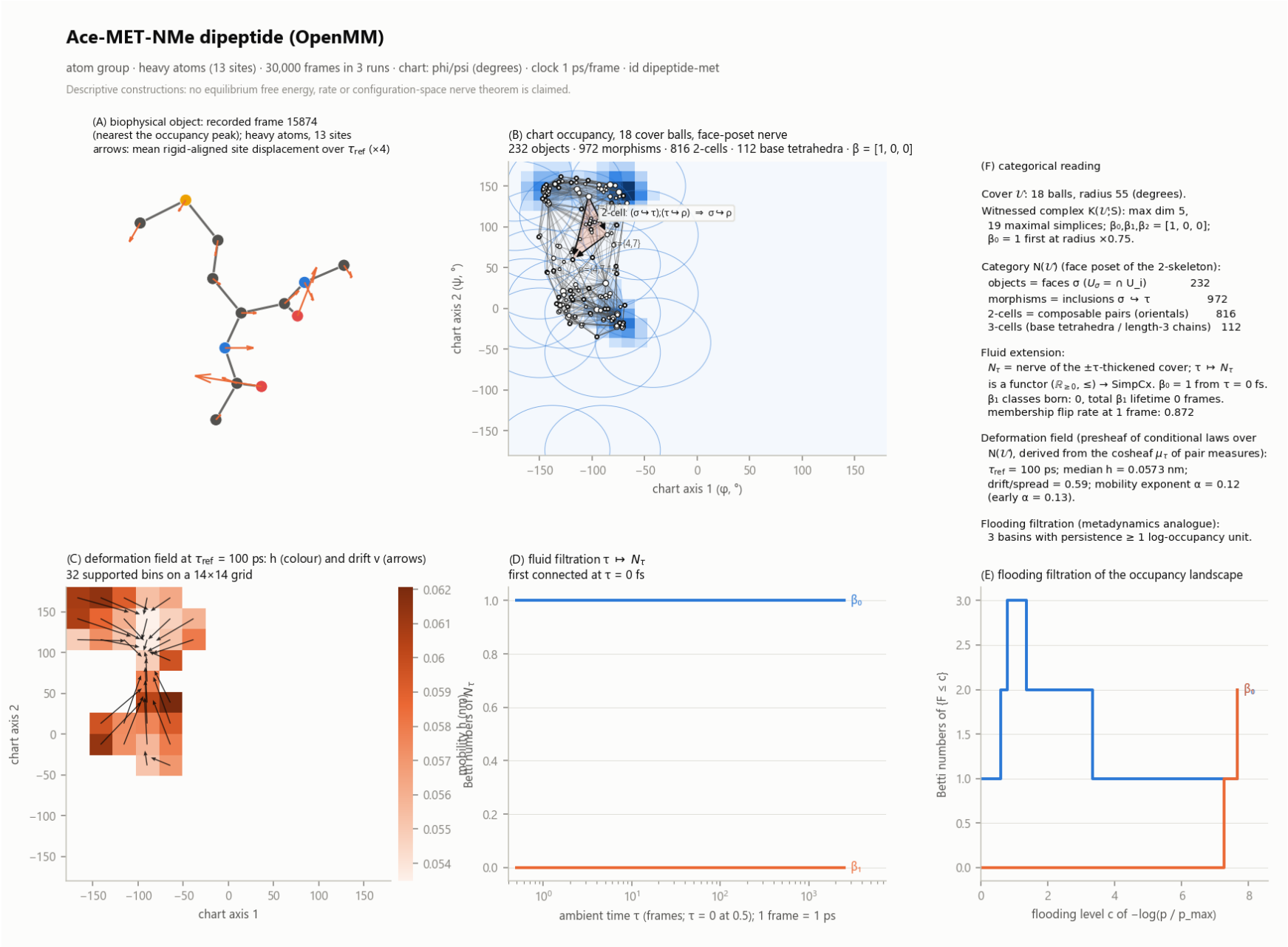

Figure 20: Ace-MET-NMe dipeptide (OpenMM) (`dipeptide-met`). Source: generate_dipeptides.py (this repository).

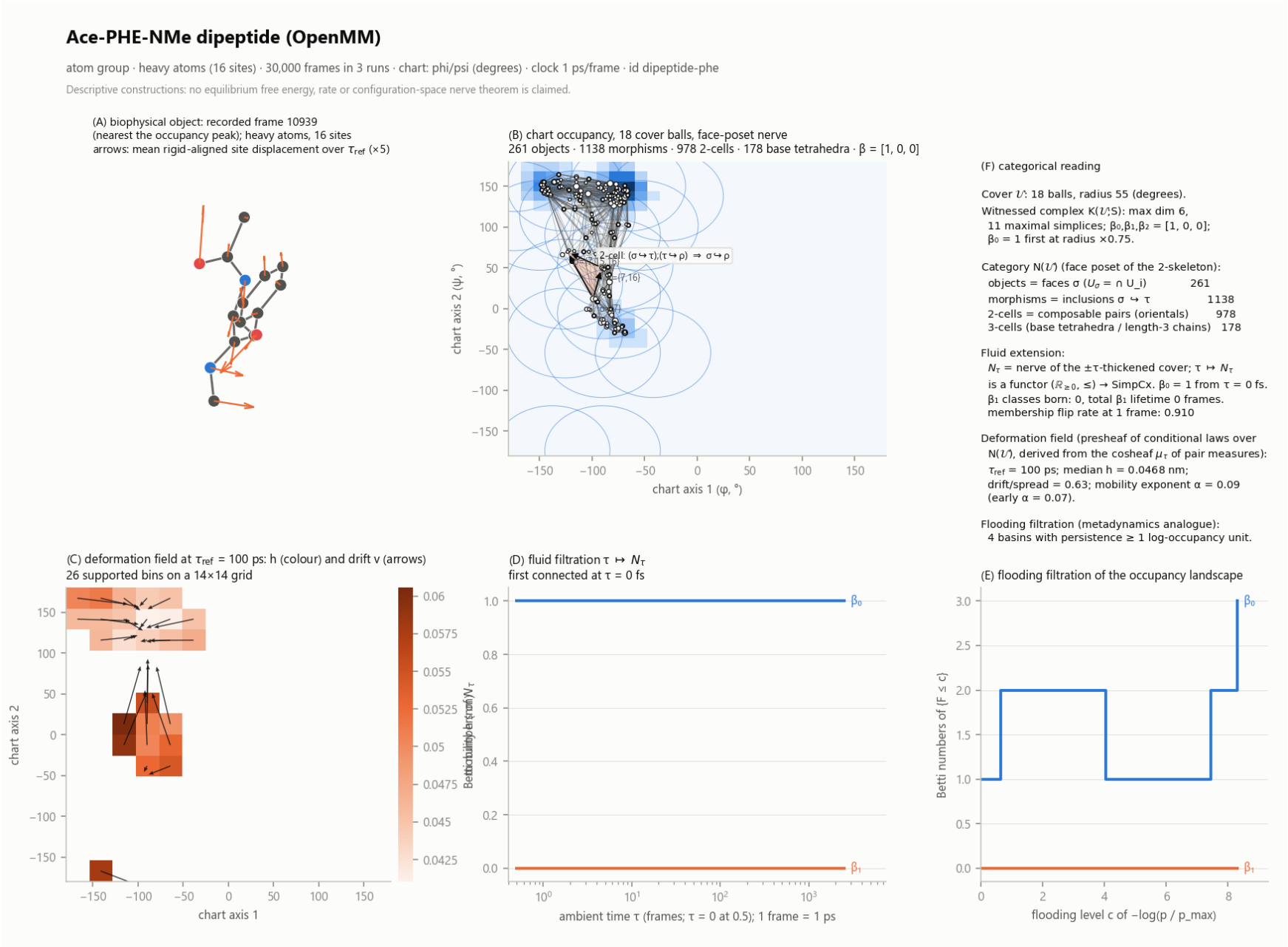

Figure 21: Ace-PHE-NMe dipeptide (OpenMM) (`dipeptide-phe`). Source: generate_dipeptides.py (this repository).

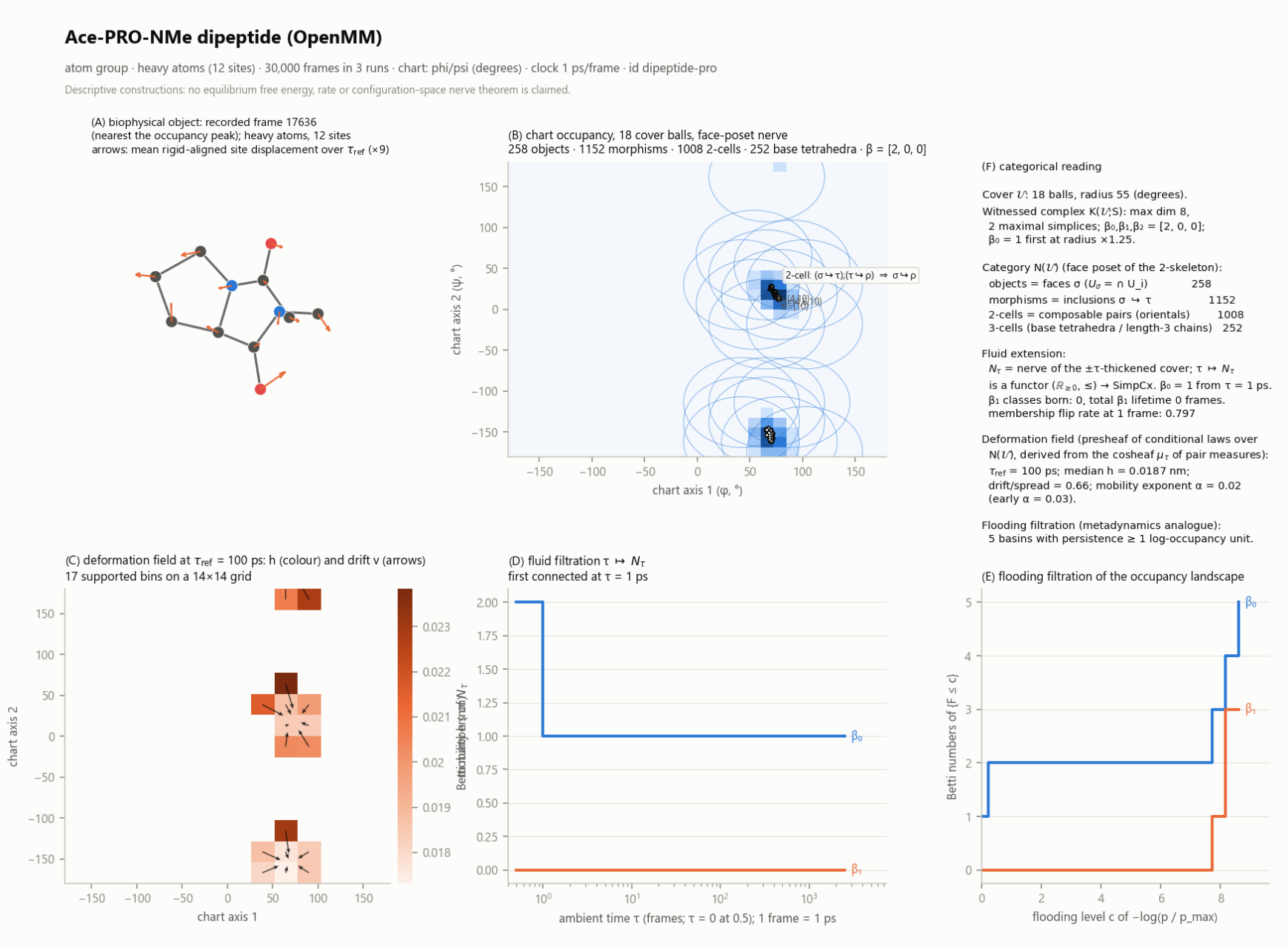

Figure 22: Ace-PRO-NMe dipeptide (OpenMM) (`dipeptide-pro`). Source: generate_dipeptides.py (this repository).

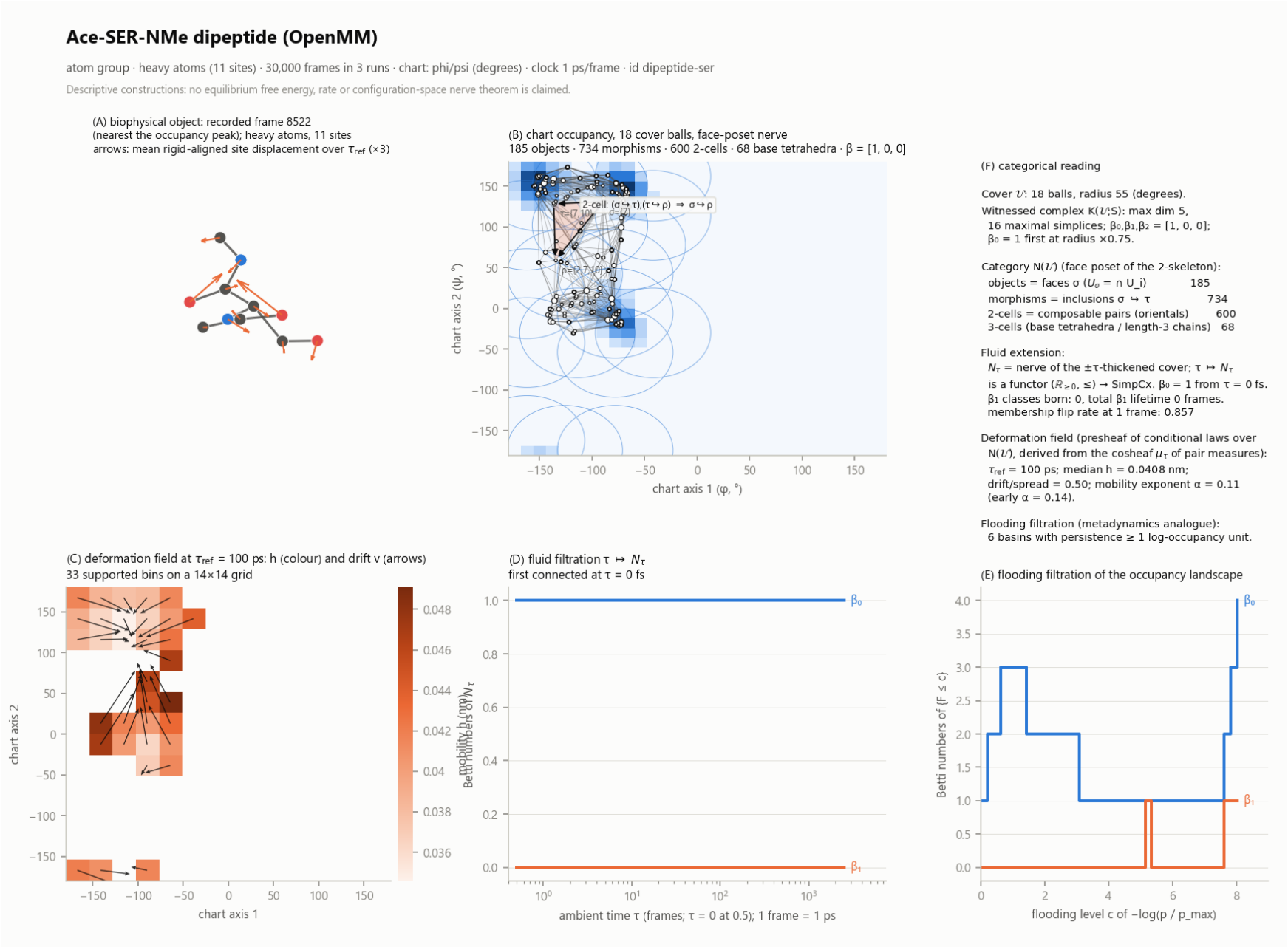

Figure 23: Ace-SER-NMe dipeptide (OpenMM) (`dipeptide-ser`). Source: generate_dipeptides.py (this repository).

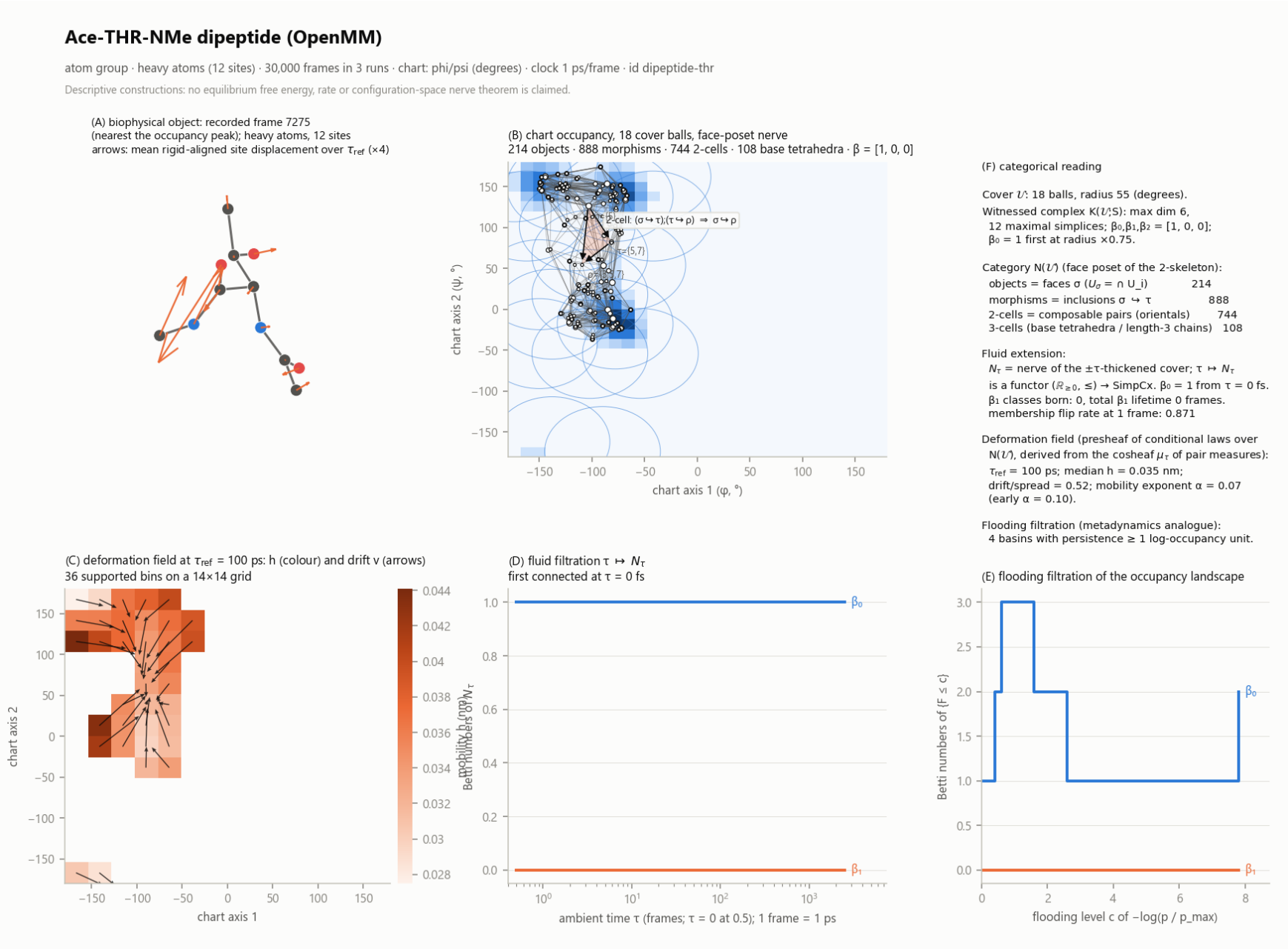

Figure 24: Ace-THR-NMe dipeptide (OpenMM) (`dipeptide-thr`). Source: generate_dipeptides.py (this repository).

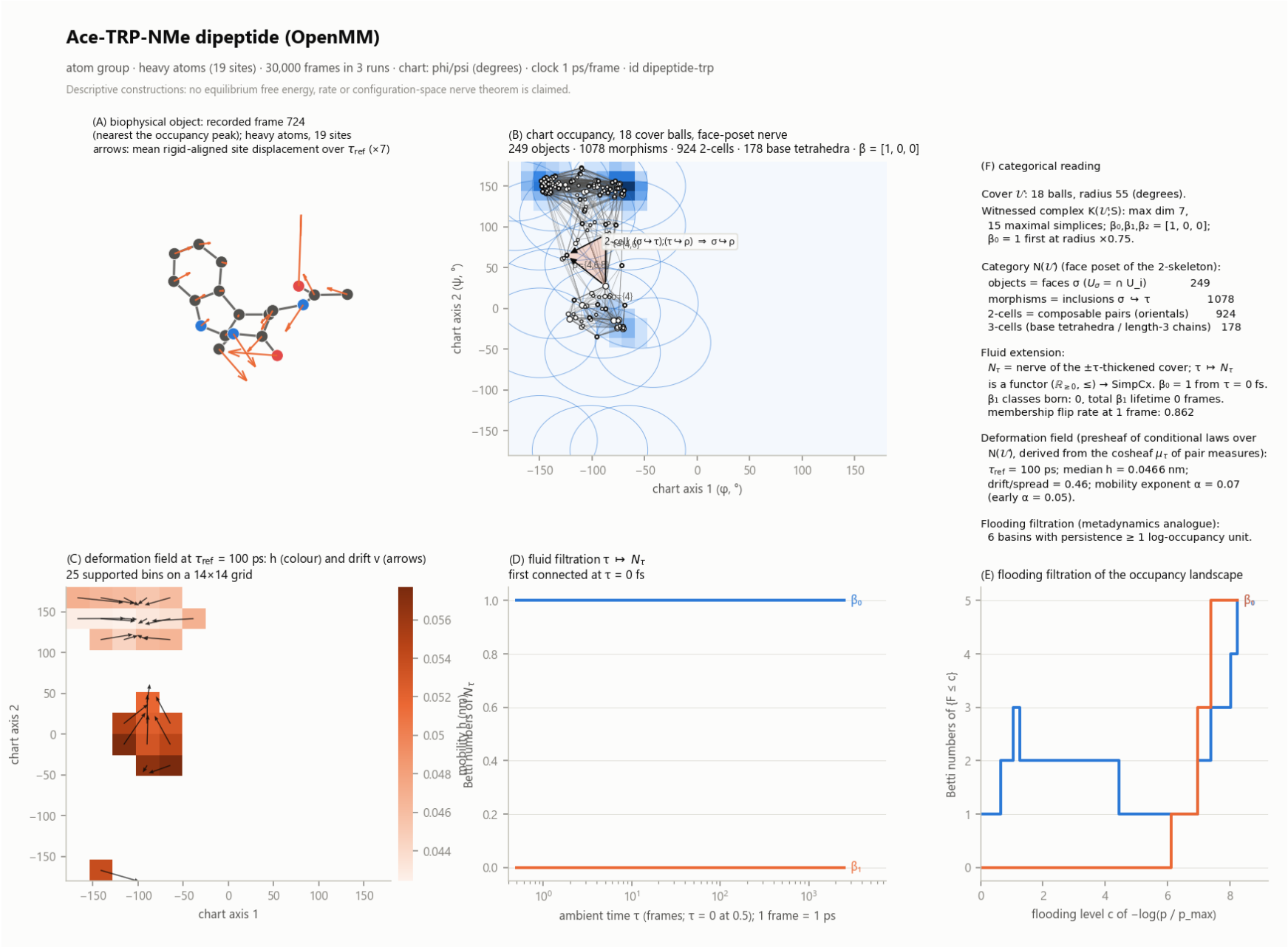

Figure 25: Ace-TRP-NMe dipeptide (OpenMM) (`dipeptide-trp`). Source: generate_dipeptides.py (this repository).

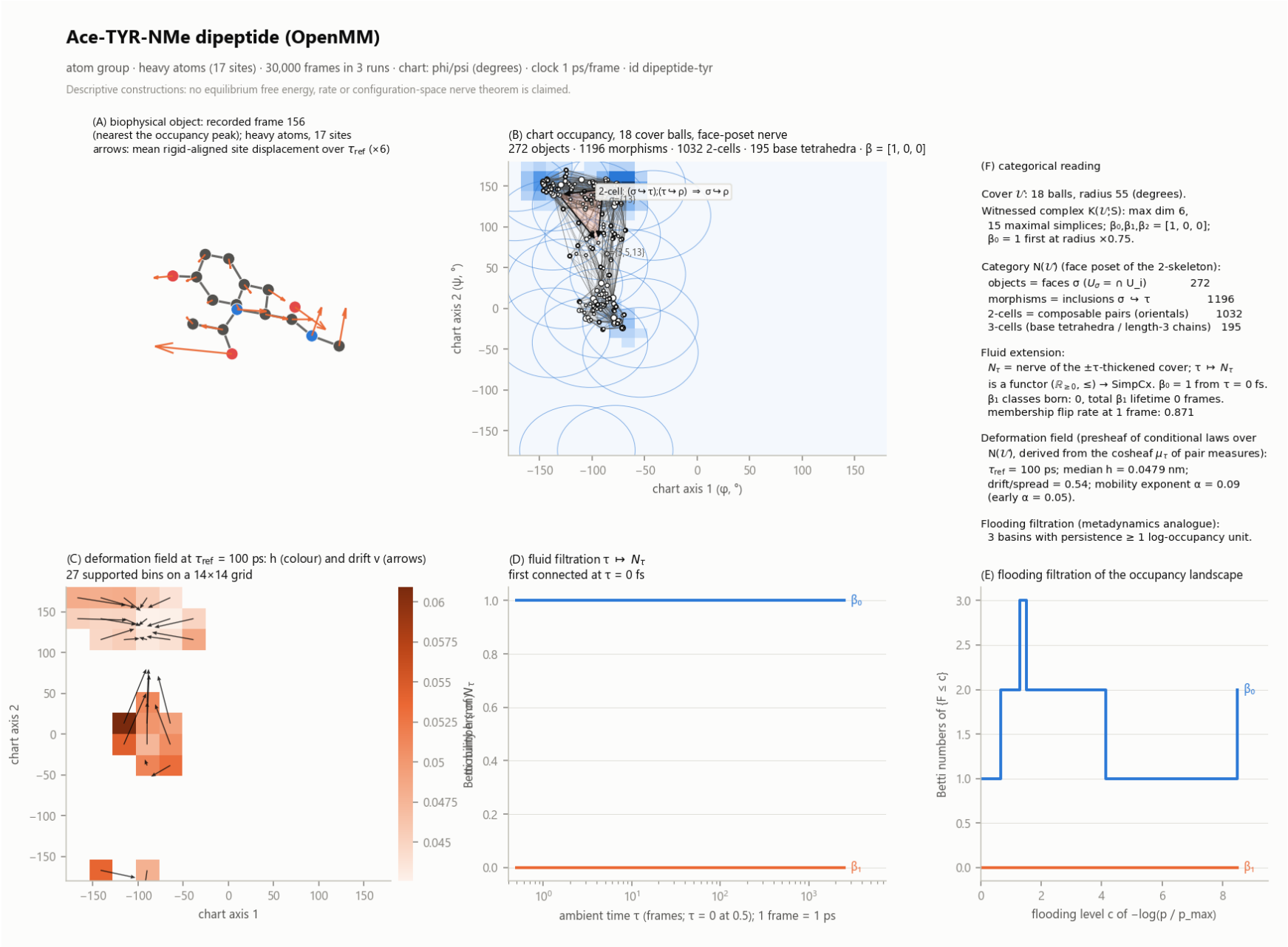

Figure 26: Ace-TYR-NMe dipeptide (OpenMM) (`dipeptide-tyr`). Source: generate_dipeptides.py (this repository).

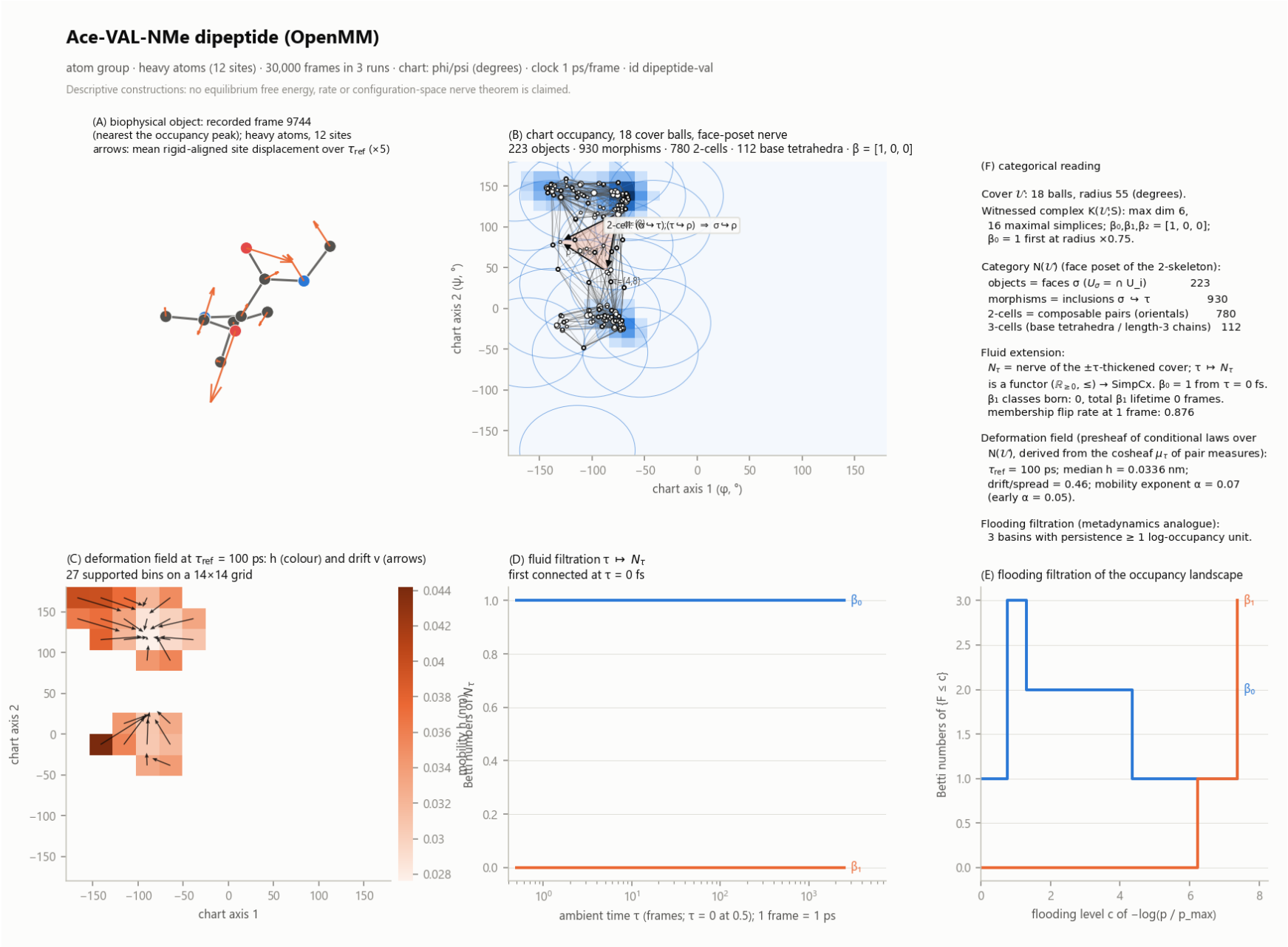

Figure 27: Ace-VAL-NMe dipeptide (OpenMM) (`dipeptide-val`). Source: generate_dipeptides.py (this repository).

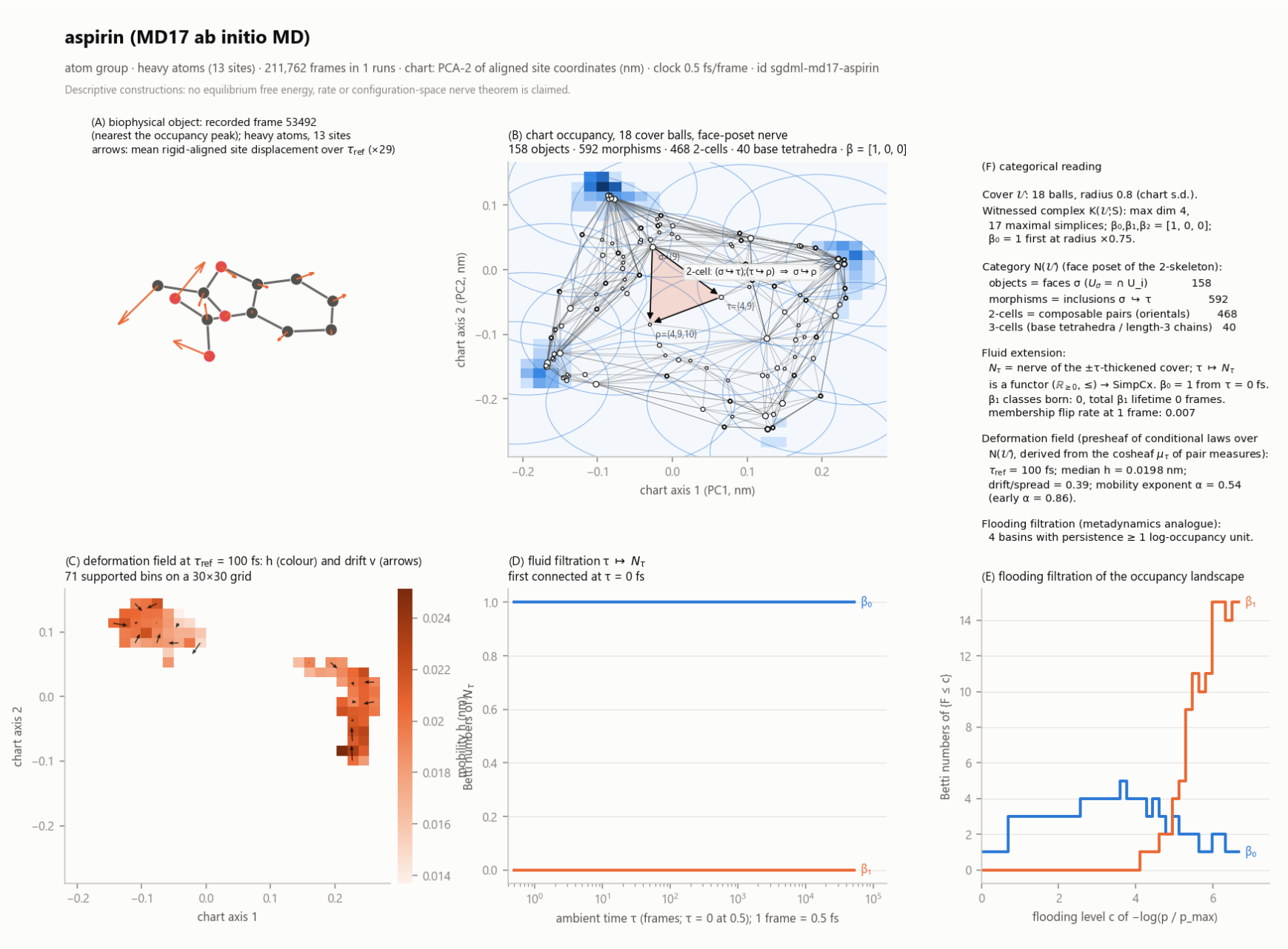

Figure 28: aspirin (MD17 ab initio MD) (`sgdml-md17-aspirin`). Source: https://www.quantum-machine.org/gdml/data/npz/md17_aspirin.npz.

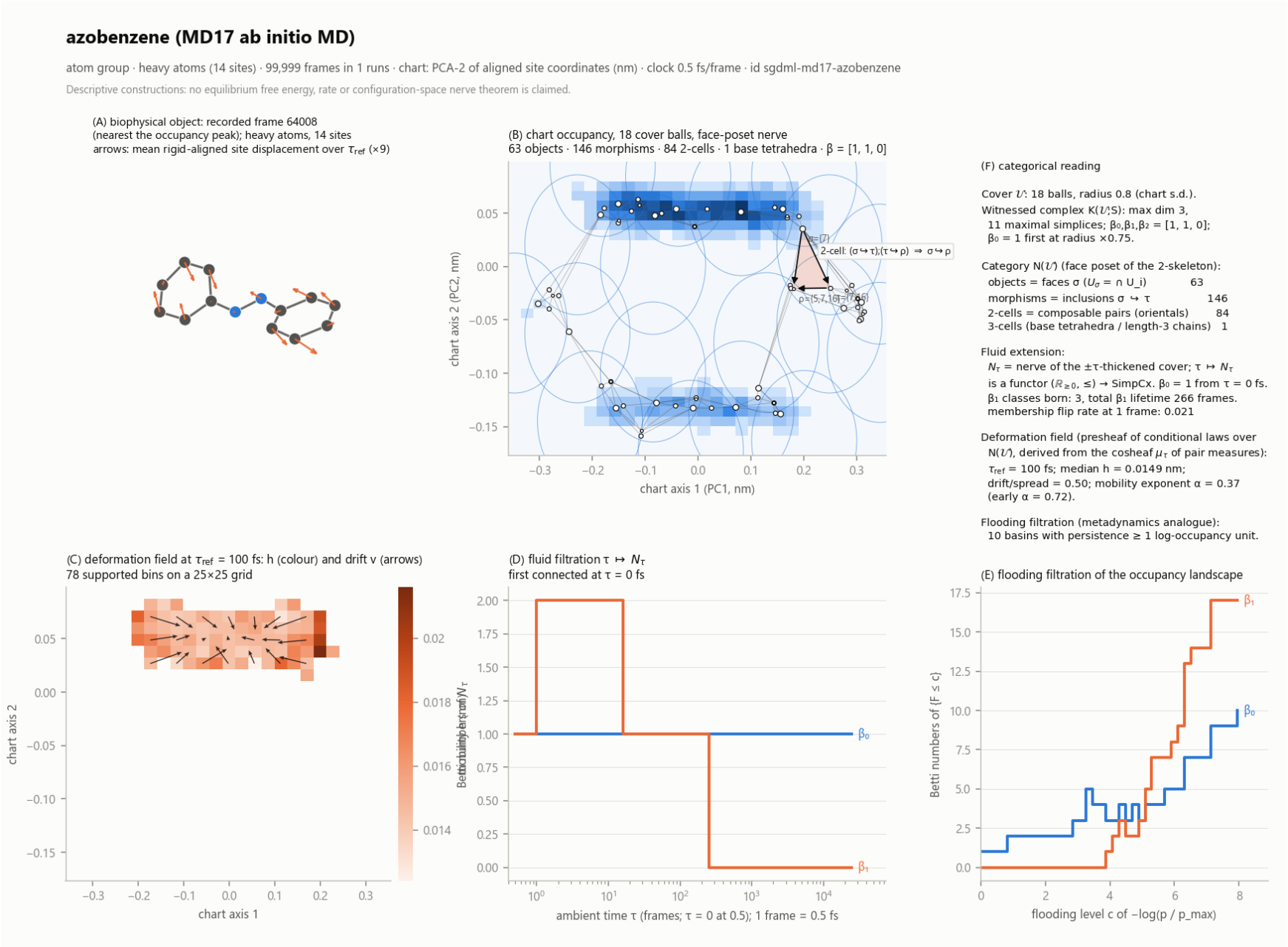

Figure 29: azobenzene (MD17 ab initio MD) (`sgdml-md17-azobenzene`). Source: https://www.quantum-machine.org/gdml/data/npz/azobenzene_dft.npz.

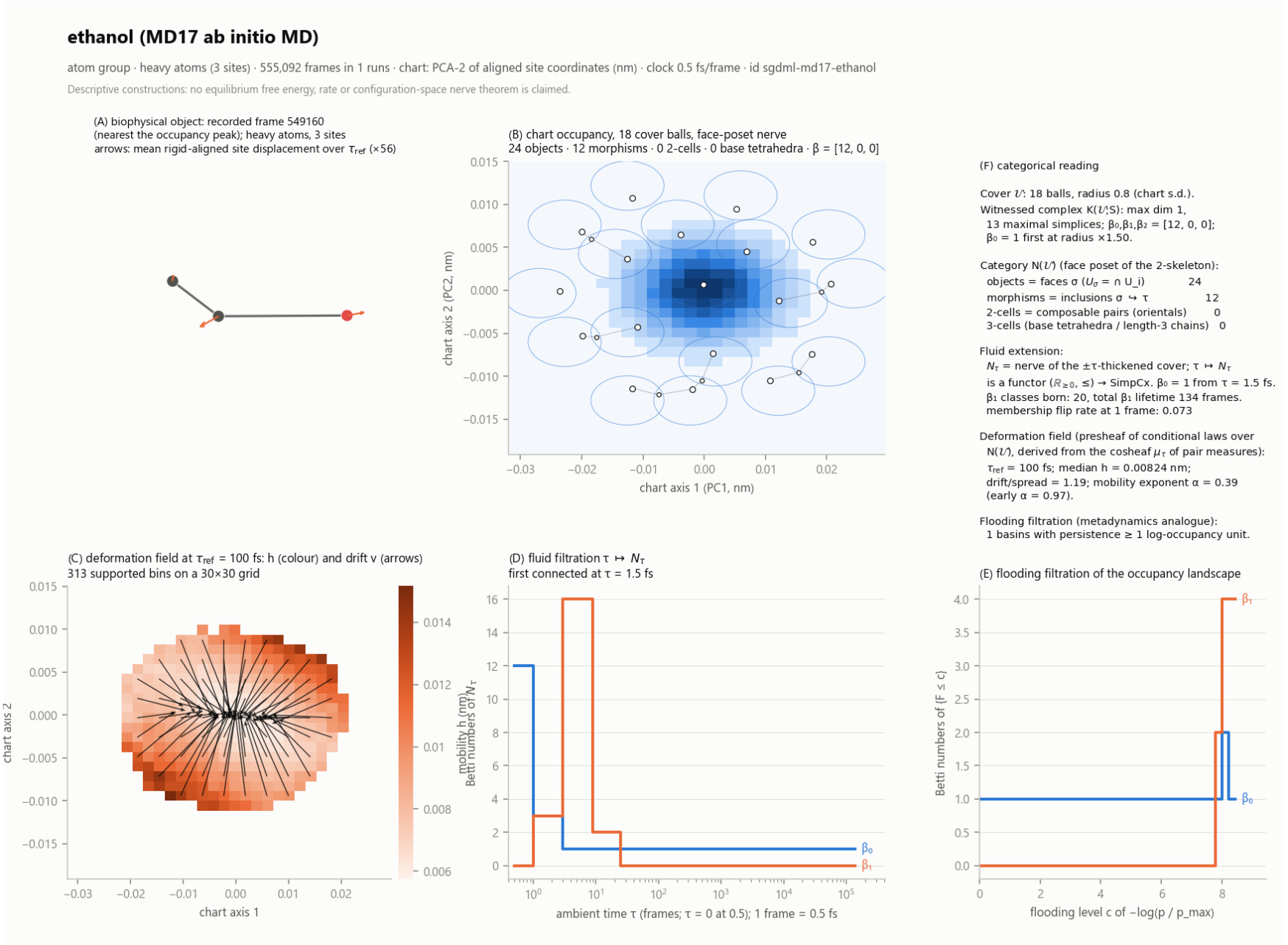

Figure 30: ethanol (MD17 ab initio MD) (`sgdml-md17-ethanol`). Source: https://www.quantum-machine.org/gdml/data/npz/md17_ethanol.npz.

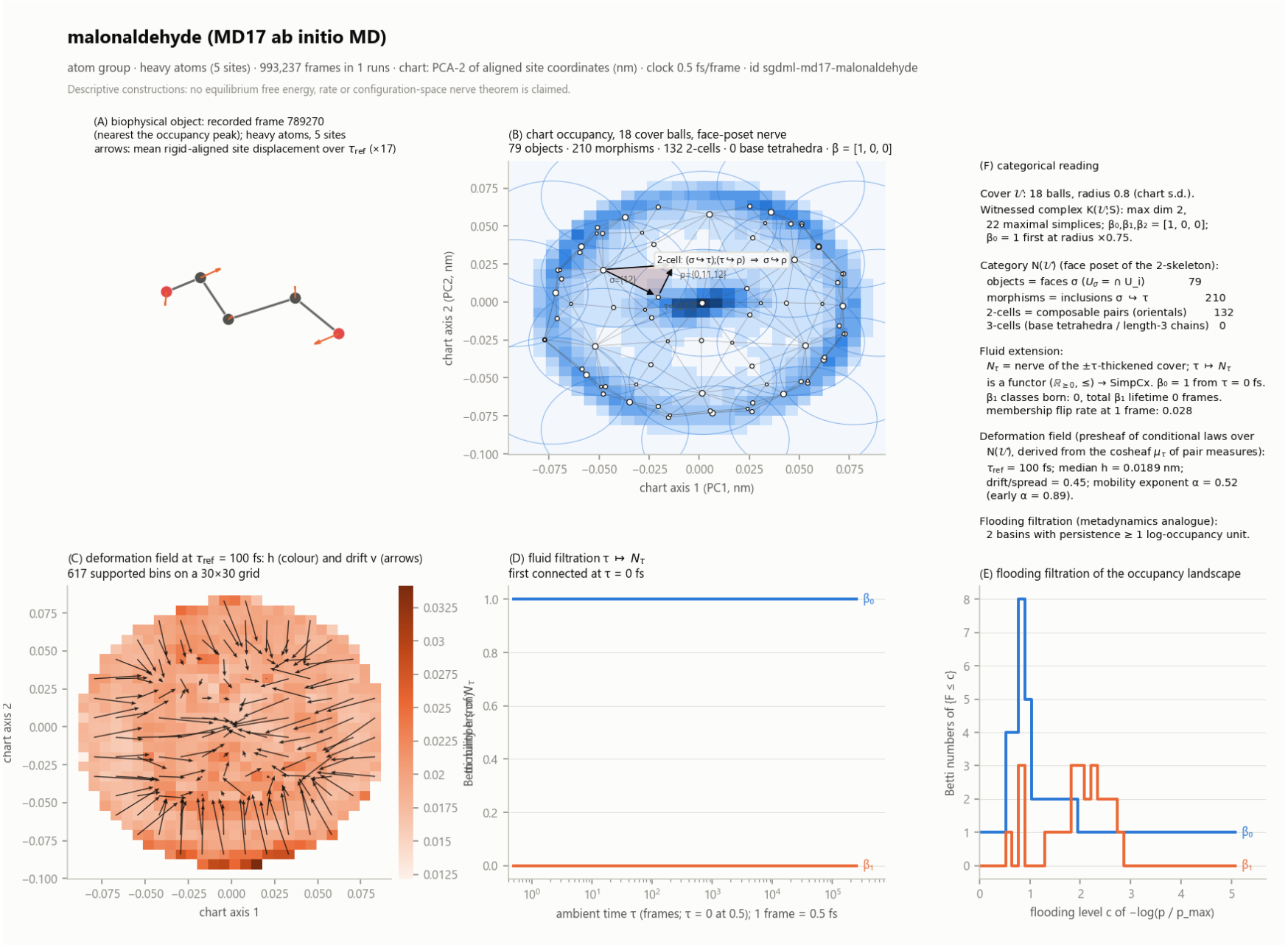

Figure 31: malonaldehyde (MD17 ab initio MD) (`sgdml-md17-malonaldehyde`). Source: https://www.quantum-machine.org/gdml/data/npz/md17_malonaldehyde.npz.

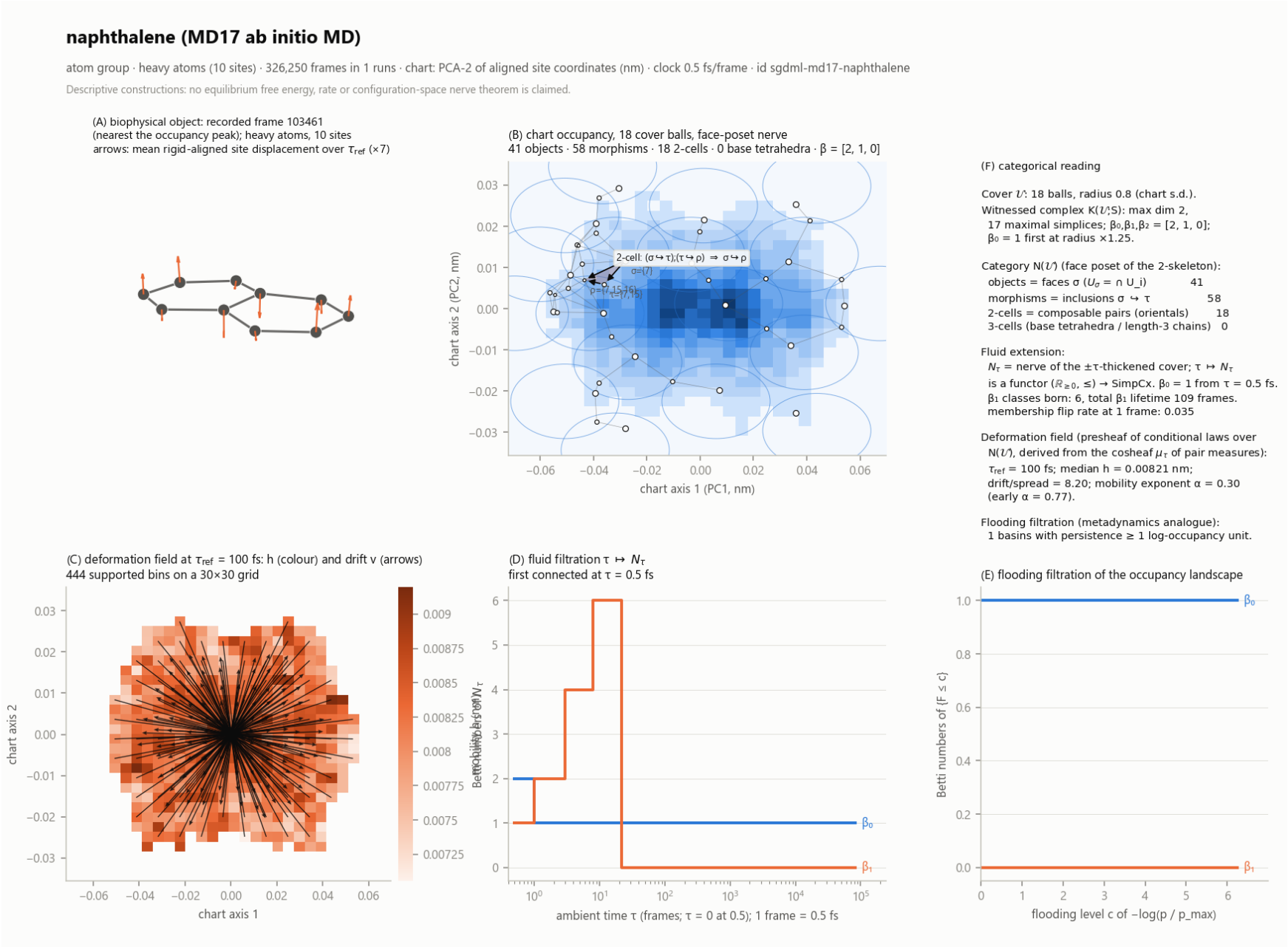

Figure 32: naphthalene (MD17 ab initio MD) (`sgdml-md17-naphthalene`). Source: https://www.quantum-machine.org/gdml/data/npz/md17_naphthalene.npz.

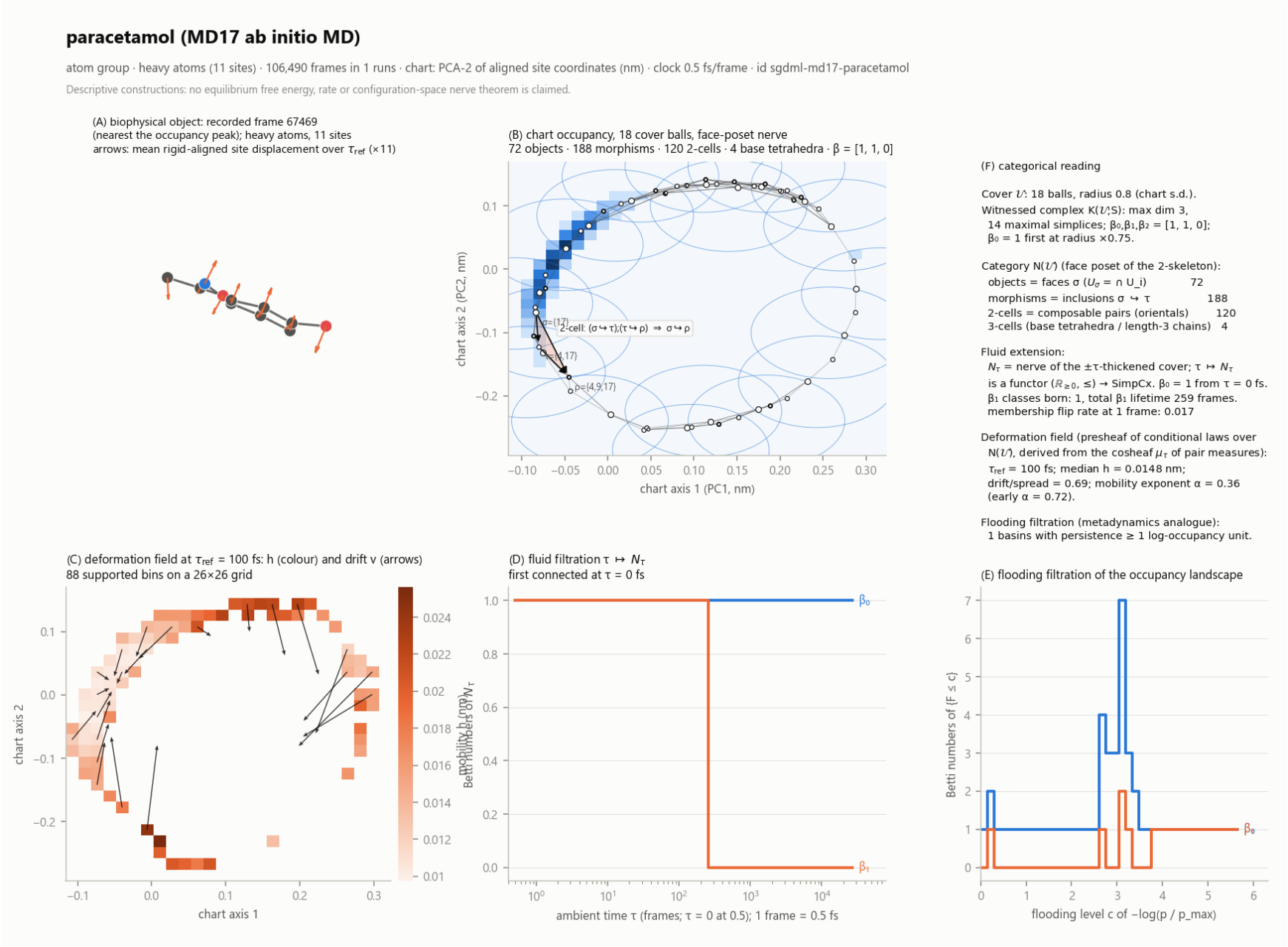

Figure 33: paracetamol (MD17 ab initio MD) (`sgdml-md17-paracetamol`). Source: https://www.quantum-machine.org/gdml/data/npz/paracetamol_dft.npz.

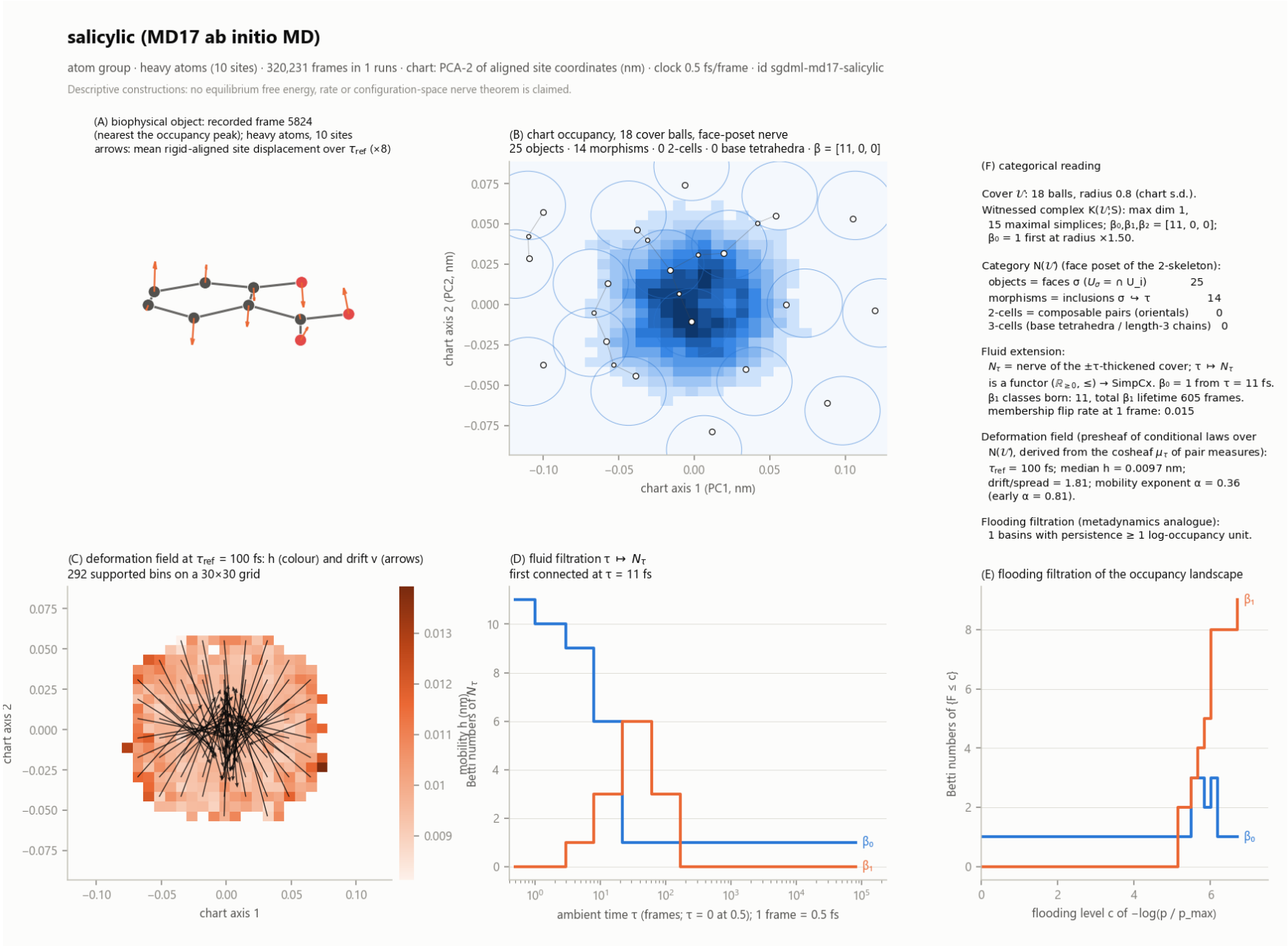

Figure 34: salicylic (MD17 ab initio MD) (`sgdml-md17-salicylic`). Source: https://www.quantum-machine.org/gdml/data/npz/md17_salicylic.npz.

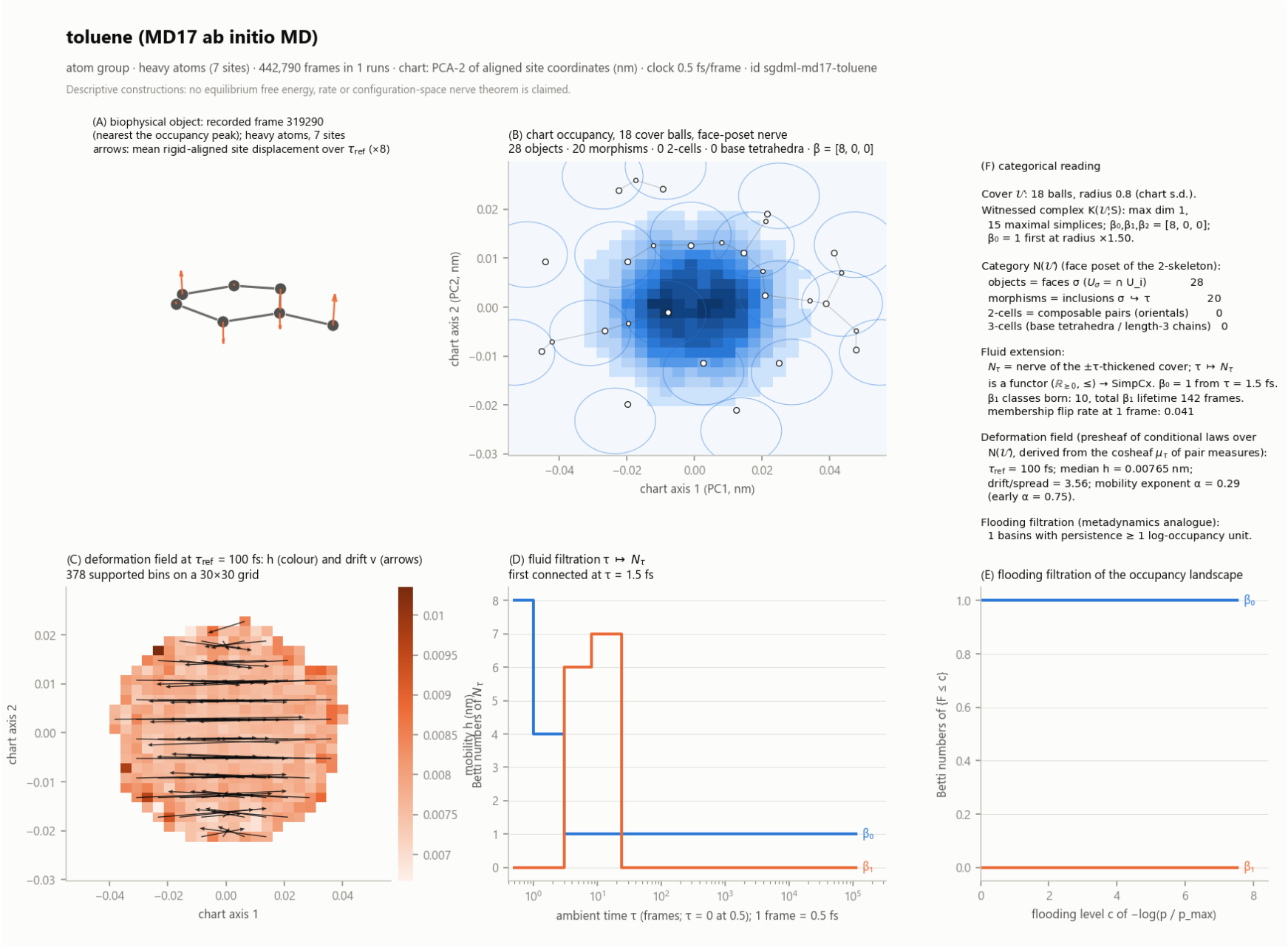

Figure 35: toluene (MD17 ab initio MD) (`sgdml-md17-toluene`). Source: https://www.quantum-machine.org/gdml/data/npz/md17_toluene.npz.

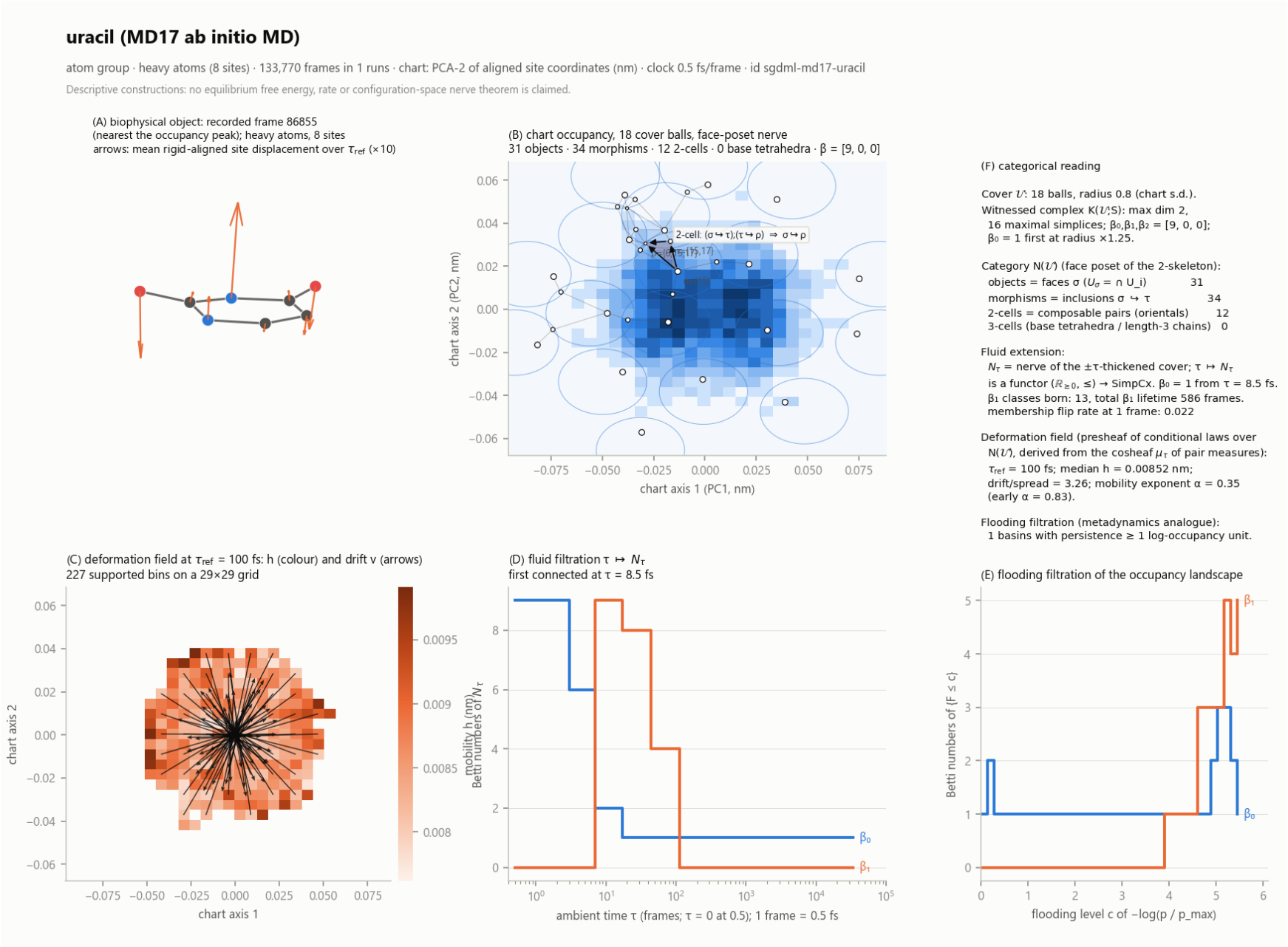

Figure 36: uracil (MD17 ab initio MD) (`sgdml-md17-uracil`). Source: https://www.quantum-machine.org/gdml/data/npz/md17_uracil.npz.

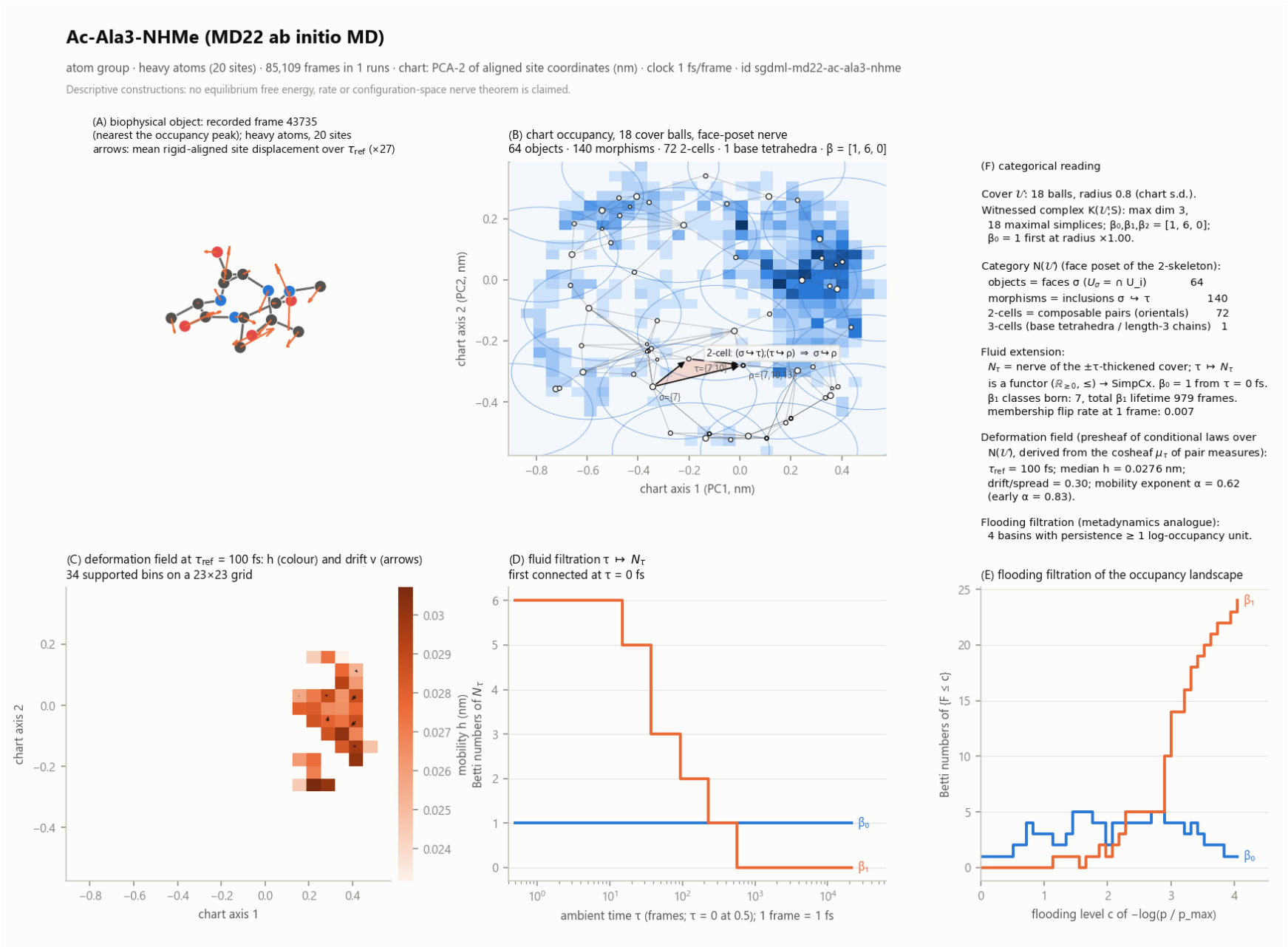

Figure 37: Ac-Ala3-NHMe (MD22 ab initio MD) (`sgdml-md22-ac-ala3-nhme`). Source: https://www.quantum-machine.org/gdml/data/npz/md22_Ac-Ala3-NHMe.npz.

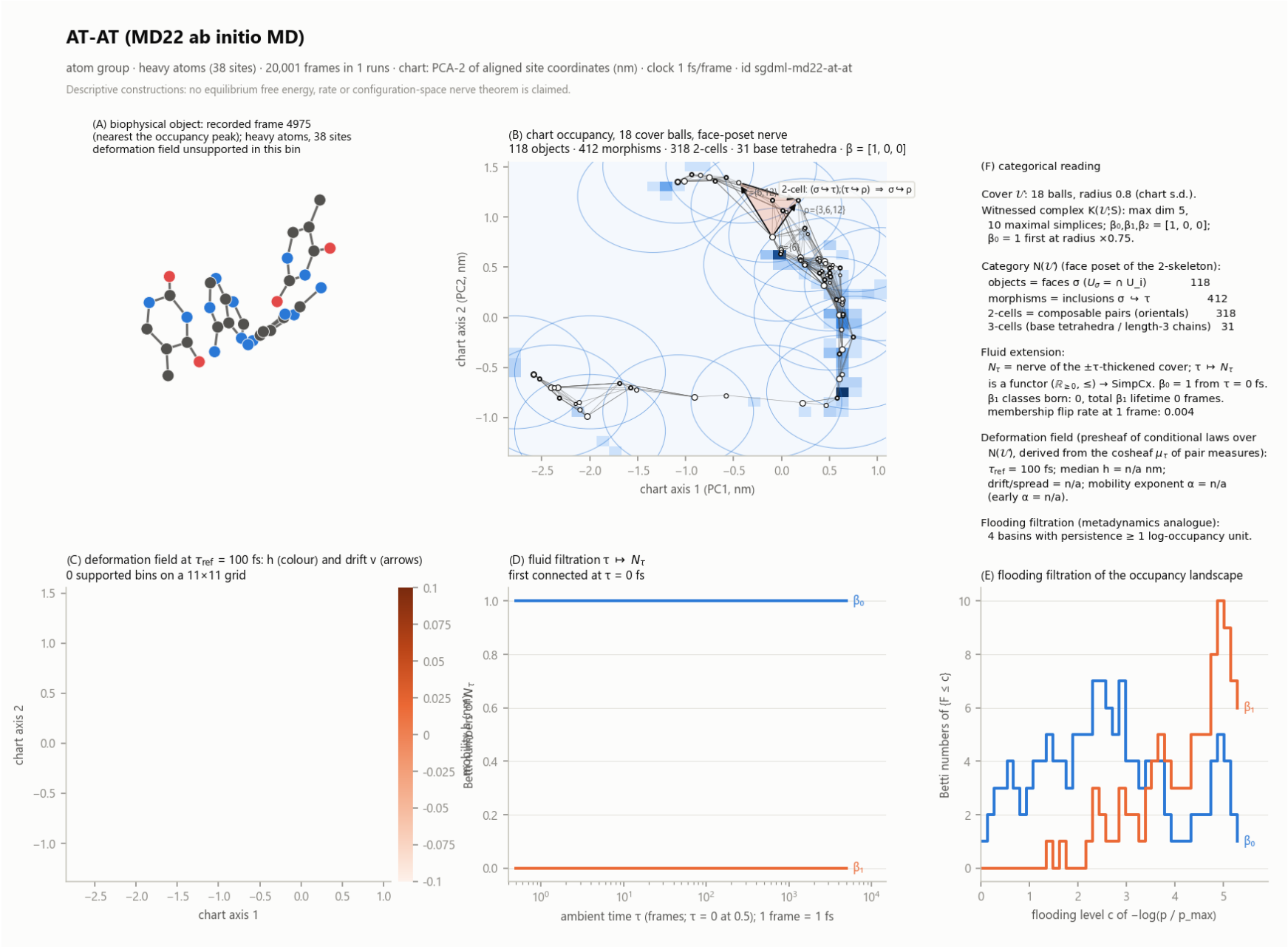

Figure 38: AT-AT (MD22 ab initio MD) (`sgdml-md22-at-at`). Source: https://www.quantum-machine.org/gdml/data/npz/md22_AT-AT.npz.

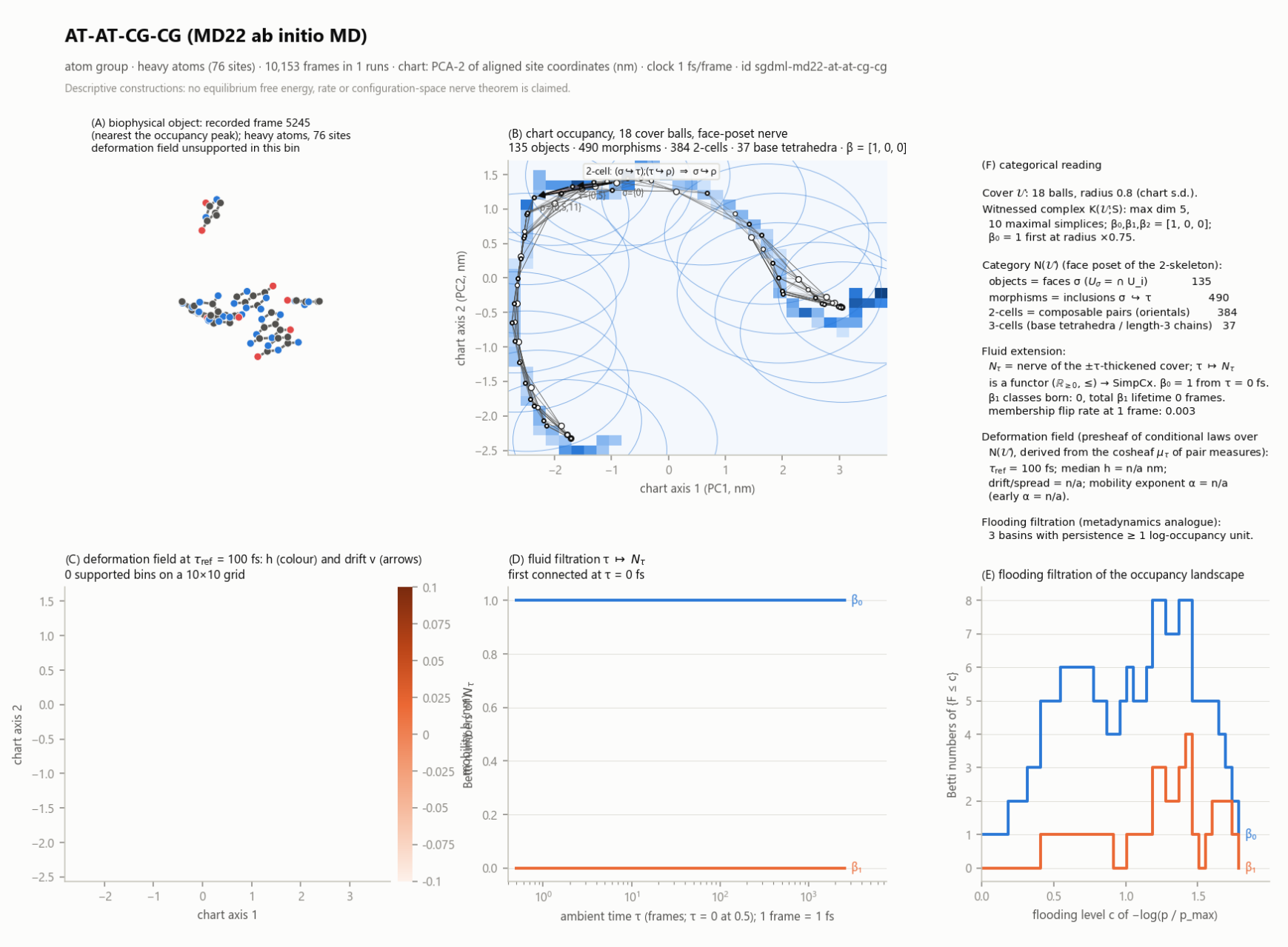

Figure 39: AT-AT-CG-CG (MD22 ab initio MD) (`sgdml-md22-at-at-cg-cg`). Source: https://www.quantum-machine.org/gdml/data/npz/md22_AT-AT-CG-CG.npz.

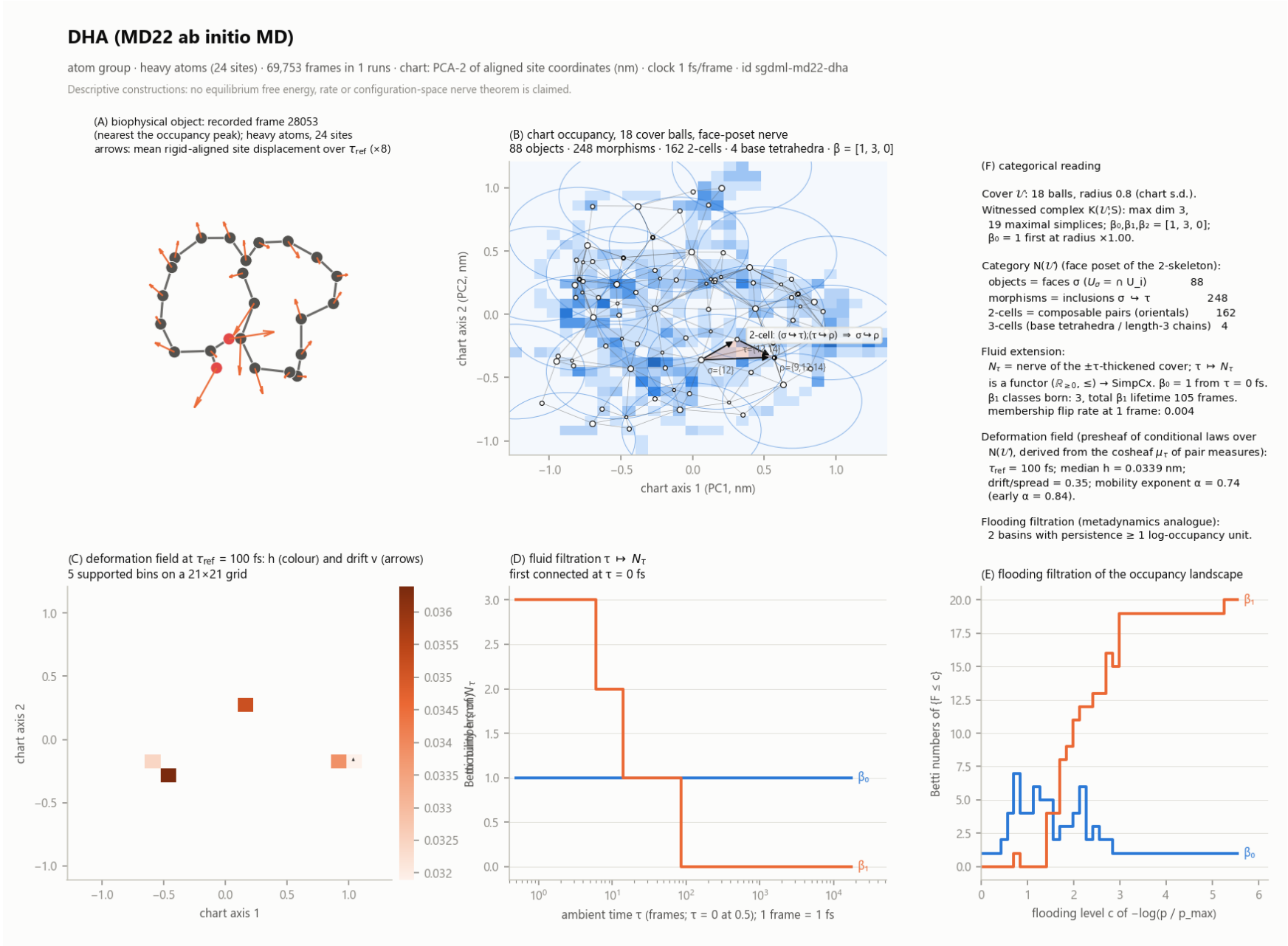

Figure 40: DHA (MD22 ab initio MD) (`sgdml-md22-dha`). Source: https://www.quantum-machine.org/gdml/data/npz/md22_DHA.npz.

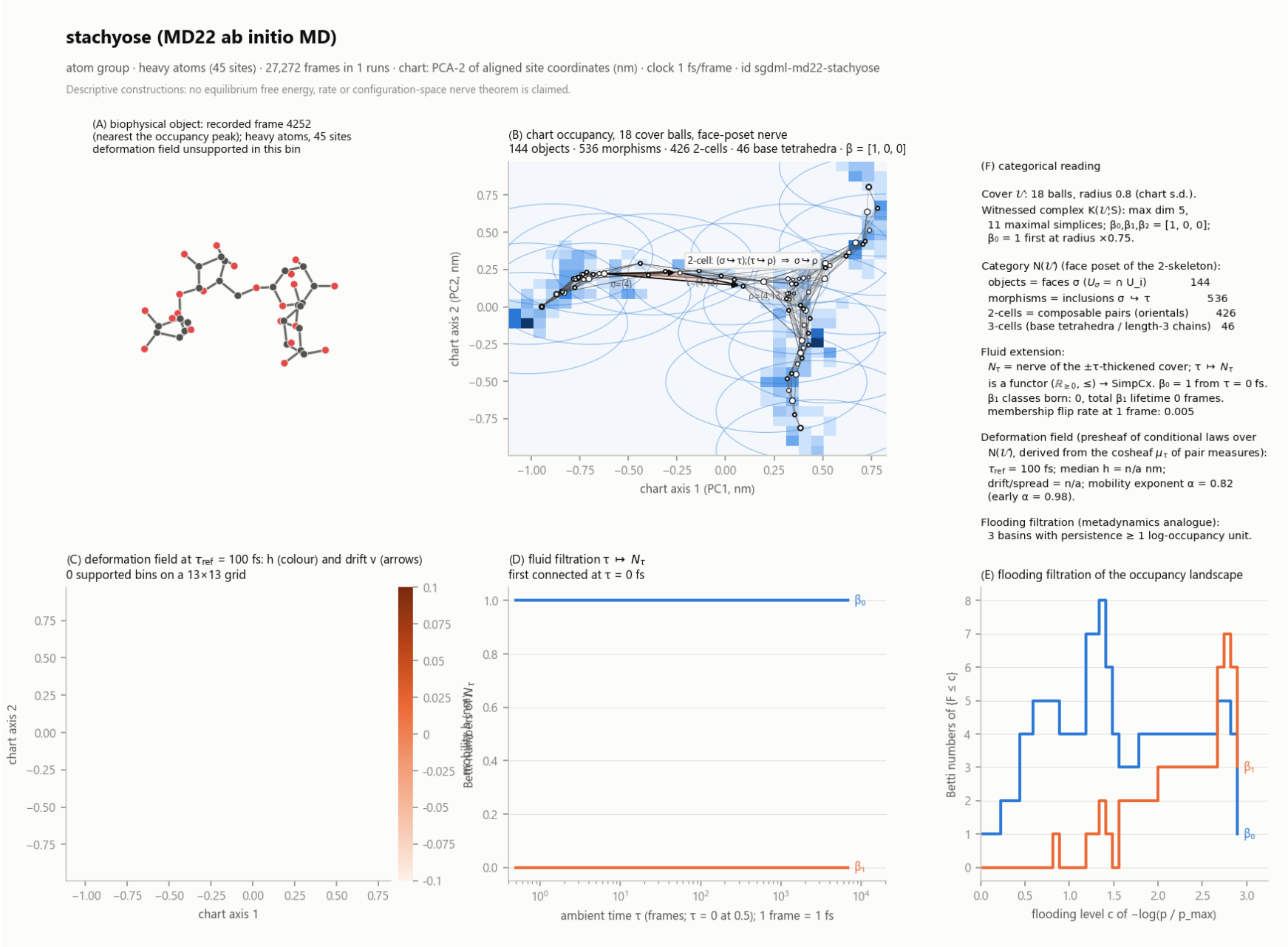

Figure 41: stachyose (MD22 ab initio MD) (`sgdml-md22-stachyose`). Source: https://www.quantum-machine.org/gdml/data/npz/md22_stachyose.npz.

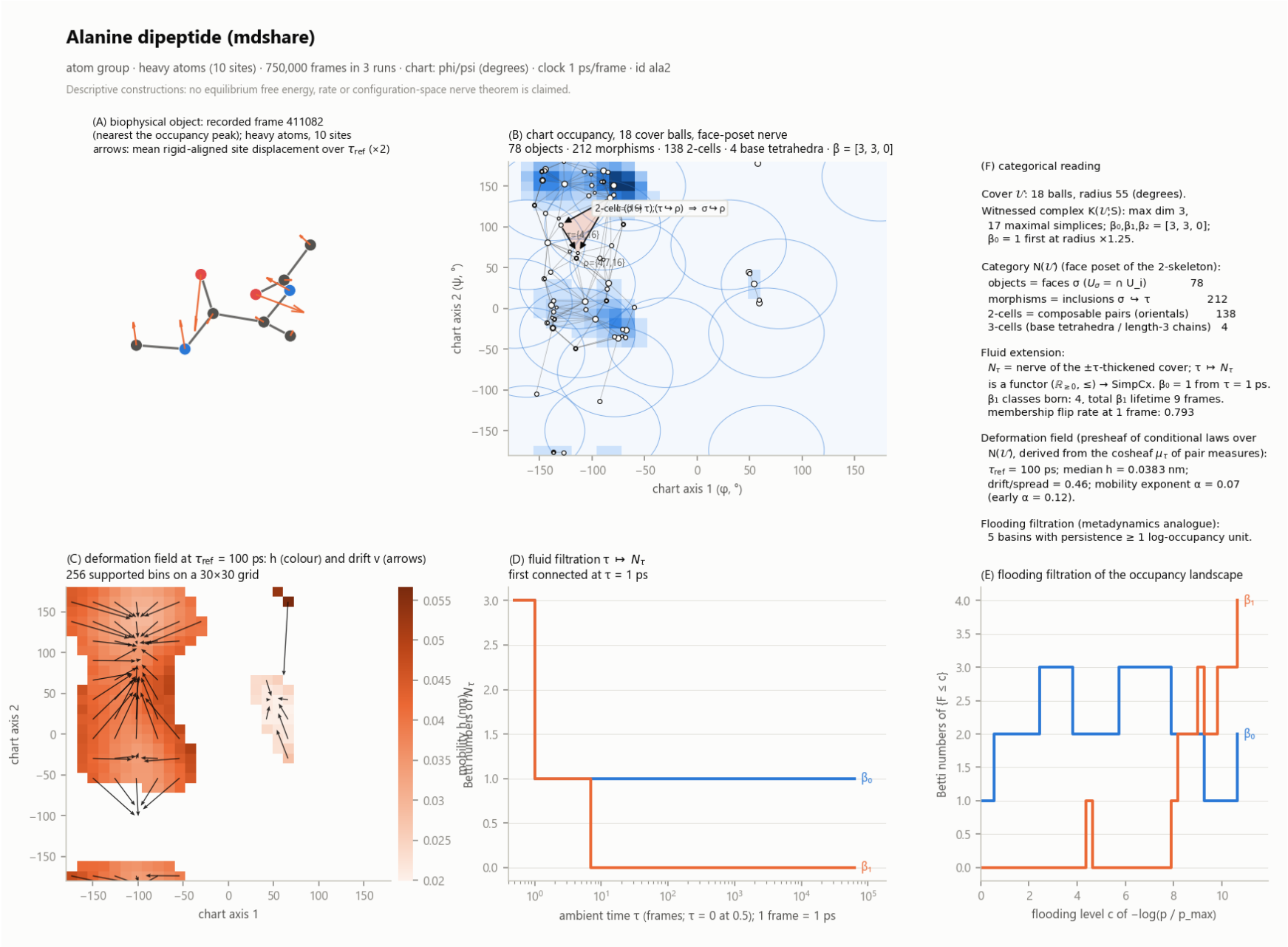

Figure 42: Alanine dipeptide (mdshare) (`ala2`). Source: https://markovmodel.github.io/mdshare/ALA2/.

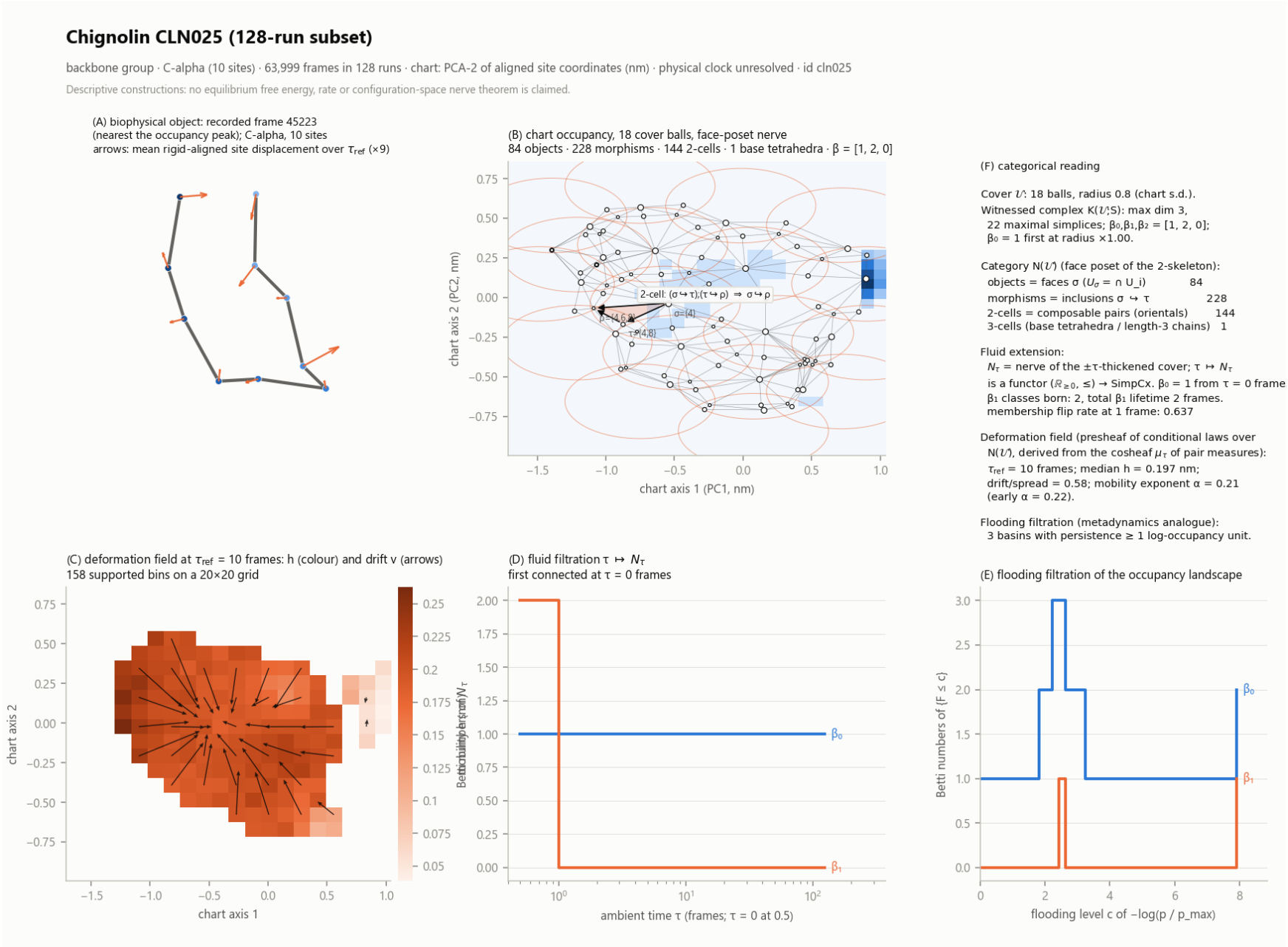

Figure 43: Chignolin CLN025 (128-run subset) (`cln025`). Source: https://ndownloader.figshare.com/files/26424614.

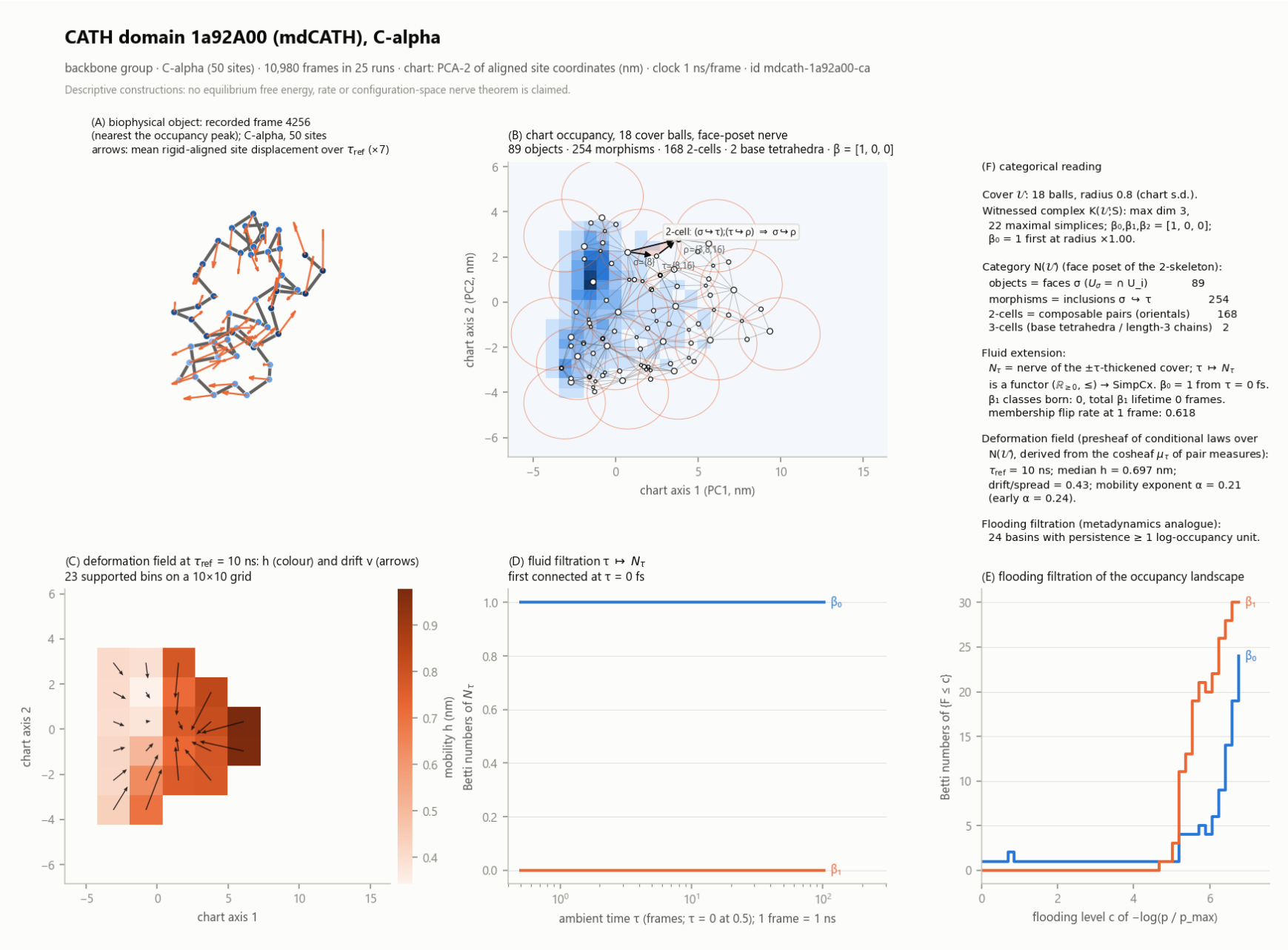

Figure 44: CATH domain 1a92A00 (mdCATH), C-alpha (`mdcath-1a92a00-ca`). Source: https://huggingface.co/datasets/compsciencelab/mdCATH/resolve/main/data/mdcath_dataset_1a92A00.h5.

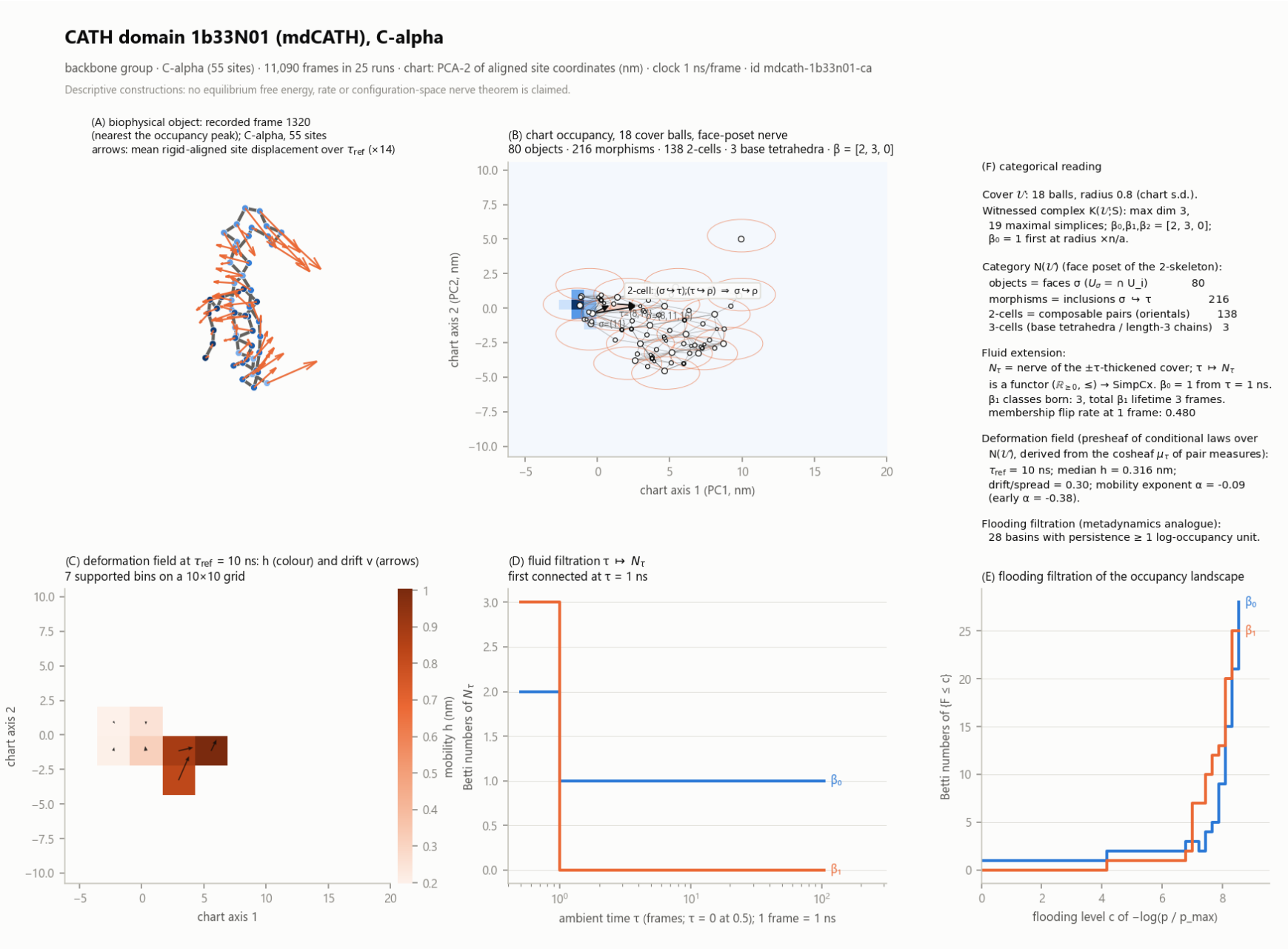

Figure 45: CATH domain 1b33N01 (mdCATH), C-alpha (`mdcath-1b33n01-ca`). Source: https://huggingface.co/datasets/compsciencelab/mdCATH/resolve/main/data/mdcath_dataset_1b33N01.h5.

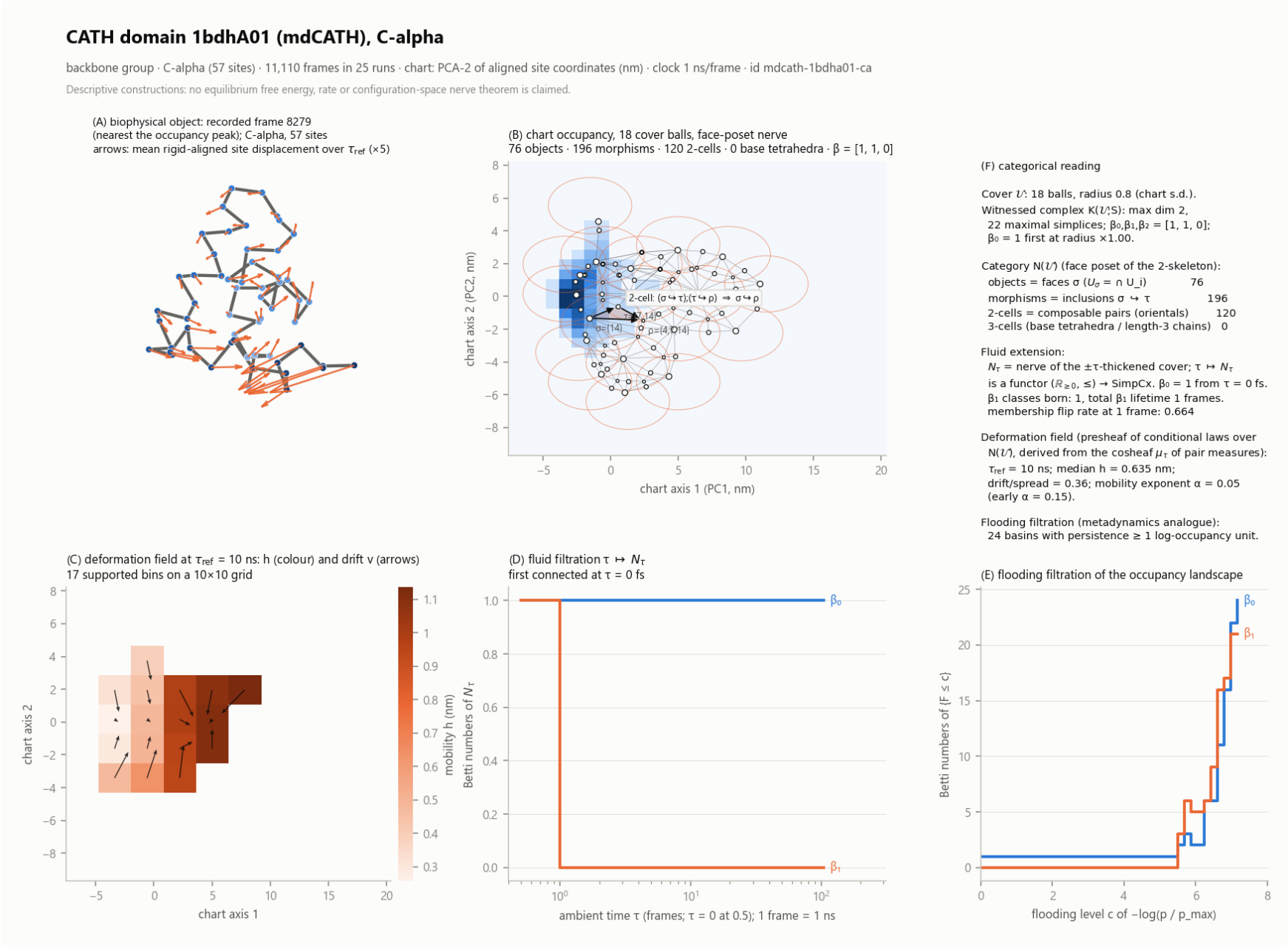

Figure 46: CATH domain 1bdhA01 (mdCATH), C-alpha (`mdcath-1bdha01-ca`). Source: https://huggingface.co/datasets/compsciencelab/mdCATH/resolve/main/data/mdcath_dataset_1bdhA01.h5

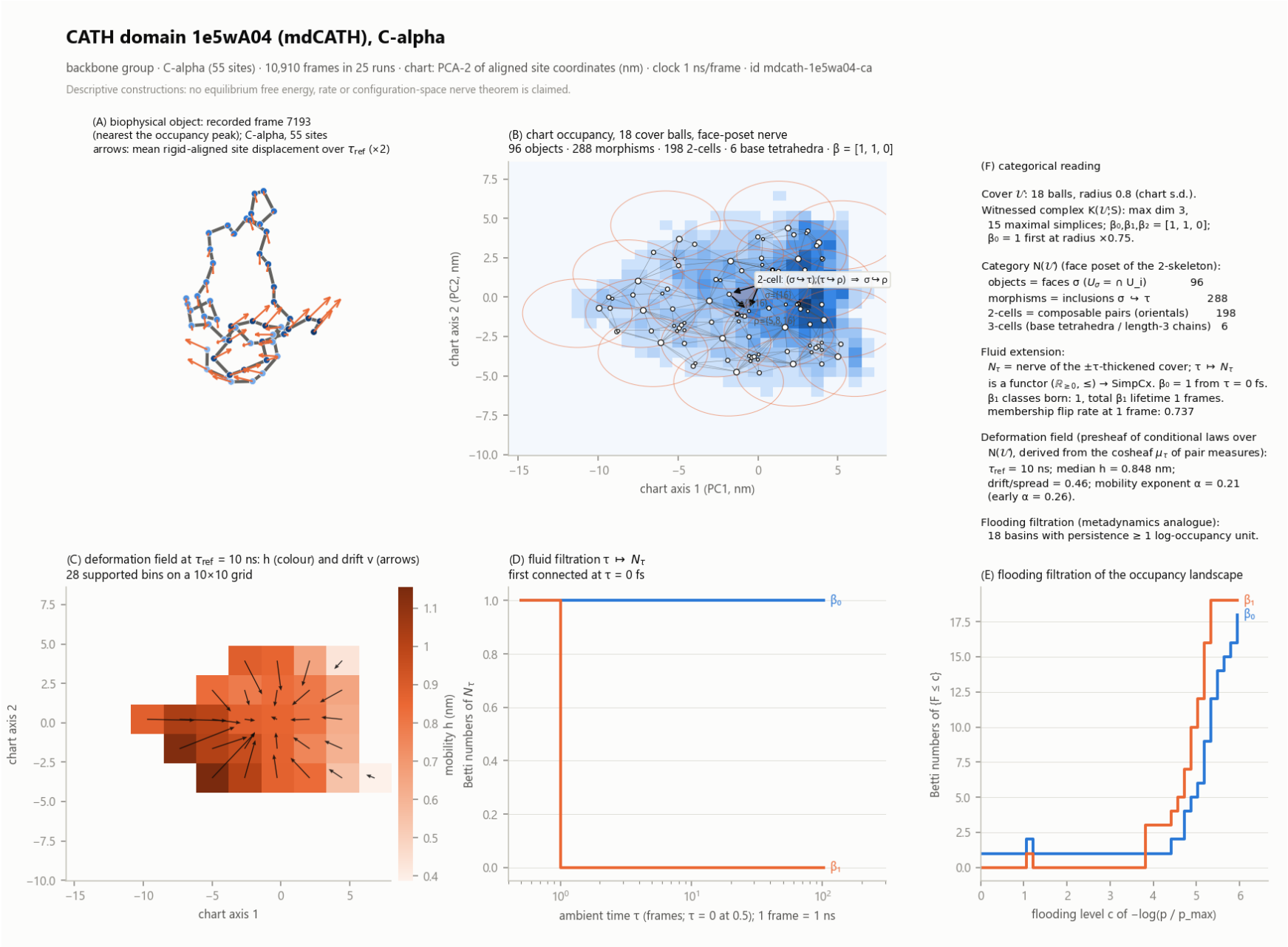

Figure 47: CATH domain 1e5wA04 (mdCATH), C-alpha (`mdcath-1e5wa04-ca`). Source: https://huggingface.co/datasets/compsciencelab/mdCATH/resolve/main/data/mdcath_dataset_1e5wA04.h5

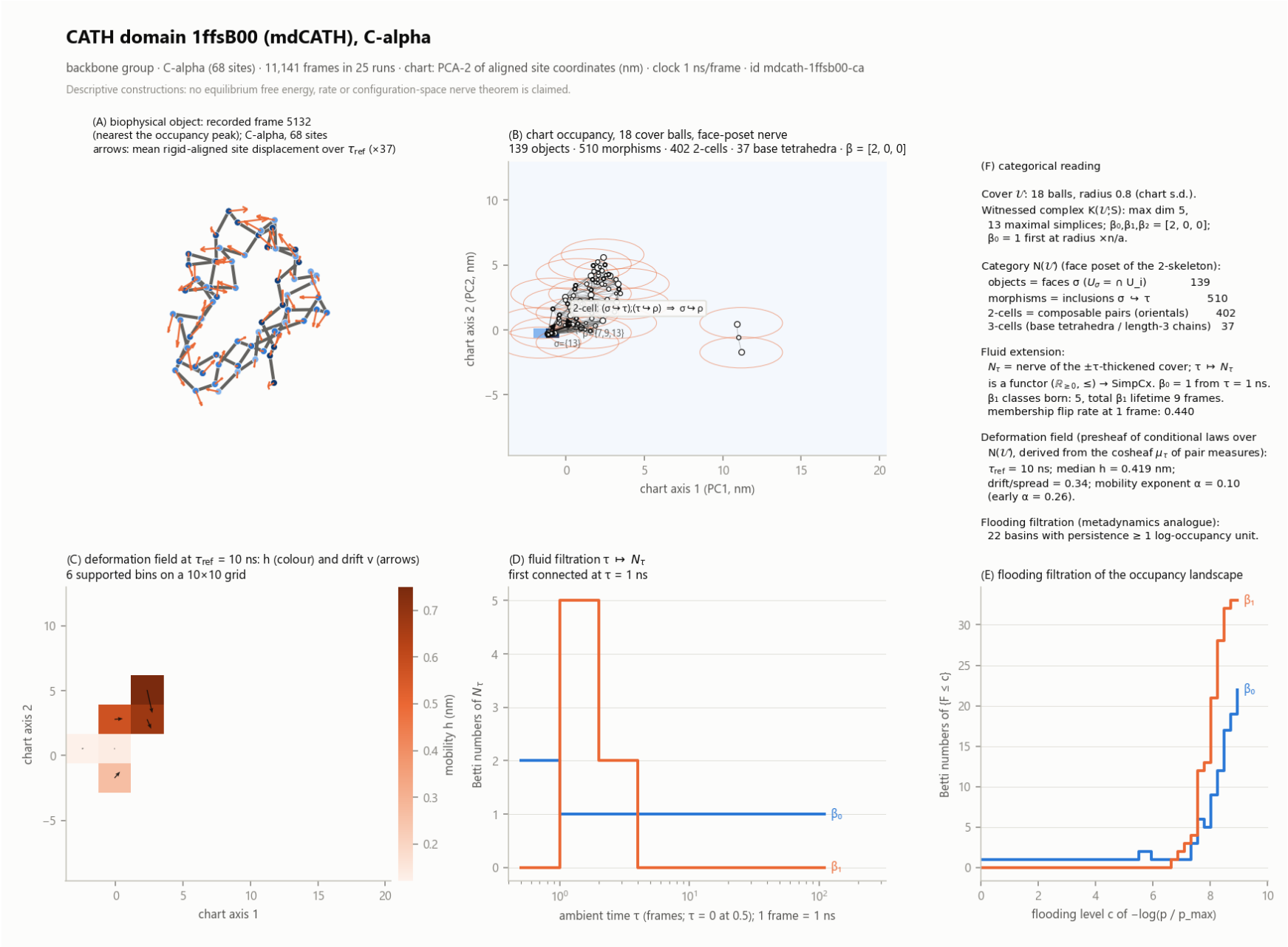

Figure 48: CATH domain 1ffsB00 (mdCATH), C-alpha (`mdcath-1ffsb00-ca`). Source: https://huggingface.co/datasets/compsciencelab/mdCATH/resolve/main/data/mdcath_dataset_1ffsB00.h5.

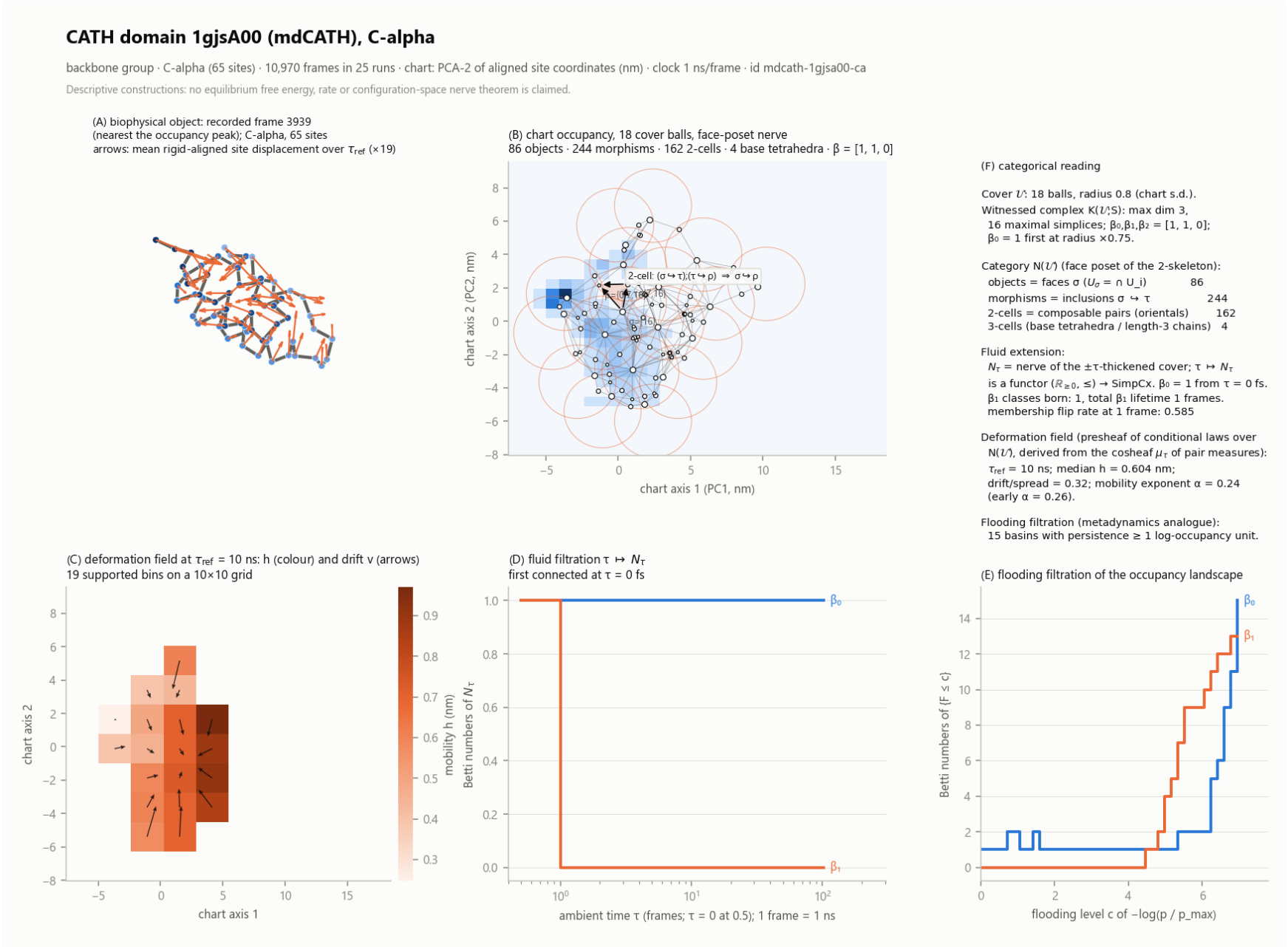

Figure 49: CATH domain 1gjsA00 (mdCATH), C-alpha (`mdcath-1gjsa00-ca`). Source: https://huggingface.co/datasets/compsciencelab/mdCATH/resolve/main/data/mdcath_dataset_1gjsA00.h5.

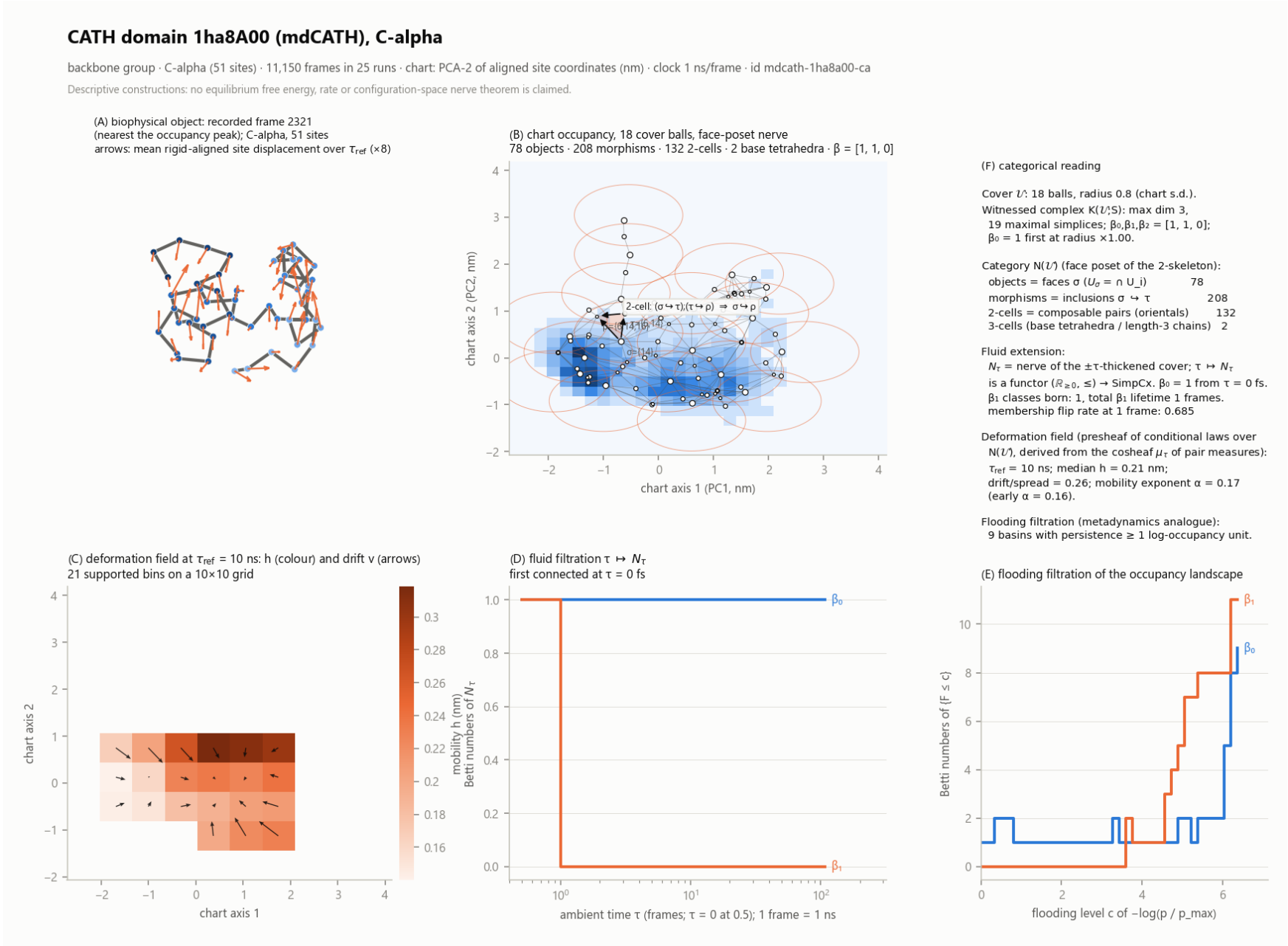

Figure 50: CATH domain 1ha8A00 (mdCATH), C-alpha (`mdcath-1ha8a00-ca`). Source: https://huggingface.co/datasets/compsciencelab/mdCATH/resolve/main/data/mdcath_dataset_1ha8A00.h5.

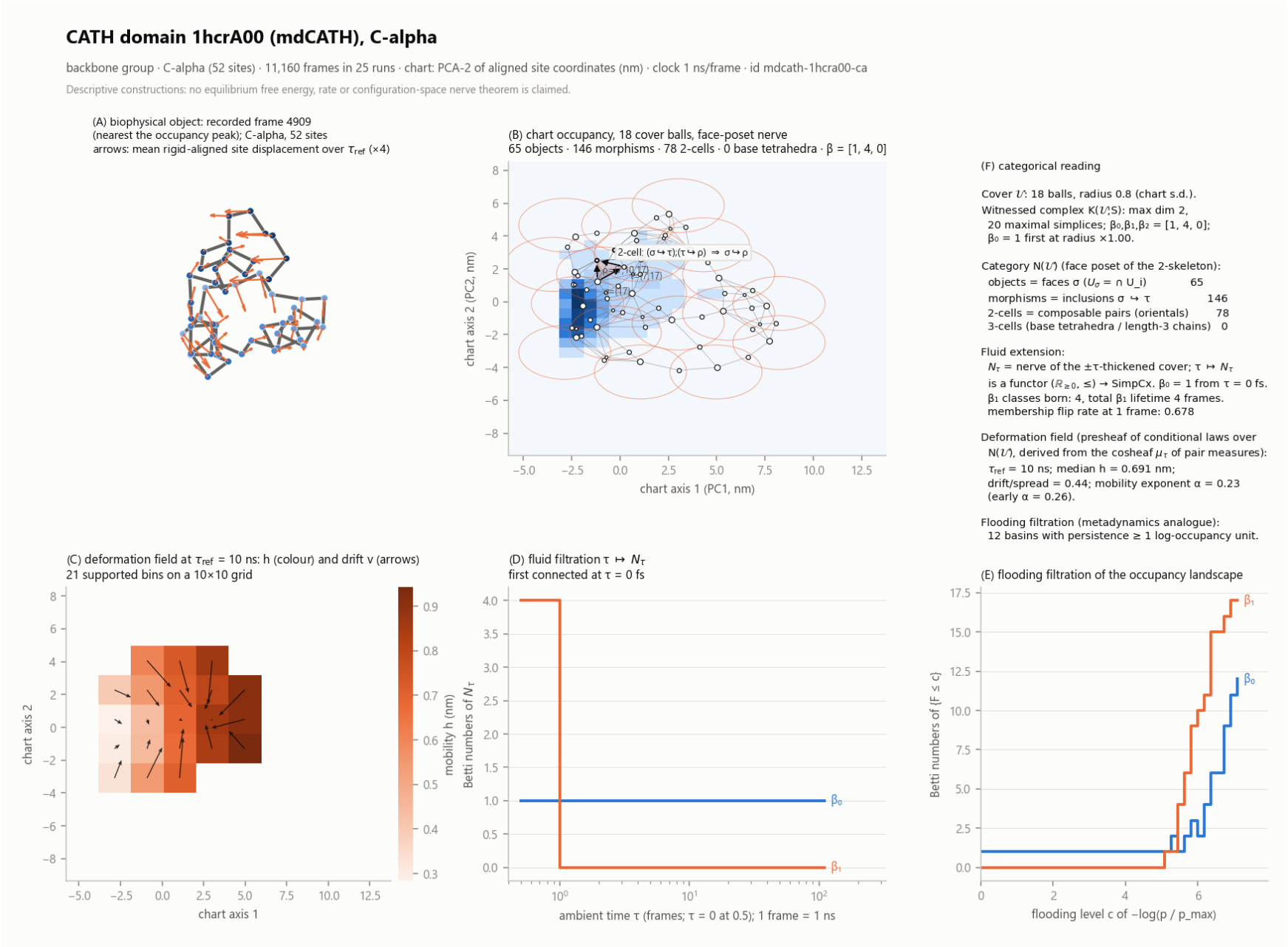

Figure 51: CATH domain 1hcrA00 (mdCATH), C-alpha (`mdcath-1hcra00-ca`). Source: https://huggingface.co/datasets/compsciencelab/mdCATH/resolve/main/data/mdcath_dataset_1hcrA00.h5.

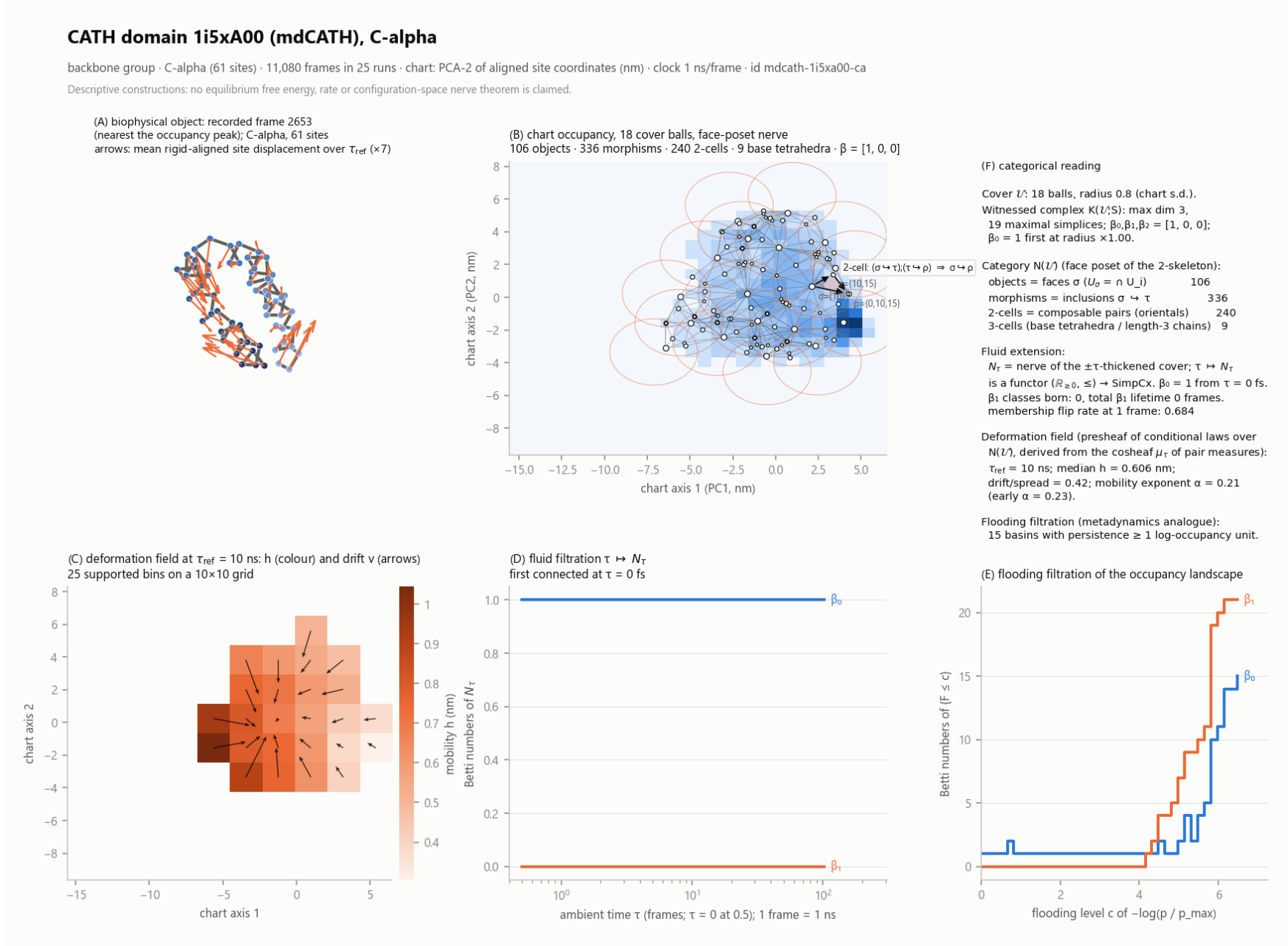

Figure 52: CATH domain 1i5xA00 (mdCATH), C-alpha (`mdcath-1i5xa00-ca`). Source: https://huggingface.co/datasets/compsciencelab/mdCATH/resolve/main/data/mdcath_dataset_1i5xA00.h5.

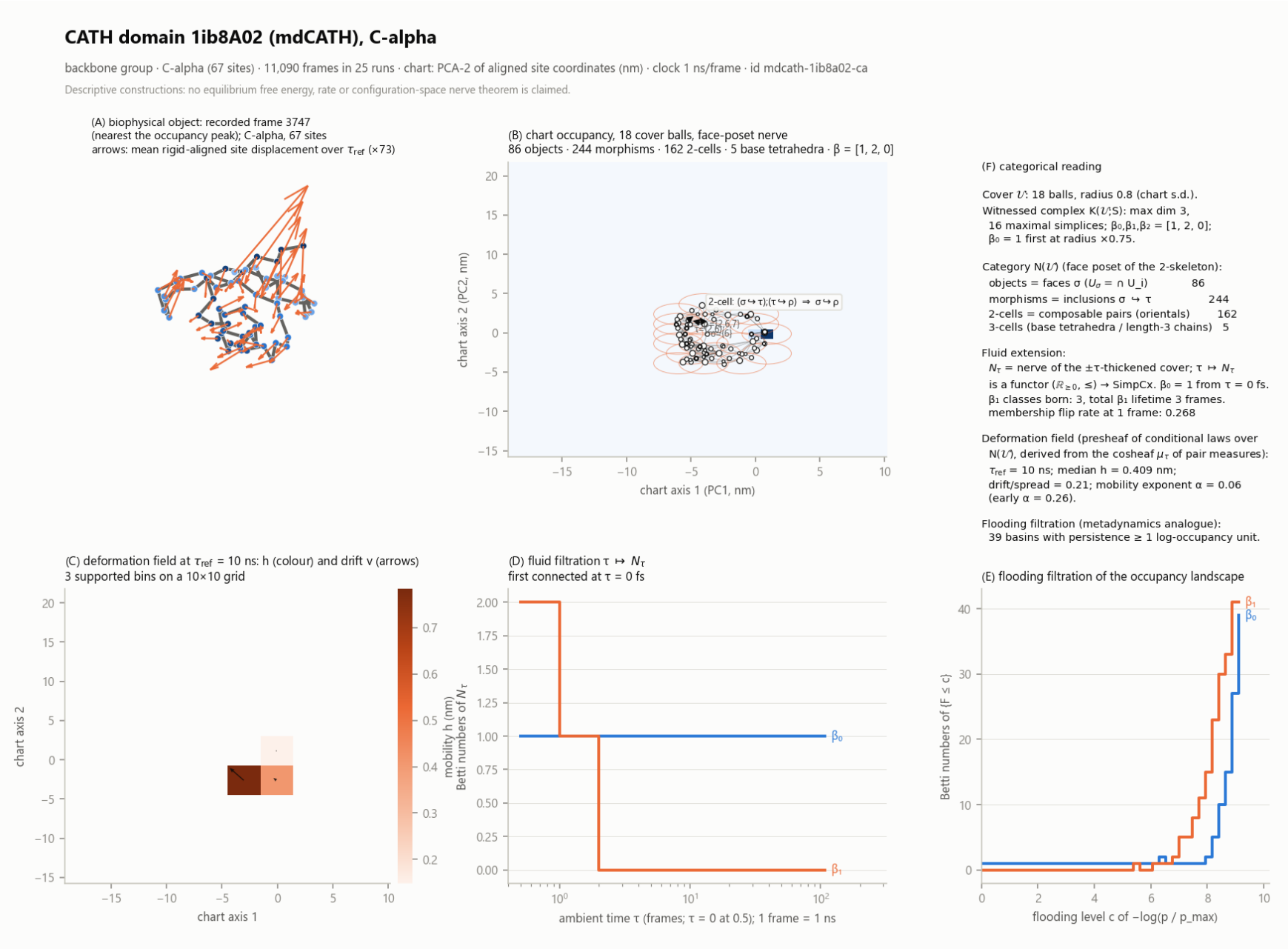

Figure 53: CATH domain 1ib8A02 (mdCATH), C-alpha (`mdcath-1ib8a02-ca`). Source: https://huggingface.co/datasets/compsciencelab/mdCATH/resolve/main/data/mdcath_dataset_1ib8A02.h5.

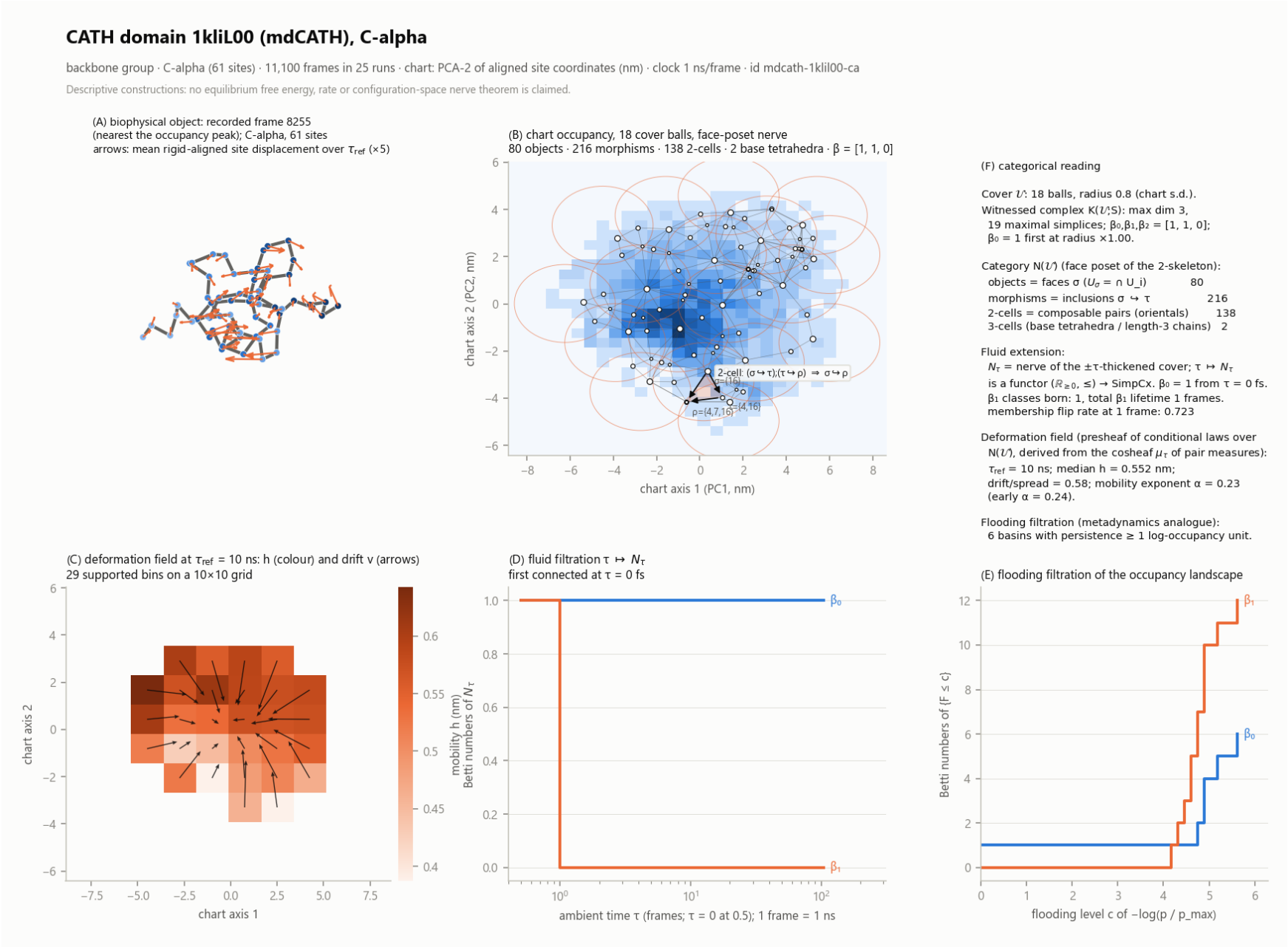

Figure 54: CATH domain 1kliL00 (mdCATH), C-alpha (`mdcath-1klil00-ca`). Source: https://huggingface.co/datasets/compsciencelab/mdCATH/resolve/main/data/mdcath_dataset_1kliL00.h5.

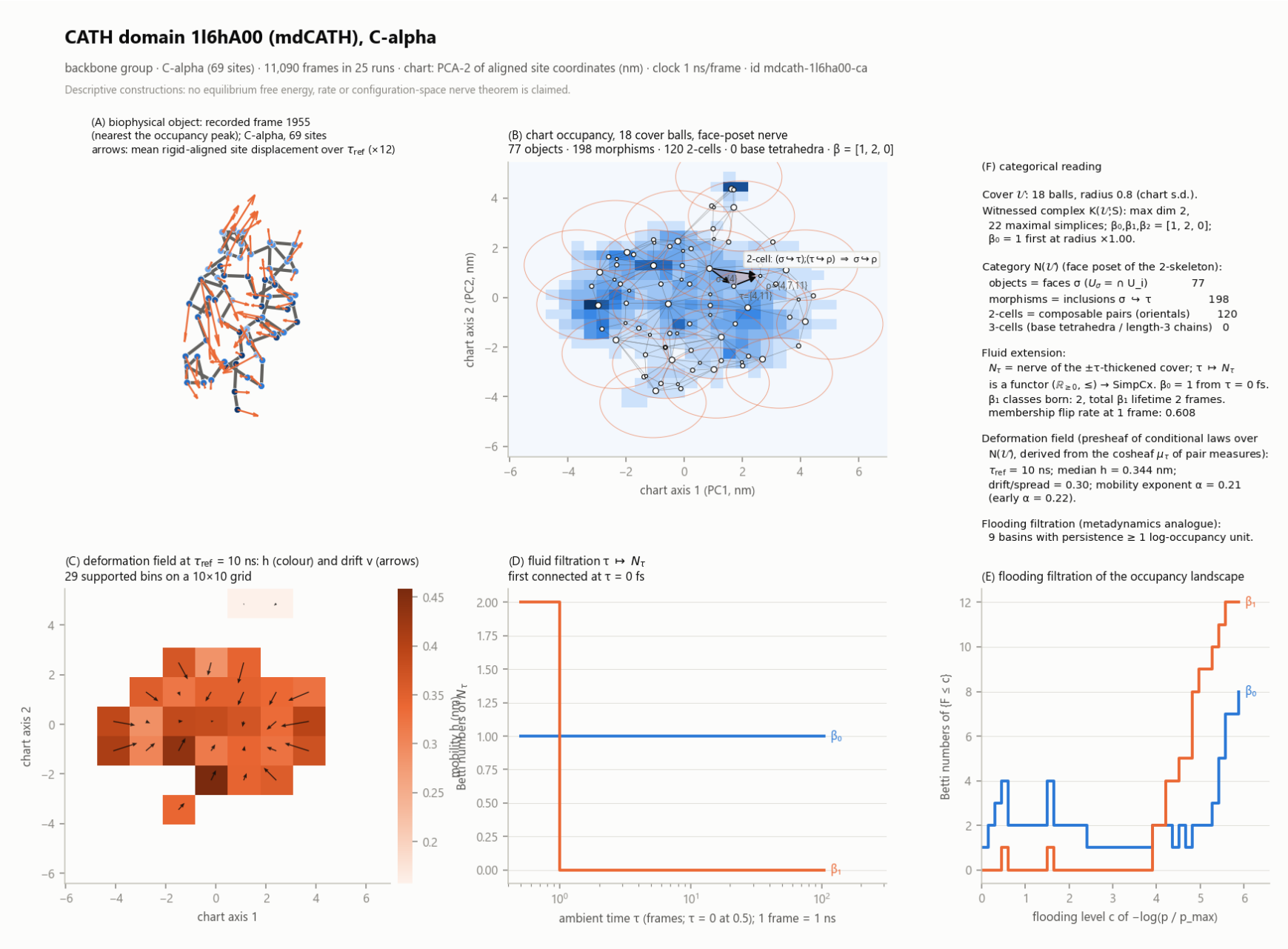

Figure 55: CATH domain 1l6hA00 (mdCATH), C-alpha (`mdcath-1l6ha00-ca`). Source: https://huggingface.co/datasets/compsciencelab/mdCATH/resolve/main/data/mdcath_dataset_1l6hA00.h5.

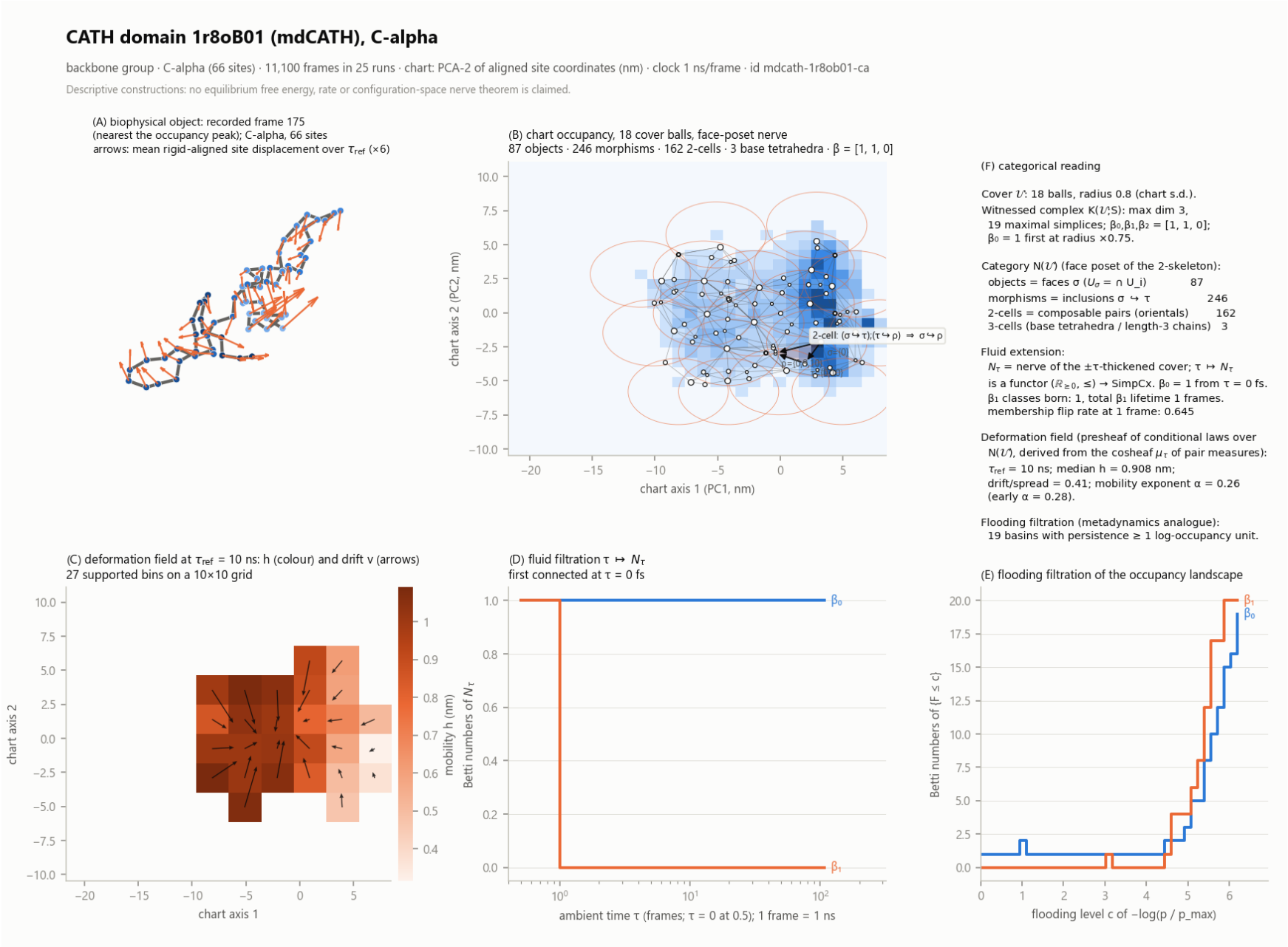

Figure 56: CATH domain 1r8oB01 (mdCATH), C-alpha (`mdcath-1r8ob01-ca`). Source: https://huggingface.co/datasets/compsciencelab/mdCATH/resolve/main/data/mdcath_dataset_1r8oB01.h5.

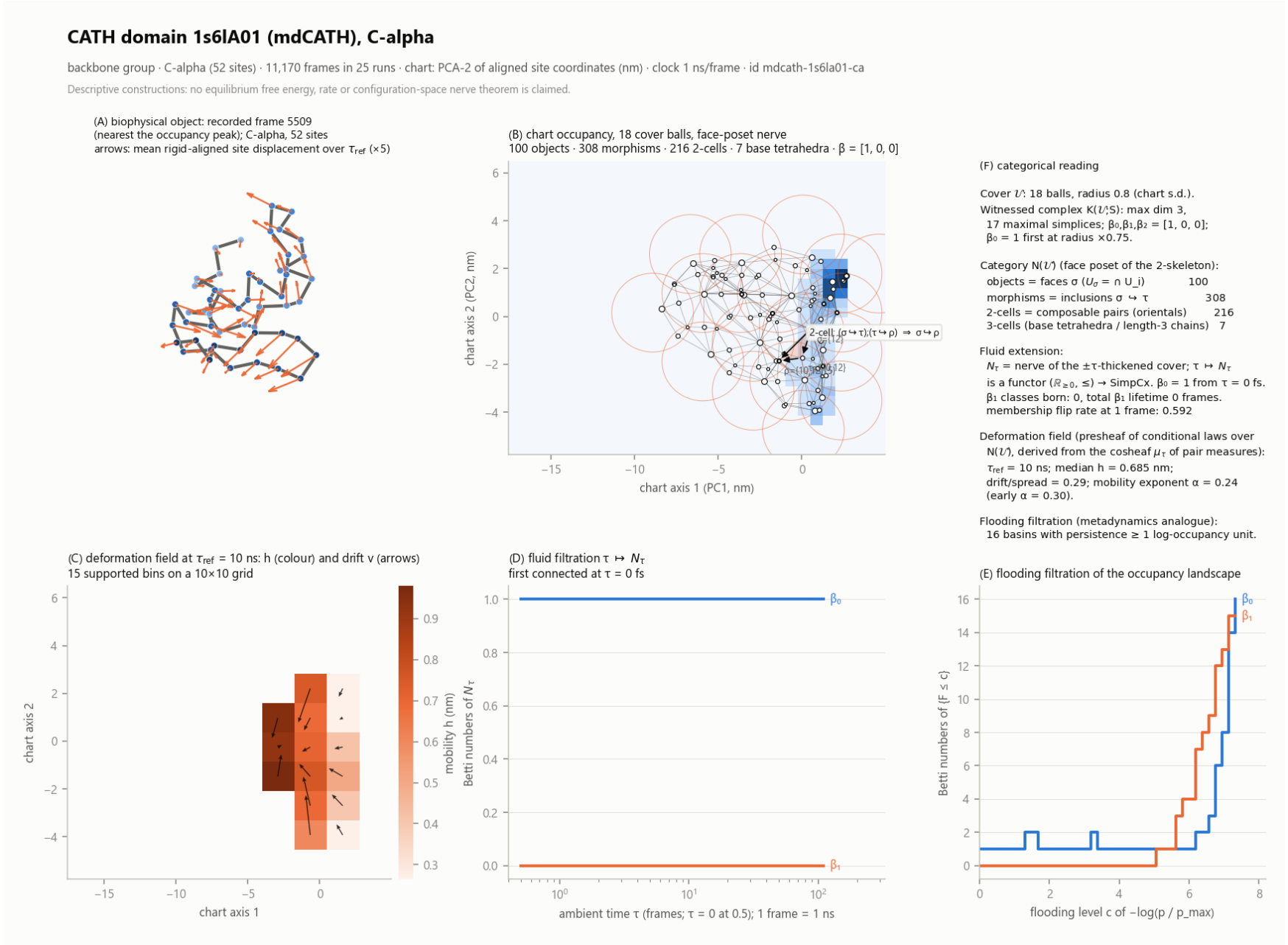

Figure 57: CATH domain 1s6lA01 (mdCATH), C-alpha (`mdcath-1s6la01-ca`). Source: https://huggingface.co/datasets/compsciencelab/mdCATH/resolve/main/data/mdcath_dataset_1s6lA01.h5.

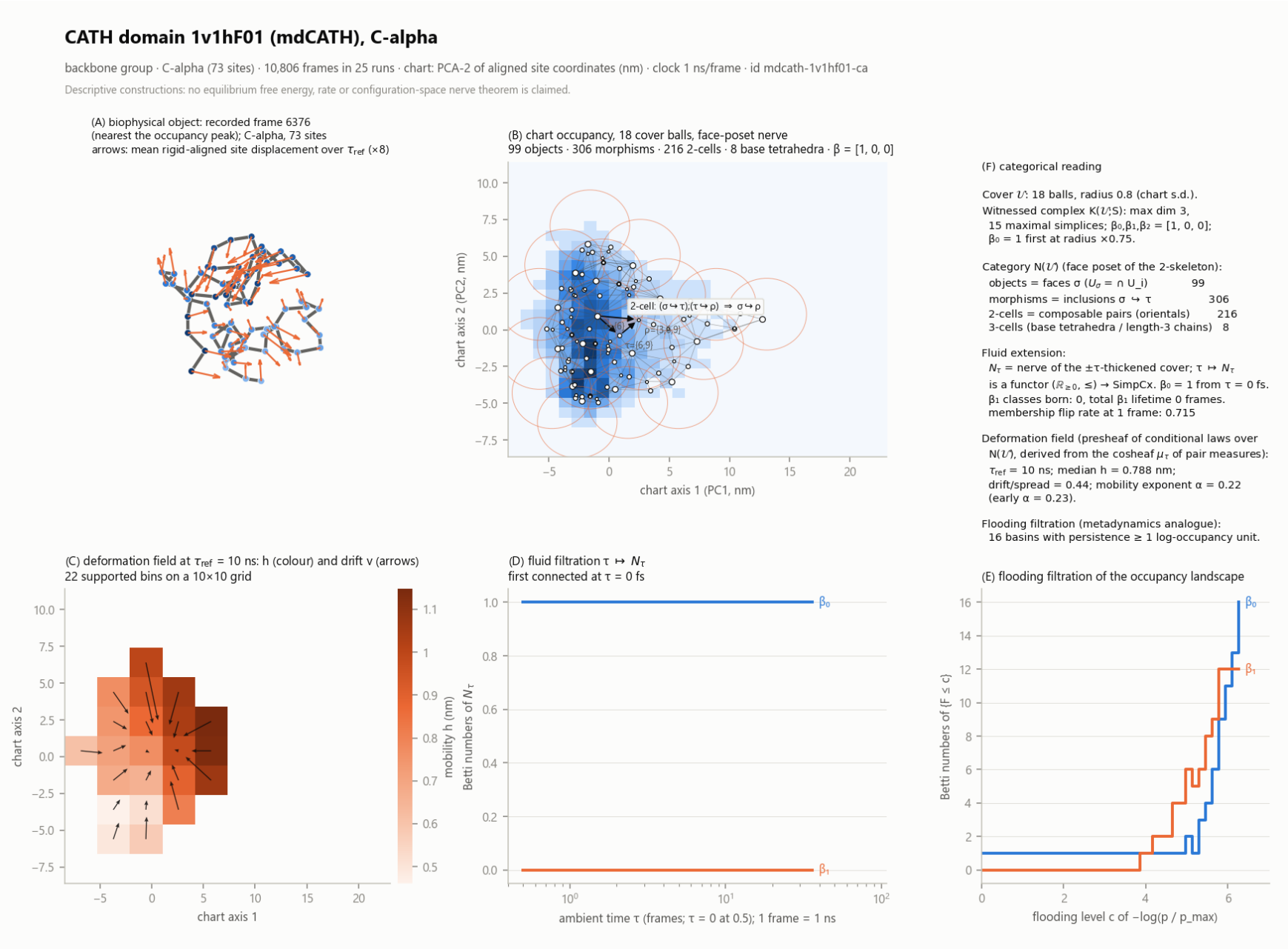

Figure 58: CATH domain 1v1hF01 (mdCATH), C-alpha (`mdcath-1v1hf01-ca`). Source: https://huggingface.co/datasets/compsciencelab/mdCATH/resolve/main/data/mdcath_dataset_1v1hF01.h5.

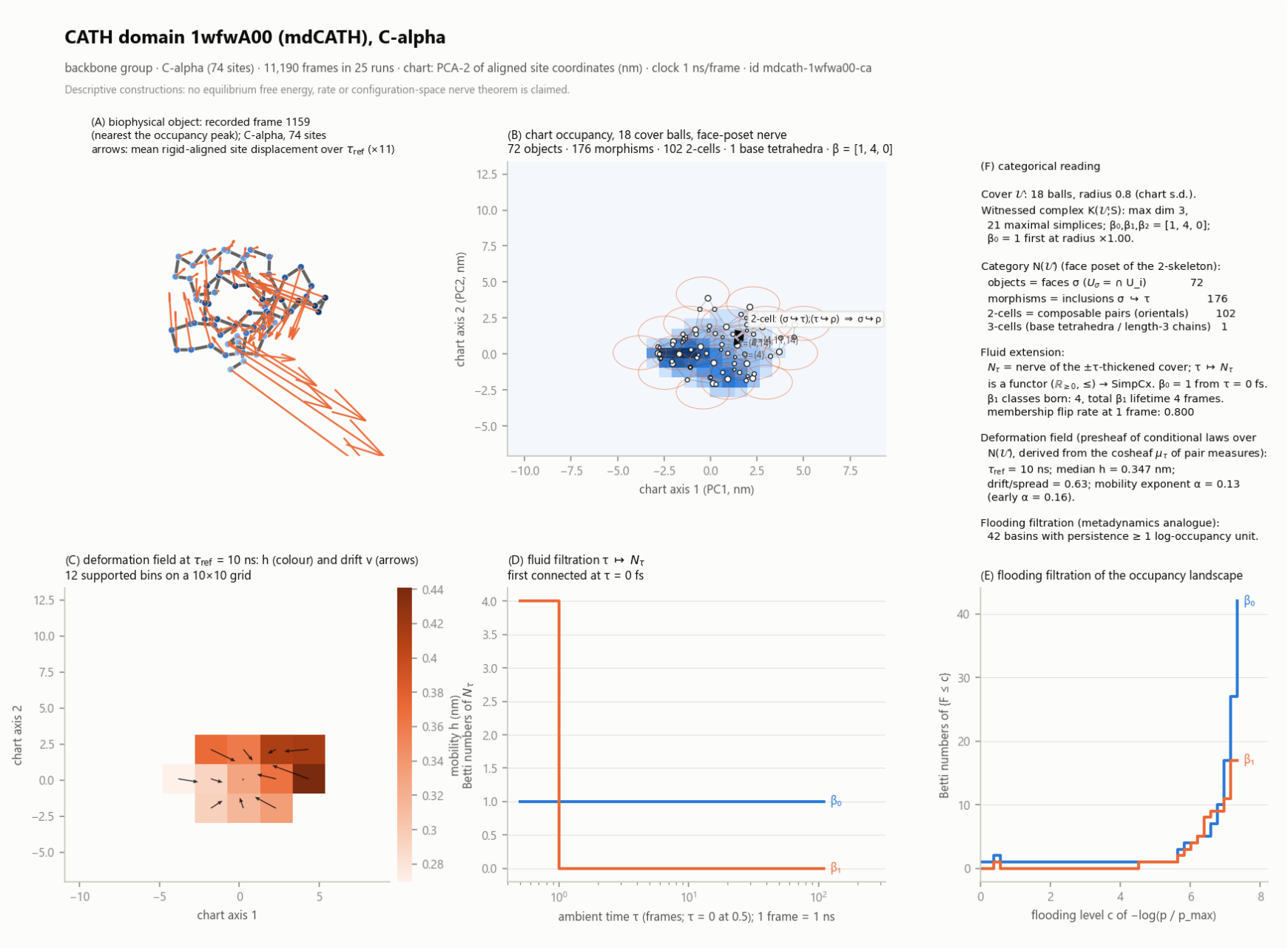

Figure 59: CATH domain 1wfwA00 (mdCATH), C-alpha (`mdcath-1wfwa00-ca`). Source: https://huggingface.co/datasets/compsciencelab/mdCATH/resolve/main/data/mdcath_dataset_1wfwA00.h5

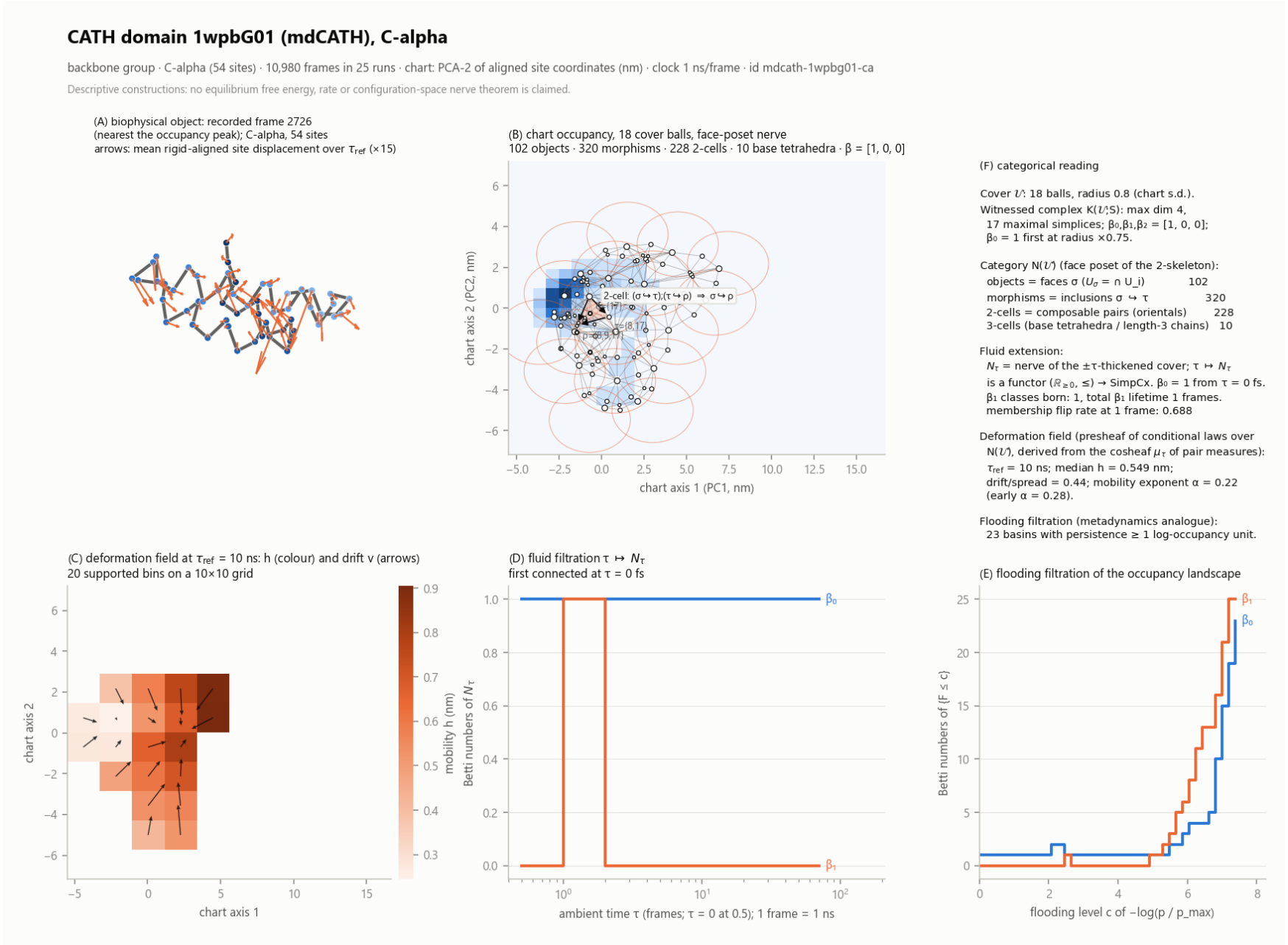

Figure 60: CATH domain 1wpbG01 (mdCATH), C-alpha (`mdcath-1wpbg01-ca`). Source: https://huggingface.co/datasets/compsciencelab/mdCATH/resolve/main/data/mdcath_dataset_1wpbG01.h

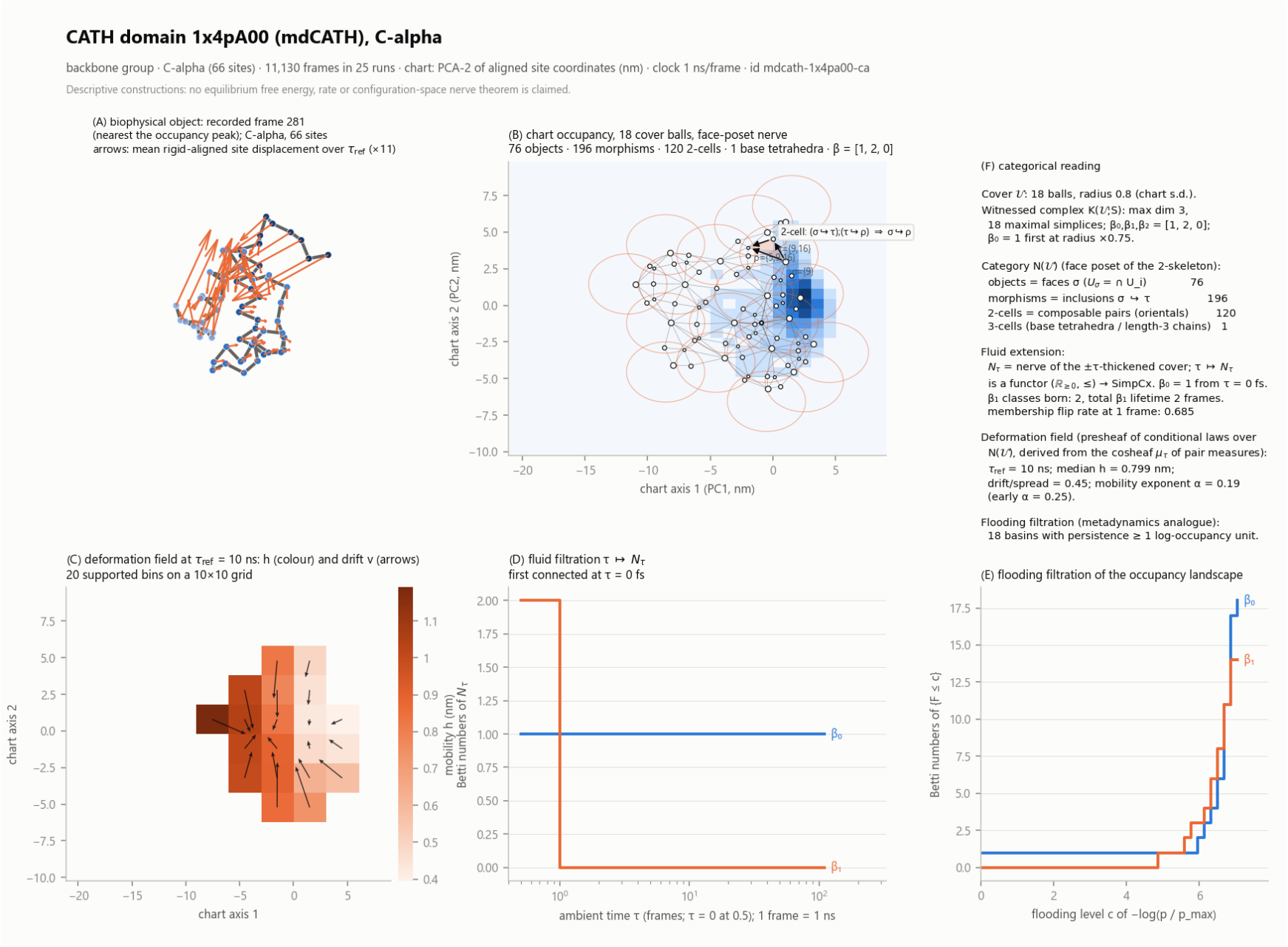

Figure 61: CATH domain 1x4pA00 (mdCATH), C-alpha (`mdcath-1x4pa00-ca`). Source: https://huggingface.co/datasets/compsciencelab/mdCATH/resolve/main/data/mdcath_dataset_1x4pA00.h5

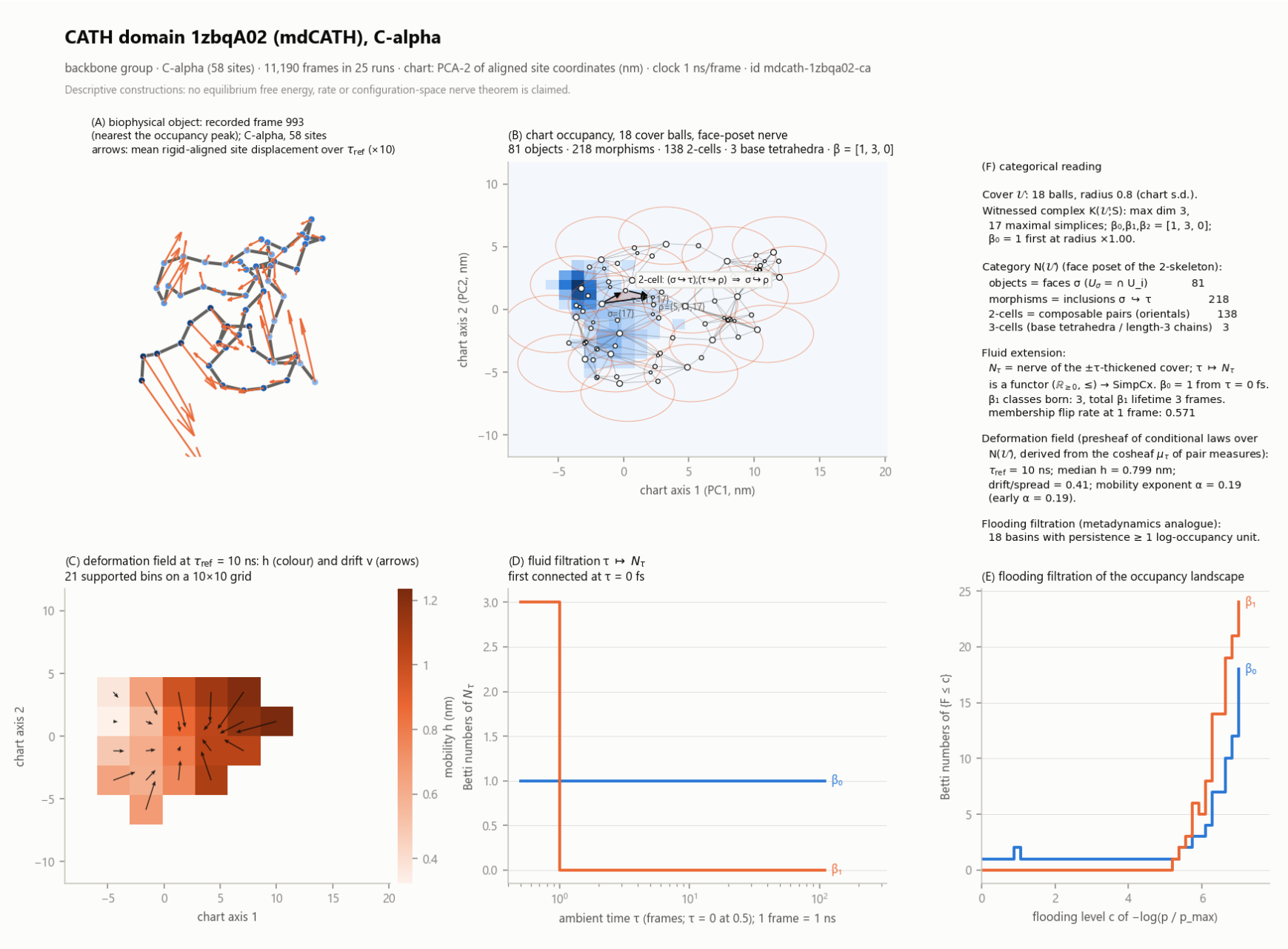

Figure 62: CATH domain 1zbqA02 (mdCATH), C-alpha (`mdcath-1zbqa02-ca`). Source: https://huggingface.co/datasets/compsciencelab/mdCATH/resolve/main/data/mdcath_dataset_1zbqA02.h5.

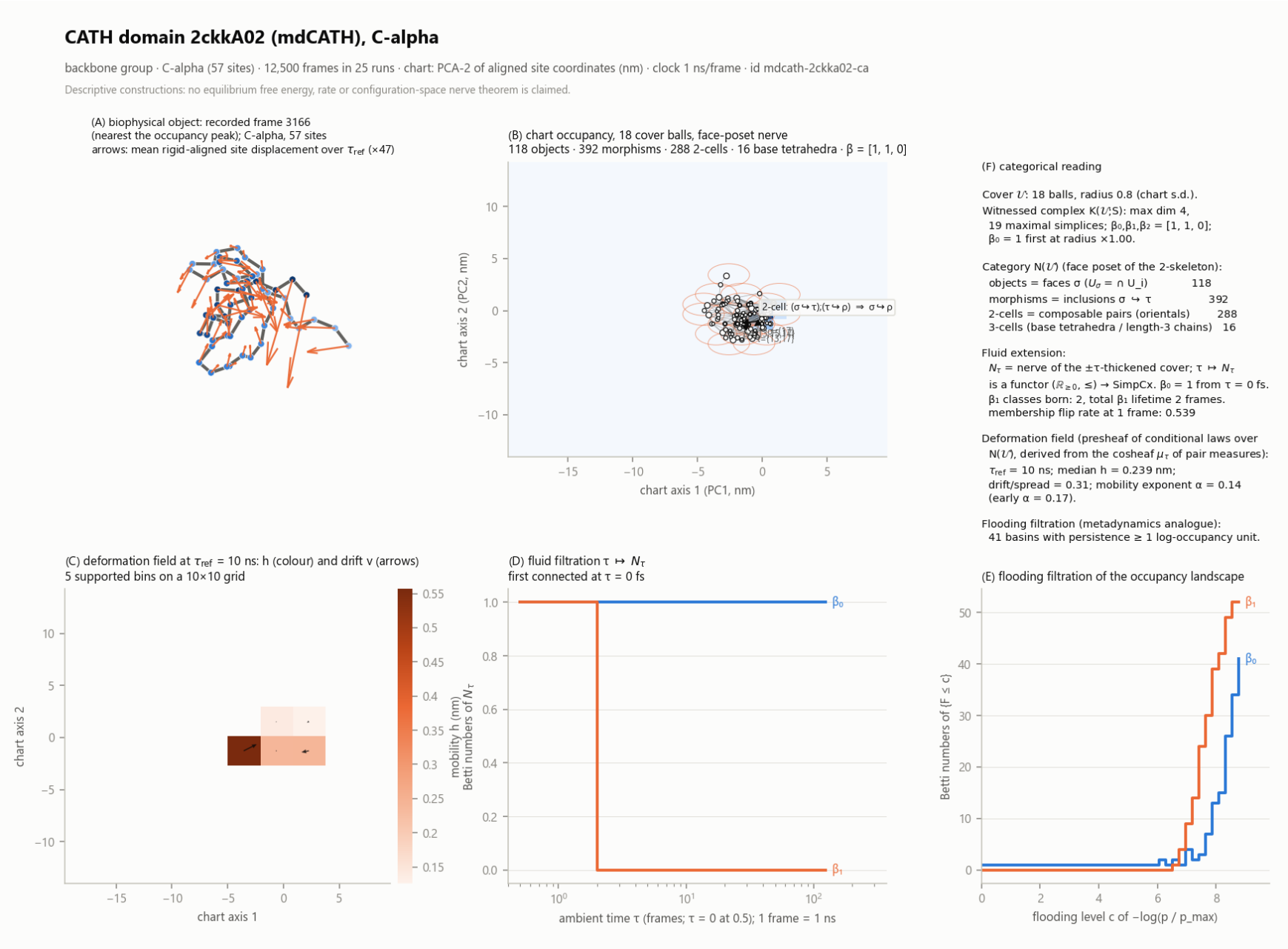

Figure 63: CATH domain 2ckkA02 (mdCATH), C-alpha (`mdcath-2ckka02-ca`). Source: https://huggingface.co/datasets/compsciencelab/mdCATH/resolve/main/data/mdcath_dataset_2ckkA02.h5.

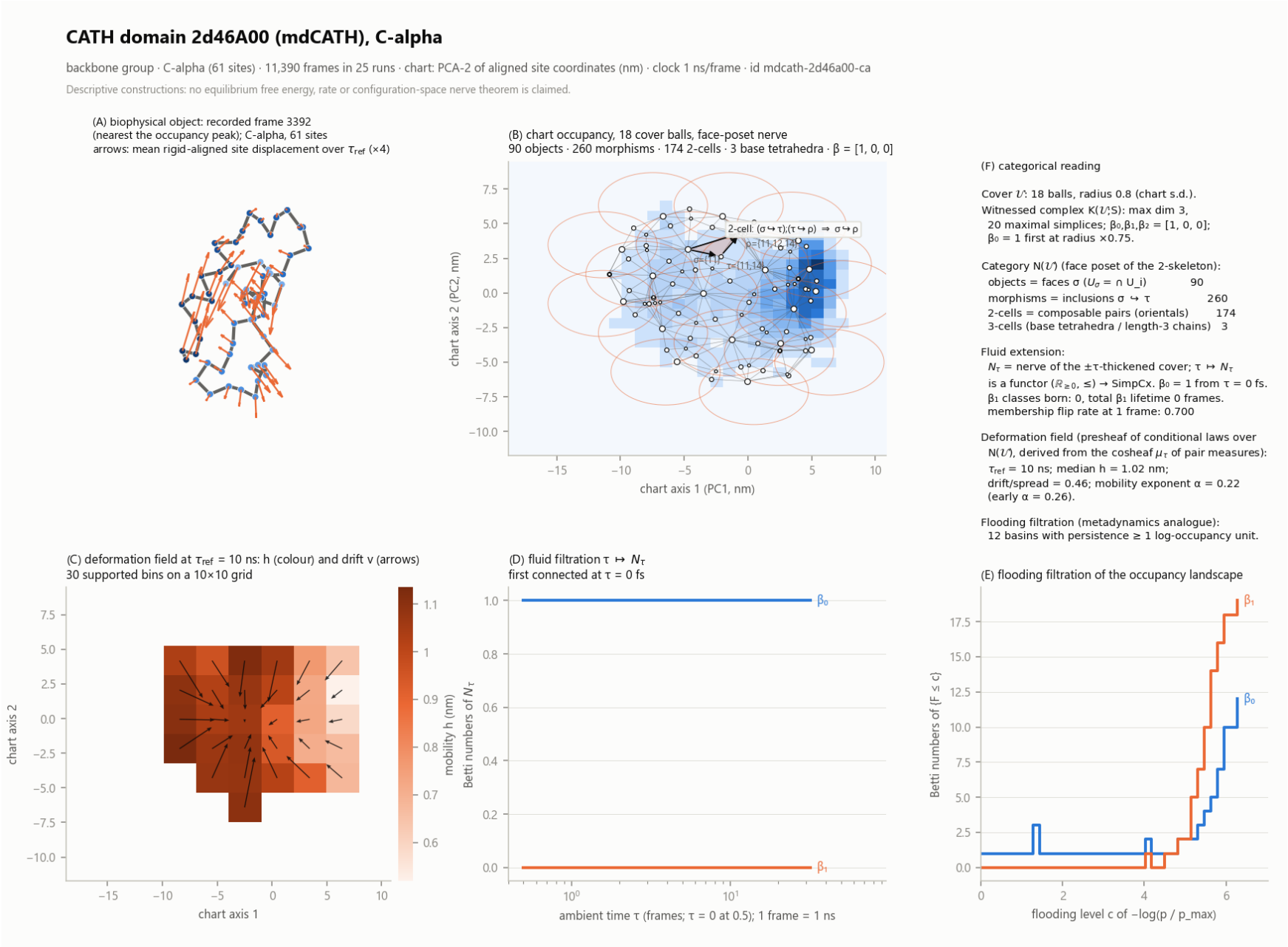

Figure 64: CATH domain 2d46A00 (mdCATH), C-alpha (`mdcath-2d46a00-ca`). Source: https://huggingface.co/datasets/compsciencelab/mdCATH/resolve/main/data/mdcath_dataset_2d46A00.h5.

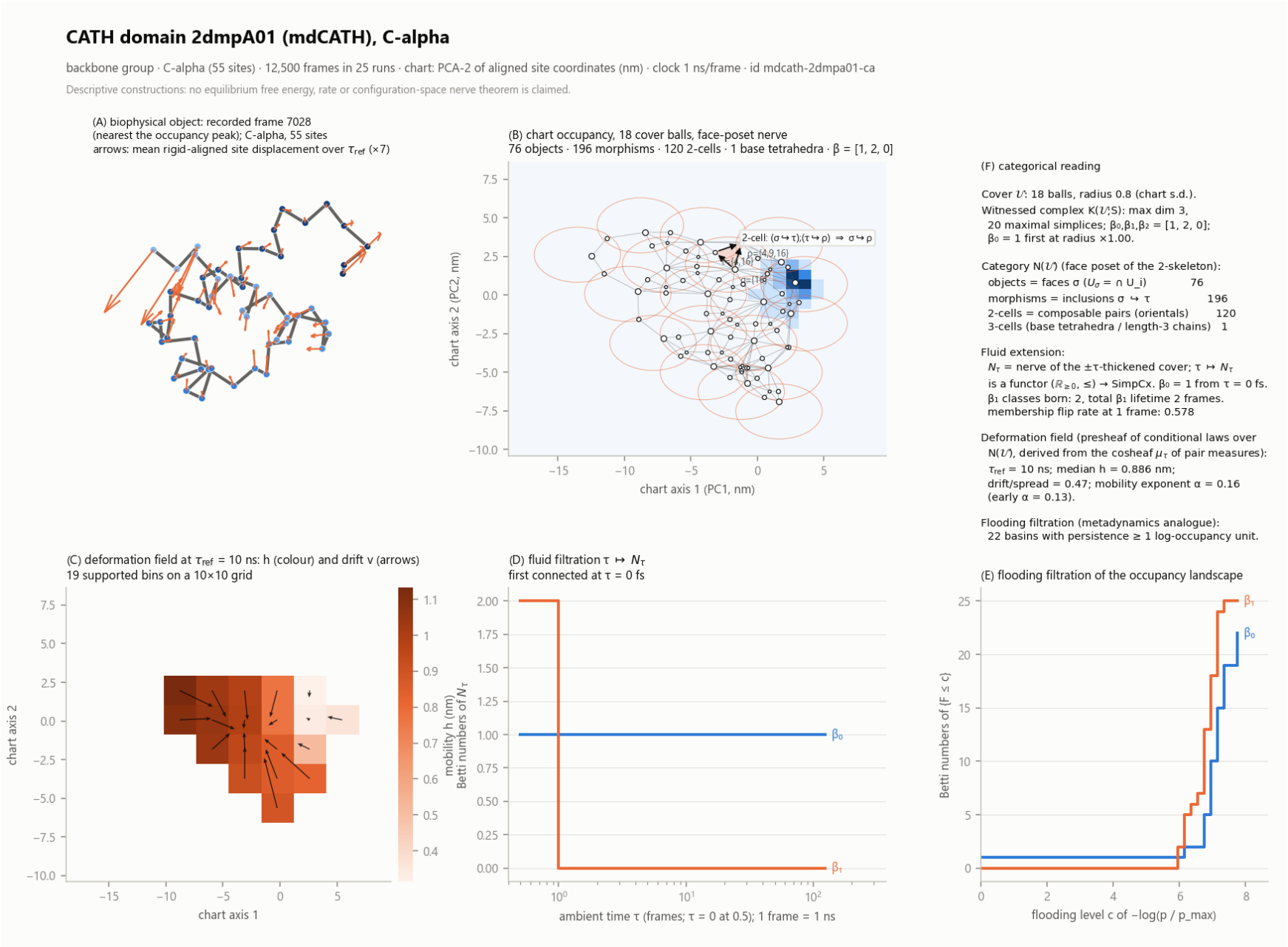

Figure 65: CATH domain 2dmpA01 (mdCATH), C-alpha (`mdcath-2dmpa01-ca`). Source: https://huggingface.co/datasets/compsciencelab/mdCATH/resolve/main/data/mdcath_dataset_2dmpA01.h

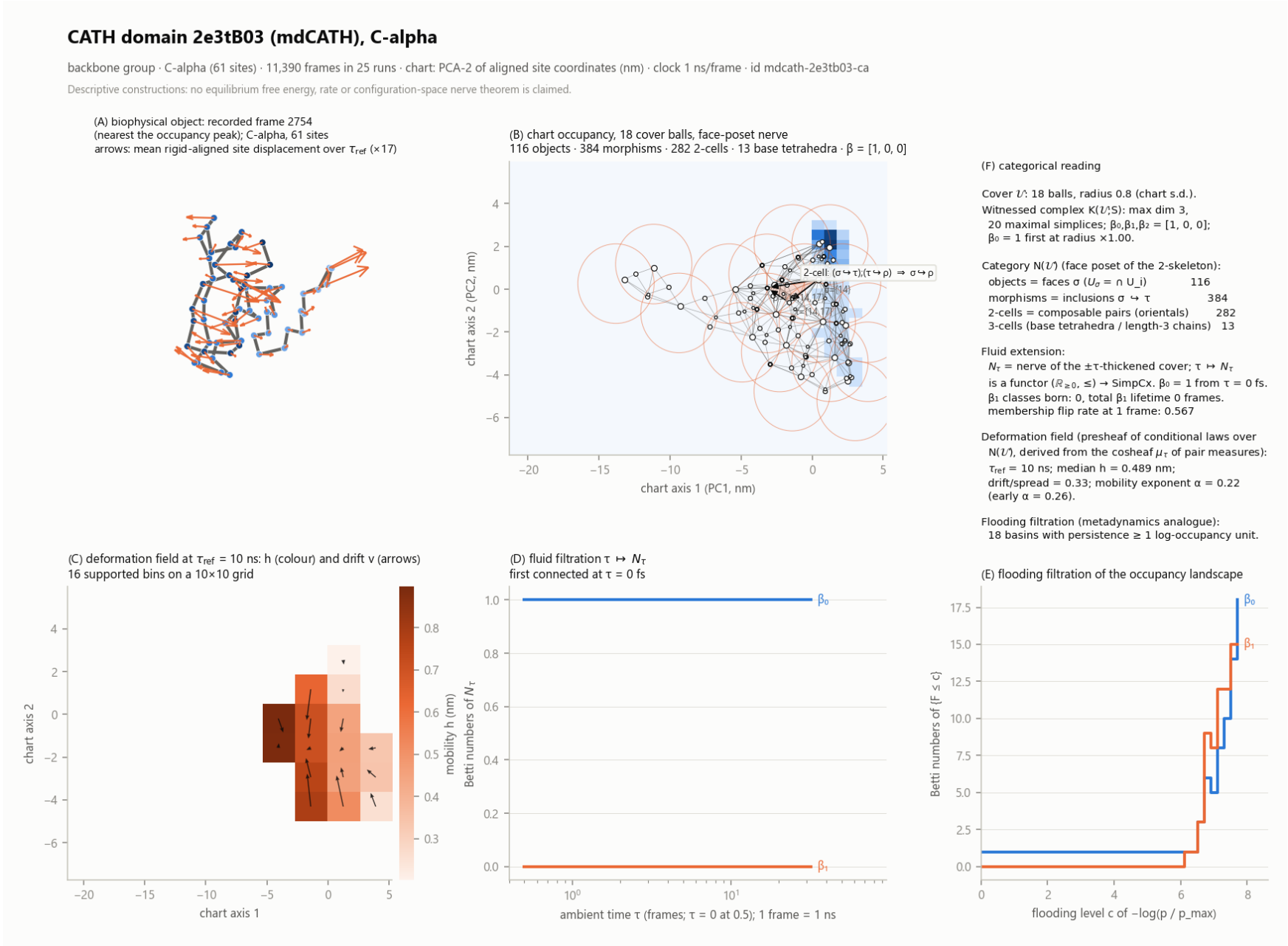

Figure 66: CATH domain 2e3tB03 (mdCATH), C-alpha (`mdcath-2e3tb03-ca`). Source: https://huggingface.co/datasets/compsciencelab/mdCATH/resolve/main/data/mdcath_dataset_2e3tB03.h5.

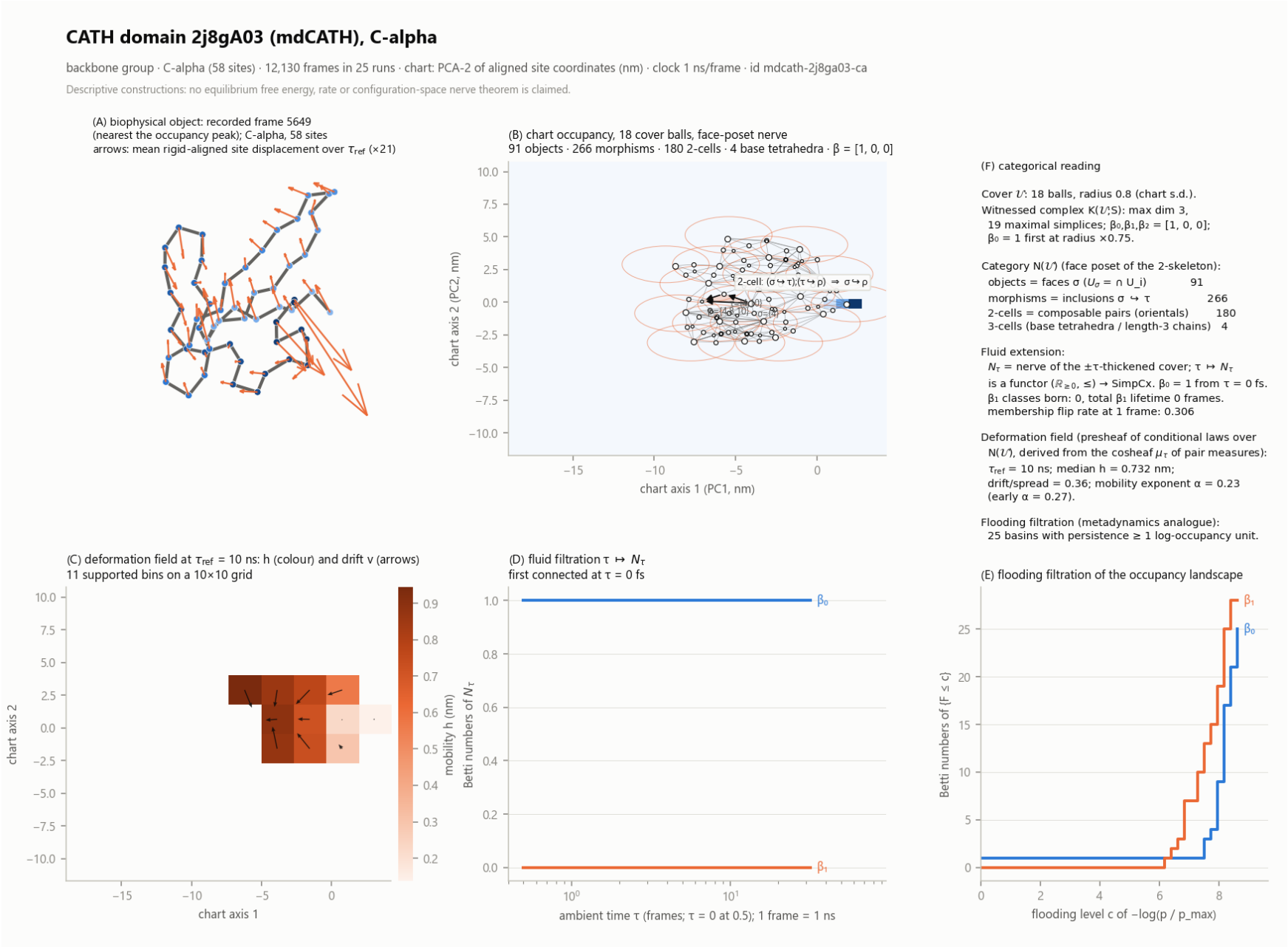

Figure 67: CATH domain 2j8gA03 (mdCATH), C-alpha (`mdcath-2j8ga03-ca`). Source: https://huggingface.co/datasets/compsciencelab/mdCATH/resolve/main/data/mdcath_dataset_2j8gA03.h5.

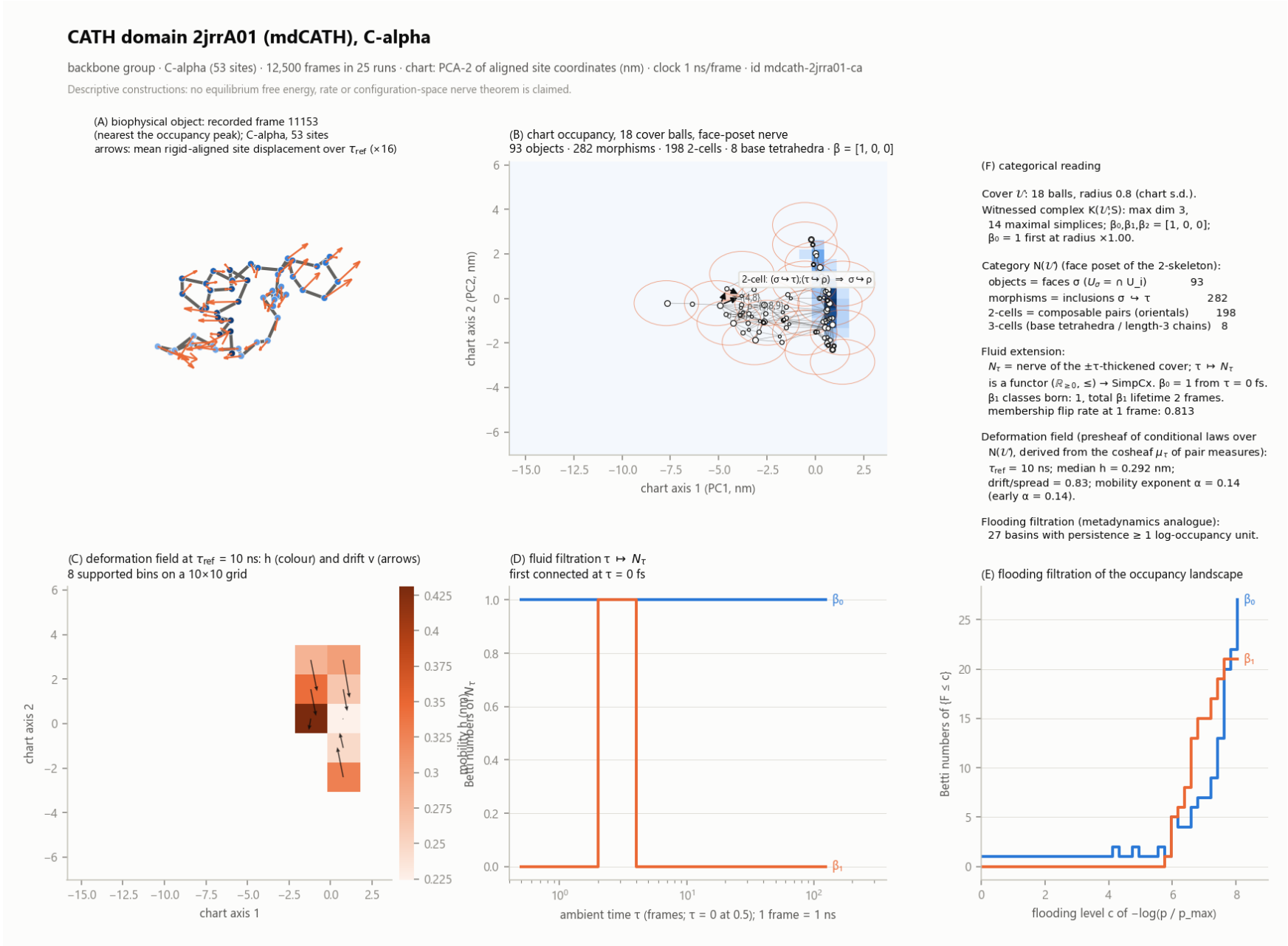

Figure 68: CATH domain 2jrrA01 (mdCATH), C-alpha (`mdcath-2jrra01-ca`). Source: https://huggingface.co/datasets/compsciencelab/mdCATH/resolve/main/data/mdcath_dataset_2jrrA01.h5.

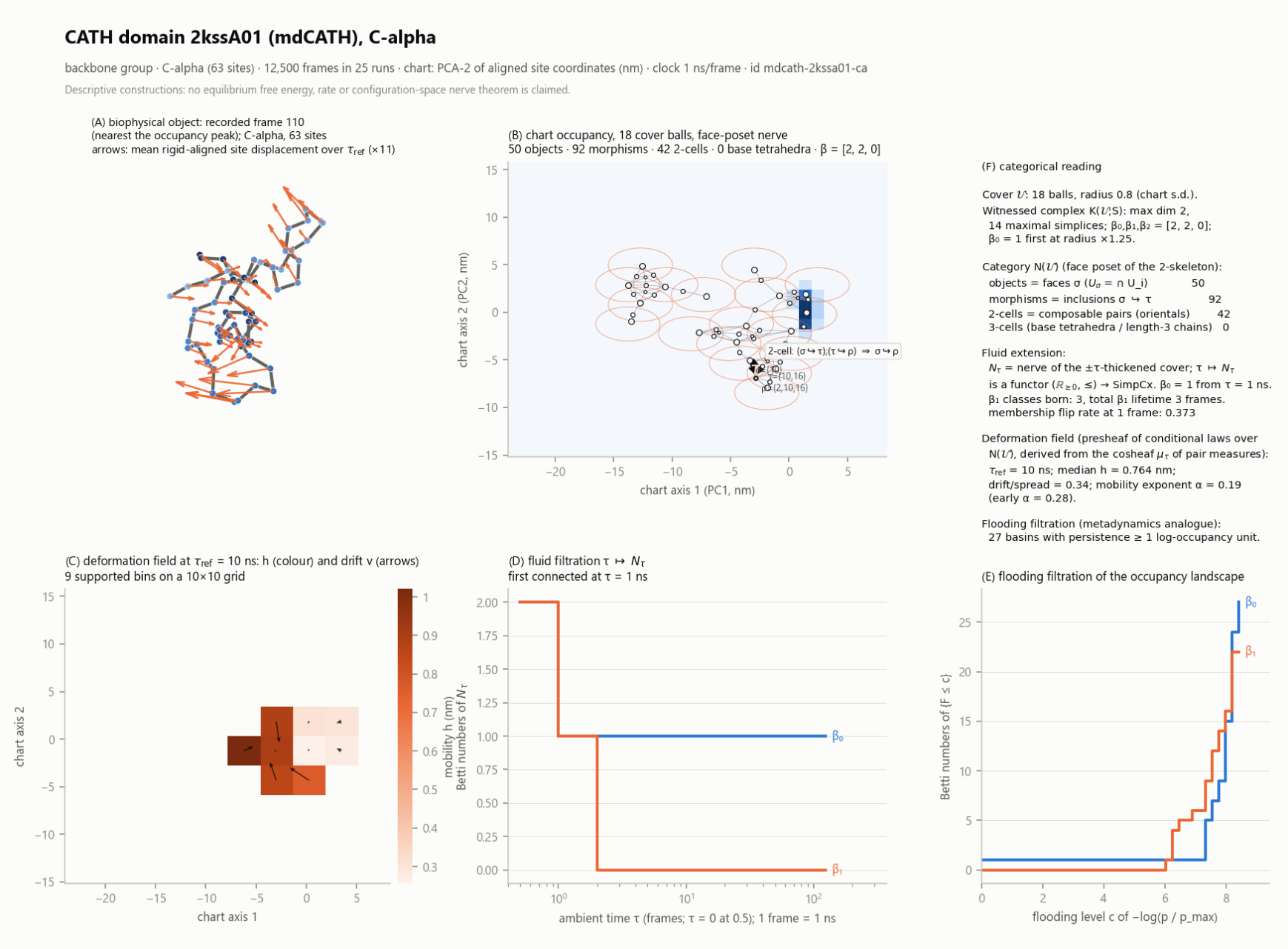

Figure 69: CATH domain 2kssA01 (mdCATH), C-alpha (`mdcath-2kssa01-ca`). Source: https://huggingface.co/datasets/compsciencelab/mdCATH/resolve/main/data/mdcath_dataset_2kssA01.h5.

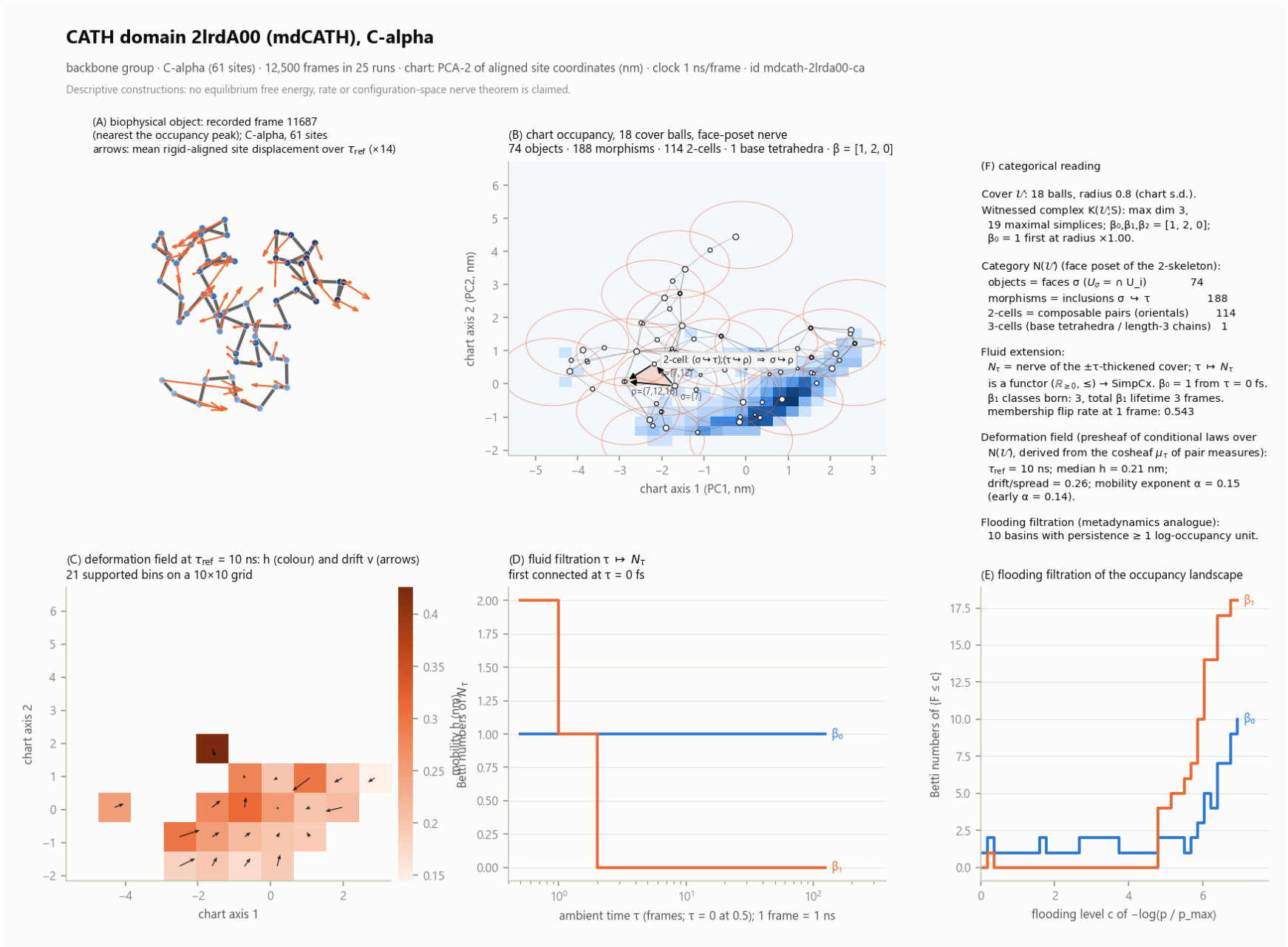

Figure 70: CATH domain 2lrdA00 (mdCATH), C-alpha (`mdcath-2lrda00-ca`). Source: https://huggingface.co/datasets/compsciencelab/mdCATH/resolve/main/data/mdcath_dataset_2lrdA00.h5.

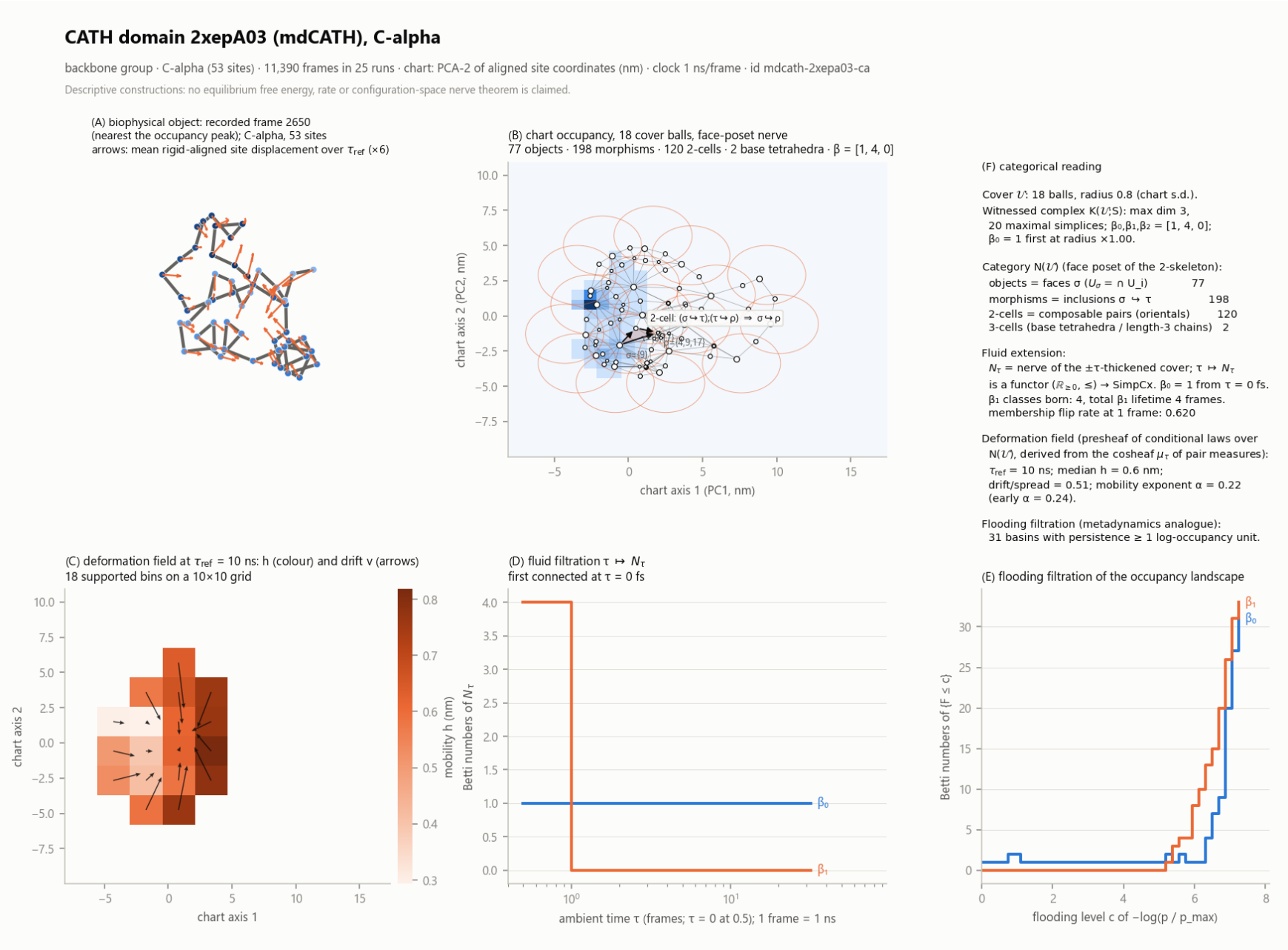

Figure 71: CATH domain 2xepA03 (mdCATH), C-alpha (`mdcath-2xepa03-ca`). Source: https://huggingface.co/datasets/compsciencelab/mdCATH/resolve/main/data/mdcath_dataset_2xepA03.h5.

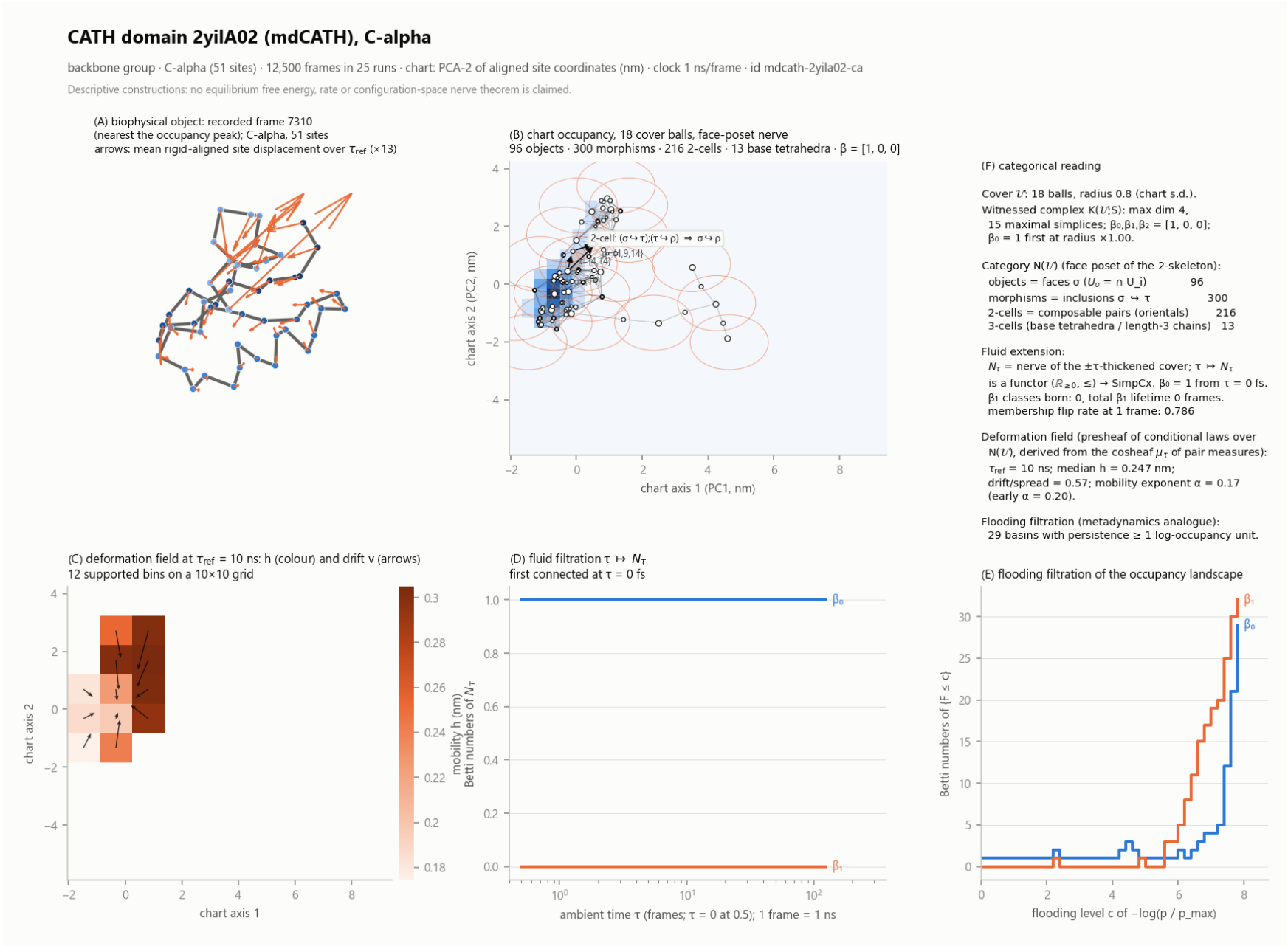

Figure 72: CATH domain 2yilA02 (mdCATH), C-alpha (`mdcath-2yila02-ca`). Source: https://huggingface.co/datasets/compsciencelab/mdCATH/resolve/main/data/mdcath_dataset_2yilA02.h5.

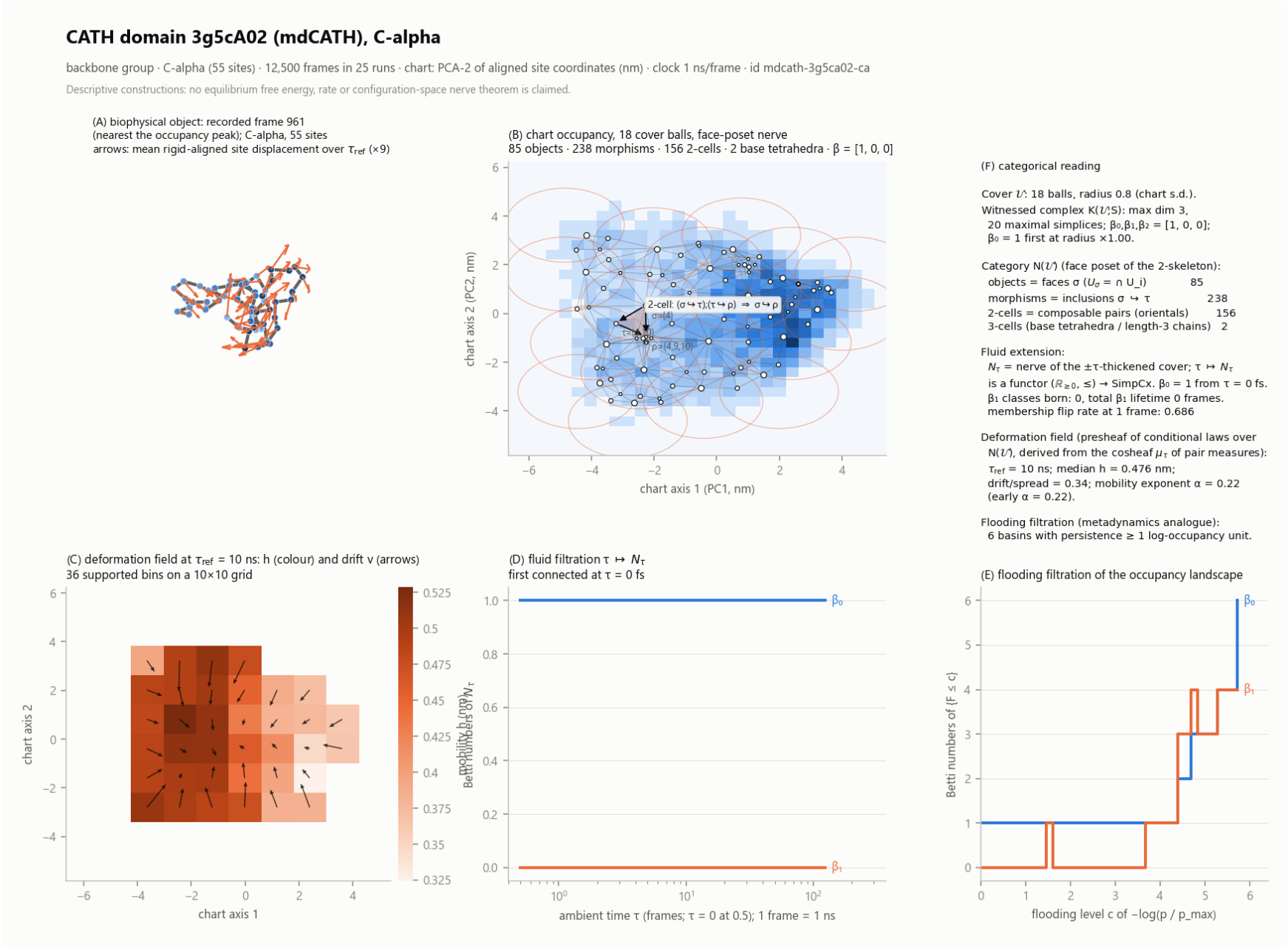

Figure 73: CATH domain 3g5cA02 (mdCATH), C-alpha (`mdcath-3g5ca02-ca`). Source: https://huggingface.co/datasets/compsciencelab/mdCATH/resolve/main/data/mdcath_dataset_3g5cA02.h5.

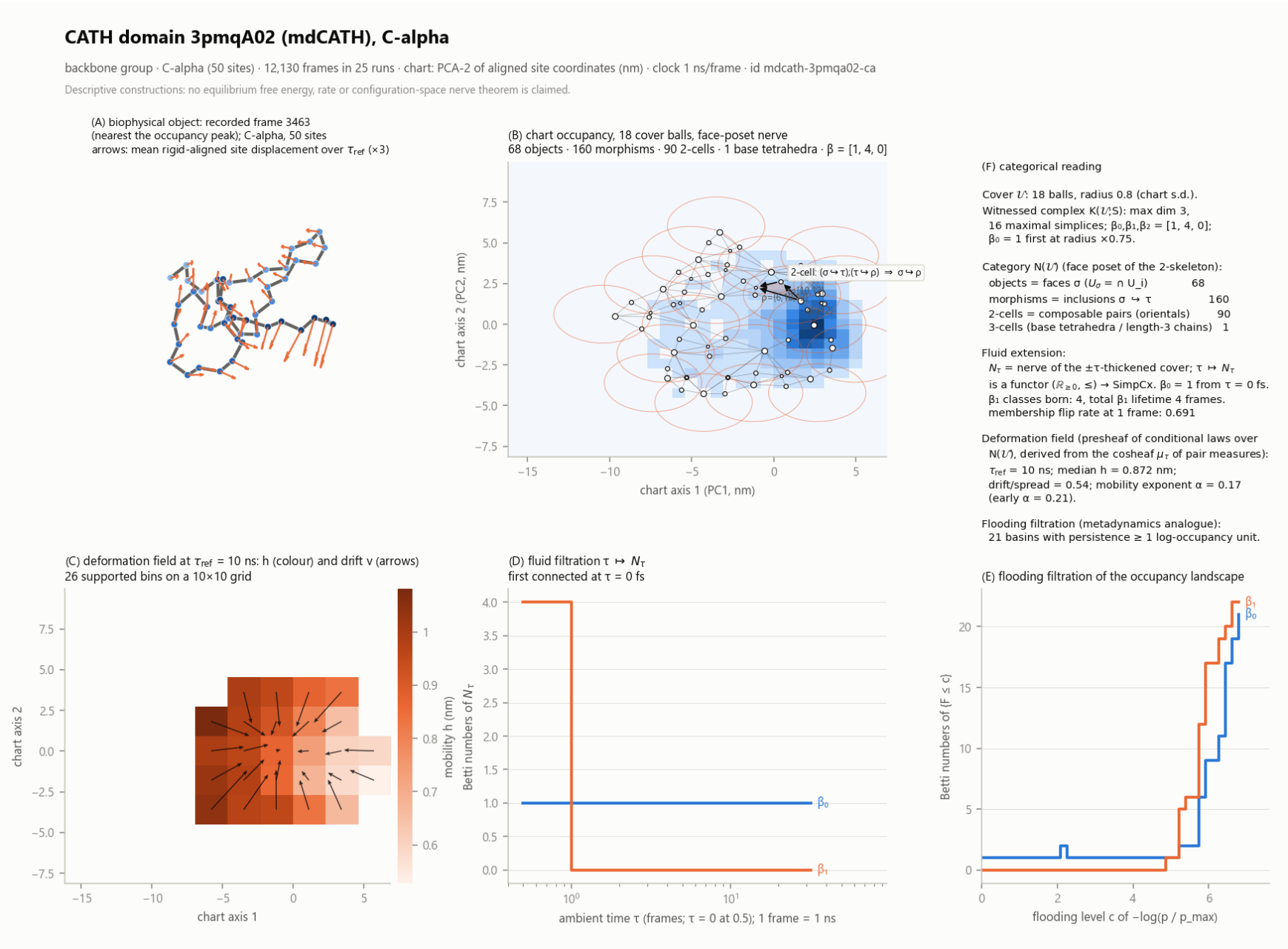

Figure 74: CATH domain 3pmqA02 (mdCATH), C-alpha (`mdcath-3pmqa02-ca`). Source: https://huggingface.co/datasets/compsciencelab/mdCATH/resolve/main/data/mdcath_dataset_3pmqA02.h

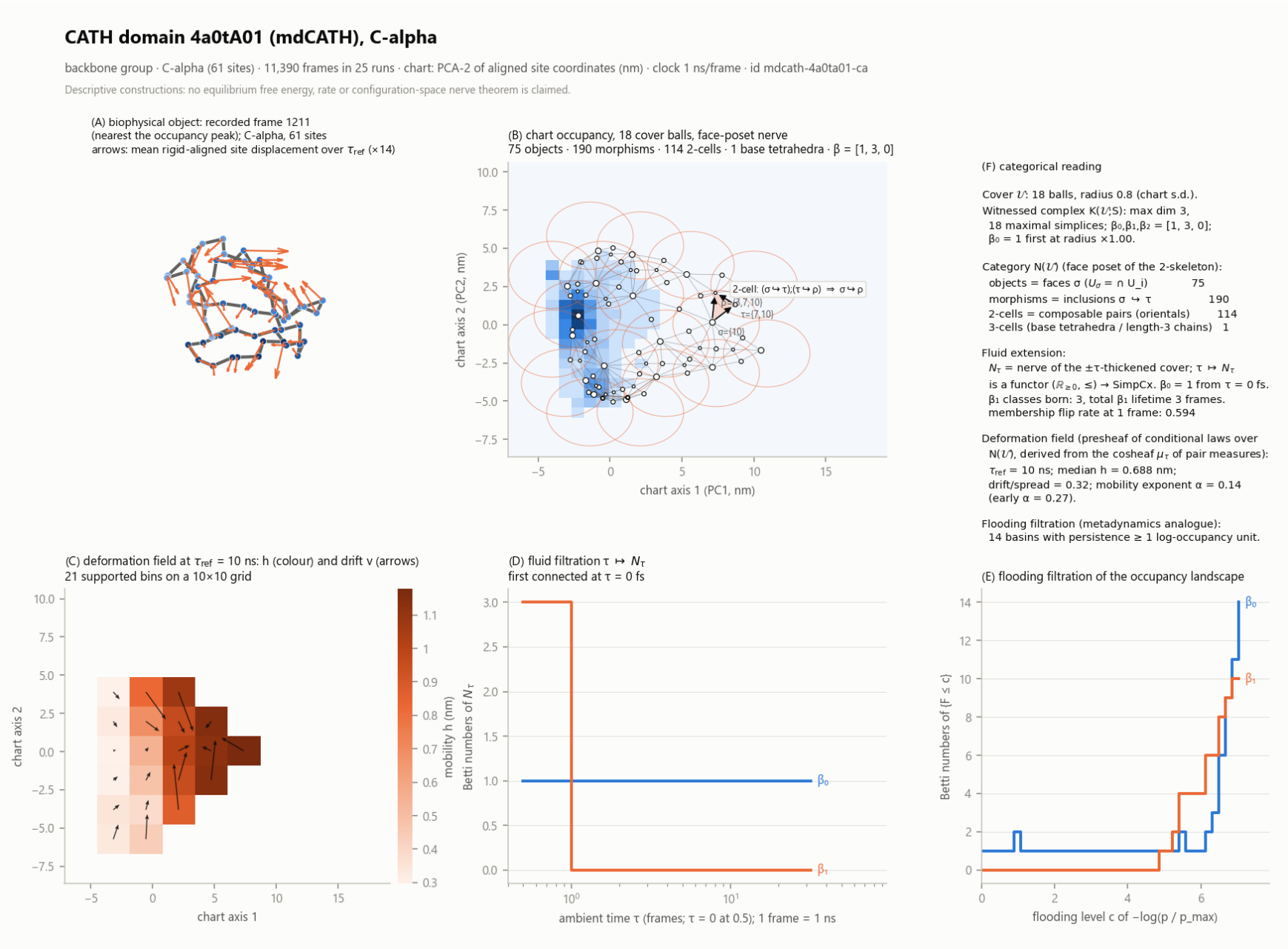

Figure 75: CATH domain 4a0tA01 (mdCATH), C-alpha (`mdcath-4a0ta01-ca`). Source: https://huggingface.co/datasets/compsciencelab/mdCATH/resolve/main/data/mdcath_dataset_4a0tA01.h5.

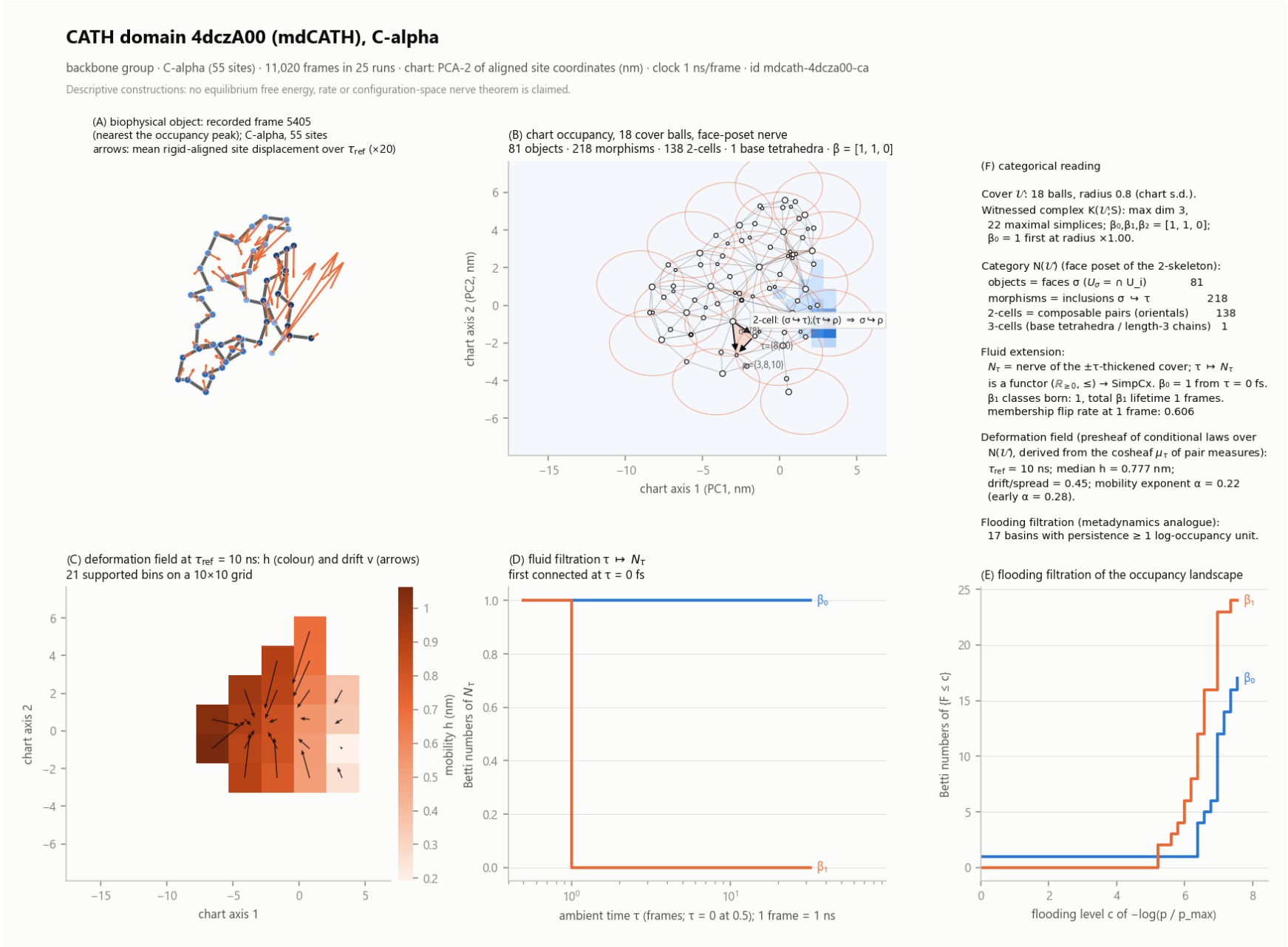

Figure 76: CATH domain 4dczA00 (mdCATH), C-alpha (`mdcath-4dcza00-ca`). Source: https://huggingface.co/datasets/compsciencelab/mdCATH/resolve/main/data/mdcath_dataset_4dczA00.h5.

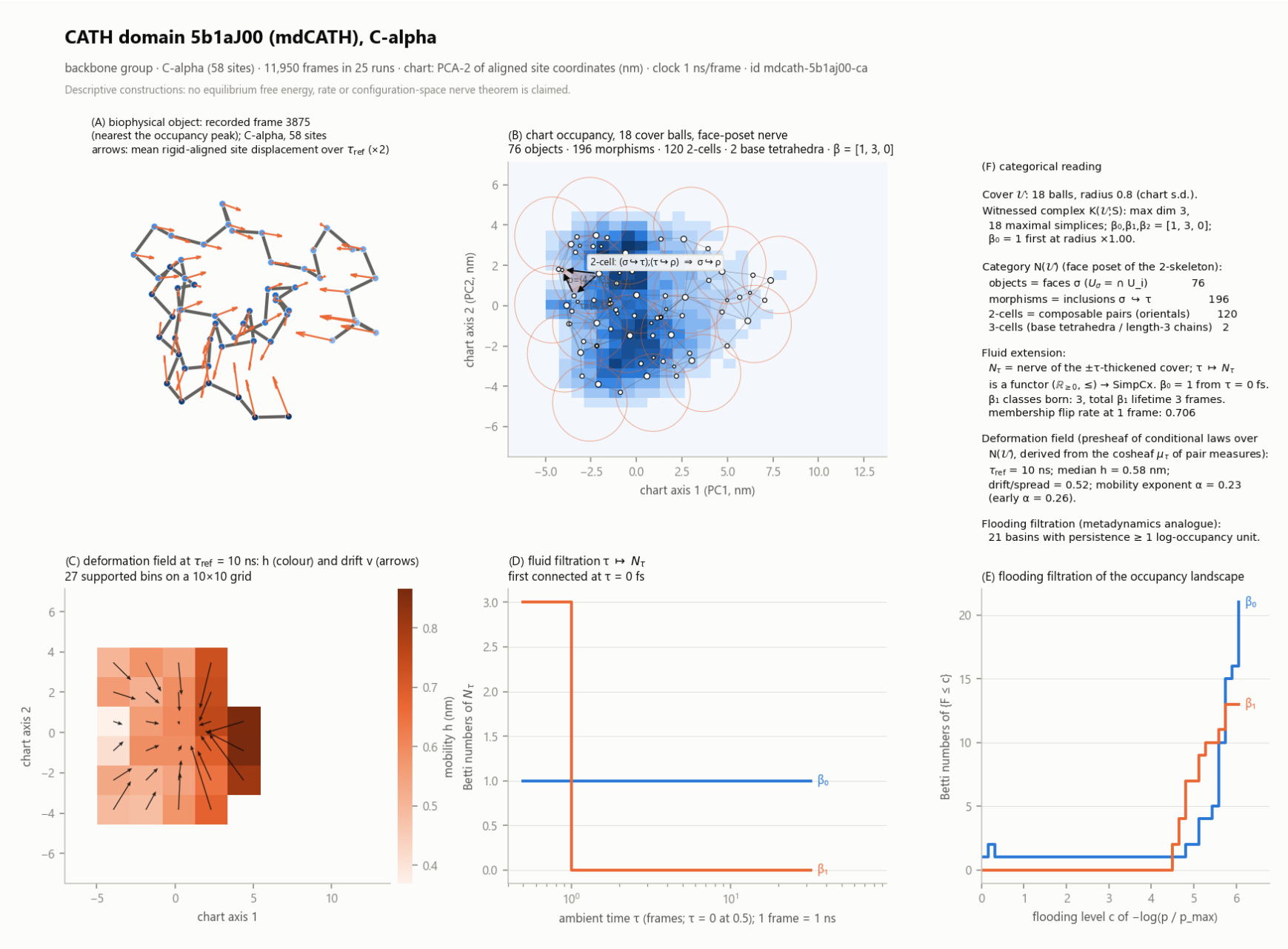

Figure 77: CATH domain 5b1aJ00 (mdCATH), C-alpha (`mdcath-5b1aj00-ca`). Source: https://huggingface.co/datasets/compsciencelab/mdCATH/resolve/main/data/mdcath_dataset_5b1aJ00.h5.

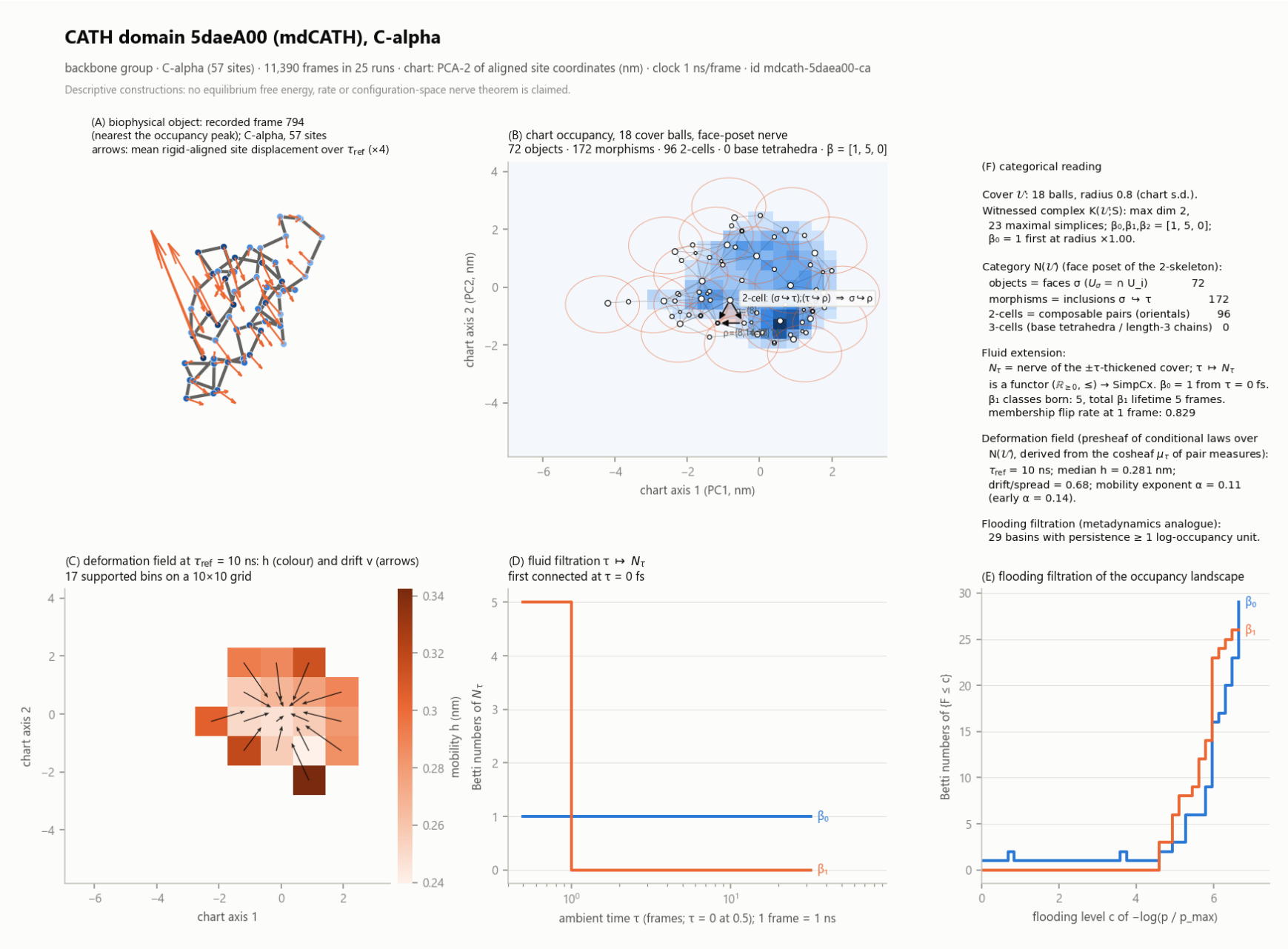

Figure 78: CATH domain 5daeA00 (mdCATH), C-alpha (`mdcath-5daea00-ca`). Source: https://huggingface.co/datasets/compsciencelab/mdCATH/resolve/main/data/mdcath_dataset_5daeA00.h5.

Paired heavy-atom representations of the backbone systems

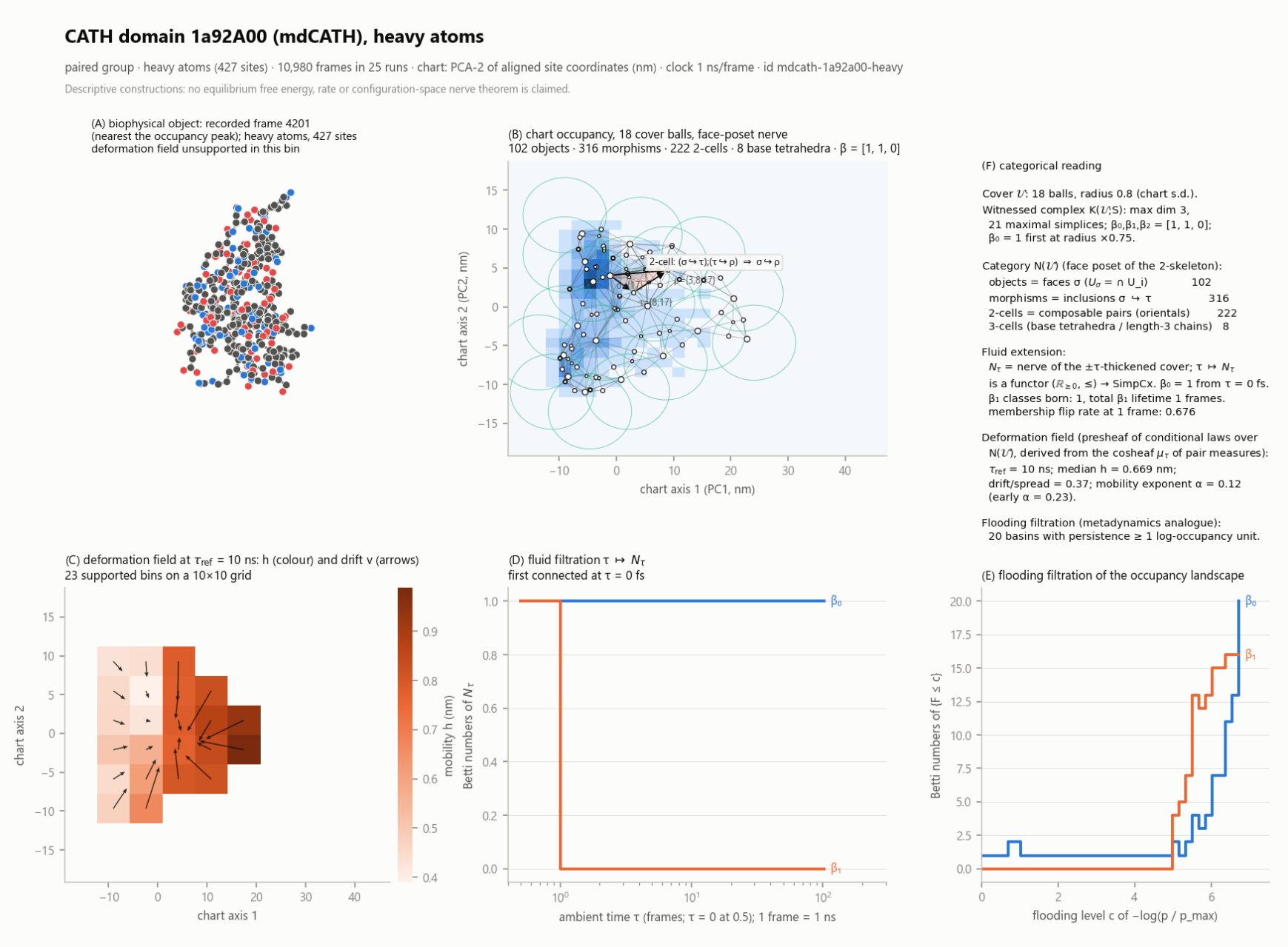

Figure 79: CATH domain 1a92A00 (mdCATH), heavy atoms (`mdcath-1a92a00-heavy`), paired with `mdcath-1a92a00-ca`.

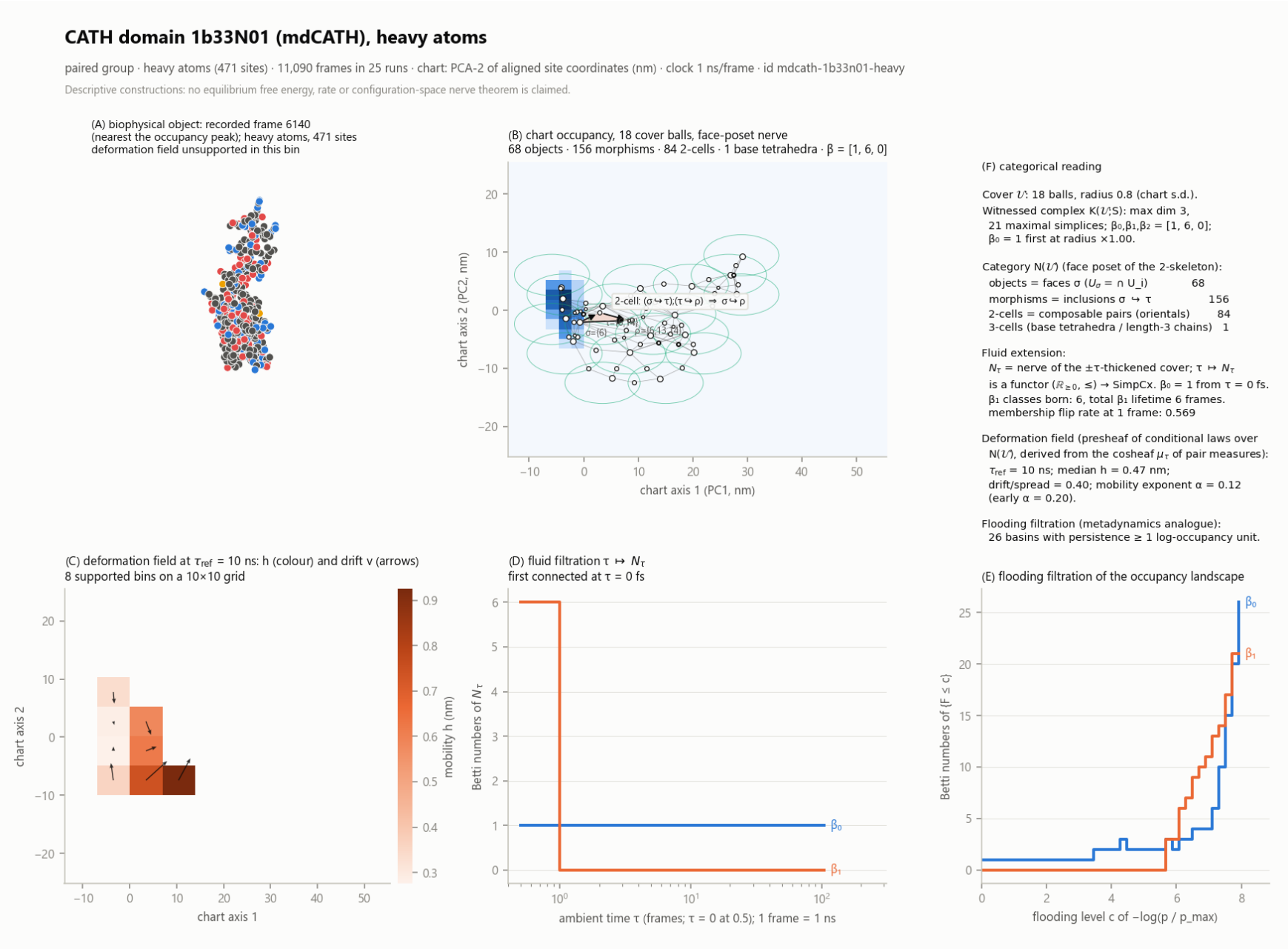

Figure 80: CATH domain 1b33N01 (mdCATH), heavy atoms (`mdcath-1b33n01-heavy`), paired with `mdcath-1b33n01-ca`.

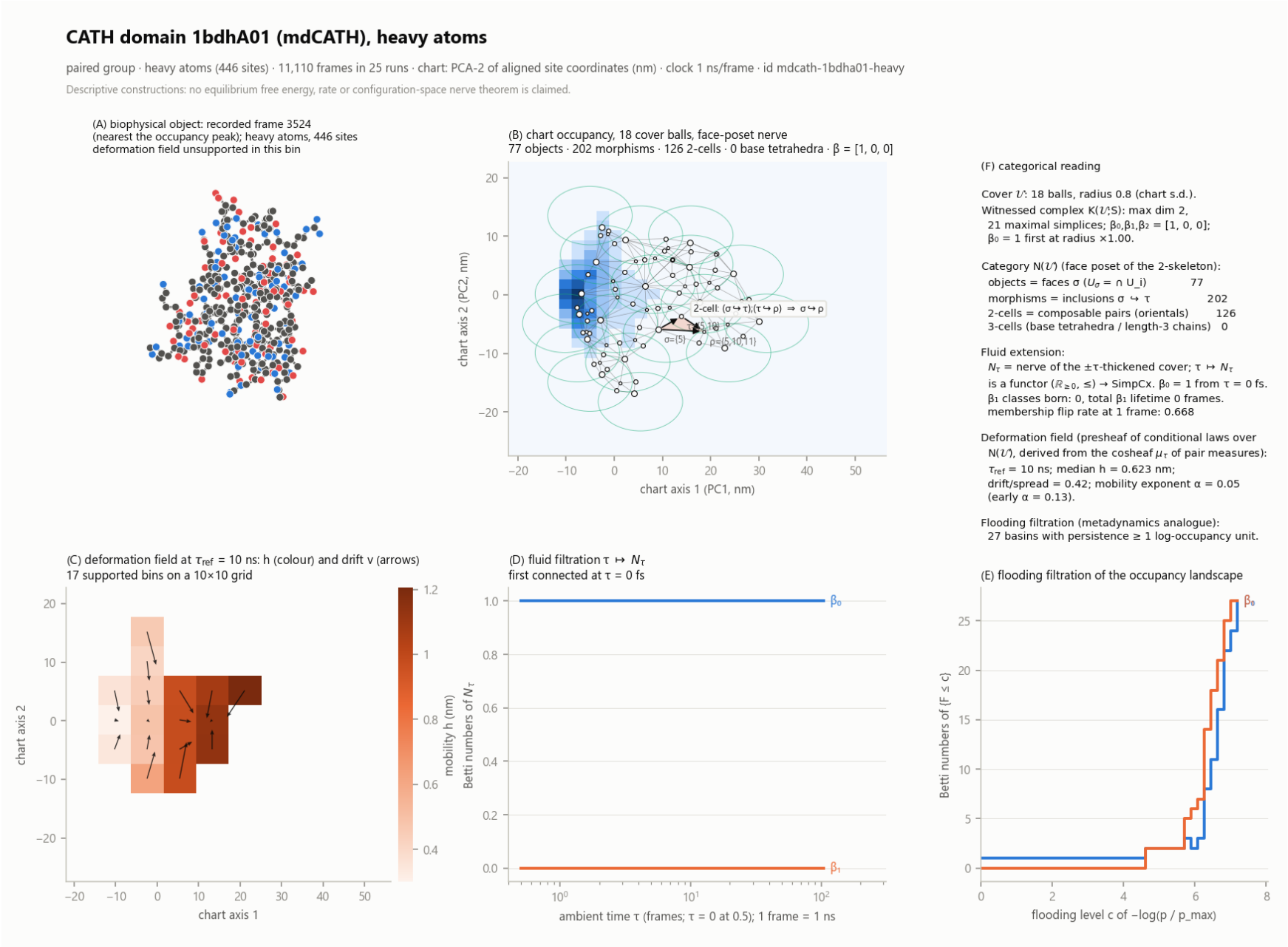

Figure 81: CATH domain 1bdhA01 (mdCATH), heavy atoms (`mdcath-1bdha01-heavy`), paired with `mdcath-1bdha01-ca`.

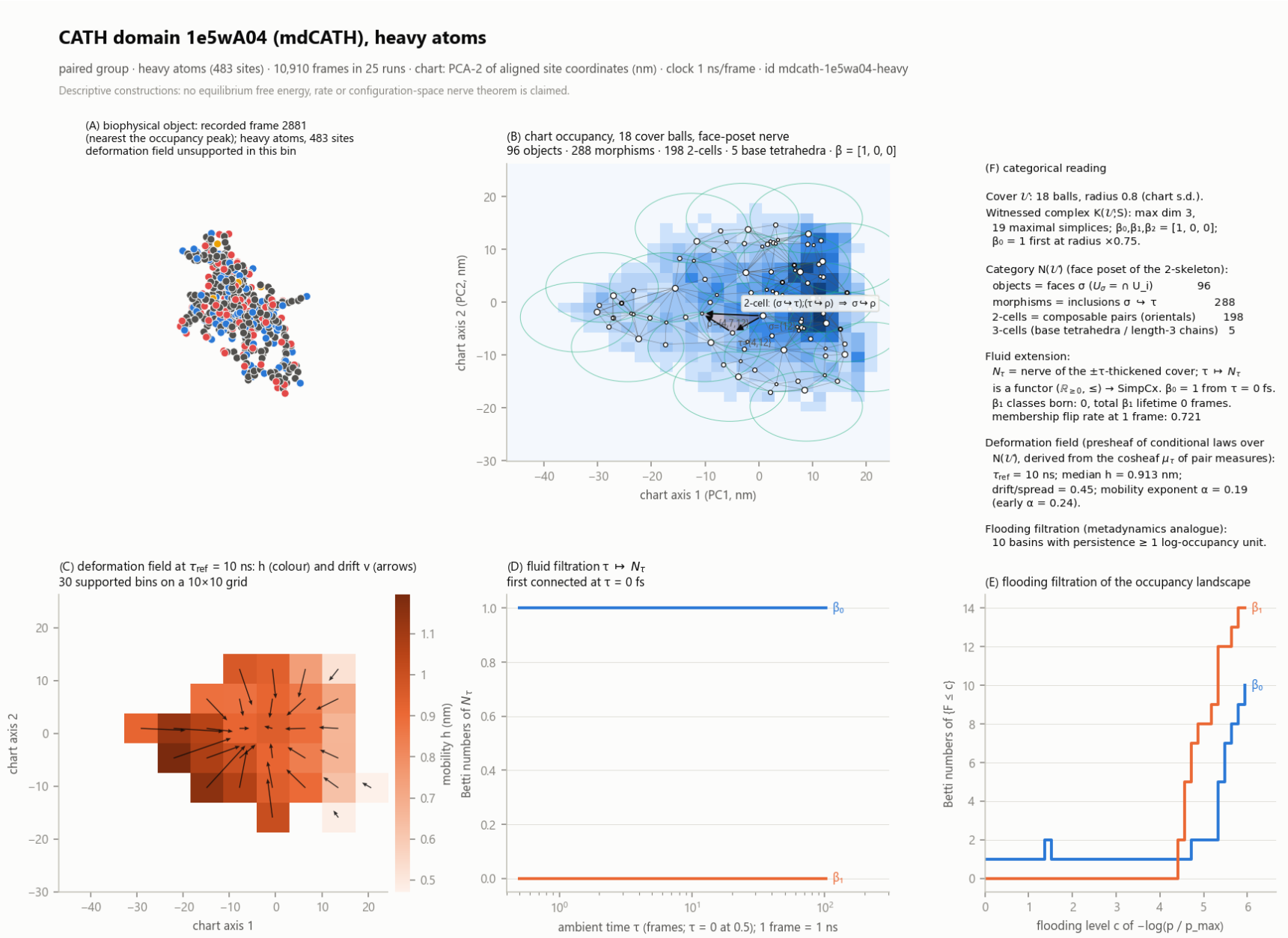

Figure 82: CATH domain 1e5wA04 (mdCATH), heavy atoms (`mdcath-1e5wa04-heavy`), paired with `mdcath-1e5wa04-ca`.

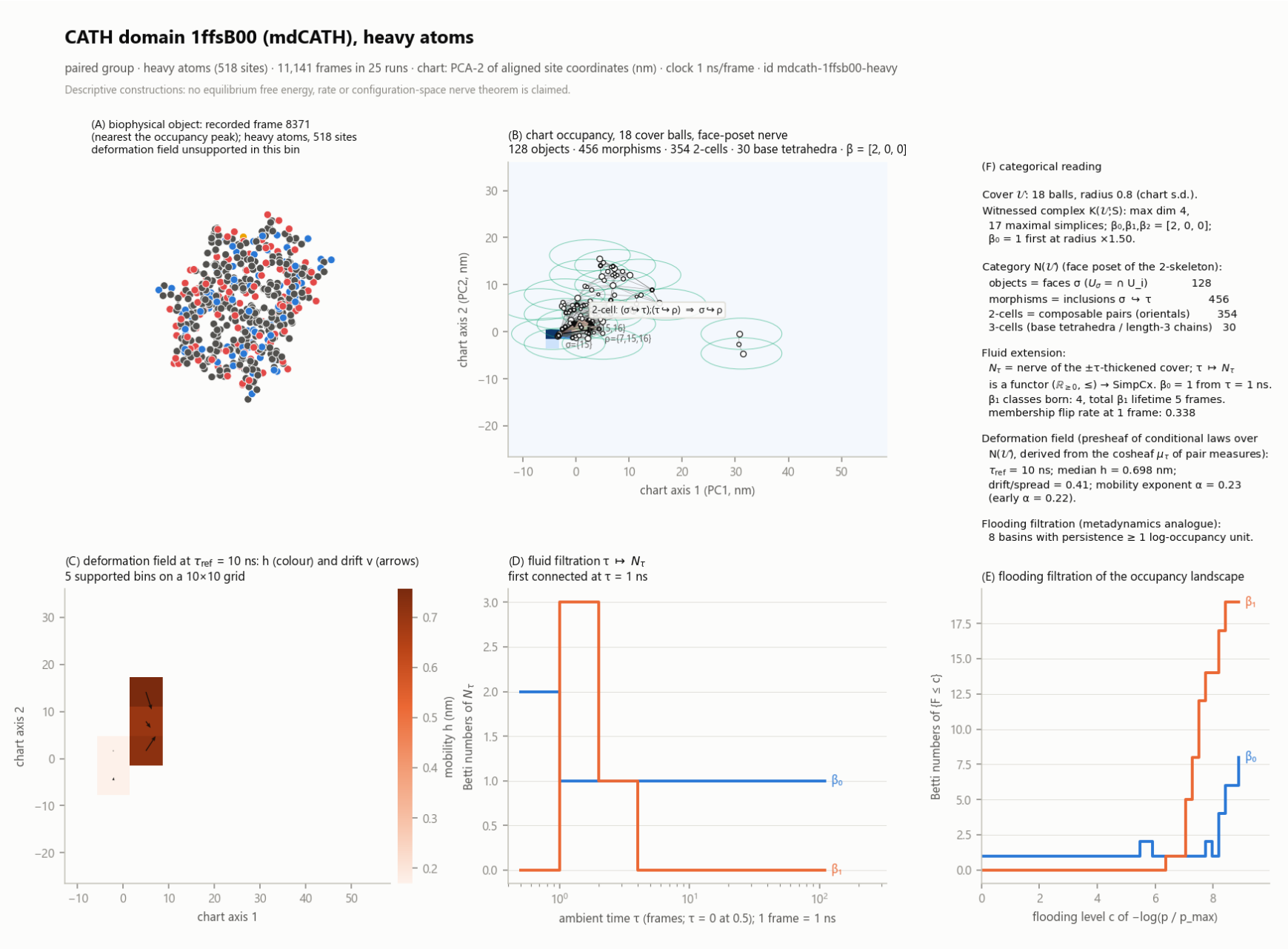

Figure 83: CATH domain 1ffsB00 (mdCATH), heavy atoms (`mdcath-1ffsb00-heavy`), paired with `mdcath-1ffsb00-ca`.

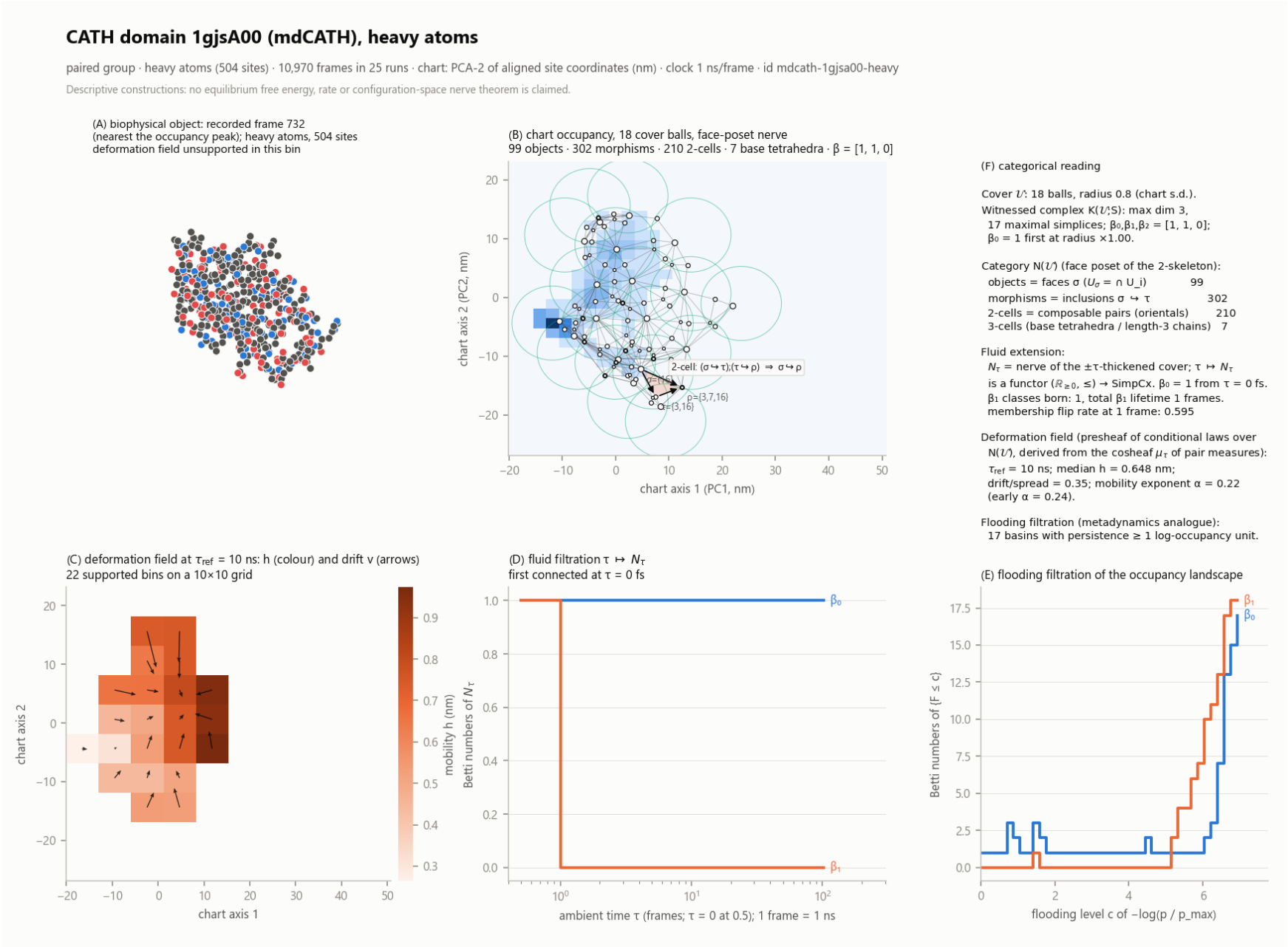

Figure 84: CATH domain 1gjsA00 (mdCATH), heavy atoms (`mdcath-1gjsa00-heavy`), paired with `mdcath-1gjsa00-ca`.

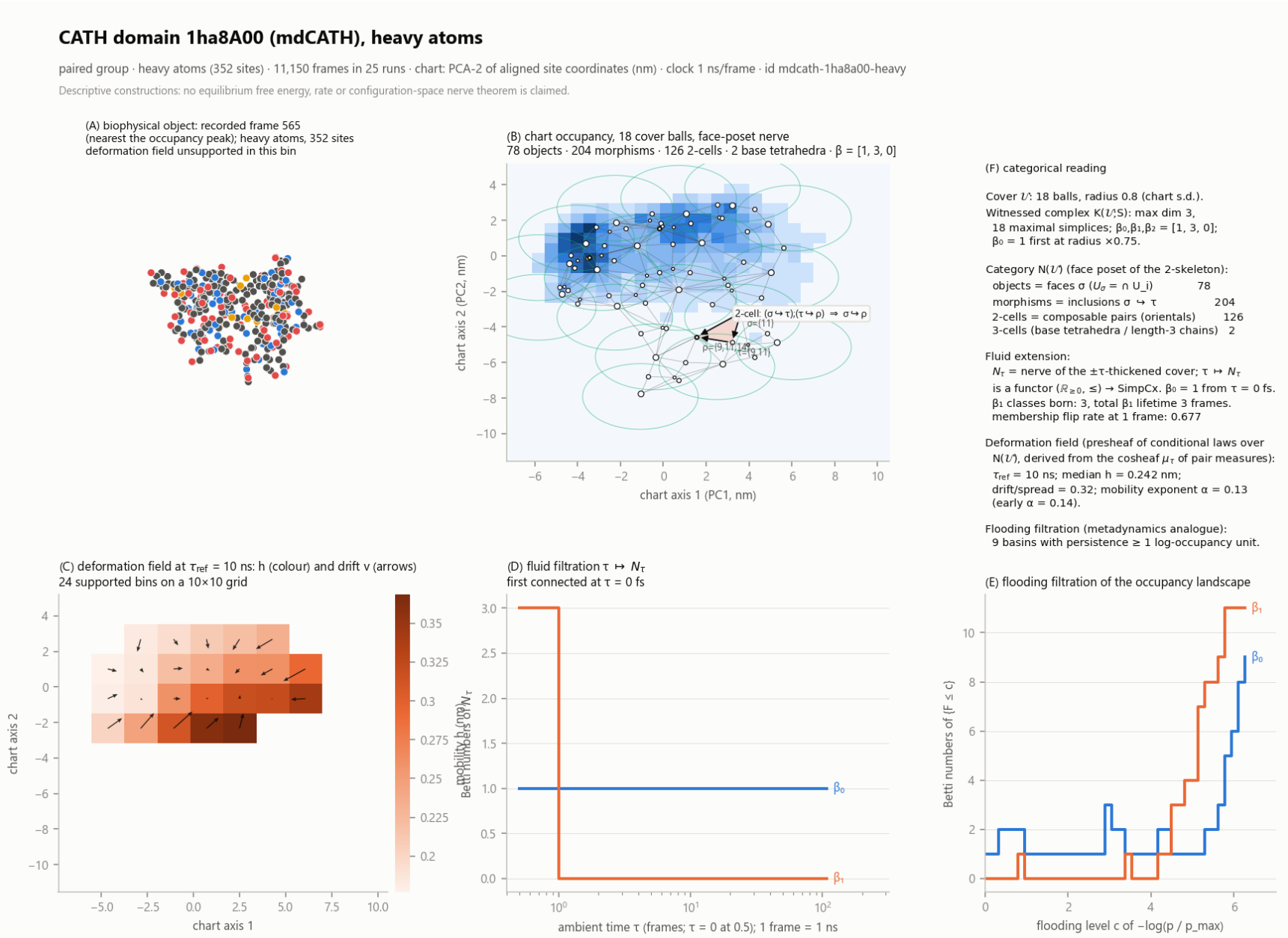

Figure 85: CATH domain 1ha8A00 (mdCATH), heavy atoms (`mdcath-1ha8a00-heavy`), paired with `mdcath-1ha8a00-ca`.

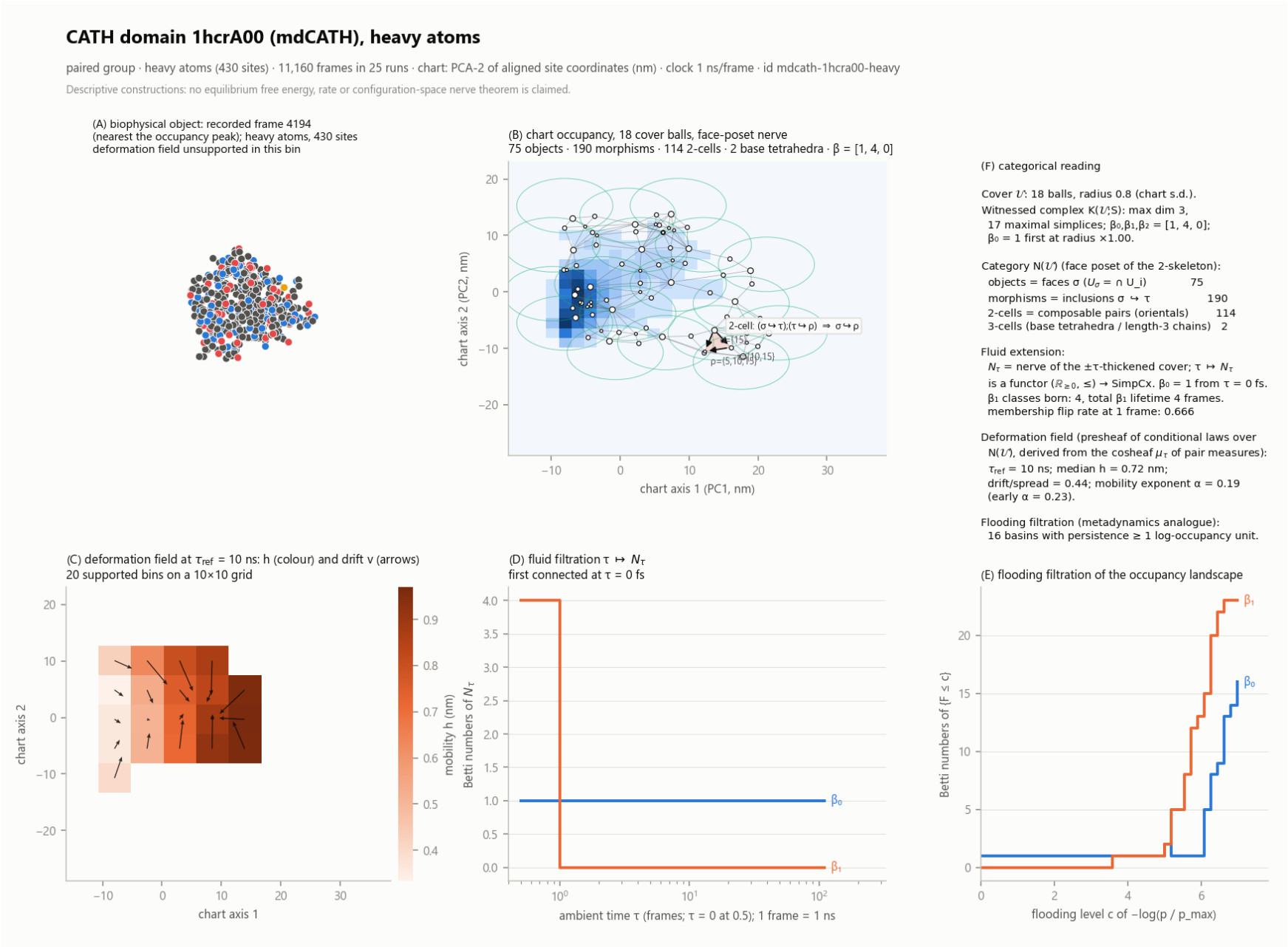

Figure 86: CATH domain 1hcrA00 (mdCATH), heavy atoms (`mdcath-1hcra00-heavy`), paired with `mdcath-1hcra00-ca`.

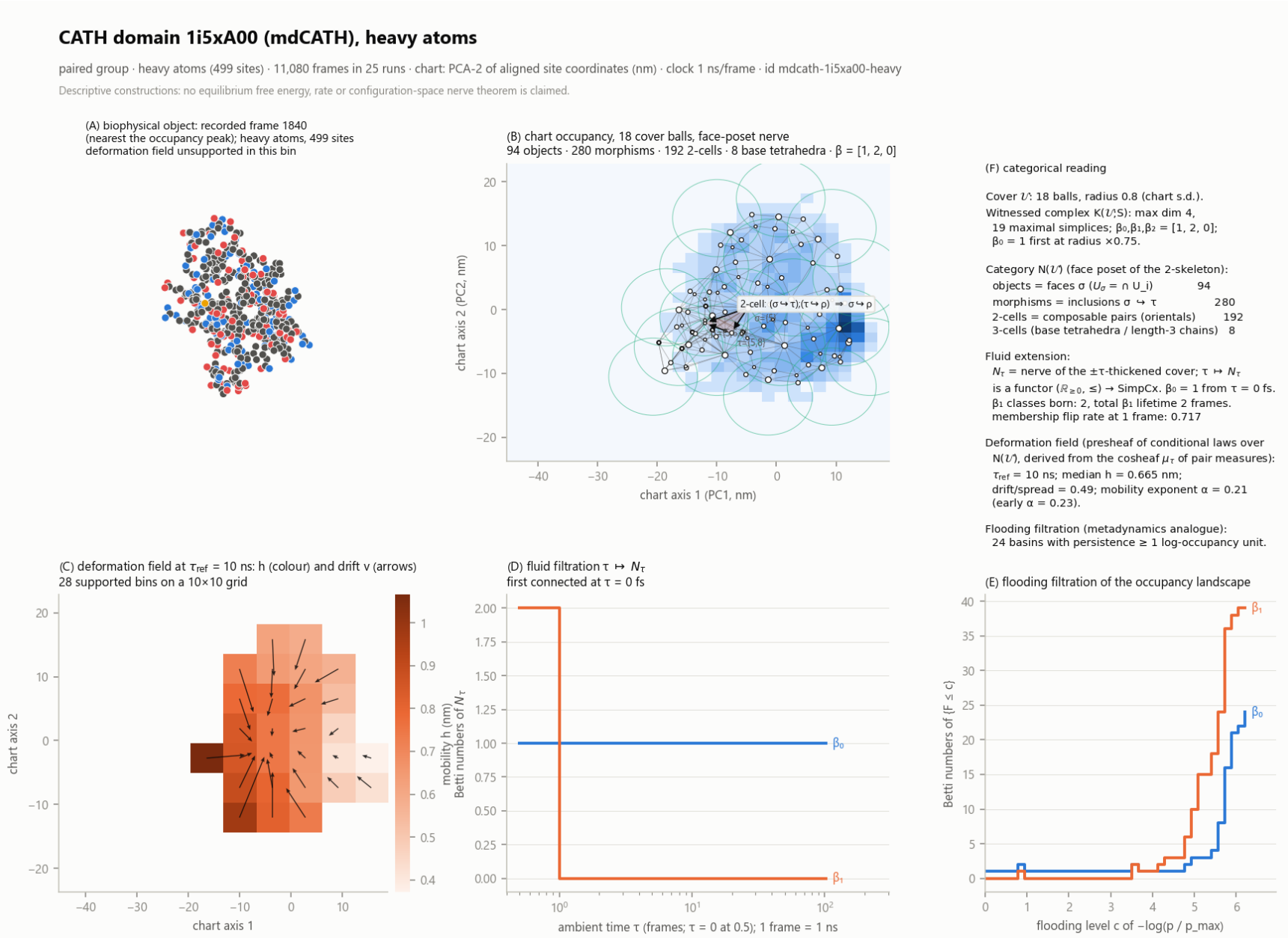

Figure 87: CATH domain 1i5xA00 (mdCATH), heavy atoms (`mdcath-1i5xa00-heavy`), paired with `mdcath-1i5xa00-ca`.

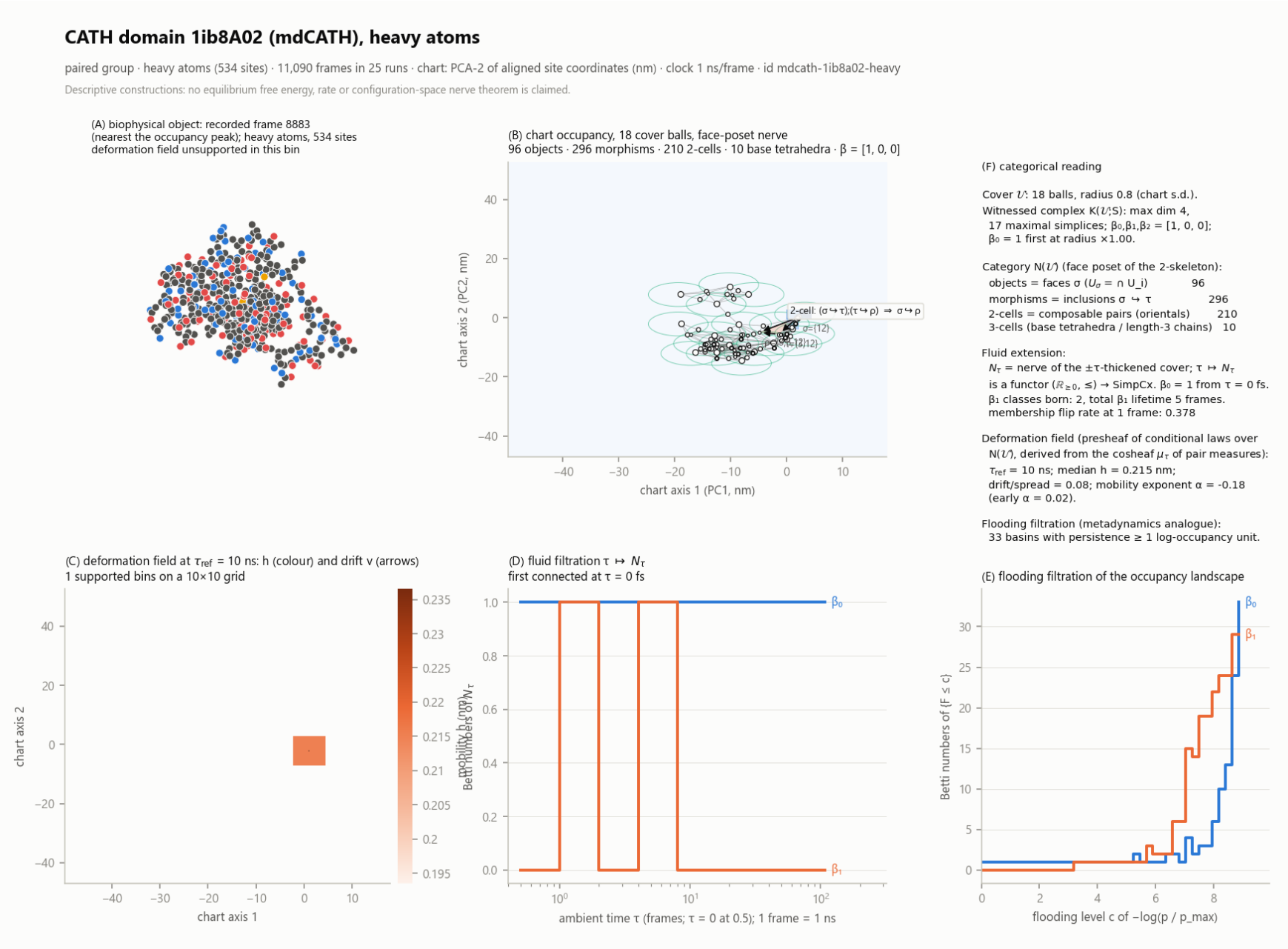

Figure 88: CATH domain 1ib8A02 (mdCATH), heavy atoms (`mdcath-1ib8a02-heavy`), paired with `mdcath-1ib8a02-ca`.

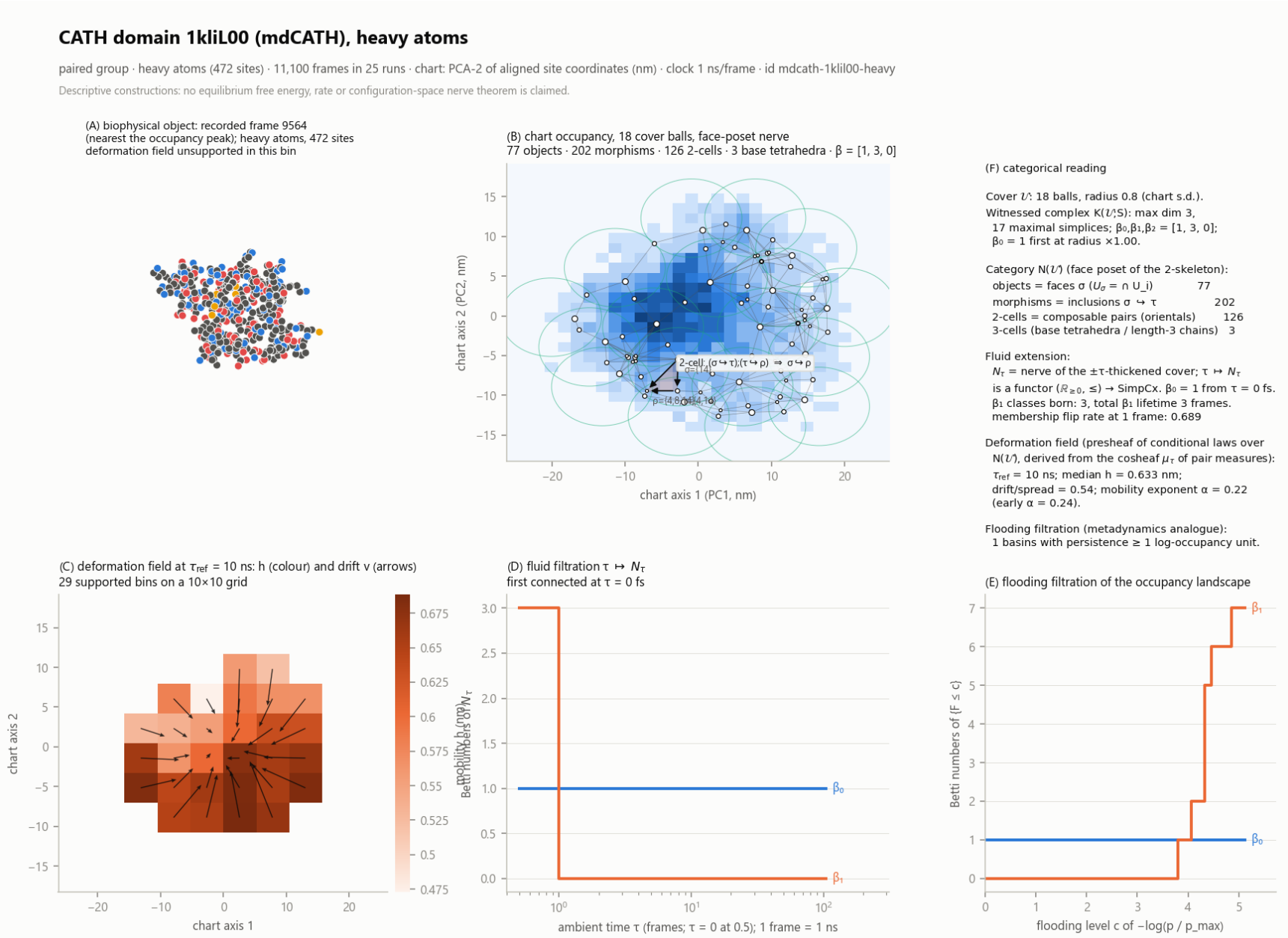

Figure 89: CATH domain 1kliL00 (mdCATH), heavy atoms (`mdcath-1klil00-heavy`), paired with `mdcath-1klil00-ca`.

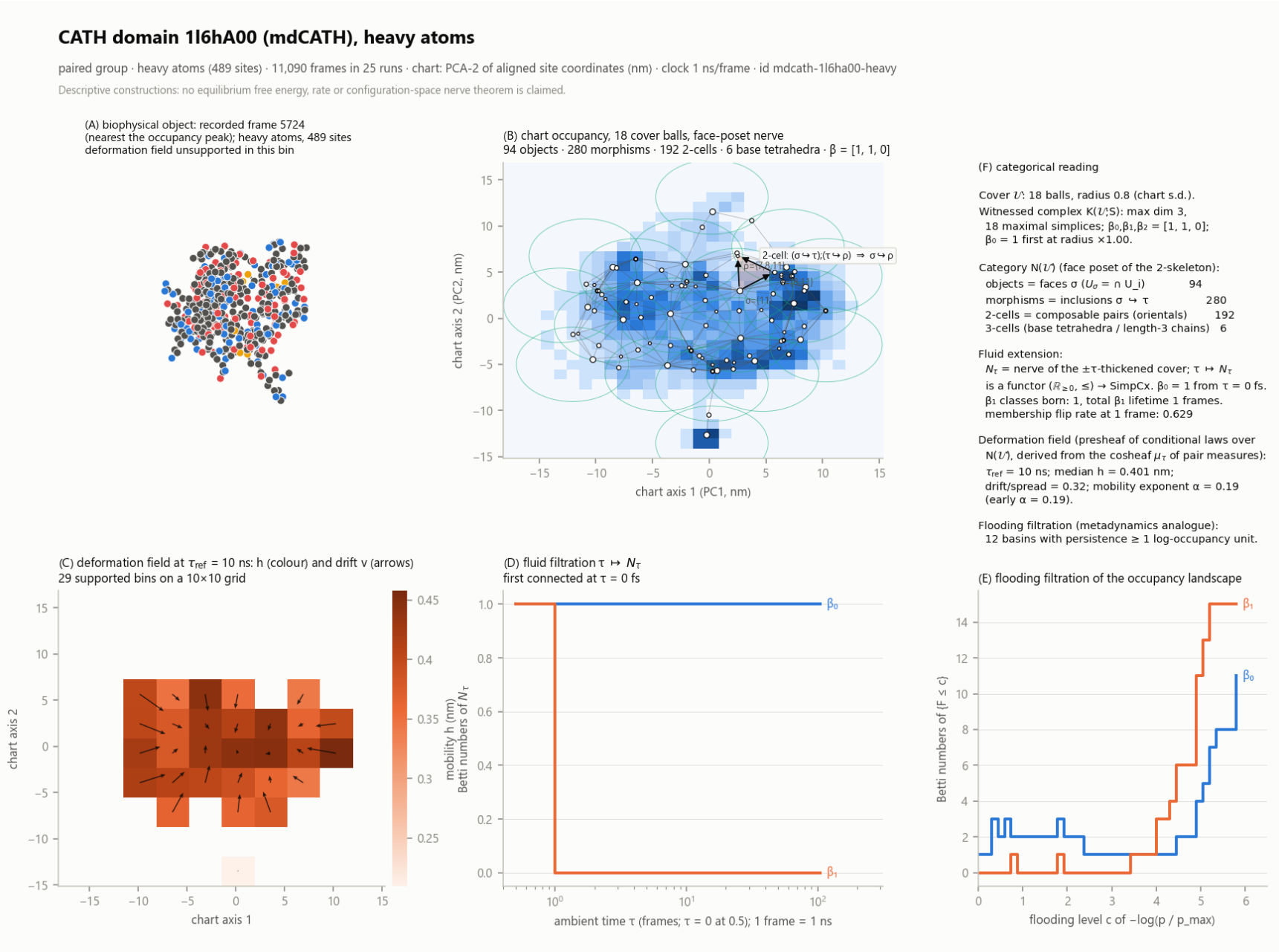

Figure 90: CATH domain 1l6hA00 (mdCATH), heavy atoms (`mdcath-1l6ha00-heavy`), paired with `mdcath-1l6ha00-ca`.

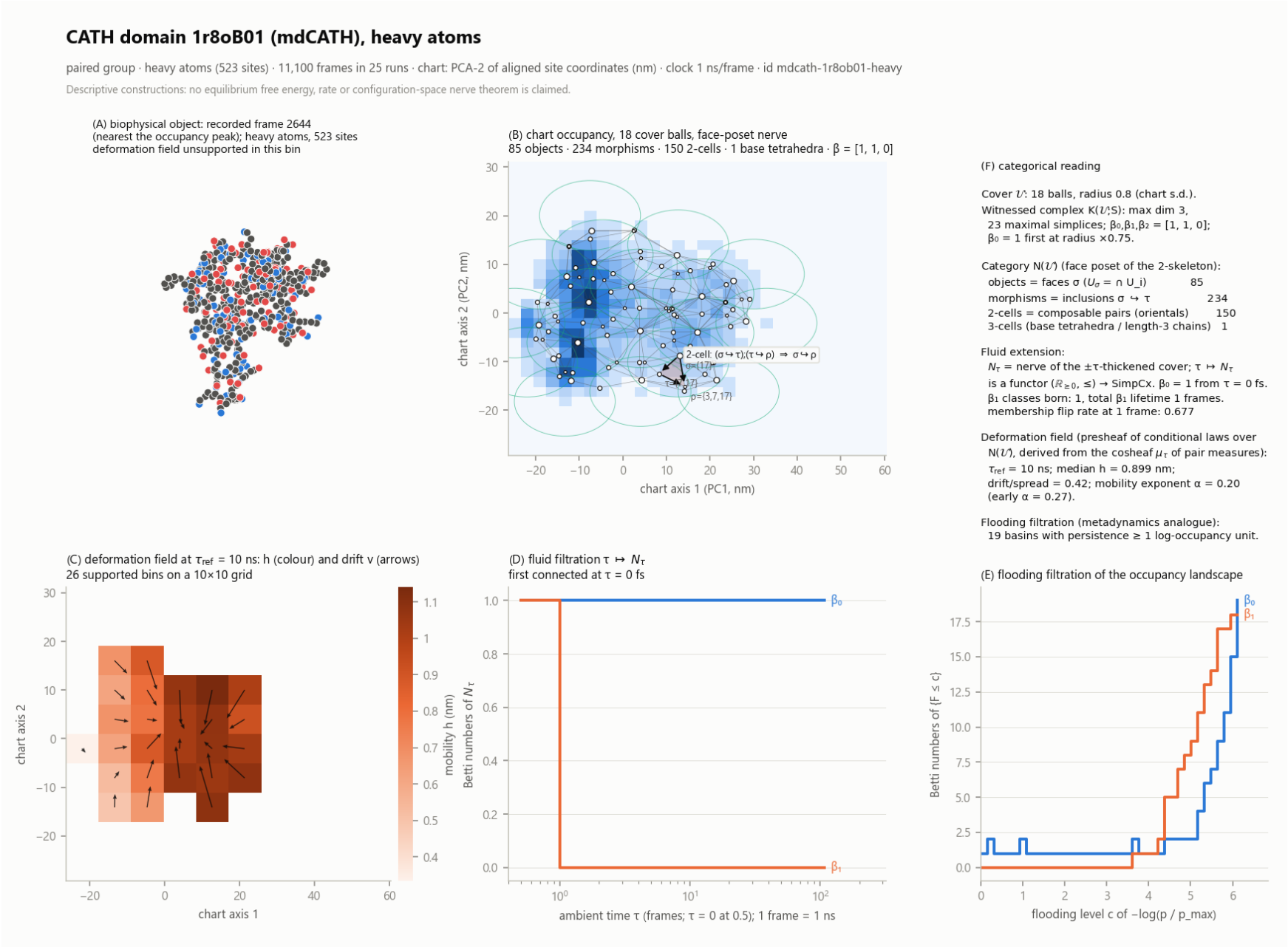

Figure 91: CATH domain 1r8oB01 (mdCATH), heavy atoms (`mdcath-1r8ob01-heavy`), paired with `mdcath-1r8ob01-ca`.

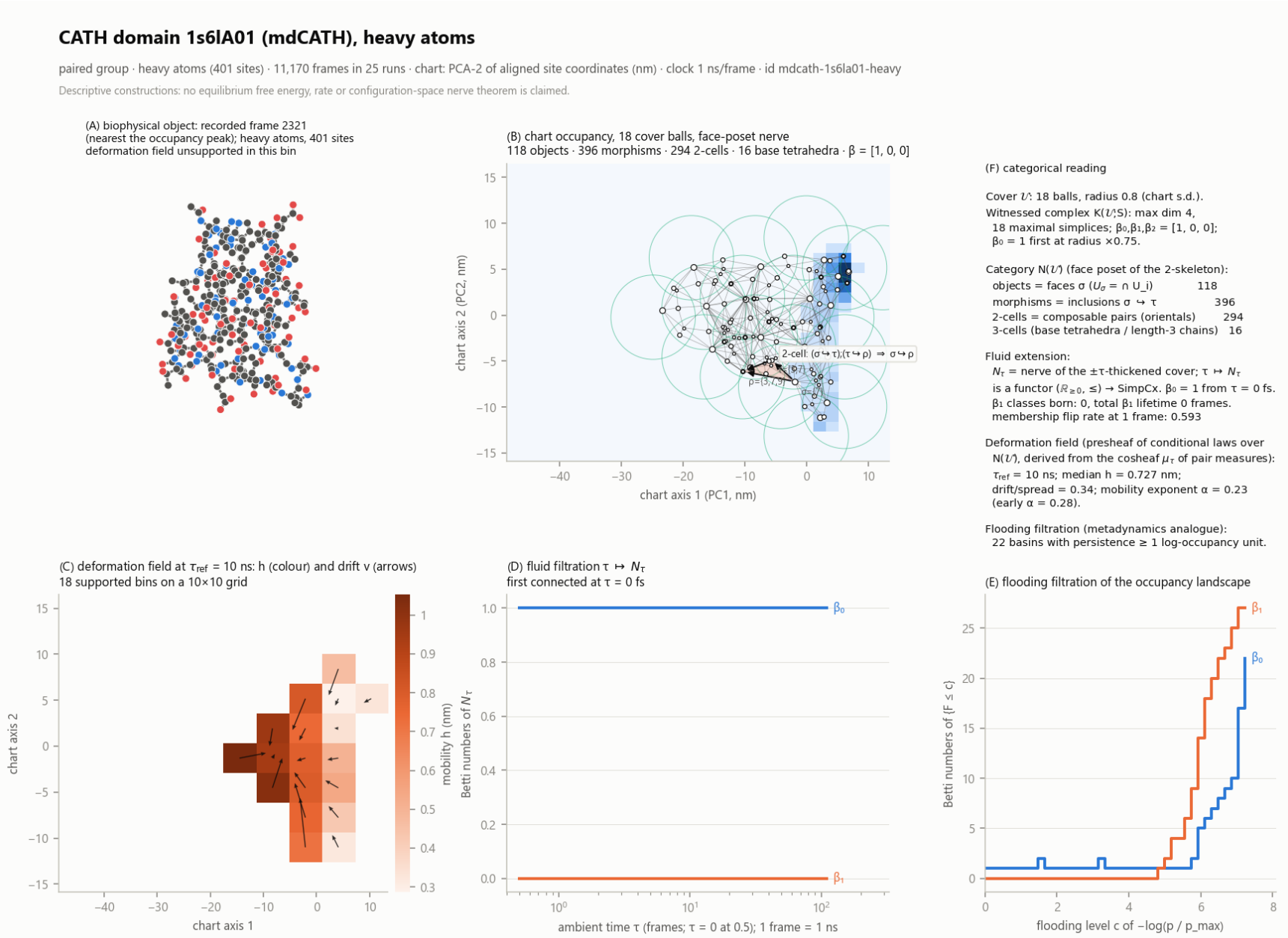

Figure 92: CATH domain 1s6lA01 (mdCATH), heavy atoms (`mdcath-1s6la01-heavy`), paired with `mdcath-1s6la01-ca`.

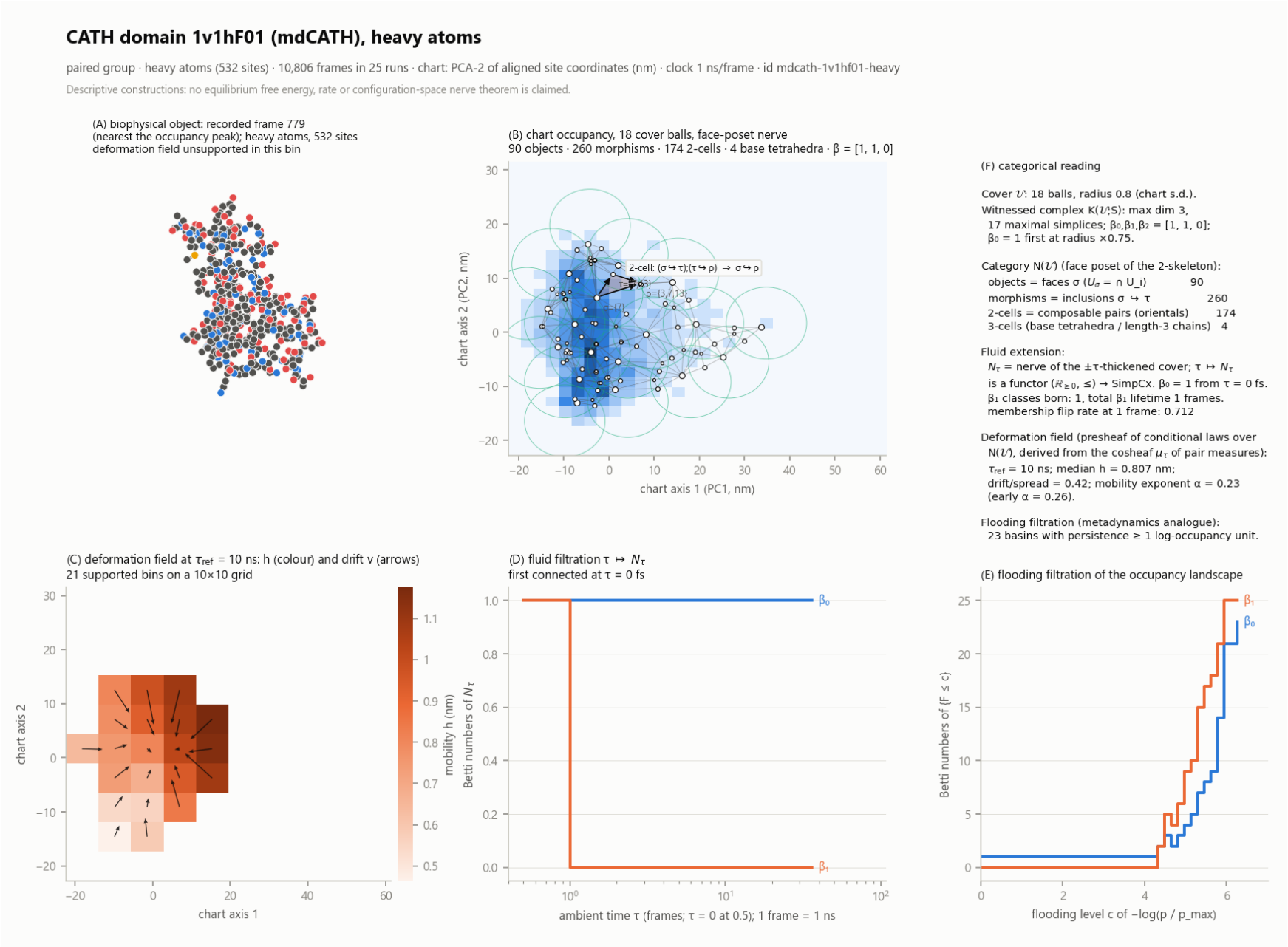

Figure 93: CATH domain 1v1hF01 (mdCATH), heavy atoms (`mdcath-1v1hf01-heavy`), paired with `mdcath-1v1hf01-ca`.

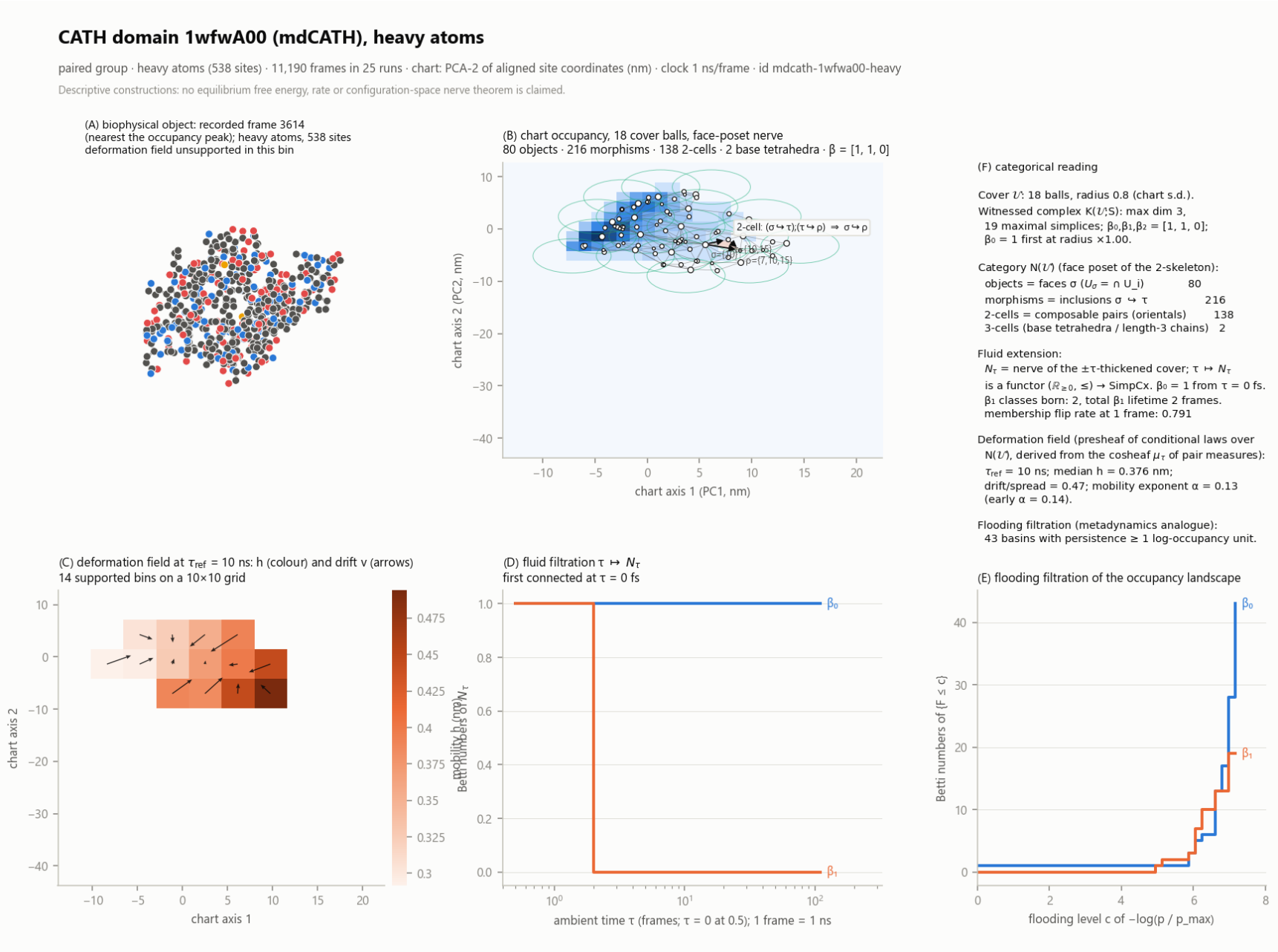

Figure 94: CATH domain 1wfwA00 (mdCATH), heavy atoms (`mdcath-1wfwa00-heavy`), paired with `mdcath-1wfwa00-ca`.

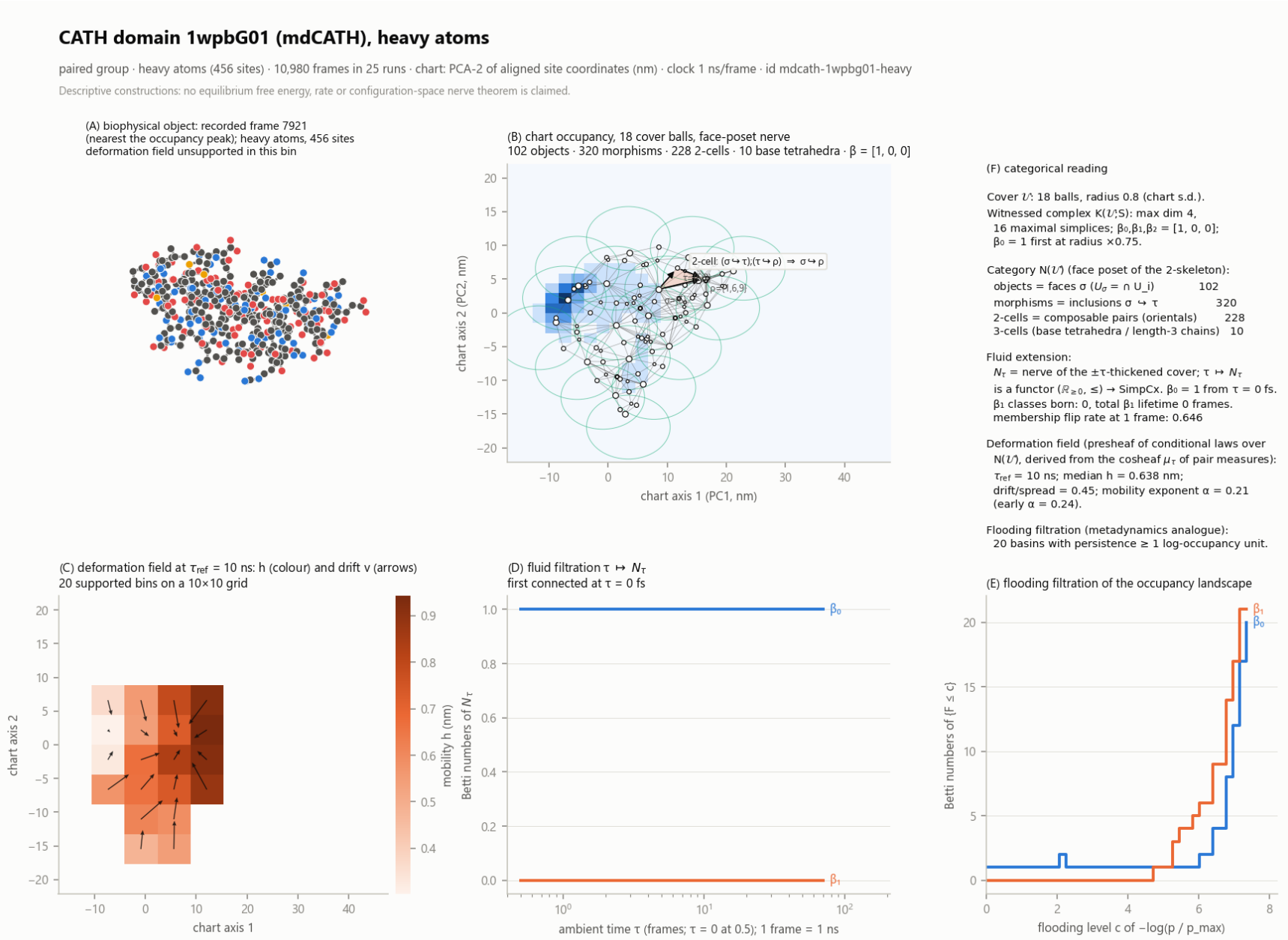

Figure 95: CATH domain 1wpbG01 (mdCATH), heavy atoms (`mdcath-1wpbg01-heavy`), paired with `mdcath-1wpbg01-ca`.

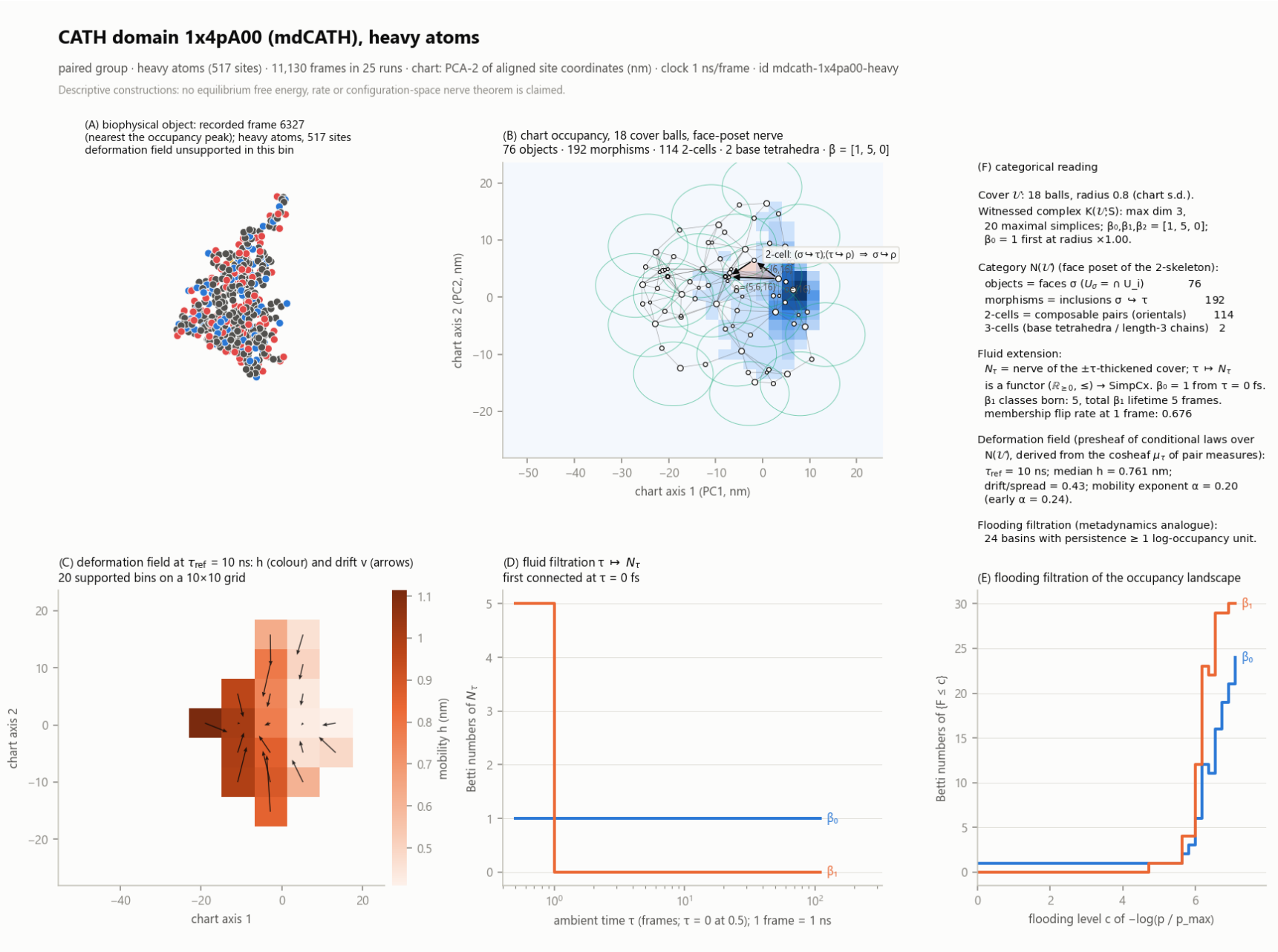

Figure 96: CATH domain 1x4pA00 (mdCATH), heavy atoms (`mdcath-1x4pa00-heavy`), paired with `mdcath-1x4pa00-ca`.

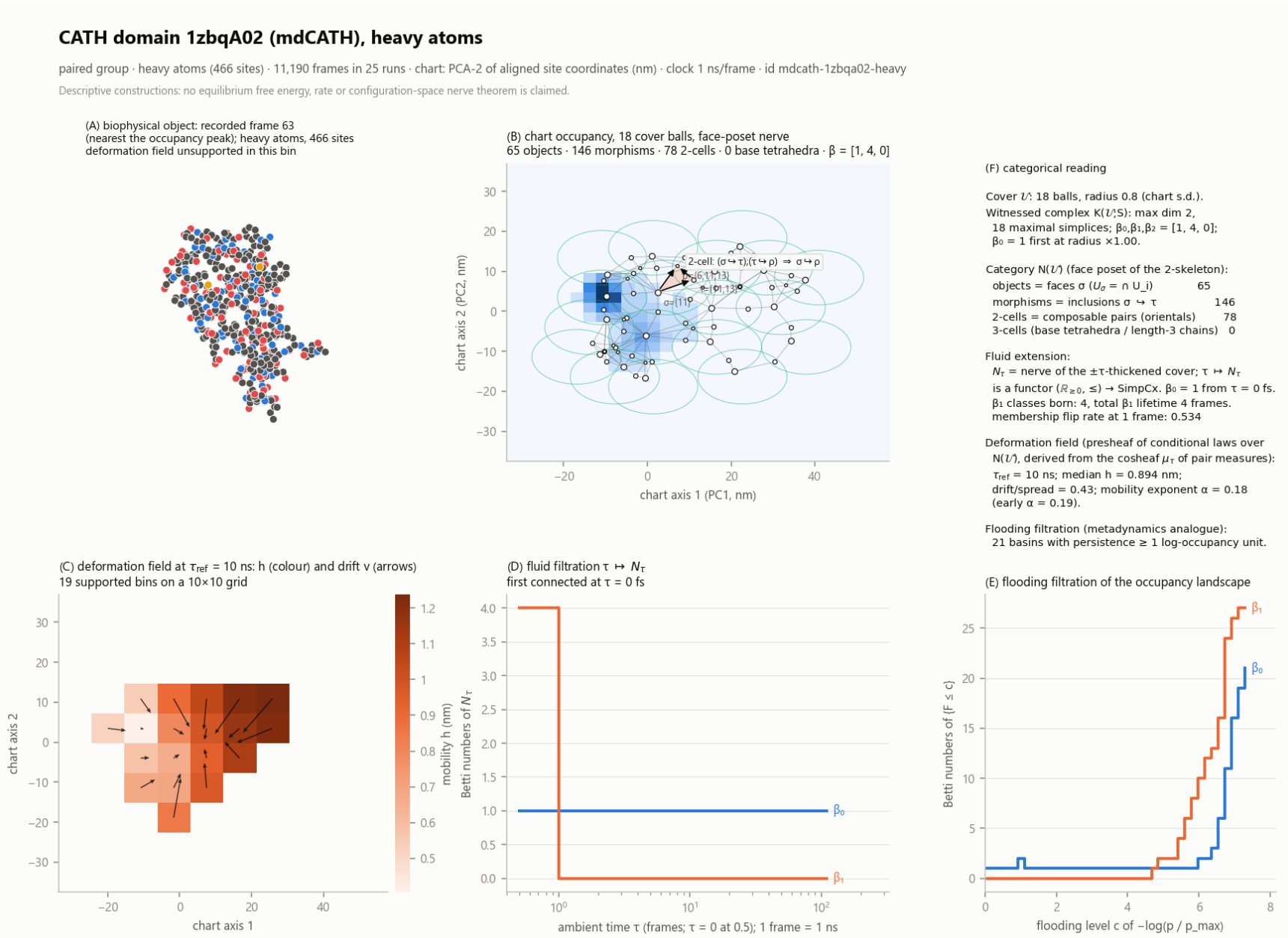

Figure 97: CATH domain 1zbqA02 (mdCATH), heavy atoms (`mdcath-1zbqa02-heavy`), paired with `mdcath-1zbqa02-ca`.

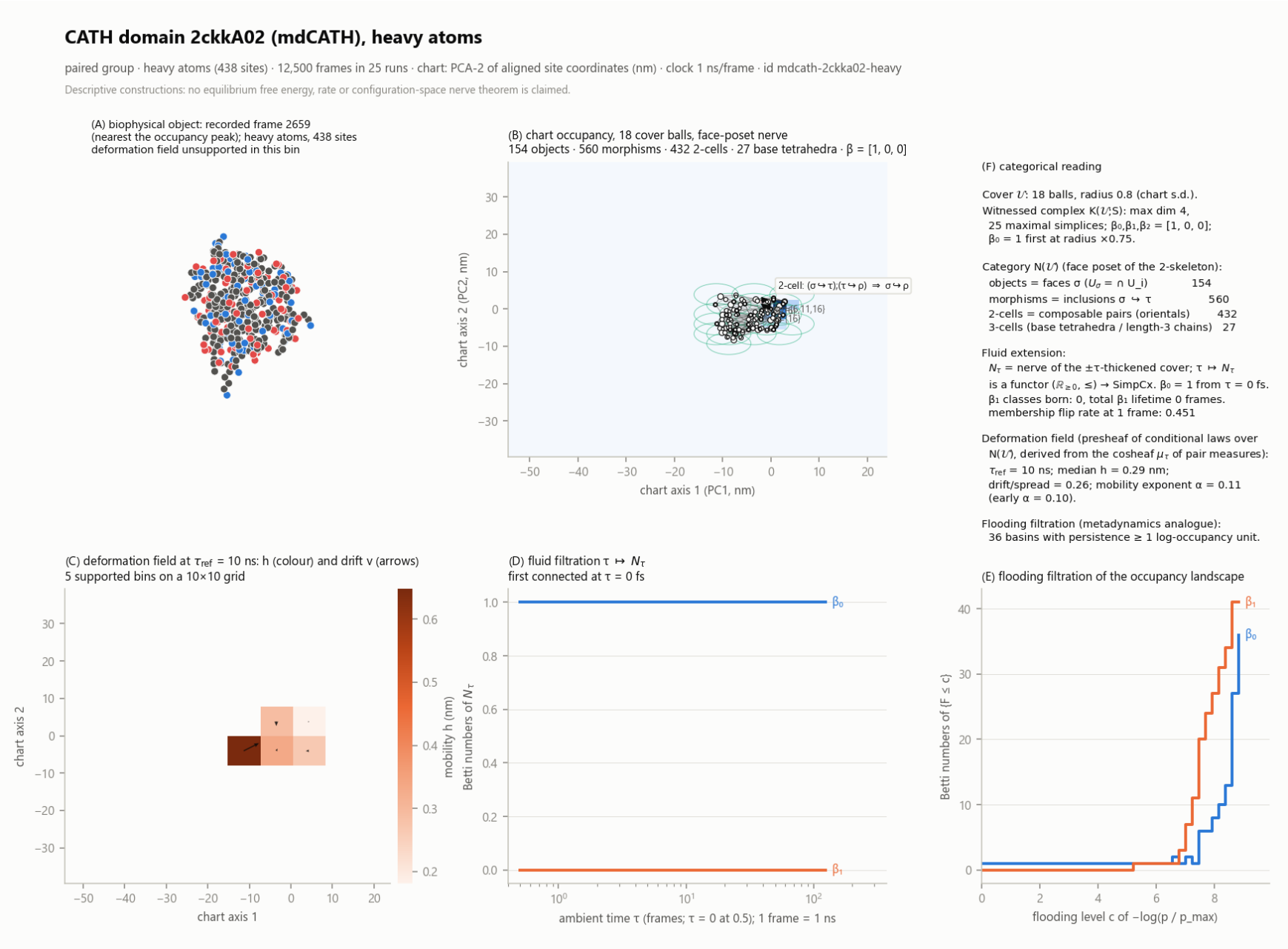

Figure 98: CATH domain 2ckkA02 (mdCATH), heavy atoms (`mdcath-2ckka02-heavy`), paired with `mdcath-2ckka02-ca`.

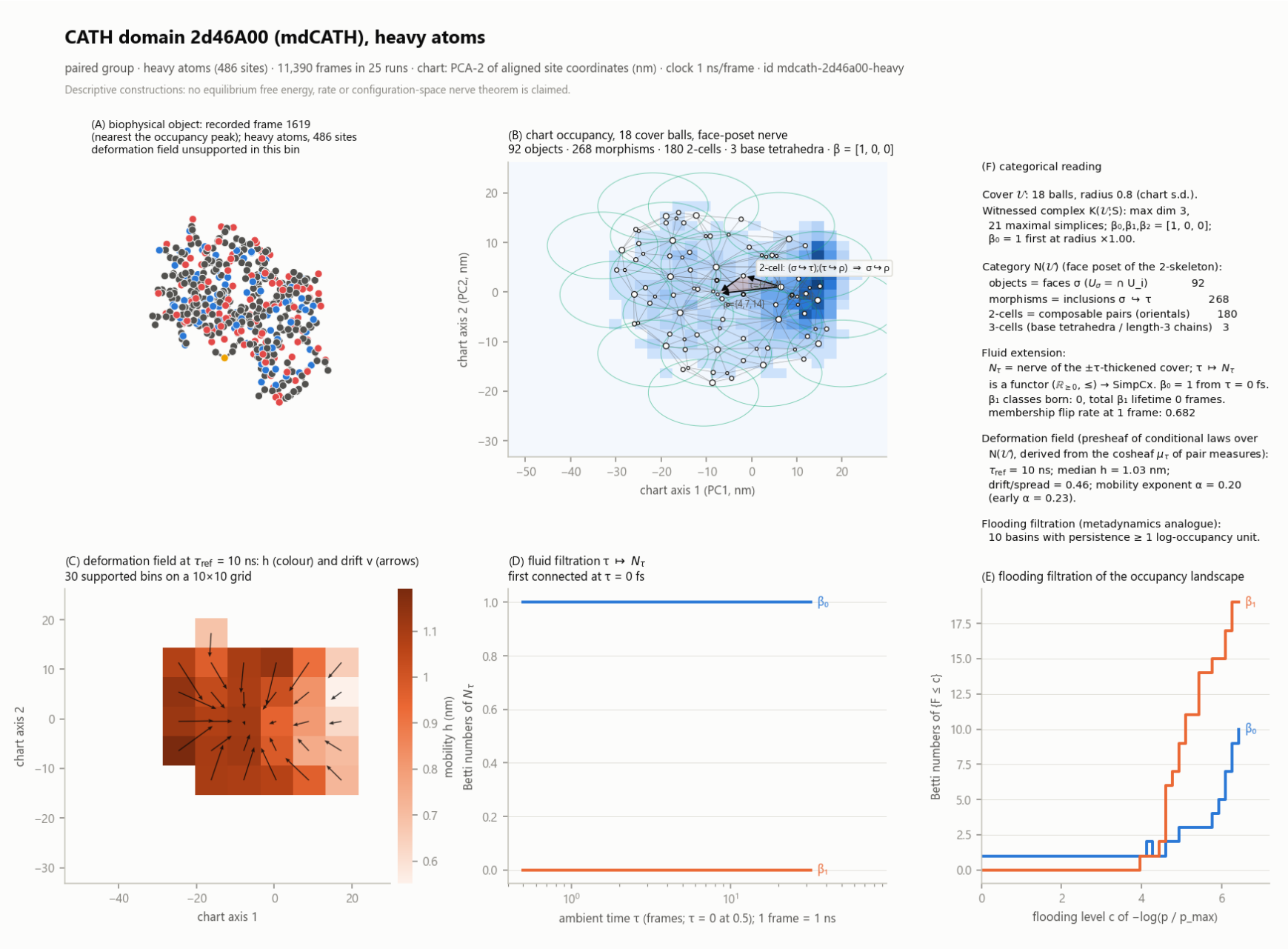

Figure 99: CATH domain 2d46A00 (mdCATH), heavy atoms (`mdcath-2d46a00-heavy`), paired with `mdcath-2d46a00-ca`.

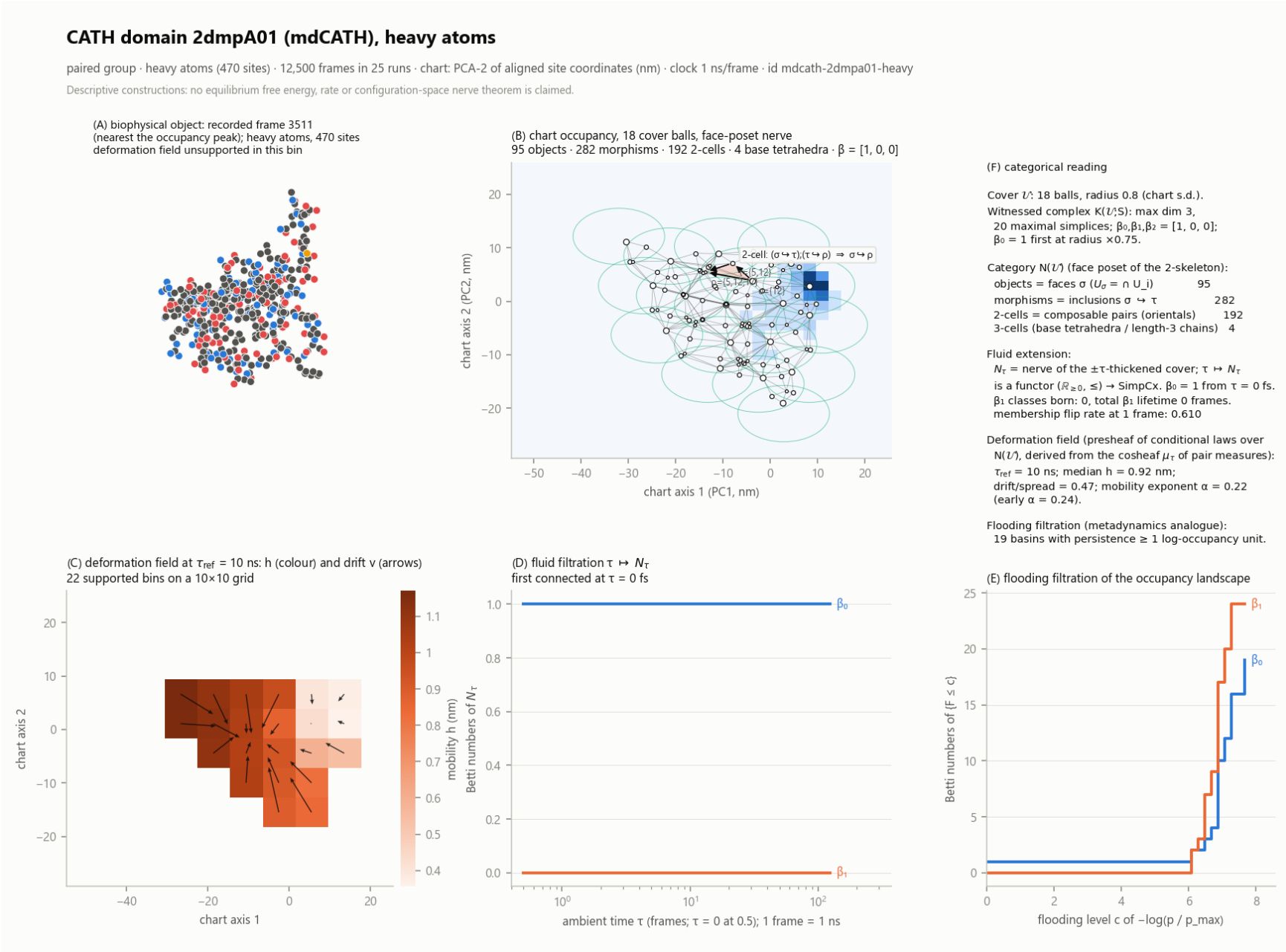

Figure 100: CATH domain 2dmpA01 (mdCATH), heavy atoms (`mdcath-2dmpa01-heavy`), paired with `mdcath-2dmpa01-ca`.

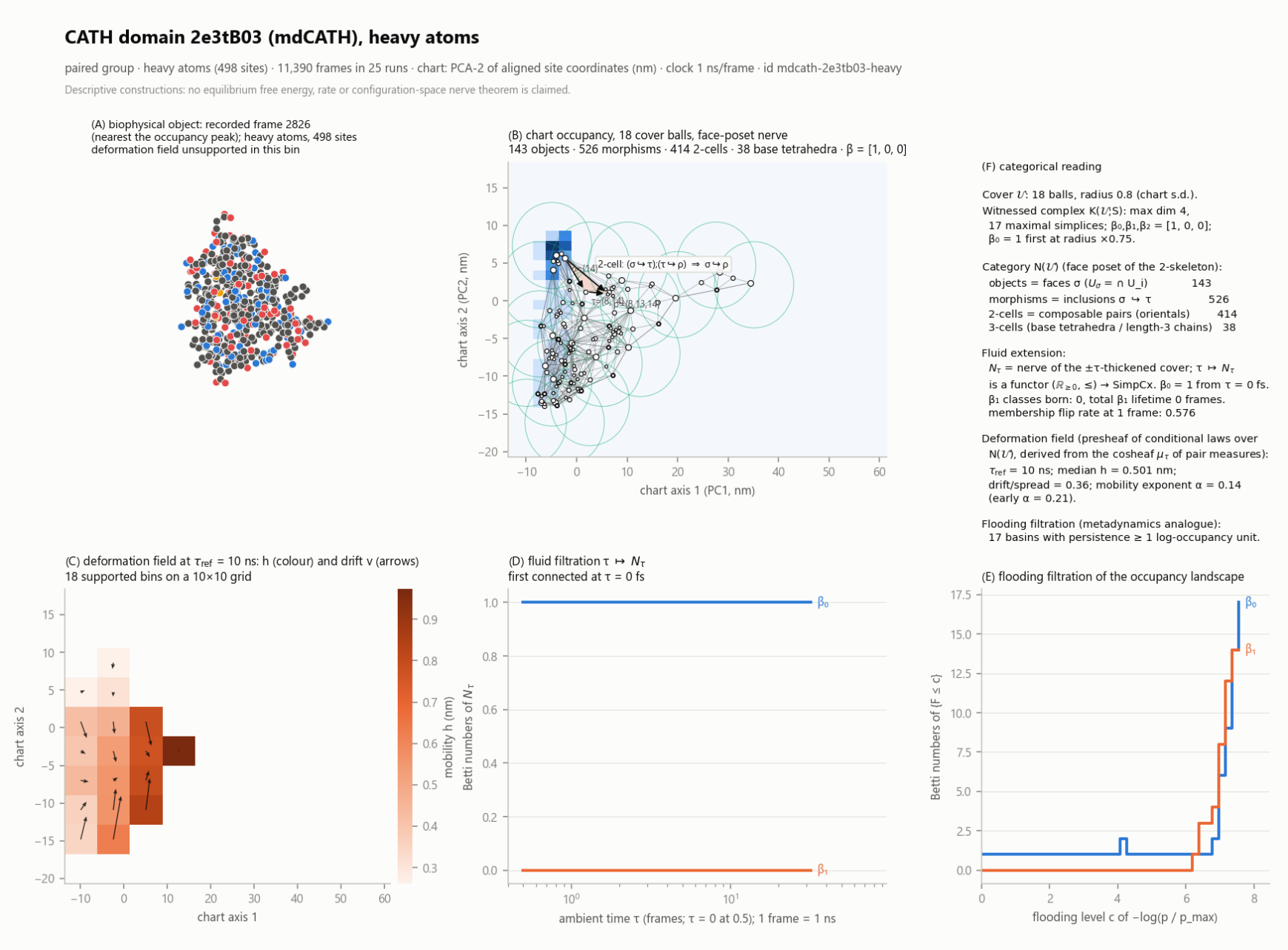

Figure 101: CATH domain 2e3tB03 (mdCATH), heavy atoms (`mdcath-2e3tb03-heavy`), paired with `mdcath-2e3tb03-ca`.

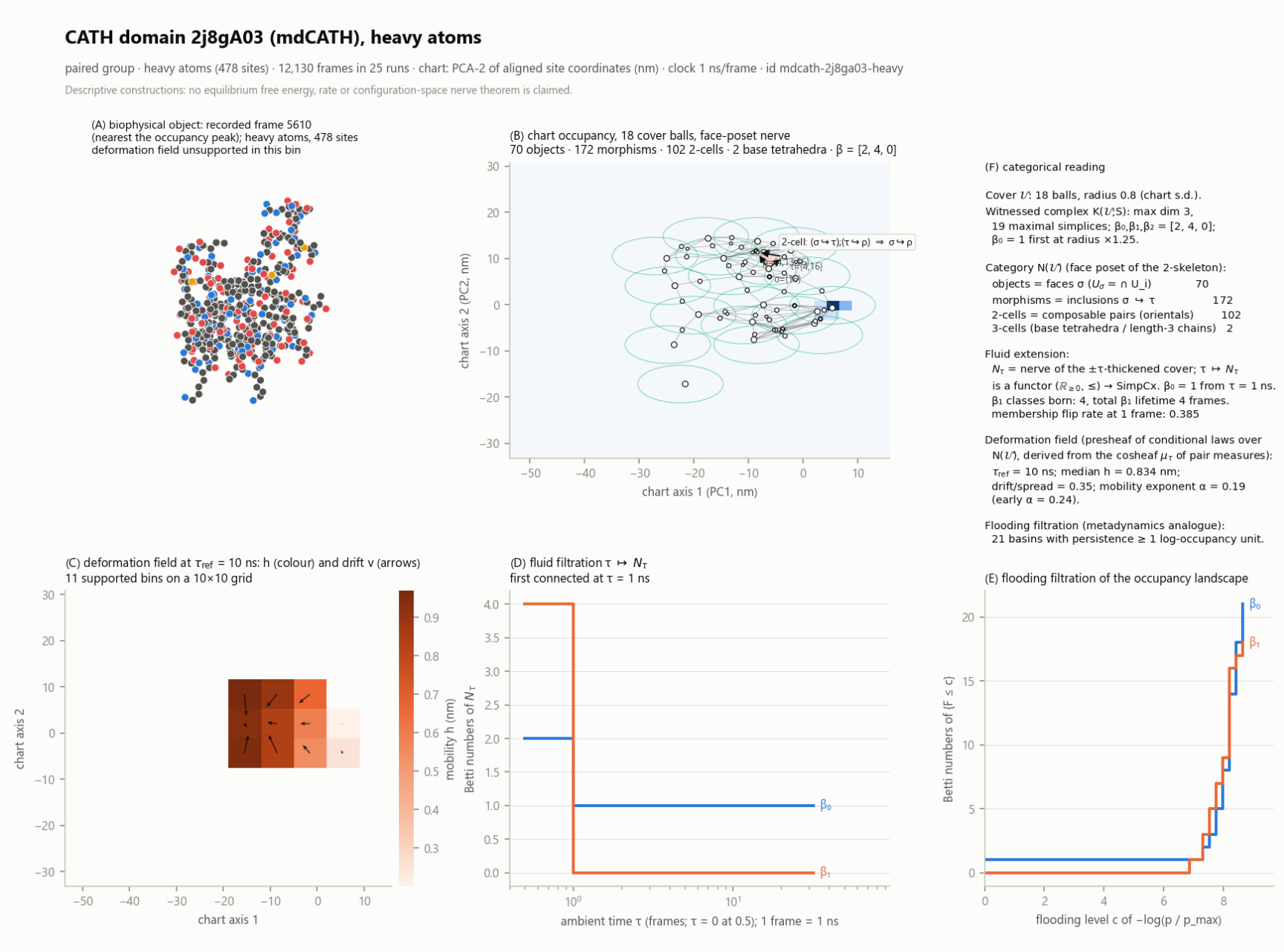

Figure 102: CATH domain 2j8gA03 (mdCATH), heavy atoms (`mdcath-2j8ga03-heavy`), paired with `mdcath-2j8ga03-ca`.

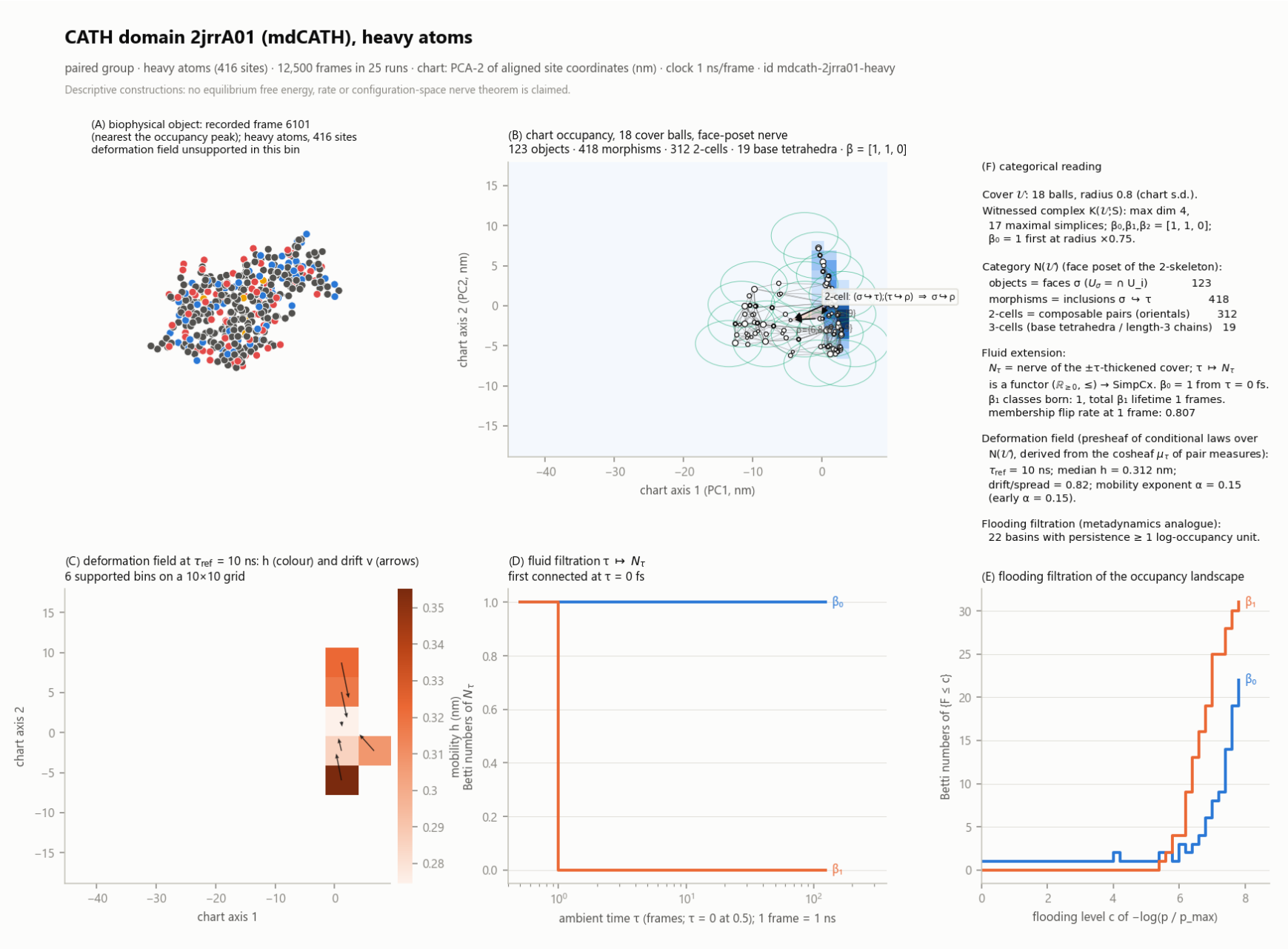

Figure 103: CATH domain 2jrrA01 (mdCATH), heavy atoms (`mdcath-2jrra01-heavy`), paired with `mdcath-2jrra01-ca`.

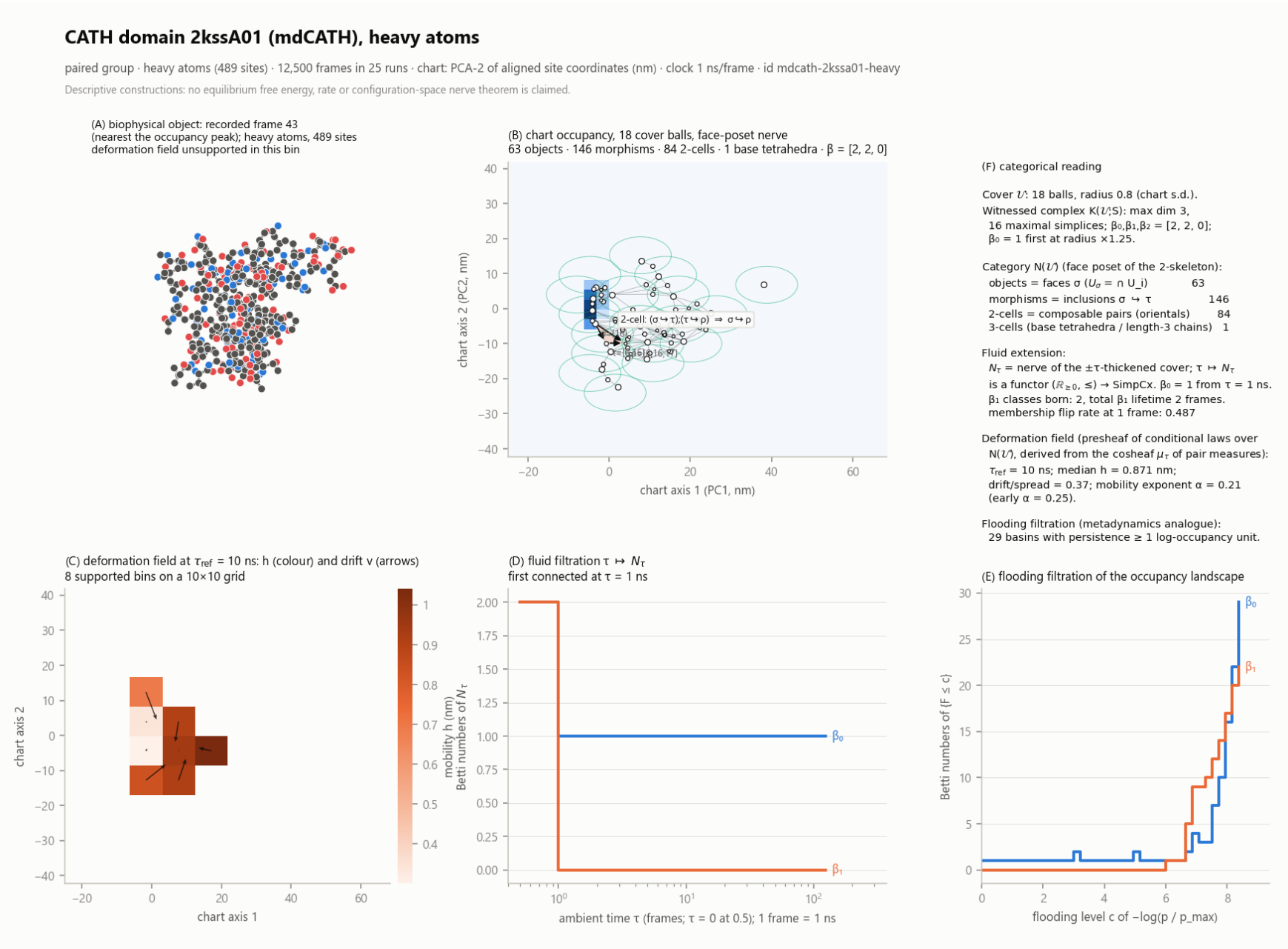

Figure 104: CATH domain 2kssA01 (mdCATH), heavy atoms (`mdcath-2kssa01-heavy`), paired with `mdcath-2kssa01-ca`.

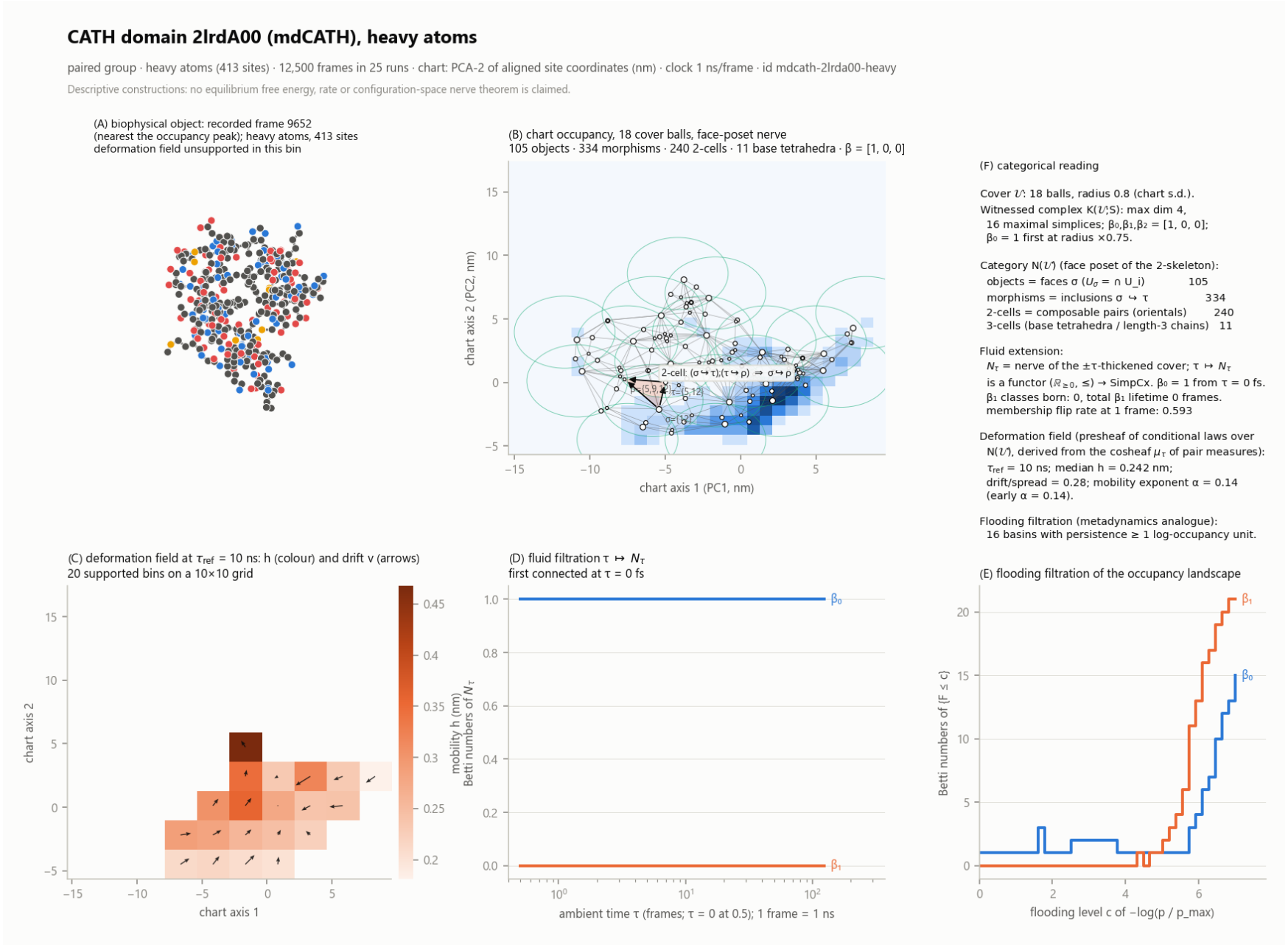

Figure 105: CATH domain 2lrdA00 (mdCATH), heavy atoms (`mdcath-2lrda00-heavy`), paired with `mdcath-2lrda00-ca`.

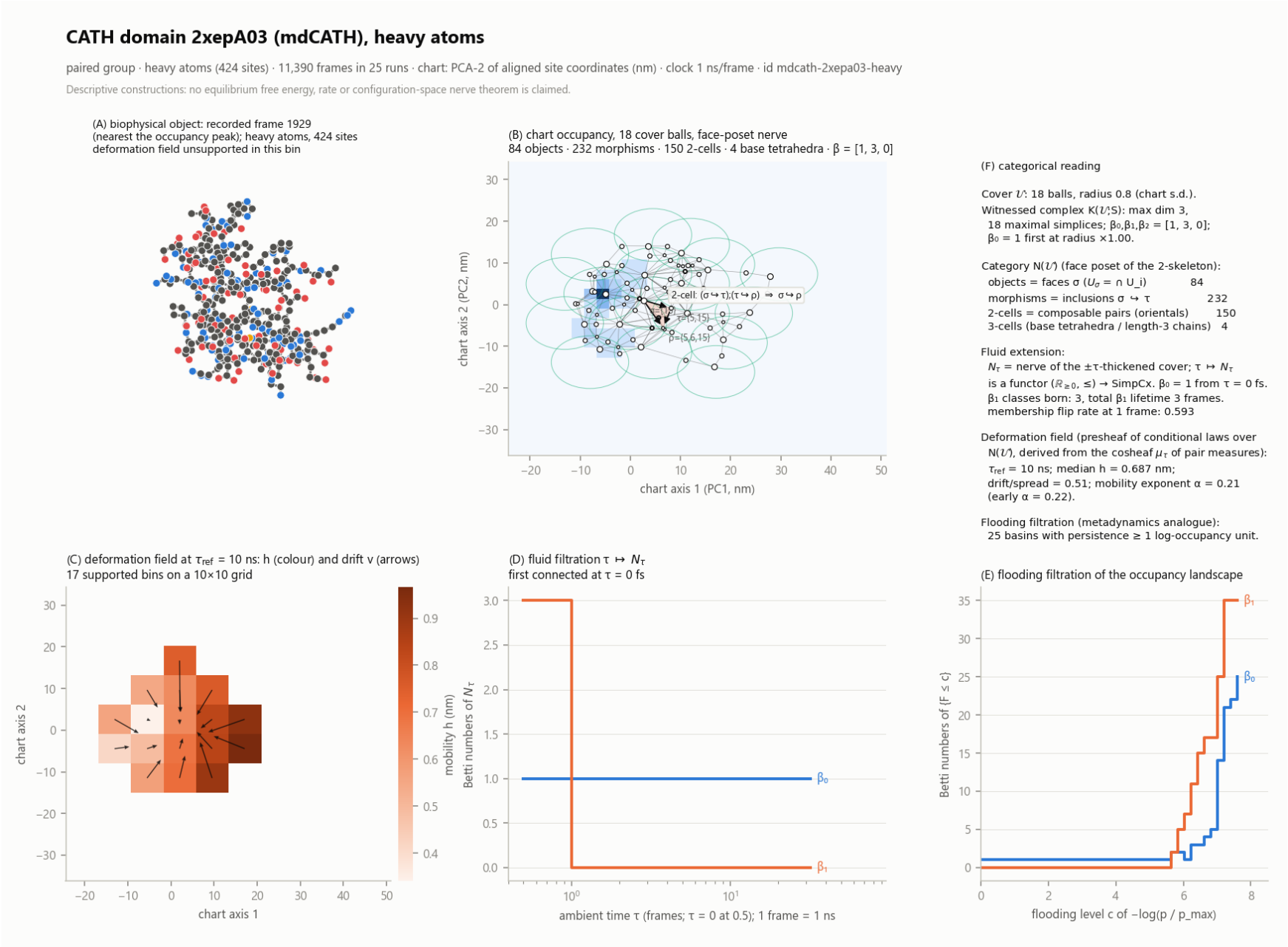

Figure 106: CATH domain 2xepA03 (mdCATH), heavy atoms (`mdcath-2xepa03-heavy`), paired with `mdcath-2xepa03-ca`.

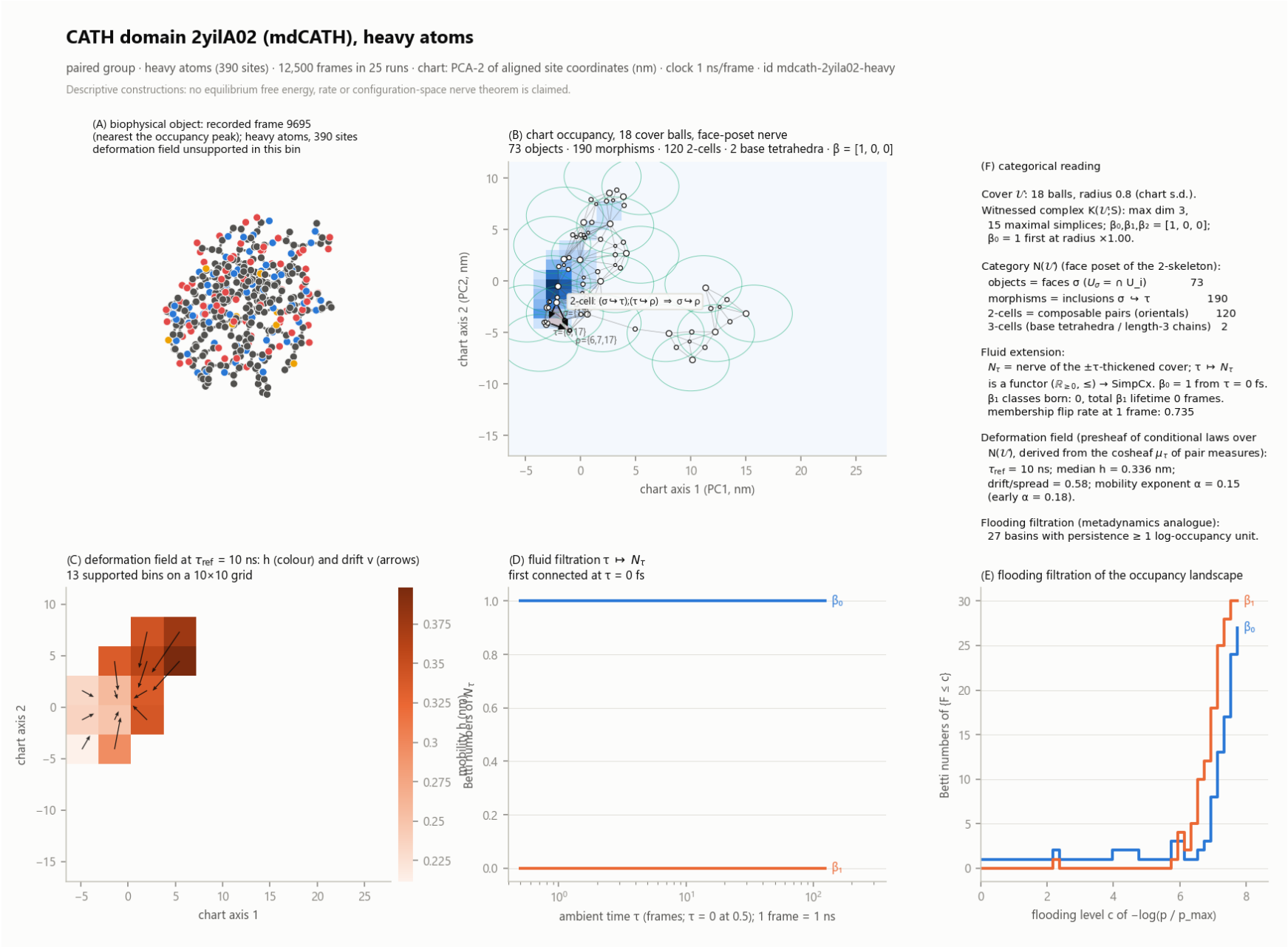

Figure 107: CATH domain 2yilA02 (mdCATH), heavy atoms (`mdcath-2yila02-heavy`), paired with `mdcath-2yila02-ca`.

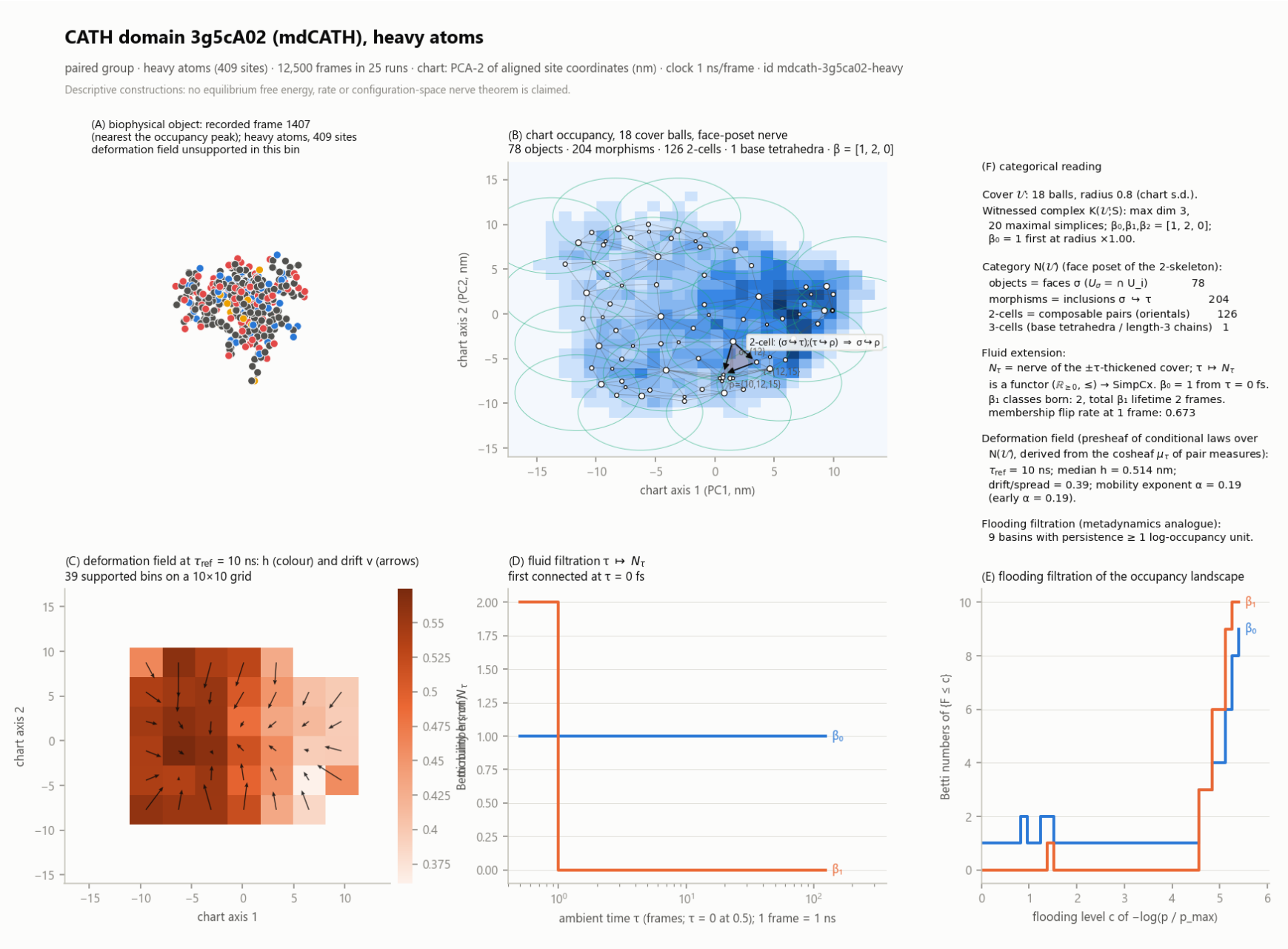

Figure 108: CATH domain 3g5cA02 (mdCATH), heavy atoms (`mdcath-3g5ca02-heavy`), paired with `mdcath-3g5ca02-ca`.

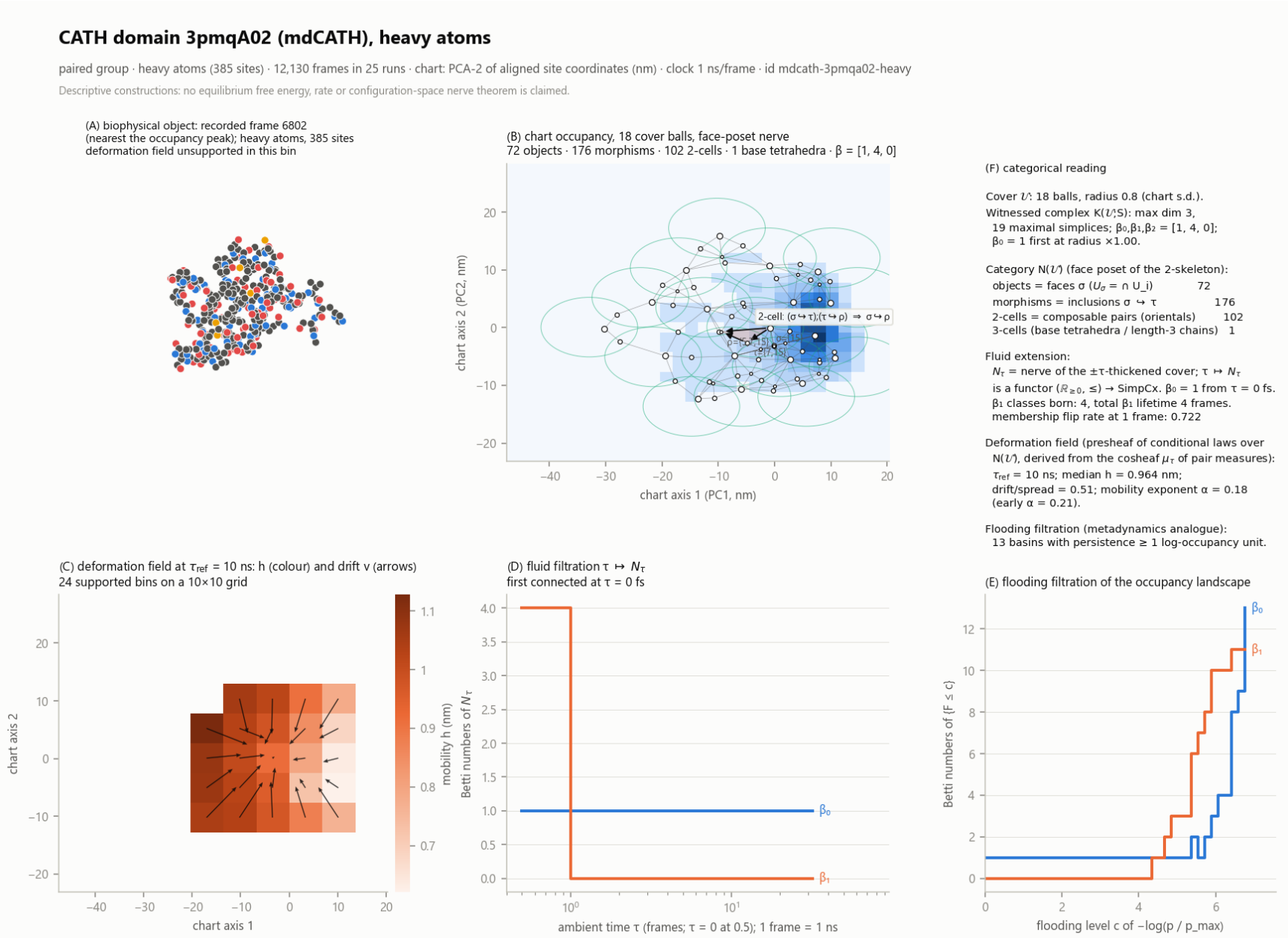

Figure 109: CATH domain 3pmqA02 (mdCATH), heavy atoms (`mdcath-3pmqa02-heavy`), paired with `mdcath-3pmqa02-ca`.

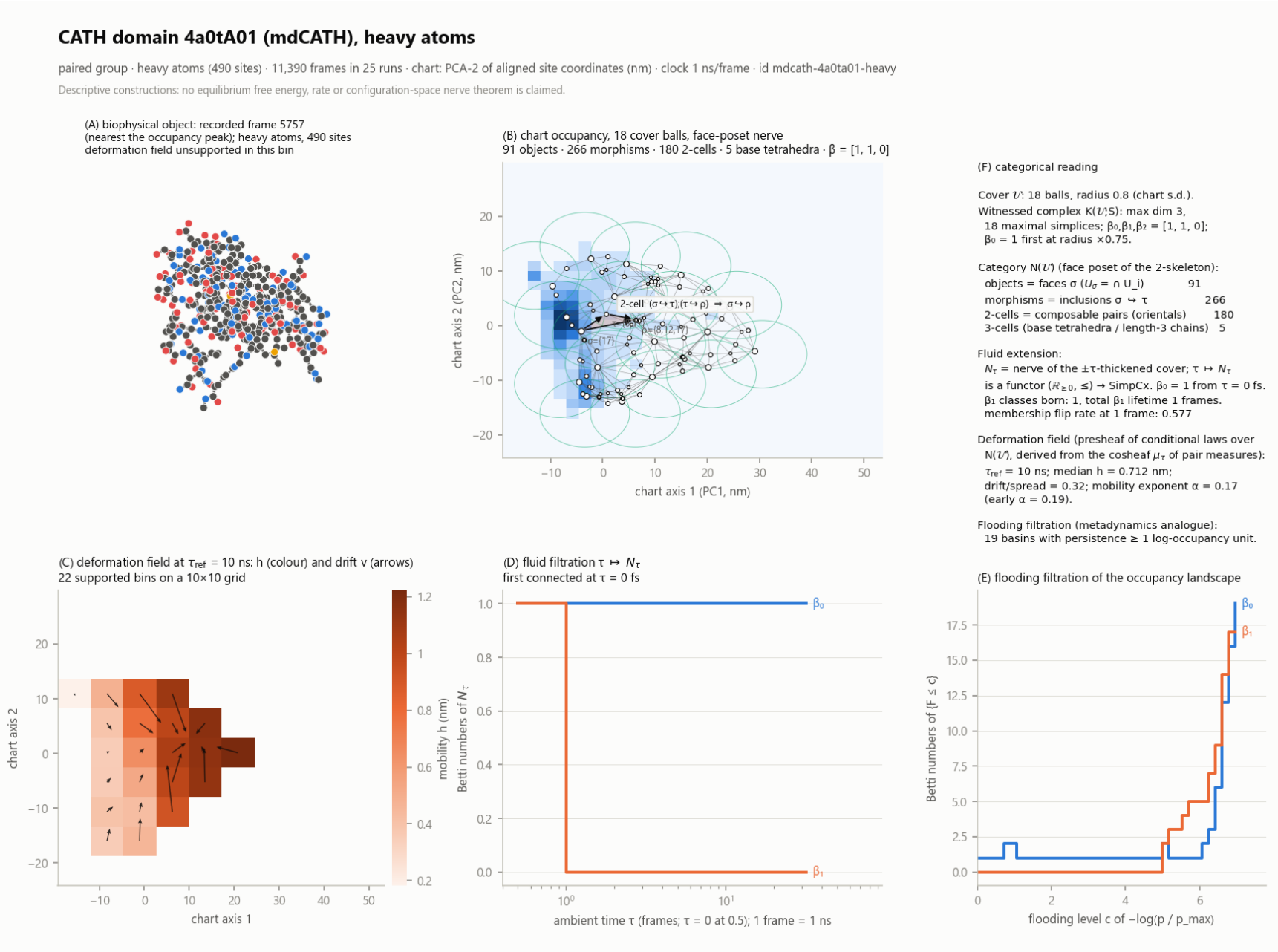

Figure 110: CATH domain 4a0tA01 (mdCATH), heavy atoms (`mdcath-4a0ta01-heavy`), paired with `mdcath-4a0ta01-ca`.

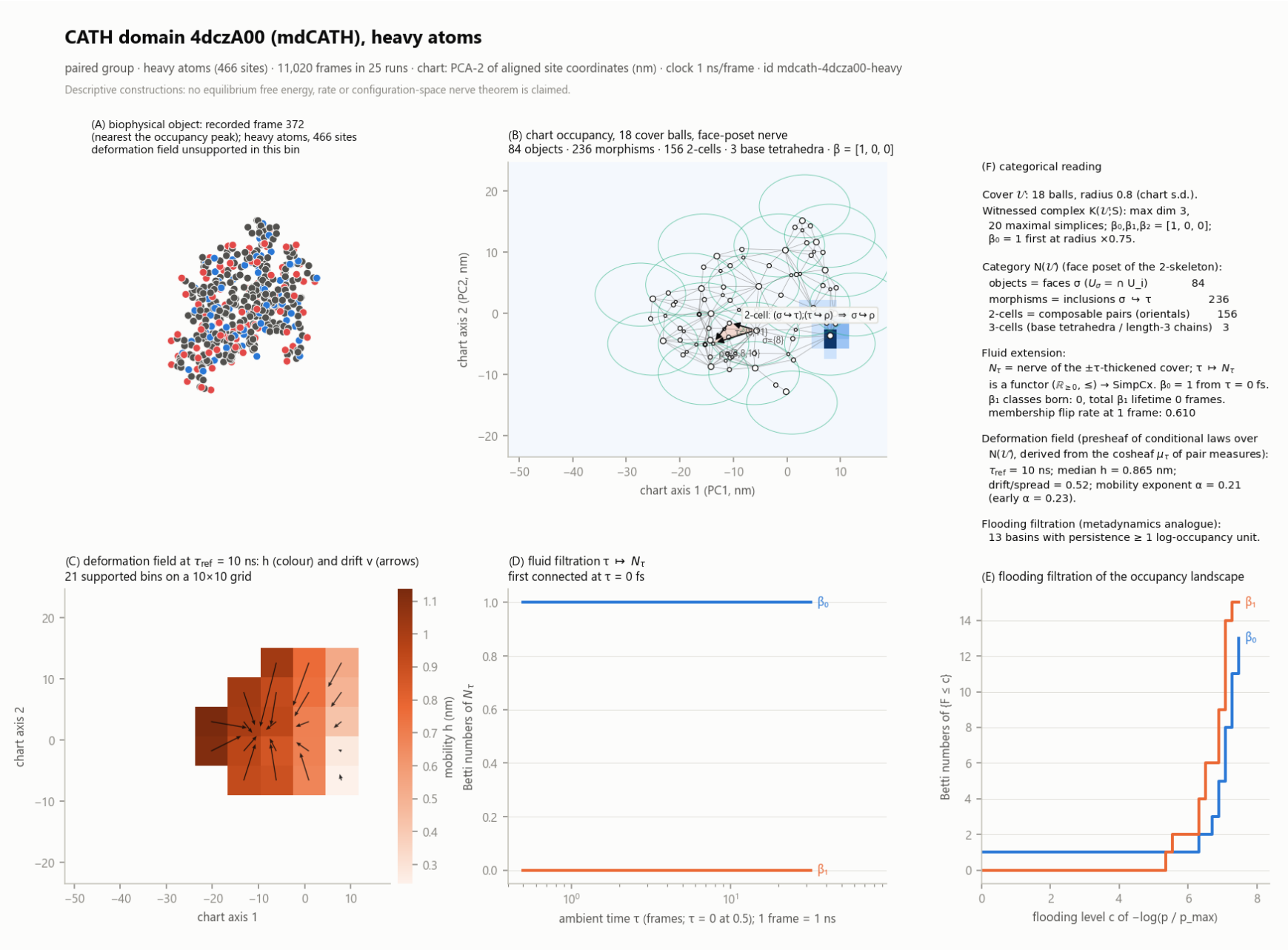

Figure 111: CATH domain 4dczA00 (mdCATH), heavy atoms (`mdcath-4dcza00-heavy`), paired with `mdcath-4dcza00-ca`.

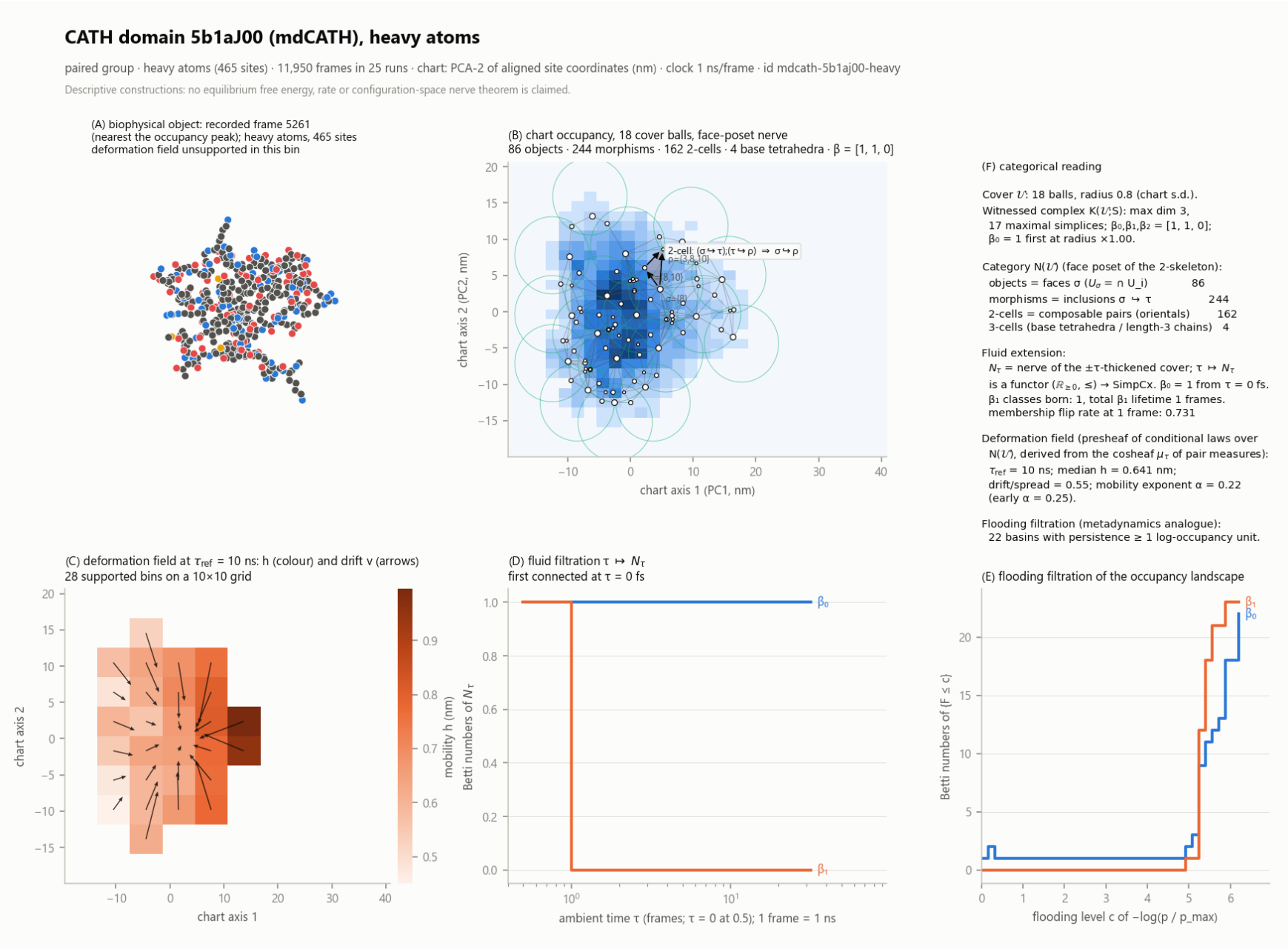

Figure 112: CATH domain 5b1aJ00 (mdCATH), heavy atoms (`mdcath-5b1aj00-heavy`), paired with `mdcath-5b1aj00-ca`.

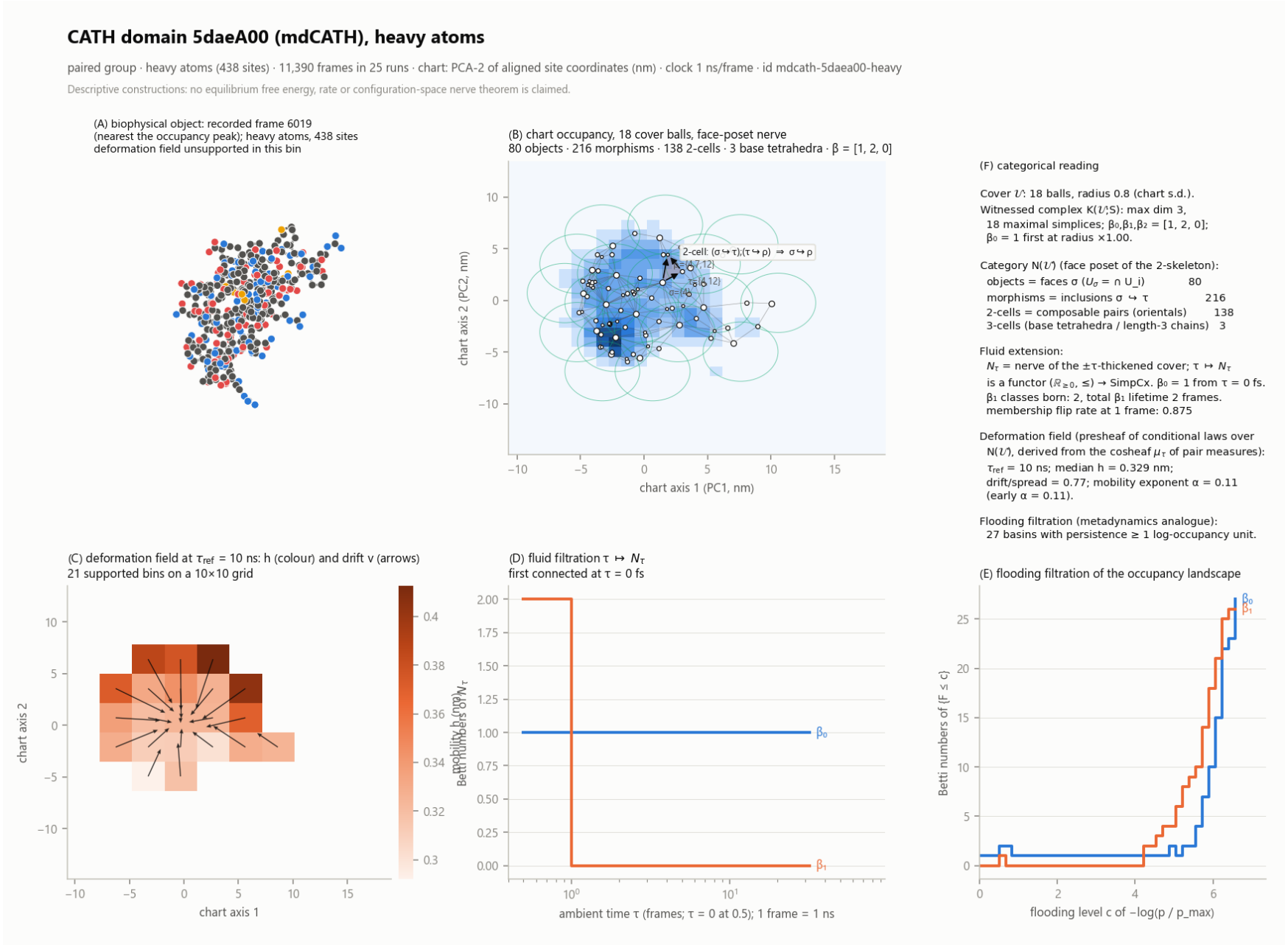

Figure 113: CATH domain 5daeA00 (mdCATH), heavy atoms (`mdcath-5daea00-heavy`), paired with `mdcath-5daea00-ca`.